The Dark Figure*

Estimating Unreported Antisemitic Hate Crime with Bayesian Record Linkage and Multiple-Systems Estimation

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Abstract

Public debate and policy on antisemitism rely on counts of incidents, but every count is a partial list. Police statistics capture only incidents that reach the police, are classified as bias-motivated and are submitted; non-governmental audits use different definitions and intake channels. This methods paper develops a framework for estimating the *dark figure* of antisemitic hate crime, the incidents that appear on no list, by combining record linkage with multiple-systems estimation (MSE). List-based MSE estimates N_V, the incidents reaching some recording channel; the remainder needs a victim survey. We retain the untestable highest-order interaction ω in the classical closed-form K-list estimator as an explicit sensitivity parameter, with sharp two-sided bounds. With three or more lists, neither positive dependence nor heterogeneity fixes the sign of the bias. We model referral between lists and show how provenance data restore identification. Under a stylised linkage-error model, linkage error acts as a linear operator on capture-history tables; we derive its first-order bias for K lists and the correction implied under that model. In a simulation study on synthetic data with known ground truth, records carry realistic attribute noise and are linked by a Fellegi–Sunter model. The resulting errors are dominated by swaps, links to the wrong partner, which the stylised model omits, and the correction then over-corrects. We therefore propagate linkage uncertainty by mixing over linkage draws and use the correction only as a diagnostic; in the simulation, mixing over approximate draws from a plug-in linkage posterior did not remove the linkage bias either. We also specify, but do not evaluate, a Bayesian latent-class and spatiotemporal small-area model. A multiplicative reporting funnel with a Shapley decomposition locates where incidents are lost between the offence and federal statistics. A pre-registered analysis plan covers harmonisation, privacy-preserving linkage and an ethics protocol. We report no empirical estimates from real data.

Keywords multiple-systems estimation; capture–recapture; record linkage; hate crime; antisemitism; sensitivity analysis; Bayesian nonparametrics.

1 Introduction

Public debate, security funding and policy on antisemitism are argued in counts. In the FBI's hate-crime statistics for 2025, law enforcement agencies reported 1,528 single-bias incidents motivated by anti-Jewish bias (Anti-Defamation League, 2026b; Federal Bureau of Investigation, 2026). For the same year, the

*Status. Methods paper; simulation study on synthetic data; no empirical estimates. Every number in Section 8 is computed from synthetic data with known ground truth; the real-data analysis of Section 9 is pre-registered and has not been carried out.

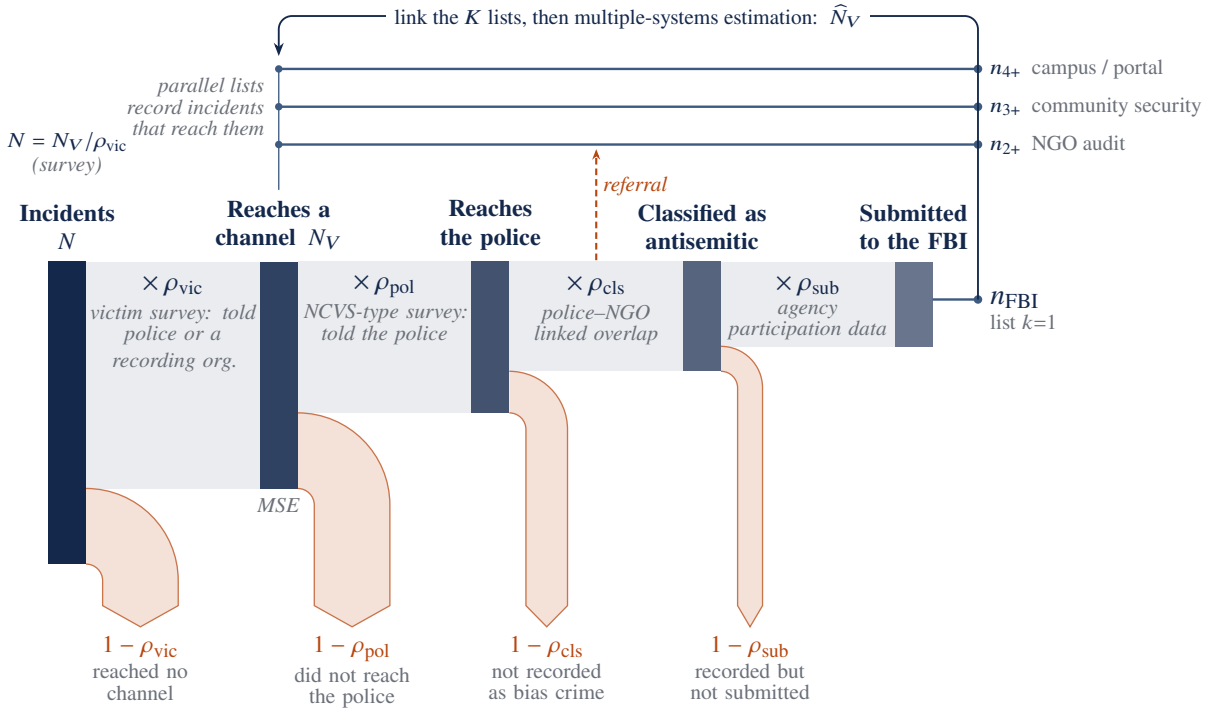


Figure 1. The reporting funnel. Each in-scope antisemitic incident passes four filters before it appears in the FBI's hate-crime count: it reaches a recording channel, by a report or by the channel's own detection (ρ_{vic}); it reaches the police (ρ_{pol}); the police record an antisemitic bias motivation (ρ_{cls}); and the agency submits the record to the FBI (ρ_{sub}), so that $\mathbb{E}[n_{FBI} \mid U_d] = \mathbb{E}[N_V \mid U_d]\,\rho_{pol}\rho_{cls}\rho_{sub}$ (Proposition 7.2). Orange streams are incidents lost from the police record at each stage; grey italics name the auxiliary data that inform each factor (Table 4). The other lists record incidents that reach them through channels other than the police, except through *referral* (dashed; Proposition 4.10). Linking all K lists, the police list included, and applying multiple-systems estimation estimates N_V. Incidents that reach no channel are on no list; a victim survey that supplies ρ_{vic} is needed to recover them. Widths are schematic, not estimates.

Anti-Defamation League's annual Audit tabulated 6,274 antisemitic incidents of harassment, vandalism and assault, criminal and non-criminal alike, after a record 9,354 in 2024 (Anti-Defamation League, 2025, 2026a). Figures like these are routinely set side by side, but they are not rival estimates of one quantity. Each is a *list*: an enumeration of the incidents that happened to pass through a particular set of filters. The number that matters, the total number of in-scope incidents including those that appear on no list, is never observed. Criminologists call it the *dark figure* (Pezzella et al., 2019). This paper is about estimating it.

How incidents go missing. The police count is the end of a chain of conditional events, sketched in Figure 1. The incident must first reach a recording channel, by a report or by the channel's own detection, and then the police. National Crime Victimization Survey data indicate that U.S. residents experienced an average of roughly 250,000 hate-crime victimizations a year over 2004–2015, the majority of which were not reported to the police (Masucci and Langton, 2017). If the incident reaches the police, an officer must recognise and record an antisemitic bias motivation, a judgement whose consistency across agencies has long been questioned (McDevitt et al., 2003). Finally, the agency must submit the record to the FBI. Coverage at this last stage is not constant: in the 2021 transition to incident-based reporting, several of the largest agencies were missing from the initial federal release (Section 2.1). Even in an ordinary year the composition of the reporting panel can move the headline. From 2022 to 2023 the published national

hate-crime total rose from 11,634 to 11,862 incidents (U.S. Department of Justice, Community Relations Service, 2024), while on a trend dataset restricted to agencies with comparable reporting in both years incidents fell by 0.6% (Federal Bureau of Investigation, 2024).

Non-governmental audits, community security organisations and campus or self-report portals see different slices, through their own intake channels and definitions (Section 2). No list is complete, each is selected by a different process, and the lists are not independent.

Why multiple-systems estimation. When a closed population is partially enumerated by several overlapping lists, the pattern of overlaps carries information about the units that every list missed. This is the logic of capture–recapture, from the two-sample estimator of Petersen (1896) and Lincoln (1930) through the multiple-recapture census of Darroch (1958) and the log-linear models of Fienberg (1972). Under the name *multiple-systems estimation* (MSE) it has become a standard tool wherever a hidden population leaves traces in several administrative or documentary sources (Bird and King, 2018): killings and disappearances in armed conflict (Ball and Price, 2018; Lum et al., 2013; Sadinle, 2018), victims of modern slavery (Silverman, 2020) and disease surveillance (Chao et al., 2001; International Working Group for Disease Monitoring and Forecasting, 1995). Antisemitic incidents fit this template: several organisations separately record overlapping subsets of the same events, and the dark figure is the unobserved cell of their cross-classification. One limit is structural. A list can record only incidents that reach it, so list-based MSE estimates N_V, the number of incidents that reach some recording channel (Section 7). The remainder needs a victim survey.

Why antisemitic incidents are hard. Five features of this domain stress the classical assumptions.

(i) *Referral dependence.* Lists feed one another. An audit that ingests law-enforcement reports is more likely to contain an incident because the police recorded it. With two lists this biases the Lincoln–Petersen estimator downward (Proposition 4.10); with more lists its effect depends on how the referral channel sits among them.

(ii) *Definitional scope mismatch.* The FBI counts criminal offences with a recorded bias motivation. NGO audits also count non-criminal harassment, under criteria that change over time. Scope must be harmonised before lists are combined.

(iii) *Record-linkage error.* The lists share no identifier. Records must be matched on sparse, noisy and sensitive attributes (date, place, incident type), and false and missed matches distort the capture-history table from which the dark figure is estimated (Ding and Fienberg, 1994).

(iv) *Heterogeneous reportability.* An assault at a synagogue and a slur on a commuter train do not have the same chance of being recorded anywhere. With two lists such heterogeneity leads to underestimation (Chao, 1987); with three or more, even the sign of the bias depends on the shape of the visibility distribution (Proposition 4.7).

(v) *Coverage shifts.* When the set of reporting agencies changes, as in 2021, a change in list coverage can masquerade as a change in incidence.

What this paper does, and does not, do. This is a methods paper. We develop the statistical framework, prove its properties, evaluate the estimators in a simulation study on synthetic data with known ground truth, and pre-register an analysis plan for real sources. We report *no* empirical estimates of the number of antisemitic incidents in any jurisdiction or year: we have not linked any real incident data, and nothing here should be read as a finding about the level or trend of antisemitism. Figure 2 summarises the pipeline from raw records to the dark figure; notation is collected in Table 1.

Table 1. **Notation.** Symbols used throughout. Bold $\boldsymbol{h}$ denotes a capture history; $\mathbf{0}$ is the all-zero history.

Population and lists

d	domain (state $s \times$ year t)
$U_d,\ N$	in-scope antisemitic incidents that reach a recording channel; unknown size (Section 7 writes U_d^V, N_V, and uses U_d, N for all incidents)
$k \in [K]$	list; $K = 4$: police (FBI), NGO audit, community security org., campus/self-report portal
Y_{ik}	1 if incident i is on list k

Capture histories and counts

$\boldsymbol{h} \in \mathcal{H} = \{0,1\}^K$	capture history; $\lvert\boldsymbol{h}\rvert = \sum_k h_k$
$n_{\boldsymbol{h}}$	incidents with history $\boldsymbol{h}$
$n = \sum_{\boldsymbol{h}\neq\mathbf{0}} n_{\boldsymbol{h}}$	distinct incidents observed
$n_{\mathbf{0}} = N - n$	**the dark figure**
$n_{k+},\ f_j$	size of list k; incidents on exactly j lists
$D_k = N/n_{k+}$	dark-figure ratio relative to list k
$s_{\boldsymbol{h}} = (-1)^{\lvert\boldsymbol{h}\rvert+1}$	alternating sign of a history
$\mathcal{A}(\boldsymbol{x}) = \prod_{\boldsymbol{h}\neq\mathbf{0}} x_{\boldsymbol{h}}^{s_{\boldsymbol{h}}}$	alternating product

Reporting funnel

ρ_{vic}	reaches a recording channel, by a report or by the channel's own detection
ρ_{pol}	reaches the police | reaches a channel
ρ_{cls}	classified as antisemitic | reaches the police
ρ_{sub}	submitted to the FBI | classified
N_V	incidents that reach a channel; what list-based MSE estimates
n_{FBI}	FBI count; $\mathbb{E}\, n_{\text{FBI}} = \mathbb{E}[N_V]\, \rho_{\text{pol}}\rho_{\text{cls}}\rho_{\text{sub}}$

Capture models

$\mu_{\boldsymbol{h}} = \mathbb{E}[n_{\boldsymbol{h}}]$	expected cell count
$u_S,\ S \subseteq [K]$	log-linear terms: $\log \mu_{\boldsymbol{h}} = \sum_{S\subseteq \text{supp}(\boldsymbol{h})} u_S$
$\omega = u_{[K]}$	highest-order interaction; sensitivity parameter
$\bar{\omega} = (-1)^K \omega$	oriented interaction: $\mu_{\mathbf{0}} = e^{\bar{\omega}} \mathcal{A}(\boldsymbol{\mu}_+)$
p_{ik}	$\mathbb{P}(Y_{ik} = 1 \mid x_i)$, capture probability
$z_i,\ \pi_j,\ \lambda_{jk}$	latent class; class weight; class-j capture probability on list k

Record linkage

$r \in R_k,\ a_r$	record of list k; its attributes
$\Lambda,\ \Lambda^\star,\ \widehat{\Lambda}$	linkage structure (partition of records into incidents); truth; estimate
$\gamma_{rr'}$	comparison vector of a record pair
$m(\gamma),\ u(\gamma)$	$\mathbb{P}(\gamma \mid \text{match})$, $\mathbb{P}(\gamma \mid \text{non-match})$
$w = \log(m/u)$	Fellegi–Sunter weight
$\delta,\ \varepsilon$	per-record probabilities of detachment and of a false merge under LE (pairwise false non-match rate $1-(1-\delta)^2$)
$\mathcal{T}(\delta, \varepsilon)$	linkage operator on capture-history tables

Small-area model

θ_{st}	incidence per unit exposure, state s, year t
$b_s,\ \tau_t,\ \chi_{st}$	state effect; national trend; state-specific trend
$c_{st,k}$	coverage of list k (police: agency participation)

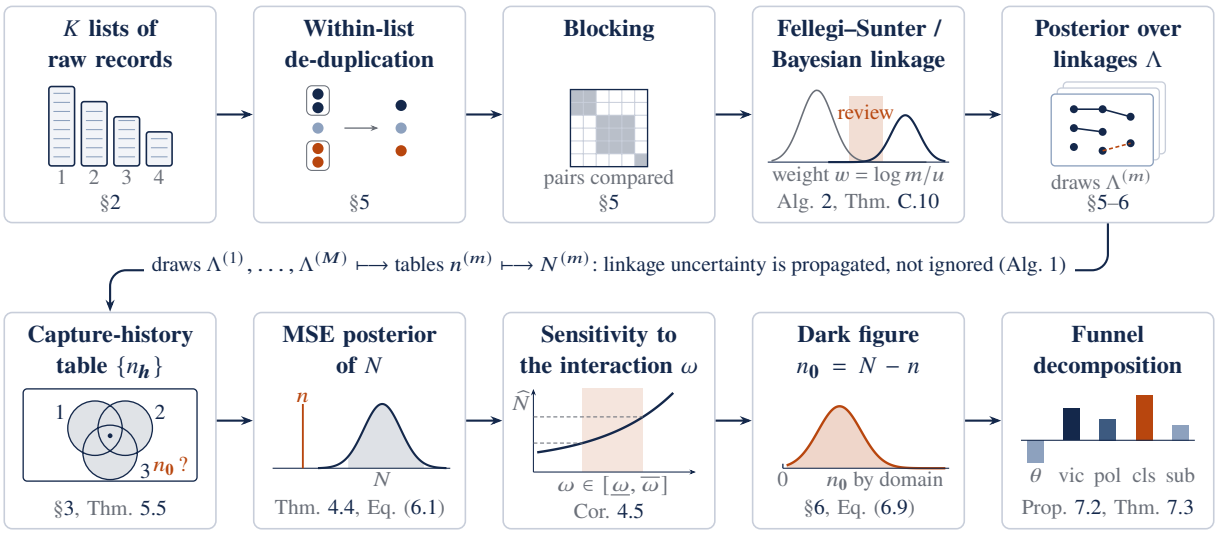


Figure 2. From records to the dark figure. Top row: records from K lists are de-duplicated within lists, restricted by blocking to plausible candidate pairs, and linked across lists with a Fellegi–Sunter or Bayesian model; uncertain pairs (orange band) can be sent to clerical review. The output is a *posterior distribution* over linkage structures Λ rather than a single matching. Bottom row: each draw $\Lambda^{(m)}$ induces a capture-history table; multiple-systems estimation yields a posterior for N; the untestable K-way interaction ω is varied over a stated range; and the dark figure $n_0 = N - n$ (the region outside every list) is reported by domain and decomposed along the reporting funnel of Figure 1 into Shapley shares of incidence (θ) and the four stages ($\rho_{\text{vic}}, \ldots, \rho_{\text{sub}}$). Labels under each tile point to the corresponding results.

Contributions. Our contributions are a synthesis, generalisation and application of existing ideas to a new and sensitive domain. Classical results are attributed where they are used.

C1 Identification and sensitivity. Without a restriction on the highest-order interaction ω of the log-linear model, the dark figure is not identified (Theorem 4.3; compare Aleshin-Guendel et al., 2024). We retain ω in the classical closed-form estimator, an alternating product of cell counts over the Boolean lattice (Bishop et al., 1975; Fienberg, 1972), as an explicit sensitivity parameter entering as the factor $e^{\pm\omega}$ (Theorem 4.4, Figure 3); this makes offset-based sensitivity analysis (Gerritse et al., 2015) analytic, with sharp bounds (Corollary 4.5). The direction of bias is the sign of an oriented interaction (Proposition 4.6), which, with three or more lists, neither positive dependence nor heterogeneous visibility fixes (Proposition 4.7); in a stylised three-list example the model without a K-way interaction overstates the dark figure more than twelvefold (Example A.7). Sensitivity analysis must therefore be two-sided. Chao's lower bound (Chao, 1987) is reversed by heterogeneity across lists (Theorem 4.8 and Proposition 4.9), so we use it as a diagnostic. We model referral dependence and show how provenance restores identification (Proposition 4.10), and give a sensitivity frontier for policy claims (Theorem 4.11).

C2 Linkage error as a linear operator. Following Ding and Fienberg (1994) and its multi-list extensions (Di Consiglio and Tuoto, 2018; Ding and Fienberg, 1996), we represent linkage error under a stylised model (LE) as an operator $\mathcal{T}(\delta, \varepsilon)$ on the capture-history table, with per-record probabilities δ of a missed and ε of a false match, and derive first-order bias coefficients for the K-list estimator (Theorem 5.5). Under independent capture, false matches do more damage when lists are sparse and missed matches when they are dense, with a closed-form crossover (Corollary 5.6 and Remark C.5); with three or more strongly dependent lists the usual direction can reverse (Remark 5.7). We derive

the correction implied under LE (Corollary 5.8), a total-variance decomposition (Proposition 5.11), and designs for clerical review and a random audit (Theorem C.10 and Proposition C.13). LE has no move for a swap, a record linked to the wrong partner (Remark 5.9), and our simulation shows that the correction over-corrects when swaps dominate; mixing over linkage draws is therefore primary, and the correction a diagnostic. Mixing is no substitute for a well-specified linkage model: in the simulation, approximate draws from a plug-in linkage posterior did not remove the linkage bias either.

C3 A Bayesian model. We assemble the latent-class MSE of Manrique-Vallier (2016) (Equation (6.1)), into which the sensitivity parameters enter explicitly (Equation (6.3)), covariate-dependent capture, and a spatiotemporal small-area model (Equation (6.9)) with police coverage as a known offset; the small-area model is specified, not evaluated. Linkage uncertainty is propagated by a modular ("cut") posterior that mixes over linkage draws (Algorithm 1), as in Sadinle (2018).

C4 Where antisemitic incidents go missing. We write the police count as a multiplicative funnel (Proposition 7.2, Figure 1) and state which data identify each stage; incidents that never reach a recording channel require a victim survey. A Shapley decomposition attributes differences in recorded rates across jurisdictions to incidence and to each stage (Theorem 7.3).

C5 A simulation study. In synthetic worlds with heterogeneous reportability, referral and noisy records, we compare estimators and naive, linkage-corrected and uncertainty-propagating pipelines on bias, error and coverage (Section 8). Simulated records carry realistic attribute noise and are linked by FS-EM (Algorithm 2); LE enters only through the correction, which is thereby tested under misspecification.

C6 A pre-registered analysis plan for a New York City pilot (Section 9), with an ethics protocol (Section 10).

Related work. Joint Bayesian models of record linkage and population size were developed by Tancredi and Liseo (2011), and Sadinle (2018) propagated linkage uncertainty into MSE for human-rights data, building on Sadinle (2017) and Fellegi and Sunter (1969). Heterogeneity in capture has been addressed through latent classes (Manrique-Vallier, 2016) and lower bounds (Chao, 1987). The hate-crime literature documents under-reporting (Pezzella et al., 2019) and the fragility of bias classification (McDevitt et al., 2003). As far as we know, these strands have not been brought together for antisemitic incidents, where referral, definitional mismatch and shifting police coverage occur together.

Roadmap. Sections 2 and 3 describe the lists and the capture model, Sections 4 to 7 develop the methods, Section 8 reports the simulation study, Sections 9 and 10 give the analysis plan and ethics protocol, and Section 11 discusses limitations. Proofs and details are in the appendices.

2 The lists: sources of antisemitic incident data

Multiple-systems estimation needs $K \geq 2$ lists, each an incomplete register of the *same* population U_d of in-scope incidents in a domain d. This section maps the candidate US lists onto the canonical design ($k = 1$ police, $k = 2$ NGO audit, $k = 3$ community security organisation, $k = 4$ campus or community portal); Table 2 summarises them. Every figure quoted is attributed to its publisher, none is an input to the simulation study, and published totals from different lists are *not* estimates of a common quantity (Section 2.5).

Table 2. Candidate sources for a US application. "List" is the role in the canonical $K = 4$ design; CST and FRA are international comparators. Entries summarise the sources' own documentation (citations in the text).

Source (role)	Unit; anti-Jewish identification	Granularity; access	Coverage issues	Dependence risks
FBI UCR/ NIBRS (list 1)	Crimes; NIBRS bias code 21, "sufficient objective facts"	Agency × year tables; records without names or addresses	Voluntary participation; NIBRS-only 2021	Feeds ADL and Clery; receives community referrals
NYPD, California DOJ (list 1, local)	Complaints; events (one or more offences); anti-Jewish category	NYPD rows public by month and precinct; CA annual report	2023 NYC state publication gap; partial-year CA submissions	Same system as the FBI list
ADL Audit, H.E.A.T. Map (list 2)	Incidents, criminal and non-criminal; Israel/Zionism rule since Oct. 2023	Incident listings with redactions	Mid-series scope change	Ingests police and partner reports
SCN (list 3)	Threats, suspicious activity; criteria not public	Aggregate summaries; records by agreement	Scope broader than crimes	ADL partner; refers to police
CST (UK)	Incidents; 3,001 reports screened out (2025)	Annual reports	Police feed disrupted since 2024	16% of 2025 incidents from police
Clery data (campus)	Crimes; religion bias only	Institution × year; public	Superset of target; campus only	Built partly from police data
NCVS (survey)	Victimisations; victim perception plus evidence	National, pooled years	Persons and households only	Auxiliary: ρ_{pol}
FRA (EU)	Respondent-reported incidents	Country; 2018, 2024 waves	Not US	Auxiliary: non-reporting

2.1 Police-recorded hate crime (list 1)

Under the Hate Crime Statistics Act of 1990 (United States Congress, 1990b), the FBI's Uniform Crime Reporting (UCR) Program attached a bias element to the offences already reported to it (Federal Bureau of Investigation, 2019). In the National Incident-Based Reporting System (NIBRS), anti-Jewish bias is code 21, reported only if the investigation reveals sufficient objective facts to lead a reasonable and prudent person to conclude that bias motivated the offender, in whole or in part (Federal Bureau of Investigation, n.d.). List 1 is therefore a list of *crimes* whose anti-Jewish motivation an agency has *classified*; for 2025 the FBI reported 1,528 single-bias anti-Jewish incidents (Anti-Defamation League, 2026b; Federal Bureau of Investigation, 2026). NIBRS holds no names or addresses, so linkage needs the agencies' own records.

Participation and the 2021 transition. Participation is voluntary at the federal level, and for data year 2021 the FBI accepted only NIBRS submissions (Congressional Research Service, 2025). The initial 2021 hate-crime release, in December 2022, covered 11,834 agencies representing 64.8% of the US population. After the FBI reopened the collection to summary-format and additional NIBRS reports from 3,025 agencies, its supplemental release of March 2023 covered 14,859 agencies and 91.1% of the population (Federal Bureau of Investigation, 2023; U.S. Department of Justice, 2023). Both formats were accepted again from 2022 (Congressional Research Service, 2025); for 2024, 16,419 participating agencies covered 95.1% of the population, and the FBI does not estimate counts for non-participants (Federal Bureau of Investigation, 2025). Participation measures *submission* (ρ_{sub} in Section 7), which can shift abruptly for reasons unrelated to victimisation; the small-area model (Equation (6.9)) therefore ties police coverage to it.

City and state data. City and state data are list 1 observed before federal submission, with richer linkage keys. The NYPD Hate Crimes dataset has one public row per confirmed complaint, with month, precinct, offence and a bias value `ANTI-JEWISH` (New York City Police Department, 2026), which makes New York City a natural pilot (Section 9); the state published no 2023 data for the city because of the NYPD's NIBRS transition (Office of the New York State Comptroller, 2024). California counts hate-crime *events* under Penal Code sections 13023 and 422.55, with 289 anti-Jewish events for 2023, 310 for 2024 and 289 for 2025, not all from full-year submissions (California Department of Justice, 2025, 2026); such units need not match federal incidents.

2.2 The NGO audit (list 2)

ADL's Audit, kept since 1979, counts criminal and non-criminal incidents of assault, harassment and vandalism, assesses each report for credibility, counts each incident once, and can treat a related series of acts as one incident (Anti-Defamation League, 2025, 2026a). Since 7 October 2023 it has included certain expressions of opposition to Zionism that could be perceived as supporting terrorism or attacks on Jews, Israelis or Zionists, while excluding legitimate political protest (Anti-Defamation League, 2025, 2026a). ADL reported that 58% of its 2024 incidents contained elements related to Israel or Zionism, and recorded 9,354 incidents for 2024, the highest in the Audit's history, and 6,274 for 2025 (Anti-Defamation League, 2025, 2026a). Totals before and after October 2023 therefore measure partly different populations.

The Audit is a *compiled* list: its largest source is victims' own reports (Anti-Defamation League, 2025), but from 2021 it incorporated reports from partner Jewish organisations, among them the Secure Community Network and Hillel International; ADL attributed 494 incidents in 2021 to these partnerships (Anti-Defamation League, 2022). Some of its records thus originate in lists 1 and 3. The H.E.A.T. Map publishes redacted incident-level listings (Anti-Defamation League, n.d.), enough for coarse linkage on date, place and category. Provenance (how ADL learned of each incident) is not, to our knowledge, public; it is the most valuable field to request, because it turns referral into an identification strategy (Proposition 4.10(c)–(d)).

2.3 Community security organisations and campus data (lists 3 and 4)

The Secure Community Network (SCN), founded in 2004, describes itself as the official safety and security organisation of the Jewish community in North America (Secure Community Network, n.d.b). Its Duty Desk shares threat and incident information with law enforcement and partners (Secure Community Network, n.d.a), and it publishes only aggregate summaries (Secure Community Network, 2024), so its use as a list requires an agreement and a mapping of its scope (threats, suspicious activity) onto the common ontology; as an Audit partner that refers to the police, it is linked to both other lists. The UK's Community Security Trust (CST) shows what a mature community list documents. It recorded 3,700 antisemitic incidents in 2025, up from 3,556 in 2024; it excluded a further 3,001 reports that it did not deem antisemitic; 587 (16%) of its 2025 incidents came from police forces under information-sharing agreements; and since March 2024 a technical problem has disrupted the flow of reports from the Metropolitan Police (Community Security Trust, 2026). A US community list needs the same documentation of screening, provenance and feed interruptions.

Campus data. The Clery Act (United States Congress, 1990a) and 34 C.F.R. § 668.46 (U.S. Department of Education, 2026) require institutions in federal student-aid programmes to disclose hate crimes in their Clery geography, published at institution level (U.S. Department of Education, Office of Postsecondary Education, n.d.). The bias categories include religion but not anti-Jewish bias, which institutions were

invited in 2024 to disclose only voluntarily and outside the statistical grids (U.S. Department of Education, 2024). Clery data thus cover a *superset* of the target, harmful in a capture list (Example 2.1), and are partly built from police statistics (U.S. Department of Education, 2026). For list 4 we therefore propose a campus or community self-report portal that identifies anti-Jewish bias, used under agreement and only within the campus stratum, where its remit is declared (Assumption A3(b), Section 9.4).

2.4 Victimisation surveys: anchors, not lists

The National Crime Victimization Survey (NCVS) counts a victimisation as a hate crime when the victim perceives bias and reports hate language, hate symbols or police confirmation, whether or not it was reported to police, and the Bureau of Justice Statistics publishes the share reported (Kena and Thompson, 2021); others use these data to study who reports (Lantz and Wenger, 2022; Pezzella et al., 2019; Zaykowski, 2010). It misses institutional victims and has, to our knowledge, no estimate for anti-Jewish bias, so it serves only for triangulation (Section 9.6). The EU Agency for Fundamental Rights has surveyed Jewish people directly: 16,395 respondents in 12 Member States in 2018, of whom 79% of those who had experienced antisemitic harassment had not reported the most serious incident to the police or to any other organisation (European Union Agency for Fundamental Rights, 2018), and almost 8,000 respondents in 13 Member States in a third survey published in 2024 (European Union Agency for Fundamental Rights, 2024). We do not transport these numbers to the United States, but non-reporting to *any* organisation means that many incidents never reach a recording channel; they lie outside N_V and are invisible to every list (Section 7).

2.5 Definitional harmonisation

The lists count different things (crimes, incidents, threats) in different units, so we code every record on five axes: (i) *conduct* (assault; threat or intimidation; property damage; other offence; non-criminal conduct); (ii) *bias* (anti-Jewish as the single bias, among multiple biases, or not established); (iii) *target* (individual, institution, public space); (iv) *setting* (campus, online, other); and (v) a *content flag* for Israel- or Zionism-related expression. The primary estimand (Section 9) counts *criminal antisemitic incidents*: conduct that is an offence under the jurisdiction's criminal law, with anti-Jewish bias meeting a stated evidentiary standard. Capture follows each list's own classification: $Y_{ik} = 1$ if list k holds a record of i that it classifies as antisemitic. An antisemitic crime that police recorded without a bias code is therefore *not* on list 1; that loss is the ρ_{cls} stage of the funnel, not a linkage failure. Scope is coded per record, blind to linkage (otherwise incidents on several lists would have more chances to be coded in scope), by a rule fixed in advance and varied in sensitivity analyses.

Every capture–recapture model assumes that each list *could* capture every member of U_d (Assumption A3). If list k cannot capture a sub-population, for example the police list and non-criminal harassment, then $\mathbb{P}(Y_{ik} = 1) = 0$ there, and this *structural zero* masquerades as heterogeneity or list interaction.

Example 2.1 (Scope mismatch as hidden heterogeneity). Let $U = A \cup B$, where A holds N_A criminal incidents and B holds N_B non-criminal ones, and suppose the target is $N = N_A + N_B$. List 1 captures only incidents in A, each with probability p_1. List 2 captures incidents independently of list 1, with probability p_{2A} in A and p_{2B} in B. Then $\mathbb{E}n_{1+} = N_A p_1$, $\mathbb{E}n_{2+} = N_A p_{2A} + N_B p_{2B}$ and $\mathbb{E}n_{11} = N_A p_1 p_{2A}$, so the Lincoln–Petersen estimator converges to

$$\frac{\mathbb{E}n_{1+}\,\mathbb{E}n_{2+}}{\mathbb{E}n_{11}} \;=\; N_A + N_B\,\frac{p_{2B}}{p_{2A}}. \tag{2.1}$$

This equals N only if $p_{2B} = p_{2A}$ and equals N_A only if $p_{2B} = 0$; in general it estimates neither. With $K > 2$ lists the same mechanism appears as spurious interactions and contaminates the extrapolation to $n_{\mathbf{0}}$.

There are two remedies: restrict every list to the common scope, so that the estimand is N_A (our primary choice), or model the strata separately with declared structural zeros, which requires at least two lists that can capture B. The Audit's October 2023 scope change is a structural zero in *time*, so domains should not straddle it. The lists differ most on Israel- or Zionism-related expression, as do the IHRA working definition (International Holocaust Remembrance Alliance, 2016) and the Jerusalem Declaration on Antisemitism (Jerusalem Declaration on Antisemitism, 2021); we do not adjudicate, but report the estimand with and without flagged incidents. For the primary, crimes-only estimand the flag matters only where an offence is also present.

2.6 Dependence between lists

Four mechanisms make the lists dependent. (a) *Referral*: police and partners feed the Audit, police feed Clery, and community organisations feed the police; Proposition 4.10 models it as forwarding a fraction q of one list's captures to another. (b) *Heterogeneity*: severity, visibility and media attention raise capture on all lists at once; ignored, this biases Lincoln–Petersen downward for two lists, while for $K \geq 3$ either sign can occur (Propositions 4.6 and 4.7). (c) *Specialisation*: lists designed to reach what others miss, such as a campus portal, produce negative dependence, which biases two-list estimates upward; for $K \geq 3$ the direction again follows the oriented top interaction. (d) *Unidentified residual dependence*: ω is not identified (Theorem 4.3) and is treated as a sensitivity parameter (Theorem 4.4 and Corollary 4.5). Linkage error interacts with all four, distorting the table before any dependence model is fitted (Theorem 5.5); its uncertainty is propagated by mixing over linkage draws (Algorithm 1).

3 Statistical framework

This section sets up the multiple-systems problem for a single domain d (for example a state–year), which we suppress from the notation until Section 6. The material is classical (Bishop et al., 1975; Darroch, 1958; Fienberg, 1972; International Working Group for Disease Monitoring and Forecasting, 1995); we present it so that the role of the single untestable parameter, the highest-order interaction ω, is explicit from the start.

3.1 Incidents, lists and the incomplete table

Let U be the set of in-scope antisemitic incidents that occur in the domain during the reference period and reach at least one recording channel, by a report or by the channel's own detection: U is U_d^V of Section 7. Incidents that reach no channel cannot be on any list, and recovering them needs a victim survey (Proposition 7.2). The size N of U (written N_V in Section 7) is unknown, and incidents are indexed by $i \in \{1, \ldots, N\}$. Incidents are recorded by K lists, indexed by $k \in [K]$. In our canonical configuration $K = 4$: police records submitted to the FBI ($k = 1$), an NGO audit ($k = 2$), a community security organisation ($k = 3$), and a campus or community self-report portal ($k = 4$); see Section 2. Let $Y_{ik} = 1$ if list k records incident i and $Y_{ik} = 0$ otherwise. The *capture history* of incident i is $\boldsymbol{h}_i = (Y_{i1}, \ldots, Y_{iK}) \in \mathcal{H} = \{0, 1\}^K$, and we write $|\boldsymbol{h}| = \sum_k h_k$ for the number of lists that record an incident with history $\boldsymbol{h}$. The cell counts are

$$n_{\boldsymbol{h}} = \#\{i : \boldsymbol{h}_i = \boldsymbol{h}\}, \qquad \boldsymbol{h} \in \mathcal{H}.$$

Let $\mathcal{H}_+ = \mathcal{H} \setminus \{\mathbf{0}\}$ denote the $2^K - 1$ non-zero histories. The 2^K table $(n_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}}$ is *incomplete*: an incident that no list records leaves no trace, so the cell $n_{\mathbf{0}}$ is missing by construction. It is not a structural zero, and it is not missing at random; it is simply unobservable. The data are $\boldsymbol{n}_+ = (n_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}_+}$. From them we

form the number of distinct observed incidents, the dark figure, the list sizes, the frequency counts and the dark-figure ratios:

$$n = \sum_{\boldsymbol{h} \in \mathcal{H}_+} n_{\boldsymbol{h}}, \qquad n_{\boldsymbol{0}} = N - n, \qquad n_{k+} = \sum_{\boldsymbol{h}:\, h_k = 1} n_{\boldsymbol{h}}, \qquad f_j = \sum_{\boldsymbol{h}:\, |\boldsymbol{h}| = j} n_{\boldsymbol{h}}, \qquad D_k = \frac{N}{n_{k+}}. \tag{3.1}$$

Here f_j is the number of incidents recorded by exactly j lists, so $f_0 = n_{\boldsymbol{0}}$. The ratio D_k is the number of in-scope incidents per incident recorded by list k; D_1 is the multiplier that converts the police count into N.

We often identify a history with its support. For $T \subseteq [K]$, let $\boldsymbol{h}_T$ denote the history with $\operatorname{supp}(\boldsymbol{h}_T) = T$, so $\boldsymbol{h}_\emptyset = \boldsymbol{0}$ and $\boldsymbol{h}_{[K]} = \boldsymbol{1} = (1, \dots, 1)$. For small K we abbreviate, e.g. $n_{101} = n_{(1,0,1)}$. Two pieces of notation recur throughout. The *alternating sign* of a history is $s_{\boldsymbol{h}} = (-1)^{|\boldsymbol{h}|+1}$, which is $+1$ if an odd number of lists record the incident and -1 if an even number do. The *alternating product* of a positive vector $\boldsymbol{x} = (x_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}_+}$ is

$$\mathcal{A}(\boldsymbol{x}) = \prod_{\boldsymbol{h} \in \mathcal{H}_+} x_{\boldsymbol{h}}^{s_{\boldsymbol{h}}} = \frac{\prod_{|\boldsymbol{h}| \text{ odd}} x_{\boldsymbol{h}}}{\prod_{|\boldsymbol{h}| \text{ even},\, \boldsymbol{h} \neq \boldsymbol{0}} x_{\boldsymbol{h}}}. \tag{3.2}$$

There are 2^{K-1} odd and $2^{K-1} - 1$ non-empty even subsets of $[K]$. Hence

$$\sum_{\boldsymbol{h} \in \mathcal{H}_+} s_{\boldsymbol{h}} = 1, \qquad \text{and} \qquad \sum_{\boldsymbol{h} \in \mathcal{H}_+:\, h_k = 1} s_{\boldsymbol{h}} = \sum_{\boldsymbol{g} \in \{0,1\}^{K-1}} (-1)^{|\boldsymbol{g}|} = 0 \quad (k \in [K],\ K \geq 2). \tag{3.3}$$

The first identity makes $\mathcal{A}$ positively homogeneous of degree one: $\mathcal{A}(c\boldsymbol{x}) = c\,\mathcal{A}(\boldsymbol{x})$ for $c > 0$.

3.2 Sampling models

Definition 3.1 (Sampling models).

(M) *Multinomial model.* N is a fixed unknown integer. The histories $\boldsymbol{h}_1, \dots, \boldsymbol{h}_N$ are independent and identically distributed with $\mathbb{P}(\boldsymbol{h}_i = \boldsymbol{h}) = \pi_{\boldsymbol{h}}$, where $\boldsymbol{\pi} = (\pi_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}}$ lies in the interior of the probability simplex. Equivalently, $(n_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}} \sim \text{Multinomial}(N, \boldsymbol{\pi})$.

(P) *Poisson model.* The counts $n_{\boldsymbol{h}}$, $\boldsymbol{h} \in \mathcal{H}$, are independent with $n_{\boldsymbol{h}} \sim \text{Poisson}(\mu_{\boldsymbol{h}})$ and $\mu_{\boldsymbol{h}} > 0$, and $N = \sum_{\boldsymbol{h} \in \mathcal{H}} n_{\boldsymbol{h}}$.

In both models we write $\mu_{\boldsymbol{h}} = \mathbb{E} n_{\boldsymbol{h}}$, so $\mu_{\boldsymbol{h}} = N\pi_{\boldsymbol{h}}$ under (M). We also write $\boldsymbol{\mu}_+ = (\mu_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}_+}$ and $\mu_{\text{obs}} = \sum_{\boldsymbol{h} \in \mathcal{H}_+} \mu_{\boldsymbol{h}} = \mathbb{E} n$.

Both models treat incidents as exchangeable and independent, which does not mean that incidents are equally easy to record. Suppose incident i has list-specific capture probabilities $\boldsymbol{p}_i = (p_{i1}, \dots, p_{iK})$ drawn independently from a distribution F, and lists record independently given $\boldsymbol{p}_i$. Then the histories are i.i.d. with $\pi_{\boldsymbol{h}} = \int \prod_k p_k^{h_k} (1 - p_k)^{1 - h_k}\, dF(\boldsymbol{p})$. Heterogeneity in reporting propensity therefore enters through $\boldsymbol{\pi}$ as interactions between lists (Proposition 4.7), the representation used by the latent-class models of Section 6. What (M) and (P) rule out is dependence *between* incidents, such as a series by one perpetrator that is reported, or missed, as a bundle.

Lemma 3.2 (Likelihood factorisation).

(a) Under (P), $N \sim \text{Poisson}(\mu_\bullet)$ with $\mu_\bullet = \sum_{\boldsymbol{h} \in \mathcal{H}} \mu_{\boldsymbol{h}}$. Conditionally on N, $(n_{\boldsymbol{h}})_{\boldsymbol{h} \in \mathcal{H}} \sim \text{Multinomial}(N, \boldsymbol{\mu}/\mu_\bullet)$.

(b) Let $\pi^c_{\boldsymbol{h}} = \pi_{\boldsymbol{h}}/(1-\pi_{\boldsymbol{0}})$ *for* $\boldsymbol{h} \in \mathcal{H}_+$*; under (P),* $\pi^c_{\boldsymbol{h}} = \mu_{\boldsymbol{h}}/\mu_{\text{obs}}$*. Write* $\mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c) = n! \prod_{\boldsymbol{h}\in\mathcal{H}_+} (\pi^c_{\boldsymbol{h}})^{n_{\boldsymbol{h}}}/n_{\boldsymbol{h}}!$*. The likelihoods of the observed table under (M) and under (P) are*

$$L_M(N, \boldsymbol{\pi}) = \frac{N!}{(N-n)! \prod_{\boldsymbol{h}\in\mathcal{H}_+} n_{\boldsymbol{h}}!}\, \pi_{\boldsymbol{0}}^{N-n} \prod_{\boldsymbol{h}\in\mathcal{H}_+} \pi_{\boldsymbol{h}}^{n_{\boldsymbol{h}}}$$

$$= \binom{N}{n} \pi_{\boldsymbol{0}}^{N-n} (1-\pi_{\boldsymbol{0}})^n\, \mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c), \qquad N \in \{n, n+1, \dots\}, \tag{3.4}$$

$$L_P(\boldsymbol{\mu}) = \prod_{\boldsymbol{h}\in\mathcal{H}_+} \frac{e^{-\mu_{\boldsymbol{h}}} \mu_{\boldsymbol{h}}^{n_{\boldsymbol{h}}}}{n_{\boldsymbol{h}}!} = \frac{e^{-\mu_{\text{obs}}} \mu_{\text{obs}}^{\,n}}{n!}\, \mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c). \tag{3.5}$$

Proof. Part (a) is the standard Poisson–multinomial relation. For (b) under (M), the observed table is the multinomial vector with the $n_{\boldsymbol{0}} = N - n$ unseen incidents summed out. Its probability is the first expression in (3.4); substituting $\pi_{\boldsymbol{h}} = (1-\pi_{\boldsymbol{0}})\pi^c_{\boldsymbol{h}}$ and $\sum_{\boldsymbol{h}\in\mathcal{H}_+} n_{\boldsymbol{h}} = n$ gives the second. Under (P), multiply the Poisson probabilities and use $\mu_{\boldsymbol{h}} = \mu_{\text{obs}}\pi^c_{\boldsymbol{h}}$. □

In (3.4) the population size enters only through the binomial factor, $n \sim \text{Binomial}(N, 1-\pi_{\boldsymbol{0}})$; the conditional multinomial factor carries all the information about the composition of the observed table and none about N, and maximising it first gives the conditional maximum likelihood estimator of Sanathanan (1972). Under (P), $\mu_{\boldsymbol{0}}$ does not appear in (3.5) at all. Information about the dark figure must therefore come from a model that ties $\mu_{\boldsymbol{0}}$ to $\boldsymbol{\mu}_+$, and the log-linear parameterisation makes that tie explicit.

3.3 Hierarchical log-linear models

The saturated log-linear model writes every cell mean as

$$\log \mu_{\boldsymbol{h}} = \sum_{S \subseteq \text{supp}(\boldsymbol{h})} u_S, \qquad \boldsymbol{h} \in \mathcal{H}, \tag{3.6}$$

so that $u_\emptyset = \log \mu_{\boldsymbol{0}}$. Under (M) the same expansion is used for $\log \pi_{\boldsymbol{h}}$, and normalisation fixes $u_\emptyset = \log \pi_{\boldsymbol{0}}$. The parameterisation is a bijection, by Möbius inversion on the Boolean lattice (Rota, 1964).

Lemma 3.3 (Möbius inversion). *For functions* f, g *on the subsets of* $[K]$*,* $g(S) = \sum_{T\subseteq S} f(T)$ *for all* S *if and only if* $f(S) = \sum_{T\subseteq S} (-1)^{|S|-|T|} g(T)$ *for all* S*. Consequently, (3.6) defines a bijection between* $\boldsymbol{u} \in \mathbb{R}^{2^K}$ *and* $\boldsymbol{\mu} \in (0,\infty)^{\mathcal{H}}$*, with inverse*

$$u_S = \sum_{T\subseteq S} (-1)^{|S|-|T|} \log \mu_{\boldsymbol{h}_T}, \qquad S \subseteq [K]. \tag{3.7}$$

Proof. If $g(S) = \sum_{R\subseteq S} f(R)$ for all S, then $\sum_{T\subseteq S} (-1)^{|S\setminus T|} g(T) = \sum_{R\subseteq S} f(R) \sum_{T:\, R\subseteq T\subseteq S} (-1)^{|S\setminus T|} = \sum_{R\subseteq S} f(R)\,(1-1)^{|S\setminus R|} = f(S)$. The converse is the same computation with the roles of f and g exchanged. Apply this with $g(S) = \log \mu_{\boldsymbol{h}_S}$ and $f = \boldsymbol{u}$. □

By (3.7), each u_S is a log contrast. The main effect $u_{\{k\}} = \log(\mu_{\boldsymbol{h}_{\{k\}}}/\mu_{\boldsymbol{0}})$ is the log-odds of being recorded by list k alone rather than by no list. The two-way term $u_{\{k,l\}}$ is the log odds ratio between lists k and l among incidents missed by every other list. In general, u_S is a generalised log odds ratio of the lists in S, computed in the stratum missed by every list outside S. Every u_S involves the unobservable cell, through the term $T = \emptyset$. The highest-order interaction

$$\omega = u_{[K]} = \sum_{T\subseteq[K]} (-1)^{K-|T|} \log \mu_{\boldsymbol{h}_T} \tag{3.8}$$

plays a special role. For $K = 2$, $\omega = \log\{\mu_{11}\mu_{00}/(\mu_{10}\mu_{01})\}$ is the log odds ratio between the two lists. For $K = 3$, $\omega = \log \mathrm{OR}_{12|h_3=1} - \log \mathrm{OR}_{12|h_3=0}$. This is the change in the police–NGO odds ratio between incidents that the community security organisation records and incidents that it misses. In general, splitting the sum in (3.8) according to whether $k \in T$ gives, for every k,

$$\omega = \psi_k(1) - \psi_k(0), \tag{3.9}$$

$$\psi_k(1) = \sum_{T \subseteq [K]\setminus\{k\}} (-1)^{K-1-|T|} \log \mu_{\boldsymbol{h}_{T\cup\{k\}}}, \qquad \psi_k(0) = \sum_{T \subseteq [K]\setminus\{k\}} (-1)^{K-1-|T|} \log \mu_{\boldsymbol{h}_T}.$$

Here $\psi_k(x)$ is the $(K-1)$-way interaction of the other lists within the stratum $h_k = x$. The stratum $h_k = 1$ is fully observed. The unobservable cell lies in the stratum $h_k = 0$, where $\log \mu_{\mathbf{0}}$ enters $\psi_k(0)$ with coefficient $(-1)^{K-1}$. Since ω is symmetric in the lists, any list may play the role of k.

Isolating the term $T = \emptyset$ in (3.8) and using $(-1)^K(-1)^{K-|T|} = (-1)^{|T|}$ gives the identity on which Section 4 rests. It holds for every $\boldsymbol{\mu} \in (0,\infty)^{\mathcal{H}}$:

$$\mu_{\mathbf{0}} = e^{\bar{\omega}}\, \mathcal{A}(\boldsymbol{\mu}_+), \qquad \bar{\omega} = (-1)^K \omega. \tag{3.10}$$

We call $\bar{\omega}$ the *oriented* top interaction. It is the log of the factor by which the expected dark figure exceeds the alternating-product extrapolation $\mathcal{A}(\boldsymbol{\mu}_+)$ from the observed cells. Under (M), applying (3.10) to $\boldsymbol{\pi}$ and using the homogeneity of $\mathcal{A}$ gives the equivalent form

$$\frac{\pi_{\mathbf{0}}}{1-\pi_{\mathbf{0}}} = e^{\bar{\omega}}\, \mathcal{A}(\boldsymbol{\pi}^c). \tag{3.11}$$

A *hierarchical* log-linear model is specified by a generating class $\mathcal{S}$: a family of subsets of $[K]$ that contains $\emptyset$ and all singletons and is closed under taking subsets. The model sets $u_S = 0$ for $S \notin \mathcal{S}$; graphical and decomposable models (Madigan and York, 1997) are special cases. As Fienberg (1972) observed, the K-way term cannot be estimated from an incomplete 2^K table, so the models used to project into the missing cell exclude it, $[K] \notin \mathcal{S}$. Every such model satisfies $\omega = 0$ and hence, by (3.10), $\mu_{\mathbf{0}} = \mathcal{A}(\boldsymbol{\mu}_+)$. We use corner-point coding; the effect coding $x_k = 2h_k - 1$ of Bishop et al. (1975) spans the same hierarchical models, and its top-order coefficient is $2^{-K}\omega$, so statements such as "$\omega > 0$" do not depend on the coding.

3.4 Assumptions

Assumption A1 (Closed population). For each domain, U is a fixed set of incidents defined by incident date and location. Every list refers to the same reference period, and each incident belongs to exactly one domain.

Assumption A2 (Perfect linkage). The capture histories are observed without error. Records of the same incident are linked, and records of different incidents are not: $\widehat{\Lambda} = \Lambda^\star$.

Assumption A3 (Harmonised scope). (a) Every retained record of every list refers to an incident in U, under a common definition, reference period and domain. (b) Every incident in U is eligible to be recorded by every list: no list has a remit that structurally excludes part of U.

Assumption A4 (Dependence restriction). The highest-order interaction satisfies either (a) $\omega = \omega_0$ for a known ω_0 (a point restriction; $\omega_0 = 0$ is the conventional model without a K-way interaction), or (b) $|\omega| \le \Omega$ for a known $\Omega \ge 0$ (a sensitivity restriction).

Assumption A1 requires domains to be defined by the date of the incident, not of the record, and the analysis to wait until late records have arrived. Assumption A2 is never exactly true: the lists share

Table 3. A hypothetical $K = 3$ capture table for one domain. **The numbers are invented for illustration and are not data.** Lists: police (1), NGO audit (2), community security organisation (3). The last row gives the exponent $s_{\boldsymbol{h}}$ with which each cell enters the alternating product (3.2).

History $\boldsymbol{h} = (h_1, h_2, h_3)$	100	010	001	110	101	011	111	000
Count $n_{\boldsymbol{h}}$ (illustrative)	240	180	120	90	40	60	30	?
Lists recording, $\lvert\boldsymbol{h}\rvert$	1	1	1	2	2	2	3	0
Exponent $s_{\boldsymbol{h}}$	+1	+1	+1	−1	−1	−1	+1	—

no identifier, and linkage errors move mass between cells; Section 5 replaces it with an explicit error model. Assumption A3(a) requires harmonised definitions (Section 2.5), and its failure is not innocuous. Out-of-scope records seen by one list only fall in cells with $|\boldsymbol{h}| = 1$, which enter (3.2) with exponent +1: adding c_k of them to the singleton cell of list k multiplies the closed-form estimate of Theorem 4.4 by $\prod_k (1 + c_k / n_{\boldsymbol{h}_{\{k\}}})$. Assumption A3(b) fails if, say, a campus portal can record only campus incidents; the portal must then be used only in the stratum where it operates. Assumption A4 is the identifying assumption, which Aleshin-Guendel et al. (2024) argue should be stated separately from the model for the observed cells. The data cannot test it (Theorem 4.3), and Section 4 studies what it buys and what happens when it fails, including under referral (Proposition 4.10) and heterogeneous visibility (Proposition 4.7).

3.5 An illustrative table

Example 3.4 (Illustrative numbers, not data). Table 3 shows a hypothetical table with $n = 760$ observed incidents, list sizes $n_{1+} = 400$, $n_{2+} = 360$, $n_{3+} = 250$, and frequency counts $f_1 = 540$, $f_2 = 190$, $f_3 = 30$. Applying the Lincoln–Petersen estimator to two lists at a time, ignoring the third, gives 1200 for lists $(1, 2)$, 1429 for $(1, 3)$ and 1000 for $(2, 3)$. Such disagreement is a symptom of dependence or heterogeneity. The model with $\omega = 0$ uses all three lists. Among incidents that the community security organisation records, the police–NGO odds ratio is $n_{111} n_{001} / (n_{101} n_{011}) = 30 \cdot 120 / (40 \cdot 60) = 1.5$. The restriction $\omega = 0$ asserts the same odds ratio of 1.5 among incidents that it misses, which pins down the missing cell:

$$\widehat{n}_{000} = 1.5 \times \frac{n_{100}\, n_{010}}{n_{110}} = 1.5 \times \frac{240 \cdot 180}{90} = 720 = \frac{n_{100}\, n_{010}\, n_{001}\, n_{111}}{n_{110}\, n_{101}\, n_{011}} = \mathcal{A}(\boldsymbol{n}_+).$$

The implied total is $\widehat{N} = 1480$, with a police dark-figure ratio of $\widehat{D}_1 = 3.7$. We return to this table in Section 4 to illustrate interval estimation, the sensitivity bounds and the sensitivity frontier. None of these numbers describes any real jurisdiction.

4 Identification and sensitivity

This section asks what the data can and cannot say about the dark figure under Assumptions A1 to A3, and what Assumption A4 adds. Without a restriction on ω the dark figure is not identified (Theorem 4.3). With ω fixed it has a closed form in which ω is an explicit sensitivity parameter (Theorem 4.4). This yields sharp bounds, the direction of bias, a reversal of Chao's lower bound under list heterogeneity, a model of referral between lists, and the sensitivity frontier of a policy claim. Proofs and further remarks are in Appendix A.

Several of these results are classical: the non-estimability of the K-way interaction and the closed-form estimator without it, an alternating product of cell counts valid for every K (Bishop et al., 1975; Fienberg,

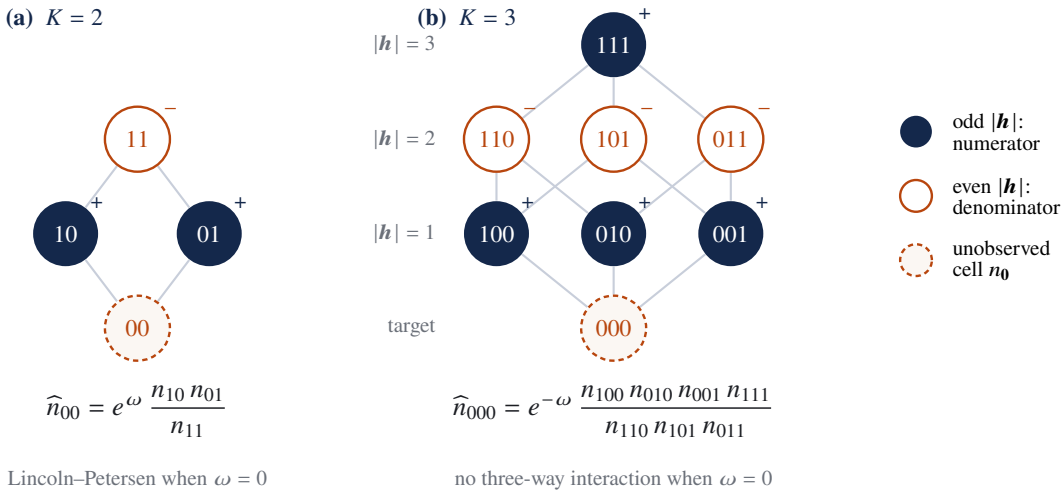


Figure 3. The closed form as an alternating product over the Boolean lattice $\{0,1\}^K$. Nodes are capture histories, ordered by the number of lists $|\boldsymbol{h}|$; edges add one list. Under a log-linear model whose only unidentified parameter is the K-way interaction ω, the unobserved cell is $\widehat{n}_{\mathbf{0}} = e^{\bar{\omega}} \mathcal{A}(\boldsymbol{n}_+)$ with $\bar{\omega} = (-1)^K \omega$ and $\mathcal{A}(\boldsymbol{n}_+) = \prod_{\boldsymbol{h} \neq \mathbf{0}} n_{\boldsymbol{h}}^{s_{\boldsymbol{h}}}$, $s_{\boldsymbol{h}} = (-1)^{|\boldsymbol{h}|+1}$ (Equation (3.10) and Theorem 4.4). Histories at odd rank enter the numerator and those at even rank the denominator. Panel (a) is the Lincoln–Petersen estimator; panel (b) the three-list case. Because ω enters only through the factor $e^{\bar{\omega}}$, an interval for ω maps monotonically to an interval for $n_{\mathbf{0}}$ (Corollary 4.5).

1972); the Lincoln–Petersen estimator (Lincoln, 1930; Petersen, 1896); and Chao's bound (Chao, 1987). Aleshin-Guendel et al. (2024) treat the absence of the highest-order interaction as an explicit identifying assumption for general K, which overlaps with Lemma 4.2 and Theorem 4.3, and retaining ω in the closed form makes explicit the offset-based sensitivity analysis of Gerritse et al. (2015). Our contribution is to draw out the consequences of keeping ω as a sensitivity parameter: sharp bounds with inference, the sensitivity frontier, and an interpretation of the sign of ω through heterogeneous visibility and referral. All asymptotic statements refer to the following regime.

> **Definition 4.1** (Large-count regime). Under (P), $\mu_{\boldsymbol{h}} = \tau m_{\boldsymbol{h}}$ for a fixed $\boldsymbol{m} \in (0,\infty)^{\mathcal{H}}$ and $\tau \to \infty$. Under (M), $\boldsymbol{\pi}$ is fixed and $N \to \infty$; we then set $\tau = N$ and $\boldsymbol{m} = \boldsymbol{\pi}$. In both cases $\pi_{\mathbf{0}} = \mu_{\mathbf{0}} / \sum_{\boldsymbol{h} \in \mathcal{H}} \mu_{\boldsymbol{h}}$ is the probability that an incident is missed by every list.

4.1 The dark figure is not identified without a restriction on ω

The engine of this section is a reparameterisation. Let $\Theta_\omega = \{\boldsymbol{u} \in \mathbb{R}^{2^K} : u_{[K]} = \omega\}$, and let $\boldsymbol{\mu}_+(\boldsymbol{u}) = (\exp \sum_{S \subseteq \mathrm{supp}(\boldsymbol{h})} u_S)_{\boldsymbol{h} \in \mathcal{H}_+}$ be the observed-cell means implied by $\boldsymbol{u}$.

> **Lemma 4.2** (Saturation given ω). *For every $\omega \in \mathbb{R}$, the map $\boldsymbol{u} \mapsto \boldsymbol{\mu}_+(\boldsymbol{u})$ restricted to Θ_ω is a bijection onto $(0,\infty)^{\mathcal{H}_+}$. Its inverse sends $\boldsymbol{\mu}_+$ to the vector $\boldsymbol{u}$ given by (3.7) with $\mu_{\mathbf{0}} = e^{(-1)^K \omega} \mathcal{A}(\boldsymbol{\mu}_+)$.*

Once ω is fixed, the model for the observed cells is therefore saturated, and (3.10) turns the choice of ω into a multiplicative choice of the dark figure.

> **Theorem 4.3** (Non-identification). *Let the log-linear model be saturated, with no restriction on ω.*
>
> *(a)* Poisson model. *Let $\boldsymbol{u}$ have top interaction ω. For every $\omega' \in \mathbb{R}$ there is exactly one $\boldsymbol{u}(\omega') \in \Theta_{\omega'}$ with $\boldsymbol{\mu}_+(\boldsymbol{u}(\omega')) = \boldsymbol{\mu}_+(\boldsymbol{u})$. Along this curve,*
>
> $$\mu_{\mathbf{0}}\big(\boldsymbol{u}(\omega')\big) = e^{(-1)^K(\omega' - \omega)}\, \mu_{\mathbf{0}}(\boldsymbol{u}), \qquad u_S(\omega') = u_S + (-1)^{K-|S|}(\omega' - \omega) \quad (S \subseteq [K]). \tag{4.1}$$

Under (P), every point of the curve induces the same law of the observed table, and the observed-data likelihood (3.5) is constant in ω'. As ω' ranges over $\mathbb{R}$, $\mu_{\mathbf{0}}$ ranges over $(0,\infty)$ and $\mathbb{E}N = \mu_{\text{obs}} + \mu_{\mathbf{0}}$ over $(\mu_{\text{obs}},\infty)$. No u-term is identified, not even a main effect.

(b) Multinomial model. *Under (M), the conditional law of $\boldsymbol{n}_+$ given n depends on $\boldsymbol{\pi}$ only through $\boldsymbol{\pi}^c$. For each ω, the map $\boldsymbol{\pi} \mapsto \boldsymbol{\pi}^c$ is a bijection from $\{\boldsymbol{\pi} : u_{[K]} = \omega\}$ onto the interior of the simplex on $\mathcal{H}_+$. Moreover, N is not consistently estimable: no estimator $\widehat{N} = \widehat{N}(\boldsymbol{n}_+)$ satisfies $\widehat{N}/N \to 1$ in probability as $N \to \infty$ for every fixed $\boldsymbol{\pi}$.*

Part (a) follows from Lemma 4.2: fix $\boldsymbol{\mu}_+$, vary ω, and read off $\mu_{\mathbf{0}}$ from (3.10). For (b), the observed table informs $(N, \pi_{\mathbf{0}})$ only through the count $n \sim \text{Binomial}(N, 1-\pi_{\mathbf{0}})$ (Lemma 3.2). Two parameter points with the same $\boldsymbol{\pi}^c$ and matched binomial means but different N cannot be separated, by the local limit theorem (Appendix A.2).

No goodness-of-fit statistic, information criterion or posterior predictive check can test the restriction on ω, because the law of the observed table is constant along the curve in Theorem 4.3(a); model search chooses among restrictions on lower-order terms *given* a restriction on ω (Remark A.3).

4.2 A closed form with a sensitivity parameter

For $\omega = 0$, the closed form below is classical for every K (Bishop et al., 1975; Fienberg, 1972). We state it with ω retained.

Theorem 4.4 (Closed-form estimator). *Suppose Assumptions A1 to A3 hold and Assumption A4(a) holds with known $\omega_0 = \omega$.*

(a) Under (P), the maximum likelihood estimator of $\mu_{\mathbf{0}}$ exists if and only if $n_{\boldsymbol{h}} > 0$ for every $\boldsymbol{h} \in \mathcal{H}_+$. It is then

$$\widehat{n}_{\mathbf{0}}(\omega) = e^{(-1)^K \omega}\, \mathcal{A}(\boldsymbol{n}_+) = e^{(-1)^K \omega}\, \frac{\prod_{|\boldsymbol{h}| \text{ odd}} n_{\boldsymbol{h}}}{\prod_{|\boldsymbol{h}| \text{ even}, \boldsymbol{h} \neq \mathbf{0}} n_{\boldsymbol{h}}}, \tag{4.2}$$

the fitted observed cells equal the observed counts, $\widehat{\mu}_{\boldsymbol{h}} = n_{\boldsymbol{h}}$ for $\boldsymbol{h} \in \mathcal{H}_+$, and $\widehat{N}(\omega) = n + \widehat{n}_{\mathbf{0}}(\omega)$.

(b) Under (M), the conditional maximum likelihood estimator of $\boldsymbol{\pi}^c$ exists in the interior of the simplex if and only if $n_{\boldsymbol{h}} > 0$ for every $\boldsymbol{h} \in \mathcal{H}_+$, and it is then $\boldsymbol{n}_+/n$. Substituting it into (3.11) gives $\widehat{\pi}_{\mathbf{0}}$ with $n/(1-\widehat{\pi}_{\mathbf{0}}) = \widehat{N}(\omega)$. Over integers $N \geq n$, the binomial factor in (3.4) at $\pi_{\mathbf{0}} = \widehat{\pi}_{\mathbf{0}}$ attains its maximum at $\lfloor \widehat{N}(\omega) \rfloor$, which is the conditional maximum likelihood estimator of Sanathanan (1972).

(c) For $K = 2$, $\widehat{n}_{\mathbf{0}}(\omega) = e^{\omega} n_{10} n_{01}/n_{11}$, where e^{ω} is the odds ratio between the two lists. At $\omega = 0$, $\widehat{N}(0) = n_{1+}n_{2+}/n_{11}$, the Lincoln–Petersen estimator.

(d) In the regime of Definition 4.1, let $v_+ = \sum_{\boldsymbol{h} \in \mathcal{H}_+} \mu_{\boldsymbol{h}}^{-1}$ and $v = v_+ + \mu_{\mathbf{0}}^{-1} = \sum_{\boldsymbol{h} \in \mathcal{H}} \mu_{\boldsymbol{h}}^{-1}$. Then:

(i) under (P), $(\log \widehat{n}_{\mathbf{0}}(\omega) - \log \mu_{\mathbf{0}})/v_+^{1/2} \Rightarrow \text{N}(0,1)$, so that $\text{Var}\{\log \widehat{n}_{\mathbf{0}}(\omega)\} \approx \sum_{\boldsymbol{h} \in \mathcal{H}_+} \mu_{\boldsymbol{h}}^{-1}$; under (M) the same holds with v_+ replaced by $v_+ - N^{-1}$;

(ii) under (P) and under (M), $(\log \widehat{n}_{\mathbf{0}}(\omega) - \log n_{\mathbf{0}})/v^{1/2} \Rightarrow \text{N}(0,1)$. Equivalently, the prediction error $\widehat{N}(\omega) - N = \widehat{n}_{\mathbf{0}}(\omega) - n_{\mathbf{0}}$ has asymptotic variance $\mu_{\mathbf{0}}^2 v_+ + \mu_{\mathbf{0}}$;

(iii) $\widehat{v}_+ = \sum_{\boldsymbol{h} \in \mathcal{H}_+} n_{\boldsymbol{h}}^{-1}$ and $\widehat{v} = \widehat{v}_+ + \widehat{n}_{\mathbf{0}}(\omega)^{-1}$ satisfy $\widehat{v}_+/v_+ \to 1$ and $\widehat{v}/v \to 1$ in probability.

By Lemma 4.2, with ω fixed the model for the observed cells is the full positive orthant, so the likelihood is maximised cell by cell at $\widehat{\mu}_{\boldsymbol{h}} = n_{\boldsymbol{h}}$, and (3.10) carries this maximiser to $\mu_{\mathbf{0}}$. Part (d) holds because $\log \widehat{n}_{\mathbf{0}}$ is linear in the cell logarithms with coefficients $s_{\boldsymbol{h}} = \pm 1$ (Appendix A.3).

Theorem 4.4(d) gives Wald intervals on the log scale for $\mu_{\mathbf{0}}$ and for the realised total, whose lower end never falls below n (Remark A.4); for Table 3 the 95% interval for N is $[1141, 2121]$. Empty cells, the rule in small domains, are handled as in Remark A.5, and Remark A.6 shows that every hierarchical log-linear estimator is the alternating product $\mathcal{A}$ of its *fitted* cells (all three remarks are in Appendix A.10).

4.3 Sharp bounds and the direction of bias

Corollary 4.5 (Sharp sensitivity bounds). *Suppose Assumptions A1 to A3 hold and Assumption A4(b) holds with bound $\Omega \geq 0$.*

(a) Identified set. *Under (P), the set of values of $\mu_{\mathbf{0}}$ compatible with the law of the observed table and with $|\omega| \leq \Omega$ is*

$$\big[\, e^{-\Omega}\mathcal{A}(\boldsymbol{\mu}_+),\ e^{\Omega}\mathcal{A}(\boldsymbol{\mu}_+)\,\big].$$

Each point is attained by exactly one parameter value with $|\omega| \leq \Omega$, so the bounds are sharp. Under (M), the same holds for the odds $\pi_{\mathbf{0}}/(1-\pi_{\mathbf{0}})$ of being missed by every list, given the conditional law of $\boldsymbol{n}_+$ given n, with $\mathcal{A}(\boldsymbol{\pi}^c)$ in place of $\mathcal{A}(\boldsymbol{\mu}_+)$. By Theorem 4.3(b), the remaining binomial factor cannot consistently narrow this set.

(b) Estimated bounds. *If all $n_{\boldsymbol{h}} > 0$, then $\{\widehat{n}_{\mathbf{0}}(\omega) : |\omega| \leq \Omega\} = [e^{-\Omega}\widehat{n}_{\mathbf{0}}(0),\ e^{\Omega}\widehat{n}_{\mathbf{0}}(0)]$, and the bounds for N are $n + e^{\pm\Omega}\widehat{n}_{\mathbf{0}}(0)$.*

(c) Inference. *In the regime of Definition 4.1 under (P), for every $\omega \in [-\Omega, \Omega]$ the interval*

$$\big[\,\widehat{n}_{\mathbf{0}}(0)\, e^{-\Omega - c\,\widehat{v}_+^{1/2}},\ \widehat{n}_{\mathbf{0}}(0)\, e^{\Omega + c\,\widehat{v}_+^{1/2}}\,\big], \qquad \Phi\big(c + 2\Omega\,\widehat{v}_+^{-1/2}\big) - \Phi(-c) = 1 - \alpha, \tag{4.3}$$

contains $\mu_{\mathbf{0}}$ with probability tending to at least $1-\alpha$. Replacing c by $z_{1-\alpha/2}$ gives the union of the Wald intervals over $|\omega| \leq \Omega$, which is conservative.

Part (c) is the construction of Imbens and Manski (2004) (see also Manski, 2003). It applies cleanly because the pivot $\log \widehat{n}_{\mathbf{0}}(\omega) - \log \mu_{\mathbf{0}} = \sum_{\boldsymbol{h}} s_{\boldsymbol{h}} \log(n_{\boldsymbol{h}}/\mu_{\boldsymbol{h}})$ does not involve ω. The identified set has log-width 2Ω whatever K is: adding lists changes what the top interaction means, not the sensitivity to it. For Table 3 with $\Omega = \log 2$, $N \in [1120, 2200]$, and (4.3) gives $[212, 2449]$ for $\mu_{\mathbf{0}}$. By (3.9) with $k = 3$, $\Omega = \log 2$ says that the police–NGO odds ratio among incidents missed by the community security organisation is within a factor of two of the odds ratio among those it records.

Proposition 4.6 (Direction of bias). *Suppose Assumptions A1 to A3 hold and the true top interaction is ω, with oriented value $\bar{\omega} = (-1)^K\omega$, but the analyst imposes $\omega = 0$. In the regime of Definition 4.1, under (P) or (M),*

$$\frac{\widehat{n}_{\mathbf{0}}(0)}{n_{\mathbf{0}}} \to e^{-\bar{\omega}}, \qquad \frac{\widehat{N}(0)}{N} \to 1 - \pi_{\mathbf{0}}\big(1 - e^{-\bar{\omega}}\big) \qquad \textit{in probability.} \tag{4.4}$$

Hence $\widehat{n}_{\mathbf{0}}(0)$ asymptotically underestimates the dark figure if $\bar{\omega} > 0$ and overestimates it if $\bar{\omega} < 0$. For every list k, (3.9) gives $\bar{\omega} = (-1)^{K-1}\{\psi_k(0) - \psi_k(1)\}$: imposing $\omega = 0$ replaces the unobservable interaction $\psi_k(0)$ by the observable $\psi_k(1)$. In particular:

(a) For $K = 2$, $\bar{\omega} = \omega = \log \mathrm{OR}_{12}$. Lincoln–Petersen underestimates the dark figure when the lists are positively dependent ($\mathrm{OR}_{12} > 1$), for example when incidents recorded by police are more likely to reach the NGO audit, and overestimates it under negative dependence.

(b) For $K = 3$, $\bar{\omega} = -\omega = \log \mathrm{OR}_{12|h_3=0} - \log \mathrm{OR}_{12|h_3=1}$. The estimator underestimates if and only if the odds ratio between two lists is larger among incidents the third list misses than among incidents it records. Since ω is symmetric, this holds for every choice of the third list. Positive pairwise dependence alone does not determine the sign, because the $\omega = 0$ model estimates all pairwise interactions freely.

(c) For general K, a given sign of ω has opposite consequences at odd and even K. We therefore report $\bar{\omega}$ rather than ω: $e^{\bar{\omega}}$ is the factor by which the $\omega = 0$ estimate of the dark figure must be multiplied.

The direction in (a) is the classical warning about dependent lists (International Working Group for Disease Monitoring and Forecasting, 1995; Wolter, 1986). With three or more lists, however, "the lists are positively dependent" says nothing about the sign of the bias. Heterogeneous visibility is a natural source of dependence in incident data: violent or publicised incidents are likely to be recorded by many lists, inconspicuous ones by few. The next result shows that the shape of the visibility distribution governs the sign.

Proposition 4.7 (Heterogeneous visibility and the sign of ω). *Suppose that, under (M), each incident has a capture probability p_i drawn independently from a non-degenerate distribution F supported in $[\eta, 1-\eta]$ for some $\eta \in (0, \frac{1}{2})$. Suppose also that, given p_i, the K lists record incident i independently, each with probability p_i; this is model M_h of Otis et al. (1978). Let $X = \mathrm{logit}(p)$. For $t \in \mathbb{R}$, let G_t be the law of X when p has density proportional to $p^t(1-p)^{K-t}$ with respect to F, and let $\kappa_r(t)$ be the rth cumulant of G_t. Then*

$$\omega = \mathbb{E}\{\kappa_K(S_K)\}, \qquad S_K = U_1 + \cdots + U_K, \quad U_j \overset{\text{iid}}{\sim} \mathrm{Uniform}(0,1), \tag{4.5}$$

so S_K has the Irwin–Hall distribution. In particular:

(a) for $K = 2$, $\omega = \mathbb{E}\{\mathrm{Var}_{G_{S_2}}(X)\} > 0$, so Lincoln–Petersen underestimates the dark figure under every non-degenerate heterogeneity of this kind;

(b) for $K \geq 3$, the sign of ω is that of an average of Kth cumulants of the tilted laws G_t, and either sign can occur. By Proposition 4.6, the $\omega = 0$ estimator underestimates if and only if $(-1)^K \mathbb{E}\{\kappa_K(S_K)\} > 0$.

Part (a) recovers the classical result that heterogeneity biases Lincoln–Petersen downward (Chao et al., 2001; International Working Group for Disease Monitoring and Forecasting, 1995). For general K the sign is governed by higher cumulants: skewness of the tilted logit-visibility distribution at $K = 3$, and kurtosis at $K = 4$. The bias can be large and of either sign. If 5% of incidents are highly visible ($p = 0.9$) and the rest are not ($p = 0.1$), Lincoln–Petersen recovers a fifth of the dark figure, whereas with $K = 3$ the $\omega = 0$ model, which has all two-way interactions, overstates it 12.5-fold (Example A.7). "Heterogeneity" fixes neither the sign nor the size of the bias, which is why Section 6 models it directly through latent classes.

4.4 Chao's bound and its reversal

Chao's estimator is often described as robust to heterogeneity. It is a lower bound under one kind of heterogeneity, across incidents, and it can fail in the other direction under heterogeneity across lists.

Theorem 4.8 (Chao's lower bound under M_h; Chao, 1987). *Suppose that, conditionally on $p_1, \dots, p_N \in [0,1]$, the indicators Y_{ik} are independent with $\mathbb{P}(Y_{ik} = 1) = p_i$ for every list k (model M_h). Write $\mathbb{E}$ for expectation conditional on $\boldsymbol{p}$, and suppose $\mathbb{E}f_2 > 0$. Then*

$$\mathbb{E}f_0 \;\geq\; \frac{K-1}{K}\,\frac{(\mathbb{E}f_1)^2}{2\,\mathbb{E}f_2}, \qquad \textit{equivalently} \qquad N \;\geq\; \mathbb{E}n + \frac{K-1}{K}\,\frac{(\mathbb{E}f_1)^2}{2\,\mathbb{E}f_2}. \tag{4.6}$$

If every $p_i \in (0,1)$, equality holds if and only if $p_1 = \cdots = p_N$. If, in addition, the p_i are i.i.d. from a distribution F on $(0,1)$, then the estimator

$$\widehat{N}_{\mathrm{C}} = n + \frac{K-1}{K}\,\frac{f_1^2}{2f_2}$$

satisfies $\widehat{N}_{\mathrm{C}}/N \to \lambda_{\mathrm{C}} \leq 1$ almost surely as $N \to \infty$, with $\lambda_{\mathrm{C}} = 1$ if and only if F is a point mass.

The bound uses only the singletons f_1 and doubletons f_2, the incidents that are hardest to record, and its proof is a single application of Cauchy–Schwarz to $\sum_i p_i(1-p_i)^{K-1}$. It holds whatever the distribution of the incident-level p_i, but only under M_h, which requires the lists to be exchangeable for each incident. That is implausible when a police list and a community portal differ greatly in reach, and heterogeneity *across lists* pushes Chao's estimator in the opposite direction.

Proposition 4.9 (List heterogeneity reverses Chao's inequality). *Suppose instead that the Y_{ik} are independent with $\mathbb{P}(Y_{ik} = 1) = a_k \in (0,1)$ for every incident (model M_t). Then*

$$\mathbb{E}f_0 \;\leq\; \frac{K-1}{K}\,\frac{(\mathbb{E}f_1)^2}{2\,\mathbb{E}f_2},$$

with equality if and only if $a_1 = \cdots = a_K$. Consequently, $\widehat{N}_{\mathrm{C}}/N$ converges almost surely to a limit ≥ 1: under M_t, Chao's estimator is an asymptotic upper *bound.*

Under M_t the independence model of Darroch (1958) holds, so $\omega = 0$ and Theorem 4.4 is consistent. When incidents and lists are both heterogeneous, the two effects compete and $\widehat{N}_{\mathrm{C}}$ can err in either direction. Chao's estimator is therefore a lower bound under M_h, not under heterogeneity in general, and we treat it as a diagnostic. The only floor that needs no assumption is the observed count n. For Table 3, $\widehat{N}_{\mathrm{C}} = 760 + \frac{2}{3}\cdot 540^2/380 = 1272$, below the $\omega = 0$ estimate of 1480. Since the three illustrative lists differ in size, Proposition 4.9 warns against reading 1272 as a hard floor.

4.5 Referral dependence

An NGO audit may ingest incidents that were first reported to police, for example through public police statements, data-sharing agreements, or victims who report to both and say so. This creates a specific and directional form of dependence, which we model explicitly.

Proposition 4.10 (Referral dependence). *Let $K = 2$ (police $k = 1$, NGO audit $k = 2$), and let incidents be i.i.d. as follows. Police record an incident with probability $p_1 \in (0,1)$, so $Y_1 \sim \mathrm{Bernoulli}(p_1)$. The NGO captures it directly (through a victim report, media monitoring or outreach) with probability $p_2 \in (0,1)$, so $D \sim \mathrm{Bernoulli}(p_2)$. If police record it, the NGO also ingests the police record with probability $q \in [0,1)$, so $R \sim \mathrm{Bernoulli}(q)$. The variables Y_1, D and R are independent, and $Y_2 = \max(D, RY_1)$.*

(a) Induced dependence. *The cell probabilities are* $\pi_{11} = p_1\{p_2 + q(1-p_2)\}$, $\pi_{10} = p_1(1-p_2)(1-q)$, $\pi_{01} = (1-p_1)p_2$ *and* $\pi_{00} = (1-p_1)(1-p_2)$. *Hence*

$$\mathrm{OR}_{12} = \frac{\pi_{11}\pi_{00}}{\pi_{10}\pi_{01}} = 1 + \frac{q}{p_2(1-q)}, \qquad u_{12} = \omega = \log\left\{1 + \frac{q}{p_2(1-q)}\right\} \geq 0, \tag{4.7}$$

with equality if and only if $q = 0$. *The odds ratio does not depend on* p_1. *It increases in* q *and decreases in* p_2.

(b) Bias of Lincoln–Petersen. *As* $N \to \infty$, *almost surely,*

$$\frac{\widehat{N}_{\mathrm{LP}}}{N} \to \frac{p_2 + p_1 q(1-p_2)}{p_2 + q(1-p_2)} = 1 - \frac{q(1-p_1)(1-p_2)}{p_2 + q(1-p_2)}, \qquad \frac{\widehat{n}_{\mathbf{0}}^{\mathrm{LP}}}{n_{\mathbf{0}}} \to \frac{1}{\mathrm{OR}_{12}} = \frac{p_2(1-q)}{p_2 + q(1-p_2)}. \tag{4.8}$$

(c) Full provenance. *Suppose each NGO record shows whether the NGO captured the incident directly* ($D = 1$), *whether or not it also ingested a police record. Recode* $Y_2^\circ = D$. *This drops referral-only captures* ($Y_1 = 1$, $R = 1$, $D = 0$) *from list* 2 *and moves them from cell* 11 *to cell* 10. *Then* Y_1 *and* Y_2° *are independent, the recoded table has* $\omega = 0$, *and* $\widehat{N}^\circ = n_{1+}n_{2+}^\circ/n_{11}^\circ$ *satisfies* $\widehat{N}^\circ/N \to 1$ *almost surely. More generally, for any* K: *if the direct-capture histories* $(Y_1, D, Y_3, \ldots, Y_K)$ *satisfy Assumption A4(a), so does the recoded table.*

(d) Referral flags only. *Suppose only the flag* $Z = RY_1$ (*"this NGO record was ingested from police"*) *is observed, and direct capture of flagged incidents is not. Let* n_Z *be the number of flagged incidents,* m_{11} *the number of unflagged incidents recorded by both lists, and* n_{01} *the number recorded by the NGO only. Remove the flagged incidents, apply Lincoln–Petersen to the rest, and add them back:*

$$\widehat{N}^{\ddagger} = n_Z + \frac{(n_{1+} - n_Z)(m_{11} + n_{01})}{m_{11}}.$$

The reduced table has odds ratio one, and $\widehat{N}^{\ddagger}/N \to 1$ *almost surely.*

(e) A third, independent list. *Let* $K \geq 3$. *Suppose* (Y_1, Y_2) *is generated as above and* $(Y_3, \ldots, Y_K)$ *is independent of* (Y_1, Y_2). *Then* $\omega = 0$; *indeed* $u_S = 0$ *whenever* S *meets both* $\{1, 2\}$ *and* $\{3, \ldots, K\}$. *Hence* $\widehat{n}_{\mathbf{0}}(0)$ *is consistent: referral between two lists does not bias the* K*-list estimator.*

The dark-figure bias in (4.8) depends only on (p_2, q). If the NGO captures 20% of incidents directly and ingests 30% of police records, the odds ratio is $1 + 0.3/(0.2 \cdot 0.7) \approx 3.14$. Lincoln–Petersen then recovers only about 32% of the dark figure, whatever the police recording rate. Parts (c) and (d) turn a metadata field into an identification strategy, which is a reason to request provenance information from every list holder (Section 9). Part (e) shows that a third list unconnected to the referral channel neutralises it in the model without a K-way term. For $K = 2$, referral produces only $\omega \geq 0$, and bounds on the ingestion rate and the direct coverage turn (4.7) into a one-sided bound on ω that list holders can speak to (Remark A.9). With three or more lists referral can bias the estimator either way (Example A.10), so $\bar{\omega}$ must be calibrated from an explicit model of the referral channels, not assumed to have a sign.

4.6 The sensitivity frontier of a policy claim

Policy discussion of hate crime can turn on qualitative claims, such as "most incidents never reach the police", rather than on point estimates. We ask how large a departure from $\omega = 0$ such a claim can survive. For $\kappa > 0$, consider the claim

$$\mathsf{C}_\kappa : \quad \mu_{\mathbf{0}} > \kappa\, \mu_{1+},$$

where $\mu_{1+} = \mathbb{E} n_{1+}$. In words, the expected number of incidents recorded by no list exceeds κ times the expected police count. For $\kappa = 1$, the dark figure exceeds the police count.

Theorem 4.11 (Sensitivity frontier). *Define the population frontier and its estimate*

$$\Omega^*_\kappa = \log \frac{\mathcal{A}(\boldsymbol{\mu}_+)}{\kappa\, \mu_{1+}}, \qquad \widehat{\Omega}_{\kappa,\alpha} = \log \frac{\widehat{n}_{\mathbf{0}}(0)}{\kappa\, n_{1+}} - z_{1-\alpha}\, \widehat{s}_1, \qquad \widehat{s}_1^2 = \sum_{\boldsymbol{h} \in \mathcal{H}_+} \frac{1}{n_{\boldsymbol{h}}} + \frac{1}{n_{1+}}.$$

(a) Under (P), Assumptions A1 to A3 and Assumption A4(b) with bound Ω: C_κ holds for every parameter value that is observationally equivalent to the truth and has $|\omega| \le \Omega$ if and only if $\Omega < \Omega^*_\kappa$.

(b) In the regime of Definition 4.1, under (P) or (M), for every $\Omega \ge 0$ and every parameter value with $|\omega| \le \Omega$ at which C_κ fails ($\mu_{\mathbf{0}} \le \kappa \mu_{1+}$),

$$\limsup\ \mathbb{P}\big(\widehat{\Omega}_{\kappa,\alpha} > \Omega\big) \le \alpha,$$

with equality when $\mu_{\mathbf{0}} = \kappa \mu_{1+}$ and $\bar{\omega} = -\Omega$.

(c) If $\Omega < \Omega^*_\kappa$, then $\mathbb{P}(\widehat{\Omega}_{\kappa,\alpha} > \Omega) \to 1$.

Declaring "C_κ is robust to $|\omega| \le \Omega$" whenever $\Omega < \widehat{\Omega}_{\kappa,\alpha}$ is therefore an asymptotically level-α test of the negation of C_κ, valid whatever the unidentified $\omega \in [-\Omega, \Omega]$ is, and consistent. The statistic $\widehat{\Omega}_{\kappa,\alpha}$ is a breakdown point in the sense of Masten and Poirier (2020), the multiple-systems analogue of the E-value (VanderWeele and Ding, 2017) and of Rosenbaum's sensitivity parameter (Rosenbaum, 2002). Any list may replace list 1. Gerritse et al. (2015) observe that a population-size estimate is less robust when implied coverage is low; in our parameterisation this is exact, $\partial \log \widehat{N}(\omega)/\partial\bar{\omega} = \widehat{n}_{\mathbf{0}}(\omega)/\widehat{N}(\omega)$. Claims about the dark figure itself behave the other way: the larger the implied dark figure relative to the police count, the larger Ω^*_κ. For Table 3, the claim that the dark figure exceeds the police count has plug-in frontier 0.59 but is barely robust once sampling error is included ($\widehat{\Omega}_{1,0.05} = 0.05$; Example A.11).

5 Record linkage and its propagation into the dark figure

Multiple-systems estimation treats the capture table $\boldsymbol{n}_+$ as data. In our application it is not data. Police, NGO, community-security and campus records share no identifier, so the table is *constructed*: an analyst, or an algorithm, decides which records describe the same incident, and every such decision moves an incident between cells. This section replaces Assumption A2 by a model of linkage error. Section 5.1 sets up the record model, and Section 5.2 explains why we need a posterior distribution over linkage structures rather than a single linkage. Section 5.3 contains the main results. It represents linkage error as an operator on capture tables, derives the bias it induces in MSE estimators (Theorem 5.5), inverts it (Corollary 5.8), shows where that inversion fails (Remark 5.9), and decomposes the uncertainty that a single linkage hides (Proposition 5.11). Section 5.4 designs clerical review (Theorem C.10) and an audit of error rates, and

Section 5.5 points to privacy-preserving linkage. Supplementary results, remarks and implementation details are in Appendix C, and proofs in Appendix B.

Throughout this section we index cells by supports. For non-empty $h \subseteq [K]$ we write n_h for $n_{\boldsymbol{h}_h}$, so that $g \subseteq h$, $h \setminus k$ and $g \cup k$ have their set meanings, and $\{k\}$ is the history of an incident recorded by list k alone. A *table* is a vector $\boldsymbol{x} = (x_h)_{h \in \mathcal{H}_+}$, and $x_{k+} = \sum_{h \ni k} x_h$ is its k-th list margin. We write $\boldsymbol{e}_h$ for the unit vector of cell h.

5.1 Records, attributes and linkage structures

After within-list deduplication, list k delivers a set R_k of records. Record r carries an attribute vector a_r (dates, location, offence type, target and venue). The attributes are noisy and heterogeneous across lists: dates agree only up to a reporting lag that depends on the pair of lists, locations are recorded at different granularities, and offence vocabularies correspond through a compatibility relation, not a bijection (Appendix C.1). We assume that within-list duplicates have been removed before cross-list linkage, or jointly with it (Steorts et al., 2016), so that an incident has at most one record in each list; residual errors are discussed in Remark C.8.

Definition 5.1 (Linkage structure and induced table). Let $R = \bigsqcup_k R_k$. A *linkage structure* is a partition Λ of R whose blocks contain at most one record from each list. A block b has support $\mathrm{supp}(b) = \{k : b \cap R_k \neq \emptyset\} \in \mathcal{H}_+$. The *induced table* is $n_h(\Lambda) = \#\{b \in \Lambda : \mathrm{supp}(b) = h\}$. The true structure $\Lambda^\star$ gives the true table $\boldsymbol{t} = \boldsymbol{n}(\Lambda^\star)$, and an estimated structure $\widehat{\Lambda}$ gives the observed table $\boldsymbol{n} = \boldsymbol{n}(\widehat{\Lambda})$.

Every record lies in exactly one block, and a block contains at most one record per list. Hence

$$n_{k+}(\Lambda) = |R_k| \qquad \text{for every linkage structure } \Lambda \text{ and every } k \in [K]. \tag{5.1}$$

Linkage therefore never changes the list sizes. It redistributes counts among the $2^K - 1$ cells within the affine subspace fixed by the K margins. All the information about N lies in the overlaps between lists, and the overlaps are exactly what linkage determines.

5.2 From Fellegi–Sunter to Bayesian record linkage

Comparison vectors and the Fellegi–Sunter rule. For a candidate pair $\ell = (r, r')$ of records from different lists, the comparison vector $\gamma_\ell = (\gamma_\ell^1, \ldots, \gamma_\ell^F) \in \Gamma$ records a discrete level of agreement on each of F fields (Appendix C.1); a missing value is a separate level or is omitted. Let M be the set of true matches (pairs in the same block of $\Lambda^\star$) and U the set of non-matches. Define

$$m(\gamma) = \mathbb{P}(\gamma_\ell = \gamma \mid \ell \in \mathrm{M}), \qquad u(\gamma) = \mathbb{P}(\gamma_\ell = \gamma \mid \ell \in \mathrm{U}), \qquad w(\gamma) = \log \frac{m(\gamma)}{u(\gamma)}.$$

Fellegi and Sunter (1969) proposed to link a pair when $w(\gamma_\ell)$ exceeds an upper threshold, not to link it when $w(\gamma_\ell)$ falls below a lower threshold, and to send it to clerical review otherwise. Among all rules with no larger probabilities of false and missed links, this rule minimises the probability of review (Proposition C.1 in Appendix C.2). That optimality concerns one pair at a time, not the coherence of many decisions.

Blocking. Comparing all $\sum_{k<k'} |R_k||R_{k'}|$ cross-list pairs is wasteful: almost all are non-matches, and they swamp the estimation of m. Blocking restricts comparison to pairs that agree on a coarse key, such as state and a window of incident dates, possibly over several passes (Christen, 2012; Herzog et al., 2007). A true match that falls in no block can never be linked, so blocking recall is a component of the false non-match rate δ of Section 5.3, and the audit of Section 5.4 must be able to detect such misses.

EM under conditional independence. Without training data, m and u are estimated from the candidate pairs as a two-component mixture under conditional independence of the fields within each component; the EM algorithm then has closed-form steps (Algorithm 2 in Appendix C.2; Enamorado et al., 2019; Jaro, 1989; Winkler, 1988) and returns pairwise match probabilities π_ℓ and weights w_ℓ. Conditional independence fails when two distinct incidents share a venue, a county and an offence type, as happens with repeated attacks on one synagogue; a third mixture component for such "related non-matches" relaxes it (Larsen and Rubin, 2001), and the simulation of Section 8 uses it.

Why pairwise decisions are not enough. Thresholding the pairwise weights w_ℓ or probabilities π_ℓ has three defects. The decisions need not form a linkage structure: a record may be linked to two records of the same list, and links need not be transitive (Jaro, 1989; Sadinle and Fienberg, 2013). A single linkage discards uncertainty, and because MSE estimators are non-linear in the table, thresholding even well-calibrated probabilities gives biased estimates (Theorem 5.5) and overconfident intervals (Proposition 5.11). And with K lists, the evidence that a police record and a campus record match may run through an NGO record that matches both, which pairwise decisions cannot use (Sadinle and Fienberg, 2013).

Bayesian multipartite matching. The Bayesian approach treats Λ as an unknown parameter. In the likelihood (C.1), the candidate pairs that Λ places in one block, the set $\mathrm{M}(\Lambda)$, contribute $m(\gamma_\ell)$ and all other candidate pairs contribute $u(\gamma_\ell)$. Samplers over linkage structures that respect the one-to-one constraint are available for two and for several lists (Sadinle, 2017; Sadinle and Fienberg, 2013; Steorts et al., 2016), and Appendix C.2 reviews this literature. The structures of Definition 5.1 are exactly the parameter space of these models.

A posterior over Λ is what MSE needs, because the estimand depends on Λ only through $\boldsymbol{n}(\Lambda)$. The posterior induces a posterior over tables, on a set of dimension $2^K - 1 - K$ once the margins (5.1) are fixed, and population-size inference can be carried out for each table draw and averaged. This is linkage-averaging (Sadinle, 2018), which Section 6.4 uses. The posterior also supplies match probabilities $\pi_\ell = \mathbb{P}(\ell \in \mathrm{M}(\Lambda) \mid \text{data})$, which, unlike the pairwise probabilities of FS-EM, respect the one-to-one constraint; Section 5.4 uses them to target review. Section 5.3 asks what goes wrong when a single $\widehat{\Lambda}$ is plugged in, and when that can be undone.

5.3 Linkage error as an operator on capture tables

Definition 5.2 (Canonical linkage-error model, LE). Let $\delta, \varepsilon \in [0, 1)$. Starting from $\Lambda^\star$, the estimated structure $\widehat{\Lambda}$ is produced in two steps.

(LE-D) *Detachment (false non-matches).* Each record detaches independently with probability δ. Within each block of $\Lambda^\star$, the records that do not detach stay together as one cluster (if there are any), and each detached record forms a singleton cluster. Call the result $\tilde{\Lambda}$ and write $\tilde{\boldsymbol{n}} = \boldsymbol{n}(\tilde{\Lambda})$.

(LE-M) *Merging (false matches).* Each singleton cluster of $\tilde{\Lambda}$ is selected independently with probability ε. The selected singletons are processed in a uniformly random order. When a selected singleton from list k is processed, and is still a singleton, it merges with a cluster drawn uniformly from the current clusters whose support excludes k; if there is none, nothing happens. A selected singleton that has absorbed another singleton before its turn is no longer a singleton and does not merge. The result is $\widehat{\Lambda}$, and $\boldsymbol{n} = \boldsymbol{n}(\widehat{\Lambda})$.

LE is a stylised forward model: it is the model that Corollary 5.8 inverts, not a description of how linkage errors arise. In particular, it is not how the simulation of Section 8 generates errors. There, records

carry realistic attribute noise and are linked by FS-EM (Algorithm 2); LE enters only through the correction, which is thereby tested under misspecification, and Remark 5.9 identifies the kind of error that LE omits. Both steps map linkage structures to linkage structures, so (5.1) holds along the whole path: $n_{k+} = \tilde{n}_{k+} = t_{k+}$. The results below extend verbatim to list-specific rates $(\delta_k, \varepsilon_k)$ (Remark C.7).

Two kinds of elementary move generate all the errors. For $k \in h$ with $|h| \geq 2$, the *split vector*

$$\boldsymbol{d}_{h,k} = \boldsymbol{e}_{h\setminus k} + \boldsymbol{e}_{\{k\}} - \boldsymbol{e}_h$$

is the change in the table when the list-k record splits off an incident with support h. A merge of a list-k singleton into a cluster with support $g \not\ni k$ changes the table by $-\boldsymbol{d}_{g\cup k,k}$: it is a split in reverse. Every split vector has zero list margins. For a table $\boldsymbol{x}$ let

$$C_k(\boldsymbol{x}) = \sum_{g \not\ni k} x_g$$

be the number of clusters into which a list-k singleton can merge. Here and below, sums over g run over non-empty $g \subseteq [K]$.

Proposition 5.3 (The detachment operator). *Under (LE-D), $\mathbb{E}[\tilde{\boldsymbol{n}} \mid \Lambda^\star] = \mathcal{D}_\delta \boldsymbol{t}$, where $\mathcal{D}_\delta$ is the linear map with matrix*

$$(\mathcal{D}_\delta)_{gh} = \mathbf{1}\{g \subseteq h\}\,(1-\delta)^{|g|}\delta^{|h|-|g|} + \delta\,\mathbf{1}\{g = \{k\} \text{ for some } k \in h\}, \qquad g, h \in \mathcal{H}_+. \tag{5.2}$$

(i) $\mathcal{D}_\delta$ is upper triangular on the Boolean lattice: $(\mathcal{D}_\delta)_{gh} = 0$ unless $g \subseteq h$. Its diagonal entries are 1 if $|g| = 1$ and $(1-\delta)^{|g|}$ if $|g| \geq 2$, so it is invertible for $\delta < 1$.

(ii) $\mathcal{D}_\delta$ preserves list margins, $(\mathcal{D}_\delta \boldsymbol{x})_{k+} = x_{k+}$. Column h sums to $1 - \delta^{|h|} + |h|\delta$, the expected number of clusters produced by an incident with support h.

(iii) If $\boldsymbol{y} = \mathcal{D}_\delta \boldsymbol{x}$, then

$$x_h = \sum_{g \supseteq h} (-\delta)^{|g|-|h|}(1-\delta)^{-|g|}\, y_g \qquad (|h| \geq 2), \tag{5.3}$$

$$x_{\{k\}} = y_{k+} - \sum_{h \ni k,\ |h|\geq 2} x_h = y_{\{k\}} + \sum_{g \ni k,\ |g|\geq 2} \Big[\frac{(-\delta)^{|g|-1}}{(1-\delta)^{|g|}} - \frac{\delta}{1-\delta}\Big] y_g. \tag{5.4}$$

(iv) For lists $j \neq k$, the number of observed clusters that contain records of both lists, $\tilde{n}_{jk} = \sum_{h \supseteq \{j,k\}} \tilde{n}_h$, is distributed as Binomial$(t_{jk}, (1-\delta)^2)$, *where $t_{jk} = \sum_{h \supseteq \{j,k\}} t_h$.*

(v) $\tilde{\boldsymbol{n}} - \mathcal{D}_\delta \boldsymbol{t}$ is a sum of independent, mean-zero vectors, one for each true incident, and $\mathbb{E}\|\tilde{\boldsymbol{n}} - \mathcal{D}_\delta \boldsymbol{t}\|_2^2 \leq K \sum_h t_h$.

Part (iii) is a weighted Möbius inversion on the Boolean lattice (Appendix C.3). The factor $(1-\delta)^{-|g|}$ in (5.3) restores clusters that lost a record, the alternating sum removes fragments of incidents with larger support, and the singleton cells follow from the margins. For $K = 2$, $t_{11} = \tilde{n}_{11}/(1-\delta)^2$ and $t_{\{k\}} = \tilde{n}_{\{k\}} - \delta(2-\delta)\tilde{n}_{11}/(1-\delta)^2$.

Merging is not linear, but its first-order effect is explicit. Define the *merge drift*

$$\boldsymbol{q}(\boldsymbol{x}) = -\sum_{k \in [K]} \sum_{g \not\ni k} \frac{x_{\{k\}}\, x_g}{C_k(\boldsymbol{x})}\, \boldsymbol{d}_{g\cup k,k}. \tag{5.5}$$

The singleton cell $\{k\}$ loses $\varepsilon x_{\{k\}}$, and each cell $g \not\ni k$ sends $\varepsilon x_{\{k\}} x_g / C_k(\boldsymbol{x})$ to $g \cup k$. By Lemma C.2 (Appendix C.3), (LE-M) moves any table $\tilde{\boldsymbol{n}}$ by $\varepsilon \boldsymbol{q}(\tilde{\boldsymbol{n}})$, up to a mean-zero fluctuation and a remainder of order $\varepsilon^2 A^2 / C_{\min}$, where A is the number of singletons and $C_{\min} = \min_k C_k(\tilde{\boldsymbol{n}})$. This motivates the following definition.

Definition 5.4 (Linkage operator). The linkage operator is $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}) = \mathcal{D}_\delta \boldsymbol{x} + \varepsilon\, \boldsymbol{q}(\mathcal{D}_\delta \boldsymbol{x})$.

$\mathcal{T}_{\delta,0} = \mathcal{D}_\delta$ is exact. $\mathcal{T}_{\delta,\varepsilon}$ is positively homogeneous of degree one and preserves list margins, and it is the forward map that Corollary 5.8 inverts. Asymptotic statements refer to the following regime, which conditions on the true table and so covers both sampling models of Definition 3.1.

Assumption A5 (Large-population regime for linkage). There is a sequence of populations, indexed by $N \to \infty$, with true tables $\boldsymbol{t} = \boldsymbol{t}^{(N)}$ such that $\boldsymbol{\tau}_N := \boldsymbol{t}/N \to \boldsymbol{\tau}$ with $\tau_h > 0$ for every $h \in \mathcal{H}_+$. The rates (δ, ε) do not depend on N.

We write $R_N = O_p^{\mathrm{u}}(r)$ when for every $\nu > 0$ there are constants c_ν and $r_0 > 0$ such that $\mathbb{P}(|R_N| > c_\nu r) \le \nu$ for all large N and all $(\delta, \varepsilon) \in [0, r_0]^2$. The remainder is small in probability, uniformly in small error rates. The results below rest on a split–merge calculus (Lemma C.3 in Appendix C.3). To first order, LE changes a smooth estimator that is homogeneous of degree one by the sum of its derivatives along the elementary moves, weighted by δ per record for splits and by ε per singleton for merges.

Theorem 5.5 (Linkage bias of multiple-systems estimators). *Assume LE.*

(i) Two lists, every realisation. *Let $K = 2$. Then $\widehat{N}_{\mathrm{LP}}(\boldsymbol{n}) = \widehat{N}_{\mathrm{LP}}(\boldsymbol{t})\, t_{11}/n_{11}$, and*

$$\widehat{n}_{\boldsymbol{0}}(\boldsymbol{n}) = \frac{n_{10} n_{01}}{n_{11}} = \frac{(t_{1+} - n_{11})(t_{2+} - n_{11})}{n_{11}}$$

is strictly decreasing in n_{11}. Detachment can only decrease n_{11} and merging can only increase it. Consequently, if $\varepsilon = 0$ then every realisation has $\widehat{N}_{\mathrm{LP}}(\boldsymbol{n}) \ge \widehat{N}_{\mathrm{LP}}(\boldsymbol{t})$ and $\widehat{n}_{\boldsymbol{0}}(\boldsymbol{n}) \ge \widehat{n}_{\boldsymbol{0}}(\boldsymbol{t})$; if $\delta = 0$ both inequalities are reversed.

(ii) Two lists, magnitude. *Let $\hat{p}_1 = t_{11}/t_{2+}$ and $\hat{p}_2 = t_{11}/t_{1+}$ be the perfect-linkage Lincoln–Petersen capture probabilities. Under Assumption A5,*

$$\begin{aligned}\frac{\widehat{N}_{\mathrm{LP}}(\boldsymbol{n})}{\widehat{N}_{\mathrm{LP}}(\boldsymbol{t})} &= \Big[(1-\delta)^2(1-2\varepsilon) + \varepsilon\Big(\frac{1}{\hat{p}_1} + \frac{1}{\hat{p}_2}\Big)\Big]^{-1} + O_p^{\mathrm{u}}\big(\varepsilon^2 + N^{-1/2}\big) \\ &= 1 + 2\delta - \varepsilon\Big(\frac{1}{\hat{p}_1} + \frac{1}{\hat{p}_2} - 2\Big) + O_p^{\mathrm{u}}\big((\delta+\varepsilon)^2 + N^{-1/2}\big). \end{aligned} \tag{5.6}$$

If $\varepsilon = 0$ the result is exact in distribution: $n_{11} \sim \mathrm{Binomial}(t_{11}, (1-\delta)^2)$. For Lincoln–Petersen applied to two lists $j \ne k$ of a K-list system, the identity $\widehat{N}^{(jk)}_{\mathrm{LP}}(\boldsymbol{n}) = \widehat{N}^{(jk)}_{\mathrm{LP}}(\boldsymbol{t})\, t_{jk}/n_{jk}$ holds for every realisation, and if $\varepsilon = 0$ then $n_{jk} \sim \mathrm{Binomial}(t_{jk}, (1-\delta)^2)$.

(iii) K lists. *For the closed-form estimator $\widehat{n}_{\boldsymbol{0}}(\omega)$ of Theorem 4.4 and under Assumption A5,*

$$\frac{\widehat{n}_{\boldsymbol{0}}(\omega; \boldsymbol{n})}{\widehat{n}_{\boldsymbol{0}}(\omega; \boldsymbol{t})} = 1 + \delta\, \beta_\delta(\boldsymbol{t}) + \varepsilon\, \beta_\varepsilon(\boldsymbol{t}) + O_p^{\mathrm{u}}\big((\delta+\varepsilon)^2 + N^{-1/2}\big), \tag{5.7}$$

with

$$\beta_\delta(\boldsymbol{t}) = K + \sum_{k \in [K]} \big\{a_k(\boldsymbol{t}) + b_k(\boldsymbol{t})\big\}, \qquad \beta_\varepsilon(\boldsymbol{t}) = -K - \sum_{k \in [K]} \frac{t_{\{k\}}}{C_k(\boldsymbol{t})}\big\{1 + \bar{a}_k(\boldsymbol{t})\big\}, \tag{5.8}$$

$$a_k(\boldsymbol{t}) = \sum_{g \not\ni k} (-1)^{|g|+1} \frac{t_{g \cup k}}{t_g}, \qquad \bar{a}_k(\boldsymbol{t}) = \sum_{g \not\ni k} (-1)^{|g|+1} \frac{t_g}{t_{g \cup k}}, \qquad b_k(\boldsymbol{t}) = \frac{t_{k+}}{t_{\{k\}}} - 1.$$

The coefficients are scale-free, so $\beta(\boldsymbol{t}) = \beta(\boldsymbol{\tau}_N)$, and they do not depend on ω.

(iv) Direction. *Say that a positive table $\boldsymbol{x}$ satisfies condition (H) if*

$$a_k(\boldsymbol{x}) \geq 0 \quad \textit{and} \quad \bar{a}_k(\boldsymbol{x}) \geq -1 \qquad \textit{for every } k \in [K]. \tag{H}$$

If $\boldsymbol{t}$ satisfies (H), then $\beta_\delta(\boldsymbol{t}) \geq K$ and $\beta_\varepsilon(\boldsymbol{t}) \leq -K$: to first order, false non-matches inflate the dark figure by at least $K\delta$ and false matches deflate it by at least $K\varepsilon$. If $\boldsymbol{\tau}$ satisfies (H) with strict inequalities, then so does $\boldsymbol{t}$ for all large N. Every positive table satisfies (H) when $K = 2$. For general K, $\boldsymbol{\tau}$ satisfies it strictly whenever, for every k, the conditional odds of capture by list k factorise as $\tau_{g\cup k}/\tau_g = c_k \prod_{j\in g} \phi_{jk}$ $(g \not\ni k)$ with pairwise factors $\phi_{jk} \in (\frac{1}{2}, 2)$ (Lemma B.1). This includes independent capture, where $\phi_{jk} \equiv 1$.

Condition (H) involves only observed cells and can be checked directly on the corrected table $\hat{\boldsymbol{t}}$ of Corollary 5.8. Remark C.4 gives the coefficients for $K = 2$ and shows that for $K = 3$ every positive table has the factorised form required in (iv).

Corollary 5.6 (Independent capture). *Suppose $\tau_h \propto \prod_{k\in h} p_k \prod_{k\notin h}(1 - p_k)$ with $p_k \in (0, 1)$. Let $\theta_k = p_k/(1 - p_k)$, and let $q_{-k} = \prod_{j\neq k}(1 - p_j)$ be the probability of being missed by every list other than k. Then $\beta(\boldsymbol{t}) \to \beta(\boldsymbol{\tau})$, where*

$$\beta_\delta(\boldsymbol{\tau}) = \sum_k \theta_k + \sum_k \frac{1}{q_{-k}}, \qquad \beta_\varepsilon(\boldsymbol{\tau}) = -K - \sum_k \frac{q_{-k}}{(1 - p_k)(1 - q_{-k})},$$

$$\beta_\delta(\boldsymbol{\tau}) + \beta_\varepsilon(\boldsymbol{\tau}) = \sum_k \frac{(1 - 2q_{-k})(\theta_k + q_{-k}^{-1})}{1 - q_{-k}}.$$

Hence missed links have the larger first-order effect, $\beta_\delta > |\beta_\varepsilon|$, if $q_{-k} < \frac{1}{2}$ for every k, and false matches have the larger effect if $q_{-k} > \frac{1}{2}$ for every k. With equal capture probabilities p the crossover is at $p^* = 1 - 2^{-1/(K-1)}$, that is 0.5, 0.293 and 0.206 for $K = 2, 3, 4$. As $\max_k p_k \to 0$, $\beta_\delta \to K$, while $\beta_\varepsilon = -\{1 + o(1)\} \sum_k \big(\sum_{j\neq k} p_j\big)^{-1} \to -\infty$.

Corollary 5.6 ranks the two error types, and the ranking is proved only under independent capture. There, sparsity decides. When lists capture small fractions of incidents, singletons outnumber overlaps by a factor of order $1/p$, so $|\beta_\varepsilon|$ is of order $1/p$ while β_δ stays close to K, and false matches matter more; dense lists reverse the ranking. For dependent lists neither the ranking nor even the signs of the coefficients are guaranteed (Remark 5.7), so no uniform rule such as "err towards not linking" is justified. We compute β_δ and β_ε from the corrected table $\hat{\boldsymbol{t}}$ of Corollary 5.8 instead. Remark C.5 gives numerical examples and the range in which the linear expansion (5.7) is informative.

Remark 5.7 (The direction can reverse for $K \geq 3$). Condition (H) cannot simply be dropped. Take $K = 3$ and $t_{\{k\}} = 10m$, $t_{\{j,k\}} = m$ and $t_{\{1,2,3\}} = 10m$ for $m > 0$. The conditional odds ratios equal 100, and (5.8) gives $a_k = -9.8$, $b_k = 1.2$ and $\beta_\delta = -22.8$. Detachment then *deflates* the closed-form dark figure. Splitting an incident recorded by all three lists moves mass into the sparse pairwise cells, which enter $\mathcal{A}$ in the denominator. Such a table is far from the no-three-way model, and its dark-figure estimate is fragile anyway, but the example shows that "false non-matches inflate" is a theorem about $K = 2$ and about tables with moderate dependence, not a universal law.

For $K = 2$, LE shares the affine structure of the dual-system model with matching error of Ding and Fienberg (1994), and like Ding and Fienberg (1996) and Di Consiglio and Tuoto (2018) it treats linkage error as a transformation of capture-history counts. We do not claim the operator viewpoint as new (Remark C.6). To our knowledge, the closed-form inverse (5.3)–(5.4), the split–merge calculus, the coefficients (5.8) and the sign conditions of Theorem 5.5(iv) are new in this form.

Because the detachment part is linear and explicitly invertible, the bias of Theorem 5.5 can be removed whenever the errors follow LE and (δ, ε) is known or estimated.

Corollary 5.8 (Linkage-corrected estimation). *Assume LE and Assumption A5. Let F be positively homogeneous of degree one and twice continuously differentiable near $\boldsymbol{\tau}$. Examples are $\widehat{N}_{\mathrm{LP}}$, $\widehat{n}_{\mathbf{0}}(\omega)$ and $\widehat{N}(\omega) = \sum_h x_h + \widehat{n}_{\mathbf{0}}(\omega)$.*

(i) False non-matches only *($\varepsilon = 0$). The corrected table $\hat{\boldsymbol{t}} = \mathcal{D}_\delta^{-1}\boldsymbol{n}$ is given in closed form by (5.3)–(5.4) and is conditionally unbiased, $\mathbb{E}[\hat{\boldsymbol{t}} \mid \Lambda^\star] = \boldsymbol{t}$. It is consistent, $F(\hat{\boldsymbol{t}})/F(\boldsymbol{t}) \to 1$ in probability, and conditionally on $\Lambda^\star$,*

$$N^{-1/2}\big\{F(\hat{\boldsymbol{t}}) - F(\boldsymbol{t})\big\} \Rightarrow \mathrm{N}(0, v_{\mathrm{link}}), \qquad v_{\mathrm{link}} = \sum_{h \in \mathcal{H}_+} \tau_h \operatorname{Var}\big\{\nabla F(\boldsymbol{\tau})^\top \mathcal{D}_\delta^{-1} \boldsymbol{Z}_h\big\},$$

where $\boldsymbol{Z}_h = \boldsymbol{e}_S \mathbf{1}\{S \neq \emptyset\} + \sum_{k \in h \setminus S} \boldsymbol{e}_{\{k\}}$ is the cluster vector produced by detaching one incident with support h, and $S \subseteq h$ keeps each $k \in h$ independently with probability $1 - \delta$. If, in addition, the capture histories of the N incidents are independent and identically distributed with cell probabilities π_h, and $\boldsymbol{\pi}_+ = (\pi_h)_{h \in \mathcal{H}_+}$, then $N^{-1/2}\{F(\hat{\boldsymbol{t}}) - NF(\boldsymbol{\pi}_+)\} \Rightarrow \mathrm{N}(0, v_{\mathrm{cap}} + v_{\mathrm{link}})$, where v_{cap} is the asymptotic variance of $N^{-1/2}\{F(\boldsymbol{t}) - NF(\boldsymbol{\pi}_+)\}$ under perfect linkage. For $K = 2$ and $F = \widehat{N}_{\mathrm{LP}}$ under independent capture, the variance inflation factor is

$$\frac{v_{\mathrm{cap}} + v_{\mathrm{link}}}{v_{\mathrm{cap}}} = 1 + \frac{(1-\delta)^{-2} - 1}{(1 - p_1)(1 - p_2)}, \tag{5.9}$$

and the corrected estimator is $(1 - \delta)^2 n_{1+} n_{2+}/n_{11}$.

(ii) Both error types. *There are $r_0, r > 0$ such that for $(\delta, \varepsilon) \in [0, r_0]^2$, with probability tending to one, the equation $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}) = \boldsymbol{n}$ has a unique solution $\hat{\boldsymbol{t}}$ in the ball $\|\boldsymbol{x} - \boldsymbol{t}\|_1 \leq 2rN$. It satisfies*

$$\frac{F(\hat{\boldsymbol{t}}) - F(\boldsymbol{t})}{N} = O_p^{\mathrm{u}}\big(\varepsilon^2 + N^{-1/2}\big).$$

The first-order bias of Theorem 5.5 is removed. A bias of order ε^2 remains, because $\mathcal{T}_{\delta,\varepsilon}$ is exact only to first order in ε.

(iii) Estimated rates from an independent audit. *Suppose $\hat{\theta} = (\hat{\delta}, \hat{\varepsilon})$ is independent of $\boldsymbol{n}$ given $\Lambda^\star$, as when it comes from auditing an independent linkage run or a separate validation sample, and that $\sqrt{m}(\hat{\theta} - \theta) \Rightarrow \mathrm{N}(0, V_\theta)$ for an audit size m with $m/N \to \kappa \in (0, \infty)$. Then $N^{-1/2}\{F(\hat{\boldsymbol{t}}(\hat{\theta})) - F(\hat{\boldsymbol{t}}(\theta))\} \Rightarrow \mathrm{N}(0, \kappa^{-1}\boldsymbol{c}^\top V_\theta \boldsymbol{c})$, independently of the term in (i), where*

$$\boldsymbol{c} = -\Big[\nabla F(\boldsymbol{\tau})^\top \big\{\nabla_{\boldsymbol{x}} \mathcal{T}_\theta(\boldsymbol{\tau})\big\}^{-1} \partial_\theta \mathcal{T}_\theta(\boldsymbol{\tau})\Big]^\top.$$

Correction trades bias for variance. In (5.9) with $\delta = 0.05$ and $p_1 = p_2 = 0.3$, the variance grows by about 22%, whereas the uncorrected estimator has a relative bias of about +11% that does not shrink with N. When the rates come from auditing the same linkage that produced $\boldsymbol{n}$, part (iii) does not apply, and adding

its variance to that of part (i) is conservative (Remark C.9). Appendix C.5 describes how the simulation code solves $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}) = \boldsymbol{n}$. Corollary 5.8 is only as good as LE, and the next remark shows a common error that LE cannot represent.

Remark 5.9 (Swaps). LE moves records out of their incidents (LE-D) and singletons into uniformly chosen clusters (LE-M); it has no move for linking a record to the wrong partner. In a *swap*, the list-k records of two clusters are exchanged, one of the two possibly being a singleton. For example, a police record is linked to the NGO record of a different incident with the same date, place and offence type, while the NGO record of its own incident stays unlinked. A swap preserves the support of every cluster, so it leaves the table, and the naive estimator, unchanged for every K. The audit of Section 5.4 nevertheless counts the true pairs that a swap separates as missed and the edges it creates as false. It reports $\hat{\delta} > 0$ and $\hat{\varepsilon} > 0$ in (5.10), and Corollary 5.8 then removes errors that are not there. For $K = 2$ a swap separates as many true pairs as it creates false edges, and under LE each missed pair lowers n_{11} by one and each false merge raises it by one. The two spurious corrections cancel to first order, and the correction over-corrects only at second order in the swap rate. For $K \geq 3$ they do not cancel. LE attributes the missed pairs to detachments spread over the overlap cells in proportion to $|h|t_h$, and the false edges to merges aimed at clusters in proportion to their counts, mostly singletons when lists are sparse. Undoing both moves mass between cells, although none should move, by an amount of first order in the swap rate and of either sign. Swaps are common when attributes are noisy and distinct incidents share a date window, a place and an offence type. Under such noise the audited LE correction can be worse than no correction, as the simulation of Section 8 shows.

Table-effective rates and an LE diagnostic. The correction needs rates that describe what the errors do to the table, not how many errors there are. Let a reviewer resolve the true structure on a simple random sample of connected components of the candidate-pair graph, which the estimated linkage never crosses. True matches that cross components are blocking misses, which the record-level search of Section 5.4 finds. Write $\boldsymbol{t}^{\mathrm{a}}$ and $\boldsymbol{n}^{\mathrm{a}}$ for the true and the estimated tables of the records in the sampled components.

Definition 5.10 (Table-effective error rates). For a bound $\bar{r} < 1$, the *table-effective* rates are

$$(\delta_{\mathrm{eff}}, \varepsilon_{\mathrm{eff}}) = \operatorname*{arg\,min}_{(\delta,\varepsilon)\in[0,\bar{r}]^2} X^2(\delta,\varepsilon), \qquad X^2(\delta,\varepsilon) = \sum_{h\in\mathcal{H}_+} \frac{\{\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t}^{\mathrm{a}})_h - n^{\mathrm{a}}_h\}^2}{\max(n^{\mathrm{a}}_h, 1)},$$

the rates at which the forward map best reproduces the observed table from the true table on the audited components, in a chi-squared distance. We write $X^2_{\mathrm{eff}} = X^2(\delta_{\mathrm{eff}}, \varepsilon_{\mathrm{eff}})$ for the minimised distance.

Because $\mathcal{T}_{\delta,\varepsilon}$ is homogeneous of degree one, a simple random sample of components estimates the rates that the full tables would give. Under LE, the table-effective rates and the count-based rates of (5.10) estimate the same (δ, ε). Under swaps they part company: a swap changes neither $\boldsymbol{t}^{\mathrm{a}}$ nor $\boldsymbol{n}^{\mathrm{a}}$, so it does not move the table-effective rates, while it inflates the count-based ones. The table-effective rates are therefore the ones to use in Corollary 5.8. In the simulation, where the true table of every record is known, the code's `le_projection` diagnostic computes their population version, the best-fit rates $(\delta^*, \varepsilon^*)$ of Section 8, from the tables of all records, over $[0, 0.95)^2$ by Nelder–Mead from three starting values. Three checks indicate whether the errors are LE-like. (a) Call an audited missed pair *swap-like* if the estimated cluster of one of its records contains another record from the list of the other. Every missed pair that a swap creates is swap-like, whereas under LE only a subsequent merge makes a missed pair swap-like, so their share should be of order ε. (b) X^2_{eff} should be compatible with its distribution under LE, which a parametric bootstrap

obtains by regenerating LE errors at $(\delta_{\text{eff}}, \varepsilon_{\text{eff}})$ on $\boldsymbol{t}^{\text{a}}$ and refitting. (c) The count-based and table-effective rates should agree within their sampling error.

Recommendation. Because LE is a stylised model, we do not rely on the correction for inference. Our primary analysis mixes MSE posteriors over draws from the linkage posterior (Algorithm 1), which needs no error model of the LE type and propagates linkage uncertainty as in Proposition 5.11 below. The LE correction, with table-effective rates, is a diagnostic and sensitivity tool. Its coefficients $\beta(\hat{\boldsymbol{t}})$ show which error type matters, and disagreement between the corrected plug-in estimate and the mixture flags a problem with one of them. We report the corrected estimate as an estimate only when checks (a)–(c) do not reject LE.

Proposition 5.11 (Total-variance decomposition). *Let $p(N, \Lambda \mid D)$ be a posterior given the data D (records and comparison vectors). It may come from a joint model or from linkage-averaging, in which Λ is drawn from a linkage-only posterior. Let $G(\Lambda) = \mathbb{E}[N \mid \Lambda, D]$, $V(\Lambda) = \mathrm{Var}(N \mid \Lambda, D)$, and assume second moments are finite.*

(i) $\mathrm{Var}(N \mid D) = V_w + V_b$, *with* $V_w = \mathbb{E}[V(\Lambda) \mid D]$ *(within linkages) and* $V_b = \mathrm{Var}(G(\Lambda) \mid D)$ *(between linkages).*

(ii) For any fixed $\widehat{\Lambda}$, the plug-in posterior $p(N \mid \widehat{\Lambda}, D)$ reports variance $V(\widehat{\Lambda})$, while the posterior mean-squared error of its centre is

$$\mathbb{E}\big[\{N - G(\widehat{\Lambda})\}^2 \mid D\big] = V_w + V_b + \big\{\mathbb{E}[N \mid D] - G(\widehat{\Lambda})\big\}^2.$$

If $V(\Lambda)$ does not depend on Λ on the support of $p(\Lambda \mid D)$, the reported variance falls short by at least V_b.

(iii) Suppose $N \mid D$ is normal and $V(\widehat{\Lambda}) = V_w$. Then the nominal $(1-\alpha)$ plug-in interval $G(\widehat{\Lambda}) \pm z_{1-\alpha/2} V_w^{1/2}$ has posterior coverage at most $2\Phi\{z_{1-\alpha/2}(1 + V_b/V_w)^{-1/2}\} - 1$. The bound is attained when $G(\widehat{\Lambda}) = \mathbb{E}[N \mid D]$.

(iv) Rubin's rules. *Let $\Lambda^{(1)}, \ldots, \Lambda^{(M)}$ be independent draws from $p(\Lambda \mid D)$, independent of N given D. Let $\bar{Q} = M^{-1} \sum_m G(\Lambda^{(m)})$, $\bar{U} = M^{-1} \sum_m V(\Lambda^{(m)})$, and let B_M be the sample variance of the $G(\Lambda^{(m)})$. Then*

$$\mathbb{E}\big[(N - \bar{Q})^2 \mid D\big] = V_w + \big(1 + M^{-1}\big)V_b = \mathbb{E}\big[\bar{U} + (1 + M^{-1})B_M \mid D\big].$$

Rubin's total variance $T_M = \bar{U} + (1 + M^{-1})B_M$ (Rubin, 1987) is therefore unbiased, over the imputation draws given D, for the posterior mean-squared error of the multiple-imputation estimate $\bar{Q}$. This assumes exact independent posterior draws and that $V(\Lambda)$ is the exact conditional posterior variance. It is a statement about the posterior given D. It says nothing about the frequentist coverage of the resulting intervals, which requires the imputation and analysis models to be congenial (Meng, 1994).

A single linkage, however good, drops V_b: with $V_b = V_w$, a nominal 95% interval has posterior coverage of at most $2\Phi(1.96/\sqrt{2}) - 1 \approx 0.83$. The plug-in can also be off-centre, because thresholding pairwise probabilities at $\frac{1}{2}$ shifts the overlap counts by $\sum_\ell (\mathbf{1}\{\pi_\ell > \frac{1}{2}\} - \pi_\ell)$ in expectation, and Theorem 5.5 converts such shifts into bias. By part (iv), multiple imputation with Rubin's rules is a Monte Carlo, two-moment implementation of linkage-averaging (Sadinle, 2018). Whether the mixture is a joint posterior or a modular ("cut") posterior (Plummer, 2015) is discussed in Section 6.4; (i)–(iv) hold in either case. To first order,

$V_b \approx \sum_\ell g_\ell^2 \pi_\ell(1-\pi_\ell)$ when uncertain links are approximately independent, where g_ℓ is the sensitivity of N to link ℓ. This is the quantity that clerical review reduces.

5.4 Optimal clerical review and audit

Human review can resolve uncertain links, but reviewer time is scarce. It is also sensitive: reviewers read narratives about victims (Section 10). We separate two uses of review with different goals. *Targeted review* resolves the links that matter most for the dark figure. A *random audit* estimates the error rates, and its sample of fully reviewed components gives the table-effective rates and diagnostics of Definition 5.10.

Targeted review. A review unit is either a candidate pair or the set of competing candidate links of one record, and reviewing unit u reveals its outcome O_u at cost c_u. The target is the between-linkage variance $V_b = \mathrm{Var}(G(\Lambda) \mid D)$ of Proposition 5.11. If the outcomes are independent under the linkage posterior and review is error-free, the expected variance that remains after reviewing a set of units is exactly the sum of the main effects $\sigma_u^2 = \mathrm{Var}(\mathbb{E}[G \mid O_u])$ of the unreviewed units plus interaction terms (Theorem C.10 in Appendix C.6). Reviewing the units with the largest σ_u^2/c_u is then optimal up to the total interaction variance and, with unequal costs, one unit (Proposition C.12). For a pair with link indicator I_ℓ, $\sigma_\ell^2 = \mathrm{Cov}(G, I_\ell \mid D)^2/\{\pi_\ell(1-\pi_\ell)\}$ can be estimated from posterior draws. To first order $\sigma_\ell^2 \approx g_\ell^2 \pi_\ell(1-\pi_\ell)$, where g_ℓ is the sensitivity of G to link ℓ (Proposition C.11): review where the posterior is most uncertain and the link matters most, per unit of cost. For the closed form, g_ℓ depends only on the cell types that the link joins (C.3). For $K = 2$ every link has the same sensitivity $-\widehat{N}_{\mathrm{LP}}/n_{11}$, so the rule reduces to reviewing the pairs with π_ℓ closest to $\frac{1}{2}$; for $K \geq 3$, links into sparse cells deserve more attention. Independence is where the approximation lies. Candidate pairs that share a record are negatively dependent, which we absorb by grouping a record's competing candidates into one unit, and Algorithm 3 reviews in batches and refits the posterior after each batch.

Random audit of error rates. Targeted review is not a probability sample, so it cannot estimate (δ, ε); a random audit can. It samples candidate pairs within strata defined by decision status (linked edge; same cluster but not an edge; different clusters) and by bins of π_ℓ, and a reviewer determines the true status of each sampled pair. From the estimated numbers TP, FN and FE of true pairs kept together, true pairs split and false linked edges,

$$\delta = 1 - \left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}\right)^{1/2}, \qquad \varepsilon = \frac{\mathrm{FE}}{f_1 + \mathrm{FE}}, \tag{5.10}$$

where f_1 is the observed number of singleton clusters. The first equation inverts the pairwise retention probability $(1-\delta)^2$ of Proposition 5.3(iv), and the second is correct to first order in ε. The stratified expansion estimates are design-based, so they remain valid when the linkage model is miscalibrated (Belin and Rubin, 1995), and Proposition C.13 gives their Neyman allocation (Appendix C.6). Pairs resolved by targeted review can be removed from the frame. A small record-level stratum, in which a reviewer searches all lists for a sampled record's counterparts, estimates the blocking component of δ, which a pair-based audit cannot see. Finally, (5.10) counts errors, and Remark 5.9 shows that counts can mislead; the component sample of Definition 5.10 measures what the errors do to the table.

5.5 Privacy-preserving linkage

The attributes used here (dates, locations, venues, narratives) can identify victims. When lists cannot share them in the clear, linkage can use encoded attributes, such as the Bloom-filter encodings of Schnell et al. (2009), in which each attribute's q-grams are hashed into a bit array and compared by a set similarity.

Everything in this section then applies with m and u defined for encoded comparisons. Encoding typically increases both δ and ε, which makes the audit more important, and the audit itself requires a trusted party with access to clear text. Basic Bloom-filter encodings are vulnerable to cryptanalysis (Christen et al., 2019; Kuzu et al., 2011; Niedermeyer et al., 2014). Governance, threat models and reviewer access are treated in Section 10, and the linkage protocol in Section 9.

6 Bayesian models for the dark figure

This section turns the identification analysis of Section 4 into estimation, through four Bayesian components that address unobserved heterogeneity (Section 6.1), observed heterogeneity (Section 6.2), sparse domains with shifting police coverage (Section 6.3) and estimated linkage (Section 6.4). In each, the unobserved count $n_{\mathbf{0}}$ is sampled together with the parameters. None of the components is new on its own; our contribution is to assemble them for this application and to make explicit where each obtains its identifying information. As in Section 3, N counts the incidents that reach some recording channel. Computation is described in Appendix D. The small-area model of Section 6.3 is specified here but not evaluated in the simulation study.

6.1 Latent-class MSE with a Dirichlet-process prior

Consider two incidents in the same city and month. A synagogue is attacked and a congregant assaulted: the police respond, the press reports it, and the NGO audit and the community security organisation both record it. A man wearing a kippah is sworn at on a street corner: he files a report on a community portal, and no other list hears of it. Within each kind, whether one list records an incident may be nearly unrelated to whether another does, but in a population that mixes the two kinds the lists are positively dependent: a police record is evidence that the incident is of the visible kind. With two lists this biases Lincoln–Petersen downward; with three or more, heterogeneity alone does not fix the sign of the bias (Propositions 4.6 and 4.7), which is one reason to model it directly.

Latent-class models represent such heterogeneity by a discrete *reportability type* $z_i \in \{1, \dots, J\}$, within which the lists record independently. Manrique-Vallier (2016) placed a Dirichlet-process prior on the distribution of types, so that the number of types need not be fixed in advance. We use his model (often called LCMCR) with the truncated stick-breaking representation of the Dirichlet process (Ishwaran and James, 2001; Sethuraman, 1994), truncated at $J^\star$ classes:

$$
\begin{aligned}
Y_{ik} \mid z_i = j &\overset{\text{ind}}{\sim} \text{Bernoulli}(\lambda_{jk}), \quad k \in [K], & z_i &\overset{\text{iid}}{\sim} \text{Categorical}(\pi_1, \dots, \pi_{J^\star}), \quad i = 1, \dots, N, \\
\pi_j &= V_j \textstyle\prod_{l<j}(1 - V_l), & V_j \mid \alpha &\overset{\text{iid}}{\sim} \text{Beta}(1, \alpha) \;\; (j < J^\star), \quad V_{J^\star} = 1, \\
\lambda_{jk} &\overset{\text{iid}}{\sim} \text{Beta}(a_\lambda, b_\lambda), & \alpha &\sim \text{Gamma}(a_\alpha, b_\alpha), \qquad p(N) \propto 1/N.
\end{aligned}
\tag{6.1}
$$

We use $a_\lambda = b_\lambda = 1$ and $a_\alpha = b_\alpha = 0.25$ (shape and rate), whose influence we examine in Appendix D. Integrating out z_i, the histories are i.i.d. as in model (M) of Definition 3.1, with cell probabilities

$$
\pi_{\boldsymbol{h}}(\psi) = \sum_{j=1}^{J^\star} \pi_j \prod_{k=1}^{K} \lambda_{jk}^{h_k} (1 - \lambda_{jk})^{1 - h_k}, \qquad \boldsymbol{h} \in \mathcal{H}, \qquad \psi = (\pi_1, \dots, \pi_{J^\star}, \boldsymbol{\lambda}). \tag{6.2}
$$

Bold subscripts index histories and plain subscripts index classes. The probability that an incident is missed by every list, $\pi_{\mathbf{0}}(\psi) = \sum_j \pi_j \prod_k (1 - \lambda_{jk})$, comes mostly from low-visibility classes.

The next lemma, an elementary calculation, records the role of the prior on N; it underlies the sampler, the sensitivity offset below and the Bayesian reading of the Horvitz–Thompson estimator in Section 6.2.

Lemma 6.1 (The unobserved count). *Let $(n_{\boldsymbol{h}})_{\boldsymbol{h}\in\mathcal{H}} \mid N, \psi \sim \text{Multinomial}(N, \boldsymbol{\pi}(\psi))$ for any parametric capture model with $\pi_{\mathbf{0}}(\psi) < 1$. Suppose that only $\boldsymbol{n}_+$ is observed, with $n \geq 1$. Let N and ψ be independent a priori, with the improper prior $p_\kappa(N) \propto \Gamma(N+\kappa)/\Gamma(N+1)$ on $N \geq 1$, for some $\kappa \geq 0$; $\kappa = 0$ gives $p(N) \propto 1/N$ and $\kappa = 1$ the flat prior. Then:*

(i) $n_{\mathbf{0}} \mid \psi, \boldsymbol{n}_+ \sim \text{NegBin}\big(n+\kappa,\ 1-\pi_{\mathbf{0}}(\psi)\big)$, the law with probabilities $\Gamma(n+\kappa+x)\,(1-\pi_{\mathbf{0}})^{n+\kappa}\pi_{\mathbf{0}}^{x}/\{\Gamma(n+\kappa)\,x!\}$, $x = 0, 1, \ldots$; when $n+\kappa$ is an integer, this is the number of failures before the $(n+\kappa)$-th success with success probability $1-\pi_{\mathbf{0}}$. Hence $\mathbb{E}[N \mid \psi, \boldsymbol{n}_+] = \{n + \kappa\,\pi_{\mathbf{0}}(\psi)\}/\{1-\pi_{\mathbf{0}}(\psi)\}$.

(ii) $p(\psi \mid \boldsymbol{n}_+) \propto p(\psi)\,\{1-\pi_{\mathbf{0}}(\psi)\}^{-\kappa} \prod_{\boldsymbol{h}\in\mathcal{H}_+} \pi_{\boldsymbol{h}}^{c}(\psi)^{n_{\boldsymbol{h}}}$, with $\pi_{\boldsymbol{h}}^{c} = \pi_{\boldsymbol{h}}/(1-\pi_{\mathbf{0}})$ as in Lemma 3.2.

In particular, under $p(N) \propto 1/N$ the marginal posterior of ψ is the posterior obtained from the conditional likelihood alone, and $\mathbb{E}[N \mid \psi, \boldsymbol{n}_+] = n/\{1-\pi_{\mathbf{0}}(\psi)\}$.

Proof. By (3.4), as a function of $n_{\mathbf{0}} = N - n$ the likelihood is proportional to $N!\,\pi_{\mathbf{0}}^{n_{\mathbf{0}}}/n_{\mathbf{0}}!$. Multiplying by $p_\kappa(N) \propto \Gamma(N+\kappa)/N!$ gives $p(n_{\mathbf{0}}, \psi \mid \boldsymbol{n}_+) \propto p(\psi)\,\{\Gamma(n+n_{\mathbf{0}}+\kappa)/n_{\mathbf{0}}!\}\,\pi_{\mathbf{0}}^{n_{\mathbf{0}}} \prod_{\boldsymbol{h}\in\mathcal{H}_+} \pi_{\boldsymbol{h}}^{n_{\boldsymbol{h}}}$. As a function of $n_{\mathbf{0}}$, this is the kernel of the stated negative binomial, which proves (i); the mean follows from $\mathbb{E}n_{\mathbf{0}} = (n+\kappa)\pi_{\mathbf{0}}/(1-\pi_{\mathbf{0}})$. For (ii), sum over $n_{\mathbf{0}} \geq 0$ using the negative-binomial series $\sum_{x\geq 0} \Gamma(r+x)\,q^x/x! = \Gamma(r)(1-q)^{-r}$, valid for $r > 0$ and $0 \leq q < 1$, with $r = n+\kappa$. This gives $p(\psi \mid \boldsymbol{n}_+) \propto p(\psi)\,(1-\pi_{\mathbf{0}})^{-(n+\kappa)} \prod_{\boldsymbol{h}\in\mathcal{H}_+} \pi_{\boldsymbol{h}}^{n_{\boldsymbol{h}}}$. Then substitute $\pi_{\boldsymbol{h}} = (1-\pi_{\mathbf{0}})\pi_{\boldsymbol{h}}^{c}$ and use $\sum_{\boldsymbol{h}\in\mathcal{H}_+} n_{\boldsymbol{h}} = n$. □

Lemma 6.1 drives data augmentation. Given ψ, draw $n_{\mathbf{0}}$ from the negative binomial and allocate the unseen incidents to classes with probabilities proportional to $\pi_j \prod_k (1-\lambda_{jk})$; the completed data form a latent-class sample of size $N = n + n_{\mathbf{0}}$ with conjugate full conditionals (Appendix D). Part (ii) shows what the prior on N does. Under $1/N$ the capture parameters are learned exactly as from the conditional likelihood in (3.4). Under the flat prior ($\kappa = 1$) the posterior of ψ is tilted by $(1-\pi_{\mathbf{0}})^{-1}$ toward low detectability, and hence toward larger N, which matters most when the dark figure is large.

Remark 6.2 (What the latent classes identify). Local independence fixes every interaction *within* a class at zero, but the mixture generally has nonzero interactions of all orders. Each posterior draw of ψ therefore implies a value of ω, by (3.8) applied to $\boldsymbol{\pi}(\psi)$, and we report its posterior, which places the LCMCR fit on the sensitivity scale of Theorem 4.4 and Corollary 4.5. The observed table does not identify ω (Theorem 4.3), so the posterior of the implied ω is a consequence of the latent-class structure and the prior on ψ, not a measurement. Three results sharpen this caveat. First, different heterogeneity distributions can fit the observed cells equally well and imply very different N (Link, 2003). Second, the sufficient condition of Allman et al. (2009) for generic identifiability of latent-class parameters covers only $J \leq 2$ classes with $K = 4$ lists, even for the complete table. Third, a class whose capture probabilities are all zero leaves no trace in the data and has prior probability zero under the beta prior on $\boldsymbol{\lambda}$. Incidents that reach no recording channel are of this kind, so every estimate in this section refers to the incidents that reach some channel (N_V of Section 7). The Dirichlet-process prior concentrates on parsimonious class structures, and that concentration is part of the identifying assumption. We therefore never report LCMCR alone, but next to the closed-form sensitivity analysis and Chao's estimator (Theorem 4.8), which is a diagnostic rather than a floor (Proposition 4.9); the only assumption-free floor is n.

Sensitivity parameters in the latent-class model. Two sensitivity parameters of Section 4 enter LCMCR as follows; the analysis plan uses both (Section 9.5). First, a departure Δ from the dependence structure that the latent classes impose is applied draw by draw to the odds of being unrecorded:

$$\frac{\pi_{\mathbf{0}}^{\Delta}(\psi)}{1-\pi_{\mathbf{0}}^{\Delta}(\psi)} = e^{\Delta}\,\frac{\pi_{\mathbf{0}}(\psi)}{1-\pi_{\mathbf{0}}(\psi)}, \qquad n_{\mathbf{0}} \mid \psi, \boldsymbol{n}_{+} \sim \mathrm{NegBin}\big(n,\, 1-\pi_{\mathbf{0}}^{\Delta}(\psi)\big). \tag{6.3}$$

By (3.11), this keeps the conditional cell probabilities $\boldsymbol{\pi}^{c}(\psi)$ and raises the implied oriented interaction $\bar{\omega}(\psi)$ by exactly Δ; it is the latent-class analogue of the offset $e^{\bar{\omega}}$ in Theorem 4.4 and Corollary 4.5. The tilted and untilted models have the same conditional likelihood, so by Lemma 6.1(ii) with $\kappa = 0$ they have the same posterior of ψ: (6.3) is exact post-processing of the LCMCR draws, and $\Delta = 0$ is the fitted model. Second, referral from the police list to list 2 with probability q (Proposition 4.10) enters through a referral-augmented model. The class probabilities λ_{j2} refer to *direct* capture $D_i \sim \mathrm{Bernoulli}(\lambda_{z_i 2})$, and $Y_{i2} = \max(D_i, R_i Y_{i1})$ with $R_i \sim \mathrm{Bernoulli}(q)$ independent. Then $\pi_{\mathbf{0}}(\psi)$ is unchanged, and the Gibbs sampler gains one step: for an incident on both lists whose provenance is unknown, draw $D_i \sim \mathrm{Bernoulli}\{\lambda_{z_i 2}/(\lambda_{z_i 2} + q(1-\lambda_{z_i 2}))\}$; where provenance records how list 2 learned of the incident, D_i is observed, which is the recoding of Proposition 4.10(c). Setting $q = 0$ recovers (6.1).

6.2 Covariate-dependent capture

Some heterogeneity is observed through incident characteristics such as offence type, target, setting and urbanicity. Let $\boldsymbol{x}_i$ collect them, harmonised across lists (Section 9), and consider

$$\mathrm{logit}\, p_{ik} = \beta_{0k} + \boldsymbol{x}_i^{\top}\boldsymbol{\beta}_k, \qquad Y_{i1}, \ldots, Y_{iK} \text{ independent given } \boldsymbol{x}_i, \qquad p_i(\boldsymbol{\beta}) = 1 - \prod_{k=1}^{K}(1-p_{ik}), \tag{6.4}$$

where p_i is the probability that incident i is recorded by at least one list. Covariates are observed only for recorded incidents, so Huggins (1989) and Alho (1990) condition on being recorded:

$$L_c(\boldsymbol{\beta}) = \prod_{i=1}^{n} \frac{\prod_k p_{ik}^{y_{ik}}(1-p_{ik})^{1-y_{ik}}}{p_i(\boldsymbol{\beta})}, \qquad \widehat{N}_{\mathrm{HT}} = \sum_{i=1}^{n} \frac{1}{p_i(\widehat{\boldsymbol{\beta}})}, \tag{6.5}$$

where $\widehat{\boldsymbol{\beta}}$ maximises L_c. Given the true $\boldsymbol{\beta}$, the estimator is unbiased, because $\mathbb{E}\sum_{i\in U}\mathbf{1}\{\boldsymbol{h}_i \neq \mathbf{0}\}/p_i = N$. Huggins' variance estimator adds the estimation error of $\widehat{\boldsymbol{\beta}}$ to the design variance:

$$\widehat{\mathrm{Var}}(\widehat{N}_{\mathrm{HT}}) = \sum_{i=1}^{n}\frac{1-\hat{p}_i}{\hat{p}_i^2} + \widehat{\boldsymbol{D}}^{\top}\widehat{\boldsymbol{I}}^{-1}\widehat{\boldsymbol{D}}, \qquad \widehat{\boldsymbol{D}} = \sum_{i=1}^{n}\frac{\partial p_i^{-1}}{\partial \boldsymbol{\beta}}\bigg|_{\widehat{\boldsymbol{\beta}}}, \tag{6.6}$$

where $\widehat{\boldsymbol{I}}$ is the observed information of $\log L_c$.

A Bayesian reading. Give each covariate profile $\boldsymbol{x}$ that occurs among the recorded incidents its own population size $N_{\boldsymbol{x}}$ with prior $\propto 1/N_{\boldsymbol{x}}$ (for continuous covariates each recorded incident is its own stratum). Within a stratum, Lemma 6.1 applies with $\pi_{\mathbf{0}} = 1 - p(\boldsymbol{x})$, so $N_{\boldsymbol{x}} - n_{\boldsymbol{x}} \mid \boldsymbol{\beta}$ is a sum of $n_{\boldsymbol{x}}$ independent geometric variables on $\{0, 1, \ldots\}$ with success probability $p(\boldsymbol{x})$. Collecting strata,

$$N - n \mid \boldsymbol{\beta}, \text{data} \overset{d}{=} \sum_{i=1}^{n} \mathrm{Geom}\,\big(p_i(\boldsymbol{\beta})\big) \text{ (independent)}, \qquad \mathbb{E}[N \mid \boldsymbol{\beta}, \text{data}] = \sum_{i=1}^{n}\frac{1}{p_i(\boldsymbol{\beta})}, \tag{6.7}$$

and the marginal posterior of $\boldsymbol{\beta}$ is proportional to $p(\boldsymbol{\beta})L_c(\boldsymbol{\beta})$. The Horvitz–Thompson estimator is this conditional mean evaluated at $\widehat{\boldsymbol{\beta}}$, not the posterior mean. The prior is data-dependent: if every possible profile, recorded or not, received the prior $1/N_{\boldsymbol{x}}$, each unrecorded profile would multiply the posterior of $\boldsymbol{\beta}$ by $\sum_{N\geq 1} N^{-1}\{1-p(\boldsymbol{x})\}^N = -\log p(\boldsymbol{x})$, favouring low detectability. Two caveats carry over from the frequentist analysis. The posterior refers only to incidents whose profiles lie in the observed support. And a recorded incident with small $\hat{p}_i$ carries a large weight and should be inspected; a common cause is a linkage error that has split one incident into singletons.

Random effects. Lists differ across places and over time in ways that $\boldsymbol{x}_i$ does not capture:

$$\operatorname{logit} p_{ik} = \beta_{0k} + \boldsymbol{x}_i^\top \boldsymbol{\beta}_k + b_{g(i),k}, \qquad \boldsymbol{b}_g = (b_{g1}, \ldots, b_{gK})^\top \overset{\text{iid}}{\sim} N_K(\mathbf{0}, \boldsymbol{\Sigma}_b), \tag{6.8}$$

where $g(i)$ is the state, or state–year, of incident i. A positive correlation in $\boldsymbol{\Sigma}_b$ between the police and NGO effects is dependence between domains, which a pooled model would misread as dependence between lists. An incident-level random effect shared by all lists gives the logistic-normal model (Coull and Agresti, 1999), whose estimate of N depends heavily on the left tail of the random-effect distribution, where the data are silent (Link, 2003). We therefore model unobserved incident-level heterogeneity with latent classes and observed heterogeneity with covariates. Categorical covariates enter (6.1) by stratification, either fitting each stratum separately or sharing the class profiles $\lambda_{j\cdot}$ across strata with stratum-specific class weights; Lemma 6.1 then applies stratum by stratum.

6.3 A spatiotemporal small-area model

In many state–years the lists together record only a handful of incidents. A separate fit in each would be unstable, and a single national fit would hide the variation that matters for policy, so we model all state–years jointly. The model is specified here as part of the framework; it is not fitted in the simulation study, and its operating characteristics remain to be evaluated. Let $s = 1, \ldots, S$ index states and $t = 1, \ldots, T$ index years, so that the domain is $d = (s, t)$. Let P_{st} be an exposure, such as the resident population; the choice of exposure is discussed in Section 9. The model is

$$\begin{aligned}
n_{st,\boldsymbol{h}} \mid \theta_{st}, \boldsymbol{\pi}_{st} &\overset{\text{ind}}{\sim} \text{Poisson}\left(P_{st}\,\theta_{st}\,\pi_{st,\boldsymbol{h}}\right), \quad \boldsymbol{h} \in \mathcal{H}, \qquad N_{st} = \textstyle\sum_{\boldsymbol{h}\in\mathcal{H}} n_{st,\boldsymbol{h}}, \\
\log \theta_{st} &= \beta_0 + \boldsymbol{x}_{st}^\top \boldsymbol{\beta} + b_s + \tau_t + \chi_{st}, \\
b_s &= \sigma_b\left(\sqrt{\phi}\,u_s + \sqrt{1-\phi}\,v_s\right), \qquad \boldsymbol{u} \sim \text{ICAR}_\star(\mathcal{G}), \quad v_s \overset{\text{iid}}{\sim} N(0,1), \\
\tau_t - 2\tau_{t-1} + \tau_{t-2} &\overset{\text{iid}}{\sim} N(0, \sigma_\tau^2), \qquad \chi_{st} - \chi_{s,t-1} \overset{\text{iid}}{\sim} N(0, \sigma_\chi^2), \\
\pi_{st,\boldsymbol{h}} &= \sum_{j=1}^{J} \pi_j \prod_{k=1}^{K} p_{st,jk}^{h_k} (1 - p_{st,jk})^{1-h_k},
\end{aligned} \tag{6.9}$$

with the capture probabilities

$$p_{st,jk} = c_{st,k}\ \text{expit}\left(\alpha_{jk} + \eta_{ks} + \xi_{kt} + \zeta\, f_{st}\mathbf{1}\{k=1\} + \boldsymbol{x}_{st}^\top \boldsymbol{\beta}_k\right), \tag{6.10}$$

where $\boldsymbol{\eta}_s = (\eta_{1s}, \ldots, \eta_{Ks})^\top \overset{\text{iid}}{\sim} N_K(\mathbf{0}, \boldsymbol{\Sigma}_\eta)$ and $\xi_{k\cdot}$ is an independent first-order random walk for each list. By Lemma 3.2(a), the first line is equivalent to $N_{st} \sim \text{Poisson}(P_{st}\theta_{st})$ with a latent-class multinomial split across histories. The incidence model is a disease-mapping model with the BYM2 spatial term (Besag et al., 1991; Riebler et al., 2016): $\mathcal{G}$ is the graph of states that share a border, $\text{ICAR}_\star$ is the intrinsic conditional autoregression scaled to unit geometric-mean marginal variance (Sørbye and Rue, 2014), σ_b is the total

standard deviation of the state effects and ϕ its spatially structured share. The incidence random effects are b_s (space), τ_t (national trend, a second-order random walk) and χ_{st} (each state's deviation from it, the type II interaction of Knorr-Held (2000); types I and IV are sensitivity analyses), with standard deviations σ_b, σ_τ, σ_χ. The latent classes, with weights π_j and baseline profiles $\alpha_{j\cdot}$, are shared by all domains, and the capture shifts η_{ks} and ξ_{kt} move every class's capture probability for list k together on the logit scale. A finite J with a sparse Dirichlet prior on (π_j) empties superfluous classes (Rousseau and Mengersen, 2011).

Police coverage as data. The police list can record an incident only if it falls in the jurisdiction of an agency that submits hate-crime data for the year. Let $c_{st,1} \in [0, 1]$ be the share of state s's population served by such agencies in year t, computed from agency participation records and population figures (Section 2.1) and entered in (6.10) as a known offset. The offset is exact if incidents in non-participating jurisdictions cannot enter list 1, incidents are spread in proportion to population, and, given the class, falling in a covered jurisdiction is independent of capture by the other lists; urban and rural strata restore the last within strata if agencies cluster where other lists are active, and an incidence-weighted coverage $c_{st,1} \exp(\sigma_c e_{st})$, truncated at one, with $e_{st} \sim N(0, 1)$, relaxes the second. The offset matters most at structural breaks such as the NIBRS-only year 2021, when participation fell sharply (Section 2.1). Without it, a model with smoothly evolving police capture must split the fall in police counts between N_{st} and police recording in proportions set by the prior; with it, the break is absorbed by data and τ_t stays smooth. The term ζf_{st}, where f_{st} is the share of the covered population whose agencies reported through NIBRS, absorbs format effects among agencies that did participate. Since both formats were accepted again from 2022, f_{st} can rise and fall, and ζ is identified only through its variation across states within a year, assuming that the other lists' capture did not change with the police reporting format.

The other lists. For $k \geq 2$, $c_{st,k} \in \{0, 1\}$ records whether list k operates in state s in year t, replacing Assumption A3(b) by a modelled footprint with structural zeros. The covariance $\boldsymbol{\Sigma}_\eta$ pools each list's state-specific capture toward its national level, so a sparse state's capture rates are shrunk toward the national pattern, as far as its own lists allow.

Which domains are identified. Let $K_{st} = \#\{k : c_{st,k} > 0\}$ be the number of lists operating in domain (s, t). Its local unknowns are θ_{st} and one capture shift per operating list; class profiles, weights and time effects are shared. It contributes $2^{K_{st}} - 1$ observed cells.

- $K_{st} = 1$ (police only): a single Poisson count with mean $P_{st}\theta_{st} \sum_j \pi_j p_{st,j1}$, so N_{st} and $D_{1,st}$ are determined by the priors (smoothing of θ, shrinkage of η_{1s}), not by the domain's data.
- $K_{st} = 2$: three cells meet three unknowns, determined only through the latent-class structure estimated elsewhere, with no local degree of freedom to check it.
- $K_{st} \geq 3$: the cells exceed the unknowns by $2^{K_{st}} - K_{st} - 2$ (3 for $K_{st} = 3$, 10 for $K_{st} = 4$), leaving local degrees of freedom for checking fit.

Every domain with $K_{st} \leq 2$ is flagged as prior-driven, and its N_{st} is reported as a model-based prediction, not a measurement.

Borrowing strength. Information is pooled in three ways: the ICAR prior $u_s \mid \boldsymbol{u}_{-s} \sim N(\bar{u}_{\partial s}, 1/d_s)$ (before scaling) and the temporal terms pull a sparse domain's incidence toward its neighbours in space and time; $\boldsymbol{\eta}_s$ shrinks its capture toward the national profile; and the latent types are learned from the whole country. Given the parameters, $n_{st,\mathbf{0}} \sim \text{Poisson}(P_{st}\theta_{st}\pi_{st,\mathbf{0}})$ independently of the observed cells, so draws of N_{st} and of any aggregate carry the full posterior correlation across domains. Borrowing strength reduces

Algorithm 1 End-to-end estimation of the dark figure with propagated linkage uncertainty.

Require: Lists $R_1, \ldots, R_K$ with harmonised attributes; blocking rule $\mathcal{B}$; numbers of linkage draws M and MSE draws L; an MSE model from Sections 6.1 to 6.3.

Ensure: Draws $\{N_d^{(m,\ell)}\}$ from the cut posterior (6.11) for every domain d.

1: **Block**: form candidate pairs of records, within and between lists, $\{(r, r') : \mathcal{B}(r, r') = 1\}$, with comparison vectors $\gamma_{rr'}$.
2: **FS-EM**: estimate $m(\cdot)$ and $u(\cdot)$ by Algorithm 2, as starting values or as plug-in values.
3: (Optional) **Clerical review** of the most uncertain pairs by Algorithm 3; fix the reviewed pairs.
4: **for** $m = 1, \ldots, M$ **do**
5: Draw $(\Lambda^{(m)}, m^{(m)}, u^{(m)})$ from the linkage posterior by MCMC over partitions of all records, which resolves within-list duplicates and between-list links jointly. In the empirical-Bayes variant, fix (m, u) at the FS-EM values.
6: Form the capture tables $\boldsymbol{n}_d^{(m)} = \boldsymbol{n}_d(\Lambda^{(m)})$ for every domain d.
7: Run the MSE sampler on $\{\boldsymbol{n}_d^{(m)}\}_d$; keep L draws $N_d^{(m,\ell)} = n_d^{(m)} + n_{d,\mathbf{0}}^{(m,\ell)}$, $\ell = 1, \ldots, L$.
8: **Mix**: pool the ML draws; report posterior summaries and the between- and within-linkage variance components of Proposition 5.11, estimated as in Appendix D.4.

variance but does not identify ω in any domain: it transfers the latent-class restriction from the nation to each state, and a state with an atypical reporting environment, such as police referrals into the NGO list (Proposition 4.10), is shrunk toward a pattern that does not apply to it. Domain-level sensitivity analysis (Corollary 4.5) therefore remains part of the plan.

6.4 Propagating linkage uncertainty

The capture tables above are functions of a linkage structure. Let R denote all records with their attributes, and let Λ be a partition of R into incidents; here a cluster may contain several records from one list, so that Λ also resolves within-list duplicates, and its history has $h_k = 1$ when it contains a record from list k. Let $\boldsymbol{n}(\Lambda)$ be the domain capture tables that Λ induces. A joint model $p(R \mid \Lambda, \psi_L)\, p(\Lambda \mid N, \psi)\, p(N, \psi)\, p(\psi_L)$, with linkage parameters ψ_L (Tancredi and Liseo, 2011; Tancredi et al., 2020), lets the population model inform which records are linked. We use instead the modular ("cut") posterior (Bayarri et al., 2009; Plummer, 2015), as Sadinle (2018) did for human-rights data:

$$p_{\text{cut}}(N, \psi \mid R) = \int p\big(N, \psi \mid \boldsymbol{n}(\Lambda)\big)\, p_L(\Lambda \mid R)\, d\Lambda, \tag{6.11}$$

where $p_L(\Lambda \mid R)$ is the posterior of a linkage model fitted without reference to the capture model. Each module can then be checked separately, a misspecified capture model cannot distort the linkage, and one set of linkage draws serves every estimator in this paper; the cost is that the weak information the population model carries about which pairs match is discarded. The posterior variance splits into within- and between-linkage components (Proposition 5.11); a plug-in posterior $p(N \mid \boldsymbol{n}(\widehat{\Lambda}))$ drops the second, and is off-centre when $\widehat{\Lambda}$ systematically splits or merges incidents (Theorem 5.5). Mixing over linkage draws is our primary treatment of linkage uncertainty. The correction of Corollary 5.8 is not a substitute for it: it inverts the stylised model LE, which has no move for a swap (a record linked to the wrong partner; Remark 5.9), and it over-corrects when swaps dominate (Section 8). We use it only as a diagnostic (Section 9.3). Mixing has its own limit: it propagates the uncertainty that the linkage posterior expresses,

but it cannot remove a bias that every draw shares, for example from a misspecified comparison model. The random audit (Proposition C.13) and the calibration study of Section 9.6 check for this. Algorithm 1 gives the procedure, and Figure 2 shows it schematically.

Three remarks on Algorithm 1. First, line 5 needs a sampler over partitions of records from several files with duplicates; Aleshin-Guendel and Sadinle (2023) treat this multifile case with Fellegi–Sunter comparison models (Sadinle (2017) treats two files without duplicates, Sadinle (2014) one file with duplicates, and Steorts et al. (2016) model attribute distortion directly). Resolving duplicates inside the loop matters, because a leftover duplicate is an extra singleton, and singletons enter the closed form with exponent +1 (Equation (3.2)). Second, the blocking rule, the clerical-review decisions and, in the empirical-Bayes variant, the FS-EM estimates of m and u are treated as fixed; the reported interval omits their uncertainty, and we report how $\widehat{N}$ changes across blocking rules. Third, the draws $\Lambda^{(m)}$ come from a Markov chain, so the between-linkage component rests on $M_{\text{eff}} < M$ effectively independent tables (Appendix D); the MSE step is parallel across m, and its output also serves the funnel decomposition of Section 7.

7 The reporting funnel: where antisemitic incidents go missing

The police count is the end of a sequence of losses: an incident must reach a recording channel, then the police; the police must record it as motivated by antisemitic bias; and the agency must submit the record to the FBI. Each stage loses incidents for different reasons and calls for a different remedy. This section formalises the sequence as a multiplicative identity (Proposition 7.2), states which data identify which stage, and attributes differences in recorded rates between jurisdictions to the stages by a Shapley decomposition (Theorem 7.3); Figure 1 depicts it. In this section U_d denotes *all* in-scope incidents in domain d, whether or not they reach a channel, and N their number; the population of Sections 3 to 6 is the subset U_d^V defined below, of size N_V.

7.1 The funnel identity

Definition 7.1 (Funnel stages). A *recording channel* is the police or an organisation that maintains one of the lists. For an incident $i \in U_d$ let $V_i = 1$ if the incident reaches at least one recording channel, either through a report by the victim, a witness or another third party ($V_i^{\text{rep}} = 1$) or through the channel's own detection ($V_i^{\text{det}} = 1$), so that $V_i = \max(V_i^{\text{rep}}, V_i^{\text{det}})$. Let $P_i = 1$ if the incident reaches the police, by report or detection; $C_i = 1$ if the police record it as motivated by antisemitic bias; and $S_i = 1$ if the agency submits that record to the FBI's hate-crime collection. The stages are nested, $S_i \le C_i \le P_i \le V_i$, and $n_{\text{FBI}} = \sum_{i \in U_d} S_i$ is the FBI count. When list 1 is the FBI submission, $n_{\text{FBI}} = n_{1+}$; when list 1 is a local police list observed before submission, as in the pilot of Section 9, $n_{1+} = \sum_i C_i$ and the submission stage drops out. Let I be uniform on U_d, independent of the indicators, and define the stage rates

$$\rho_{\text{vic}} = \mathbb{P}(V_I = 1), \qquad \rho_{\text{pol}} = \mathbb{P}(P_I = 1 \mid V_I = 1),$$
$$\rho_{\text{cls}} = \mathbb{P}(C_I = 1 \mid P_I = 1), \qquad \rho_{\text{sub}} = \mathbb{P}(S_I = 1 \mid C_I = 1).$$

Probabilities are taken over the random incident and the reporting process, given U_d.

The first stage is reaching a recording channel, not disclosure to anyone: an incident that the victim tells only to family has $V_i = 0$ and is on no list. Where channels learn of incidents only through reports, $\rho_{\text{vic}} = \mathbb{P}(V_I^\circ = 1)\mathbb{P}(V_I = 1 \mid V_I^\circ = 1)$, with $V_i^\circ = 1$ if the incident is disclosed to anyone; a survey that asks both questions identifies both factors, which respond to victim support and to the accessibility of channels

respectively. The rates are *aggregate*: ρ_{pol} is the share that reaches the police among incidents that reach some channel, whatever their type. External sources observe a stage only on a subpopulation selected by their own mechanism. Let $W_i = 1$ if incident i is within the scope of a victim survey and reported by a sampled respondent; $G_i = 1$ if i is recorded by an NGO or community list ($k \geq 2$) whose record states that the incident was reported to the police; and $A_i = 1$ if the agency with jurisdiction over i submits hate-crime data for the period, with c_d the share of the domain's population served by such agencies.

Assumption A6 (Source ignorability).

(F1) *Survey.* $W_I \perp V_I$, and $W_I \perp P_I \mid V_I = 1$. Within the survey frame, channels learn of incidents only through reports: $V_I^{\text{det}} = 0$ whenever $W_I = 1$.

(F2) *Overlap.* $G_I = 1$ implies $P_I = 1$; $G_I \perp C_I \mid P_I = 1$; and $G_I \perp S_I \mid C_I = 1$.

(F3) *Participation.* $S_i = C_i A_i$ for every i; $A_I \perp C_I$; and $\mathbb{P}(A_I = 1) = c_d$.

(F4) *MSE target.* The capture model of Section 4 or Section 6, including its dependence restriction, holds on the incidents that reach a channel, $U_d^V = \{i : V_i = 1\}$, with size $N_V = |U_d^V|$. (A list can record only incidents known to its organisation, so $\mathbb{P}(\boldsymbol{h}_i \neq \mathbf{0} \mid V_i = 0) = 0$ by construction.)

Proposition 7.2 (Funnel identity).

(a) Accounting. $\mathbb{E}[n_{\text{FBI}} \mid U_d] = N\,\rho_{\text{vic}}\,\rho_{\text{pol}}\,\rho_{\text{cls}}\,\rho_{\text{sub}}$, *with no assumption beyond nesting. The realised analogue* $n_{\text{FBI}} = N \cdot \frac{n_V}{N} \cdot \frac{n_P}{n_V} \cdot \frac{n_C}{n_P} \cdot \frac{n_{\text{FBI}}}{n_C}$, *with* $n_V = \sum_i V_i$, $n_P = \sum_i P_i$ *and* $n_C = \sum_i C_i$, *holds whenever these counts are positive.*

(b) Heterogeneity. *Suppose the stage probabilities of incident i depend on i only through a type X_i, such as a latent class (Section 6.1) or a covariate profile (Section 6.2). Write* $\rho_{\text{vic}}(x) = \mathbb{P}(V_i = 1 \mid X_i = x)$, $\rho_{\text{pol}}(x) = \mathbb{P}(P_i = 1 \mid V_i = 1, X_i = x)$, *and similarly* $\rho_{\text{cls}}(x)$ *and* $\rho_{\text{sub}}(x)$. *Let* $X \sim G_d$, *the empirical type distribution of* U_d. *Then each aggregate rate is a selection-weighted mean, for example* $\rho_{\text{pol}} = \mathbb{E}[\rho_{\text{vic}}(X)\rho_{\text{pol}}(X)]/\mathbb{E}[\rho_{\text{vic}}(X)]$, *and* $\mathbb{E}[n_{\text{FBI}} \mid U_d] = N\,\mathbb{E}[\rho_{\text{vic}}(X)\rho_{\text{pol}}(X)\rho_{\text{cls}}(X)\rho_{\text{sub}}(X)]$. *The factorisation into* unweighted *type averages,*

$$\mathbb{E}[n_{\text{FBI}} \mid U_d] = N\,\mathbb{E}[\rho_{\text{vic}}(X)]\,\mathbb{E}[\rho_{\text{pol}}(X)]\,\mathbb{E}[\rho_{\text{cls}}(X)]\,\mathbb{E}[\rho_{\text{sub}}(X)], \tag{7.1}$$

holds if the four random variables $\rho_{\text{stage}}(X)$ *are mutually independent under* G_d, *in particular if at most one stage varies with type. If X is real-valued and all four stage probabilities are nondecreasing in X (a common severity gradient), then the left side of* (7.1) *is at least the right side.*

(c) Identification by source. *Under (F1),* $\rho_{\text{vic}} = \mathbb{P}(V_I = 1 \mid W_I = 1)$ *and* $\rho_{\text{pol}} = \mathbb{P}(P_I = 1 \mid V_I = 1, W_I = 1)$. *Under (F2),* $\mathbb{P}(S_I = 1 \mid G_I = 1) = \rho_{\text{cls}}\,\rho_{\text{sub}}$. *Under (F3),* $\rho_{\text{sub}} = c_d$. *Under (F4), MSE estimates* N_V, *and* $\mathbb{E}[N_V \mid U_d] = N\rho_{\text{vic}}$, *so that* $\mathbb{E}[n_{\text{FBI}} \mid U_d] = \mathbb{E}[N_V \mid U_d]\,\rho_{\text{pol}}\,\rho_{\text{cls}}\,\rho_{\text{sub}}$.

Proof. (a) By nesting, $\{S_I = 1\} = \{V_I = 1, P_I = 1, C_I = 1, S_I = 1\}$, so the chain rule gives $\mathbb{P}(S_I = 1) = \rho_{\text{vic}}\rho_{\text{pol}}\rho_{\text{cls}}\rho_{\text{sub}}$. Moreover, $\mathbb{E}[n_{\text{FBI}} \mid U_d] = \sum_i \mathbb{P}(S_i = 1) = N\mathbb{P}(S_I = 1)$. The realised identity telescopes. (b) Conditioning on the type, $\mathbb{P}(V_I = 1, P_I = 1) = \mathbb{E}[\rho_{\text{vic}}(X)\rho_{\text{pol}}(X)]$ and $\mathbb{P}(V_I = 1) = \mathbb{E}[\rho_{\text{vic}}(X)]$. Their ratio is ρ_{pol}, and the other stages are handled in the same way. The same argument gives $\mathbb{P}(S_I = 1) = \mathbb{E}[\prod_{\text{stage}} \rho_{\text{stage}}(X)]$. Under mutual independence the expectation of the product is the product of the expectations. For the inequality, let $f, g \geq 0$ be nondecreasing. Then $\mathbb{E}[f(X)g(X)] \geq \mathbb{E}f(X)\,\mathbb{E}g(X)$ by Chebyshev's association inequality, which follows from $\mathbb{E}[\{f(X) - f(X')\}\{g(X) - g(X')\}] \geq 0$

Table 4. What identifies each factor of the funnel (Proposition 7.2(c)), and what does not.

Factor	Source	Assumption	Not identified without further input
N_V	MSE on the linked lists (Sections 4 and 6)	(F4), Assumption A4	the top interaction ω (Theorem 4.3); a sensitivity range for it
ρ_{vic}	victim survey: was the incident reported, by the respondent or anyone else, to the police or to an organisation that records incidents?	(F1)	channel self-detection; incidents outside the survey frame (institutional targets, non-household victims); state-level variation
ρ_{pol}	victim survey: share reported to the police among incidents reported to some channel	(F1)	state-level variation; rare-event sampling error, so a prior is needed
ρ_{cls}	police–NGO overlap among incidents that NGOs code as antisemitic and flag as reported to the police	(F2), (F3), linkage correction	without the police-report flag, only $\rho_{\text{pol}}\rho_{\text{cls}}\rho_{\text{sub}}$ among NGO-recorded incidents
ρ_{sub}	agency participation data	(F3)	incidence-weighted coverage; prior on σ_c (Section 6.3)

for an independent copy X'. A product of nonnegative nondecreasing functions is nonnegative and nondecreasing, so induction over the four stages gives $\mathbb{E}[\prod \rho_{\text{stage}}(X)] \geq \prod \mathbb{E}[\rho_{\text{stage}}(X)]$. (c) Under (F1), $\mathbb{P}(V_I = 1 \mid W_I = 1) = \mathbb{P}(V_I = 1)$ and $\mathbb{P}(P_I = 1 \mid V_I = 1, W_I = 1) = \mathbb{P}(P_I = 1 \mid V_I = 1)$, and within the survey frame $V_I = V_I^{\text{rep}}$, which the survey observes. Under (F2), nesting and $G_I \Rightarrow P_I$ give $\mathbb{P}(S_I = 1 \mid G_I = 1) = \mathbb{P}(C_I = 1 \mid P_I = 1, G_I = 1)\mathbb{P}(S_I = 1 \mid C_I = 1, G_I = 1) = \rho_{\text{cls}}\rho_{\text{sub}}$. Under (F3), $\rho_{\text{sub}} = \mathbb{P}(A_I = 1 \mid C_I = 1) = \mathbb{P}(A_I = 1) = c_d$. Under (F4), every recorded incident lies in U_d^V by construction, and the capture model holds on U_d^V, so the MSE estimand is N_V. Moreover $\mathbb{E}[N_V \mid U_d] = \sum_i \mathbb{P}(V_i = 1) = N\rho_{\text{vic}}$, and the last display follows from (a). □

Part (a) is an accounting identity; the content lies in (b) and (c). Part (b) warns against assembling the funnel from rates measured on different type mixtures: under a common severity gradient, the product of unweighted stage averages understates the share of incidents that reach the FBI, so dividing n_{FBI} by it overstates N. Part (c) says what each source delivers (Table 4). With all four sources the identity is over-identified, and the comparison of n_{FBI} with $\widehat{N}_V\hat{\rho}_{\text{pol}}\hat{\rho}_{\text{cls}}\hat{\rho}_{\text{sub}}$ enters the joint model as a likelihood term rather than a constraint.

What is not identified. Five gaps remain, to be closed by priors, external data or a reported range.

(1) *The unrecorded.* An incident that reaches no recording channel cannot appear on any list, however good the linkage, so every MSE estimator in this paper estimates at most N_V, and only under (F4). Recovering $N = N_V/\rho_{\text{vic}}$ needs a victim survey. Without one we report N_V and the downstream stages and treat ρ_{vic} as a sensitivity parameter. Such incidents change only the unobservable cell: in expectation they add $N(1 - \rho_{\text{vic}})$ to $\mu_{\mathbf{0}}$ and leave $\boldsymbol{\mu}_+$ unchanged, so by (3.10) targeting N rather than N_V is equivalent to raising $\bar{\omega}$ by

$$\Delta_{\text{vic}} = \log\left\{1 + \frac{(1 - \rho_{\text{vic}})\, N_V}{\rho_{\text{vic}}\, \mu_{\mathbf{0}}}\right\} \geq 0, \tag{7.2}$$

where $\mu_{\mathbf{0}}$ is the expected number of unrecorded incidents in U_d^V. The $\bar{\omega}$ sensitivity analysis therefore covers ρ_{vic} once its upper end is extended by Δ_{vic} (Section 9.5). (F4) also demands that the dependence

restriction hold on U_d^V. A list that detects incidents itself, such as a security patrol, enlarges U_d^V, but type-dependent detection can break the restriction even when every probability is positive: with two equally common types, a patrol list detecting with probabilities 0.6 and 0.05 and an NGO list with 0.5 and 0.05, independent given type, Lincoln–Petersen recovers 59% of the truth in expectation. Detection also escapes surveys, which is why (F1) rules it out within the survey frame; where it occurs there, the survey understates ρ_{vic} and $\widehat{N}_V/\hat{\rho}_{\text{vic}}$ overstates N.

(2) *Survey scope.* Household surveys of the NCVS type cover persons aged 12 or older in sampled households (Kena and Thompson, 2021; Masucci and Langton, 2017), so they say nothing about ρ_{vic} or ρ_{pol} for incidents whose only victim is an institution, such as a synagogue or cemetery; these need priors, informed where possible by surveys of institutions.

(3) *Separating ρ_{pol} from ρ_{cls}.* Incidents reported to the police but not classified as antisemitic are invisible in police data and appear only in NGO records that note a police report. Without that flag, the overlap identifies only the product of the downstream stages among NGO-recorded incidents, which also differs from the population product under referral (Proposition 4.10).

(4) *Failures of (F2).* If an NGO learns of incidents from police statements, NGO capture depends on classification and the overlap overstates ρ_{cls}. Linkage errors distort the overlap in both directions (under LE a true pair survives with probability $(1-\delta)^2$), so the overlap is computed from the cut posterior of Section 6.4.

(5) *Participation is not submission of every incident.* (F3) treats an agency that reports zero hate crimes as participating, as the FBI's counts do: for 2024, 19.0% of participating agencies reported at least one hate-crime incident (Federal Bureau of Investigation, 2025). An agency that never classifies anything as a bias crime contributes to c_d but not to n_{FBI}, a loss correctly attributed to ρ_{cls} only if ρ_{cls} is measured.

Finally, national surveys rarely support state-level estimates of ρ_{vic} and ρ_{pol}, so in the cross-jurisdiction analysis below their variation comes from a prior or a residual. We then pool unmeasured stages into a single residual player and say so.

7.2 Attributing differences to stages

Consider jurisdictions $s = 1, \dots, S$ with weights $w_s > 0$, $\sum_s w_s = 1$ (population shares or equal weights), and recorded rates $r_s = n_{\text{FBI},s}/P_s$. By Proposition 7.2(a), in its realised or its expected form, $r_s = \theta_s\, \rho_{\text{vic},s}\rho_{\text{pol},s}\rho_{\text{cls},s}\rho_{\text{sub},s}$ with $\theta_s = N_s/P_s$. Taking logs,

$$Y_s = \log r_s = \sum_{j=1}^{m} X_{js}, \qquad (X_{1s}, \dots, X_{5s}) = (\log\theta_s, \log\rho_{\text{vic},s}, \log\rho_{\text{pol},s}, \log\rho_{\text{cls},s}, \log\rho_{\text{sub},s}), \tag{7.3}$$

with $m = 5$ players: incidence and the four stages. Let Var_w and Cov_w denote weighted moments across jurisdictions. The Shapley value (Shapley, 1953) of a game v on the players $[m] = \{1, \dots, m\}$, with $v(\emptyset) = 0$, allocates the worth $v([m])$ as

$$\varphi_j(v) = \sum_{A \subseteq [m]\setminus\{j\}} \frac{|A|!\,(m-1-|A|)!}{m!}\,\{v(A\cup\{j\}) - v(A)\}, \tag{7.4}$$

the average marginal contribution of j over all orderings of the players. Following Shorrocks (2013), a factor is "eliminated" by holding it at a reference value. For dispersion the reference is the cross-jurisdiction mean; for a comparison of two jurisdictions it is the first jurisdiction's value.

Theorem 7.3 (Shapley decomposition of the funnel).

(i) Dispersion. *For the game $v(A) = \mathrm{Var}_w\big(\sum_{j\in A} X_j\big)$, the Shapley value is*

$$\varphi_j = \mathrm{Cov}_w(X_j, Y) = \mathrm{Var}_w(X_j) + \sum_{i\neq j} \mathrm{Cov}_w(X_i, X_j).$$

The shares sum to $\mathrm{Var}_w(Y)$, and $\varphi_j < 0$ if and only if $\mathrm{Cov}_w(X_j, Y) < 0$.

(ii) Two jurisdictions, levels. *Let jurisdictions A and B have factor vectors $\boldsymbol{a}, \boldsymbol{b} \in [0,\infty)^m$, and $r_A = \prod_j a_j$, $r_B = \prod_j b_j$. For the switching game $v(T) = \prod_{j\in T} b_j \prod_{j\notin T} a_j - r_A$, the Shapley value is*

$$\begin{aligned}\varphi_j &= (b_j - a_j) \sum_{T\subseteq[m]\setminus\{j\}} \frac{|T|!\,(m-1-|T|)!}{m!} \prod_{i\in T} b_i \prod_{i\notin T\cup\{j\}} a_i \\ &= (b_j - a_j) \int_0^1 \prod_{i\neq j} \{a_i + t(b_i - a_i)\}\, dt.\end{aligned} \tag{7.5}$$

It is exact, $\sum_j \varphi_j = r_B - r_A$. It is antisymmetric under exchanging A and B, and $\varphi_j = 0$ when $a_j = b_j$. For strictly positive factors, the LMDI-I allocation $\varphi_j^{\mathrm{L}} = \mathcal{L}(r_A, r_B)\log(b_j/a_j)$, with the logarithmic mean $\mathcal{L}(x,y) = (x-y)/(\log x - \log y)$ and $\mathcal{L}(x,x) = x$, is also exact. Both allocations are line integrals of ∇f, where $f(\boldsymbol{x}) = \prod_i x_i$, from $\boldsymbol{a}$ to $\boldsymbol{b}$. Shapley integrates along the segment $\boldsymbol{a} + t(\boldsymbol{b} - \boldsymbol{a})$, and LMDI along the geometric path $\boldsymbol{a}^{1-t}\boldsymbol{b}^t$. With $d_i = \log(b_i/a_i)$,

$$\varphi_j - \varphi_j^{\mathrm{L}} = \frac{\sqrt{r_A r_B}}{12}\, d_j \sum_{i=1}^{m} d_i(d_i - d_j) + O(\|\boldsymbol{d}\|^5). \tag{7.6}$$

(iii) Posterior uncertainty. *Let the factors be uncertain, with posterior means $\mu_{js} = \mathbb{E}[X_{js} \mid \text{data}]$ and $\mu_{Ys} = \mathbb{E}[Y_s \mid \text{data}]$, and write $\bar{X}_j = \sum_s w_s X_{js}$ and $\bar{Y} = \sum_s w_s Y_s$. The posterior of φ_j in (i) is obtained by evaluating $\mathrm{Cov}_w(X_j, Y)$ at each posterior draw, and*

$$\mathbb{E}[\varphi_j \mid \text{data}] = \mathrm{Cov}_w(\mu_j, \mu_Y) + \sum_s w_s\, \mathrm{Cov}(X_{js}, Y_s \mid \text{data}) - \mathrm{Cov}(\bar{X}_j, \bar{Y} \mid \text{data}). \tag{7.7}$$

Summed over j, the correction terms equal $\sum_s w_s \mathrm{Var}(Y_s \mid \text{data}) - \mathrm{Var}(\bar{Y} \mid \text{data}) \geq 0$. If Y is observed (realised recorded rates), the correction vanishes, and the plug-in share $\mathrm{Cov}_w(\mu_j, Y)$ is the posterior mean.

Proof. (i) Let Σ be the weighted covariance matrix of $(X_1, \ldots, X_m)$. Then $v(A) = \sum_{i,k\in A} \Sigma_{ik}$, and for $j \notin A$ the marginal contribution is $v(A\cup\{j\}) - v(A) = \Sigma_{jj} + 2\sum_{i\in A}\Sigma_{ij}$. In a uniformly random ordering, each $i \neq j$ precedes j with probability $1/2$. Averaging over orderings therefore gives $\varphi_j = \Sigma_{jj} + \sum_{i\neq j}\Sigma_{ij} = \mathrm{Cov}_w(X_j, \sum_i X_i)$. Summing over j gives $\sum_{i,j}\Sigma_{ij} = \mathrm{Var}_w(Y)$.

(ii) Since $v(T\cup\{j\}) - v(T) = (b_j - a_j)\prod_{i\in T} b_i \prod_{i\notin T\cup\{j\}} a_i$, the first expression in (7.5) is (7.4). For the second, expand $\prod_{i\neq j}\{(1-t)a_i + tb_i\} = \sum_T t^{|T|}(1-t)^{m-1-|T|}\prod_{i\in T} b_i \prod_{i\notin T\cup\{j\}} a_i$ and use $\int_0^1 t^q(1-t)^{m-1-q}dt = q!(m-1-q)!/m!$. Exactness is the efficiency of the Shapley value: $\sum_j \varphi_j = v([m]) = r_B - r_A$. Antisymmetry follows from the substitution $t \mapsto 1-t$. Along $\boldsymbol{x}(t) = \boldsymbol{a} + t(\boldsymbol{b} - \boldsymbol{a})$, $\partial_j f(\boldsymbol{x}(t))\,\dot{x}_j(t) = (b_j - a_j)\prod_{i\neq j} x_i(t)$, which is the integrand in (7.5). Along $\boldsymbol{x}(t) = \boldsymbol{a}^{1-t}\boldsymbol{b}^t$, $\partial_j f(\boldsymbol{x}(t))\,\dot{x}_j(t) = f(\boldsymbol{x}(t))\,d_j = r_A^{1-t} r_B^t d_j$, whose integral over $[0,1]$ is $\mathcal{L}(r_A, r_B)\,d_j = \varphi_j^{\mathrm{L}}$. Exactness of both allocations is the gradient theorem, $\sum_j \int \partial_j f\, dx_j = f(\boldsymbol{b}) - f(\boldsymbol{a})$; for LMDI it can also be checked

directly, since $\mathcal{L}(r_A, r_B) \sum_j d_j = \mathcal{L}(r_A, r_B) \log(r_B/r_A) = r_B - r_A$. The expansion (7.6) is proved in Appendix D.7.

(iii) Each φ_j is a function of the factors, so its posterior is the image of their posterior. Write $\varphi_j = \sum_s w_s X_{js} Y_s - \bar{X}_j \bar{Y}$ and take posterior expectations, using $\mathbb{E}[UV] = \mathbb{E}U\,\mathbb{E}V + \mathrm{Cov}(U, V)$ for each product. Summing over j gives $\sum_s w_s \mathrm{Var}(Y_s \mid \text{data}) - \mathrm{Var}(\bar{Y} \mid \text{data})$. This is nonnegative, because $\mathrm{Var}(\sum_s w_s Y_s) \le (\sum_s w_s \sigma_s)^2 \le \sum_s w_s \sigma_s^2$ with $\sigma_s^2 = \mathrm{Var}(Y_s \mid \text{data})$, by the Cauchy–Schwarz and Jensen inequalities. If Y is observed, φ_j is linear in the unknowns, and every correction term is zero. □

Part (i) is elementary: the covariance shares $\mathrm{Cov}_w(X_j, Y)/\mathrm{Var}_w(Y)$ are the "natural" decomposition of a variance by factor components (Shorrocks, 1982), and the proof shows that they are also the Shapley value of the elimination game. The closed form in (ii) is an instance of Owen's multilinear extension (Owen, 1972). Shapley decompositions of multiplicative identities are standard in index decomposition analysis (Albrecht et al., 2002), as is LMDI (Ang, 2004, 2015); to our knowledge, the cubic expansion (7.6) is not stated elsewhere. *Both* decompositions are exact, with no residual; they are different allocations of the same total, agreeing when all factors differ by the same proportion and to second order in general, and differing materially when a factor differs twofold or more between jurisdictions, which is common for participation. LMDI's shares equal the log shares $d_j / \sum_i d_i$, consistent with part (i), but LMDI is undefined when a factor is zero (for example, no participating agency), whereas (7.5) is not. We report the Shapley allocation for levels and LMDI as a check; with more than two jurisdictions, (ii) is applied to each against a common reference such as the national factors.

Reading the shares. A positive φ_j in (i) means that stage j widens the spread of recorded rates; a negative share means that factor j is negatively correlated with the recorded rate across jurisdictions. The most consequential case is incidence. If $\varphi_1 < 0$, recorded rates tend to *fall* as incidence rises. In that case, places with more antisemitic incidents record fewer per capita, because their funnel loses more, and a map of recorded rates tends to rank jurisdictions in the opposite order to incidence. Restricting the game to the four stages decomposes instead the dispersion of $\log(r_s/\theta_s) = -\log D_{1,s}$. By part (iii), shares are computed draw by draw: plugging in posterior means understates dispersion when Y is latent, because posterior means are shrunk toward the national mean (Louis, 1984). We decompose realised rates where counts are large, and expected rates where counts are small or zero.

7.3 An illustration with invented numbers

Table 5 applies Theorem 7.3(ii) to two hypothetical jurisdictions. **All inputs are invented for exposition** and describe no real place. Jurisdiction B has 50% more incidents per capita than A, yet its recorded rate is less than a third of A's. Incidence contributes +0.390 per 100,000 to the difference $r_B - r_A = -0.974$, that is −40% of the gap. Classification (51%) and submission (58%) together account for more than the whole gap, police reporting for 23% and reaching a channel for 8%; with A's classification and submission rates, B's recorded rate would be 1.536, above A's. The LMDI shares differ from the Shapley shares by 0.5 to 4.3 percentage points; the largest difference, for incidence, is 0.042 per 100,000, and the cubic term of (7.6) gives 0.039.

7.4 Policy reading

Each stage corresponds to a different lever.

Table 5. Shapley and LMDI decompositions of the difference in recorded rates between two *hypothetical* jurisdictions (Theorem 7.3(ii)). **Illustrative, invented numbers; not data.** Rates are per 100,000 residents; percentages are shares of $r_B - r_A$ and may not sum to 100 because of rounding.

				Shapley		LMDI-I	
Factor	A	B	$\log(b_j/a_j)$	φ_j	%	φ_j^{L}	%
Incidence θ	8.00	12.00	+0.405	+0.390	−40.0	+0.348	−35.7
Reaching a channel ρ_{vic}	0.60	0.55	−0.087	−0.080	8.2	−0.075	7.7
Police report ρ_{pol}	0.45	0.35	−0.251	−0.228	23.4	−0.216	22.2
Classification ρ_{cls}	0.70	0.40	−0.560	−0.494	50.7	−0.481	49.3
Submission ρ_{sub}	0.95	0.50	−0.642	−0.562	57.7	−0.551	56.6
Recorded rate r	1.436	0.462	−1.134	−0.974	100	−0.974	100

Incidence (θ).
Prevention, security and education; not a reporting problem. A high recorded rate may reflect a well-functioning funnel, and a low one may hide high incidence.

Reaching a channel (ρ_{vic}).
Victim support and community trust, which raise disclosure, and accessible channels such as multilingual, anonymous and third-party reporting.

Police reporting (ρ_{pol}).
Accessible and trusted police: liaison officers, online reporting, and forwarding of reports by community organisations with the victim's consent.

Classification (ρ_{cls}).
Classification protocols and training, documented bias indicators, and periodic audits comparing police and NGO coding of the same incidents (Proposition 7.2(c)).

Submission (ρ_{sub}).
NIBRS participation and data quality: technical assistance, funding for records systems, and follow-up on agencies that persistently report zero.

Three cautions apply. Shapley shares account for observed differences; they are not causal effects of interventions, and a change that raises trust in the police may raise ρ_{vic} as well as ρ_{pol}. Improving the funnel raises recorded counts, and applied to two years instead of two places, Theorem 7.3(ii) separates a rise due to better reporting from a rise in incidence, a separation needed before a recorded increase is read as an increase in antisemitism. And a share is only as informative as the data behind that stage's variation; shares of prior-driven stages are reported as such or pooled into a residual.

8 Simulation study

Every number in this section comes from synthetic worlds with known ground truth. No real incident or record is used. The study asks six questions.

(Q1) How do the estimators of Sections 4 and 6 behave under capture heterogeneity and referral, and does provenance recoding (Proposition 4.10(c)) remove the referral bias?

(Q2) Do the sensitivity bands of Corollary 4.5 contain the true dark figure?

(Q3) Does the linkage correction of Corollary 5.8 work when its own model, LE (Definition 5.2), generates the errors?

(Q4) Does it work when records carry realistic noise and are linked by Fellegi–Sunter EM?

(Q5) Does multiple imputation over linkage draws improve on a single linkage?

(Q6) Has the latent-class sampler converged at the settings we used?

Q3 and Q4 are kept apart on purpose. In the record-level scenarios the records carry attribute noise and are linked by FS-EM, and LE is used nowhere in generating them. LE enters only through the correction, which is therefore tested under misspecification. A separate table-level scenario generates the errors by LE itself. The results are from the second run of the study, made after a code audit; Appendix E lists what changed from the first run, together with the full configurations, seeds, software versions, runtimes and reproduction commands.

8.1 Design

Synthetic worlds. Each world has $N = 2000$ incidents. An incident has a date in one year (a quarter of the incidents fall in three two-week "flashpoint" spikes), an hour, one of 30 counties on a 6×5 grid, an offence type, a target and a venue string built from generic word lists. Four lists record the incidents: police, an NGO audit, a community security organisation and a campus or self-report portal ($K = 4$). Capture follows one of three models. Under *independent capture*, list k records each incident with probability p_k; the four probabilities are 0.2725, 0.305, 0.2275 and 0.2075. Under *latent classes*, each incident belongs to one of three reportability classes, "visible", "moderate" and "low visibility", with weights $(0.15, 0.35, 0.50)$, and given the class the lists record independently, with probabilities between 0.30 and 0.80 in the visible class and between 0.06 and 0.15 in the low-visibility class (Table 12). This is (6.1) with three classes, so LCMCR is correctly specified, and the independent-capture probabilities are its marginals. Under *latent classes with covariates*, class membership depends on the offence type, and type, target and campus location shift the logit capture probabilities; campus incidents, for instance, are less likely to reach the police and more likely to reach the portal. Referral (Proposition 4.10) is added by letting the NGO ingest each police-recorded incident with probability q. Every ingested record carries a provenance flag.

Records and linkage. Every capture produces a record. At noise level ν, records carry list-specific date lags and gross date errors, county errors and coarsening, type relabelling across four list vocabularies, typos, dropped words and missing values in the venue string, hour errors, and within-list duplicates (probability 0.04ν per record). All rates scale with ν, and $\nu = 0$ gives clean records (Appendix E.2). The records are then linked as in lines 1–2 of Algorithm 1. Candidate pairs lie in the same or adjacent counties and at most 14 days apart. Comparison vectors record agreement on date, hour, county, harmonised type and venue (Jaro–Winkler), with "missing" as a level of its own. A three-class FS-EM model (match, related non-match, unrelated non-match) relaxes conditional independence for distinct incidents at the same venue (Larsen and Rubin, 2001). After within-list deduplication, a greedy point linkage merges candidate pairs in decreasing order of the posterior match probability $\pi \geq 0.5$, with at most one record per list in each cluster. The error rates (δ, ε) of the point linkage are estimated by the random audit of Proposition C.13. The audit draws $B = 300$ candidate pairs, Neyman-allocated over strata of link status and posterior match probability, reveals the truth for each sampled pair, and computes $(\hat{\delta}, \hat{\varepsilon})$ by (5.10). Its estimand is defined

Table 6. Simulation scenarios (synthetic data). R: replicates per level; in parentheses, the replicates on which LCMCR is also run.

Scen.	Capture	q	Noise ν / linkage error	R	Question
S1	independent	0	0	100 (10)	baseline
S2	latent classes	0	0	100 (25)	Q1, Q6
S3	independent	0.5	0	100 (10)	Q1
S4	latent classes, covariates	0	1	100 (25)	Q1, Q4–Q6
S5	independent	0	0, 0.25, 0.5, 1, 1.5, 2	40 per ν	Q4, Q5
S6	closed form $\widehat{n}_{\mathbf{0}}(\omega)$ on the oracle tables of S1–S4				Q2
S7	independent	0, 0.25, . . . , 1	oracle tables only	200 per q	Q1
S8	independent (tables)	0	LE errors, (δ, ε) on a 3×3 grid	200 per cell	Q3

on the candidate-pair frame, so true matches outside every block are invisible to it. It is not the targeted review of Algorithm 3, and reviewed links are not fixed.

Scenarios. Table 6 lists them. S6 reuses the oracle tables of S1–S4, and S7 analyses oracle tables only. S8 bypasses records. It draws a true table under independent capture and applies the LE model itself, incident by incident, on the grid $\delta \in \{0.05, 0.1, 0.2\} \times \varepsilon \in \{0, 0.02, 0.05\}$, so the model behind Corollary 5.8 holds exactly. The rates are then estimated by a simulated random audit of the LE linkage, with 200 records for δ and 100 linked edges for ε.

Estimators. We compare five: Lincoln–Petersen on police and NGO in Chapman's form (LP; Chapman, 1951), the closed form of Theorem 4.4 at $\omega = 0$ with the interval of Remark A.4 (CF), model averaging over the 64 hierarchical log-linear models with all main effects and any subset of two-way interactions, with BIC weights and model-averaged Wald intervals (BMA; Turek and Fletcher, 2012), Chao's estimator (Theorem 4.8), and LCMCR, that is (6.1) with $J^\star = 10$ and the priors of Section 6.1, summarised by the posterior median and the equal-tailed interval.

Pipelines. Each estimator is applied to the table of one pipeline. *Oracle* uses the true linkage and *naive* the point linkage. *LE-corrected* applies Corollary 5.8(ii) to the naive table with the audit rates $(\hat{\delta}, \hat{\varepsilon})$. Its inner interval is widened by v_{link} of Corollary 5.8(i), computed by the delta method through the Jacobian of the LE operator; for LCMCR, one chain is run on each of four imputed tables and the draws are pooled. A second interval also adds the audit variance of Corollary 5.8(iii); because the audit samples the same linkage, this sum is conservative (Remark C.9). Three diagnostic variants replace the audit rates. The *realised* rates of the point linkage separate misspecification of LE from audit error. The $\varepsilon = 0$ *variant*, $(\hat{\delta}, 0)$, was added after the first run had shown the full correction failing, so it is *post hoc*. The *best-fit* rates $(\delta^*, \varepsilon^*)$ minimise a Pearson distance between $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t})$ and the naive table, with $\boldsymbol{t}$ the oracle table; they are *infeasible* and show the best that any scalar (δ, ε) can do. *MI* uses twenty draws from a single-record Gibbs sampler over partitions with at most one record per list per cluster, targeting the plug-in Fellegi–Sunter pairwise posterior (m and u fixed at their FS-EM values, pairs with $\pi < 10^{-3}$ truncated). These are approximate draws for the empirical-Bayes variant of line 5 of Algorithm 1, not a fully Bayesian linkage posterior. LP, CF, BMA and Chao are combined by Rubin's rules (Rubin, 1987) on $\log(\widehat{N} - c)$, with a lower offset c common to all draws (Proposition 5.11(iv), Appendix E.5); LCMCR pools one chain on each of the first four draws.

Metrics. The primary metric is the median over replicates of $(\widehat{N} - N)/N$. Means are also reported, but a few blow-ups ($\widehat{N} > 5N$) can dominate them, so blow-ups are counted. Coverage is that of nominal 95%

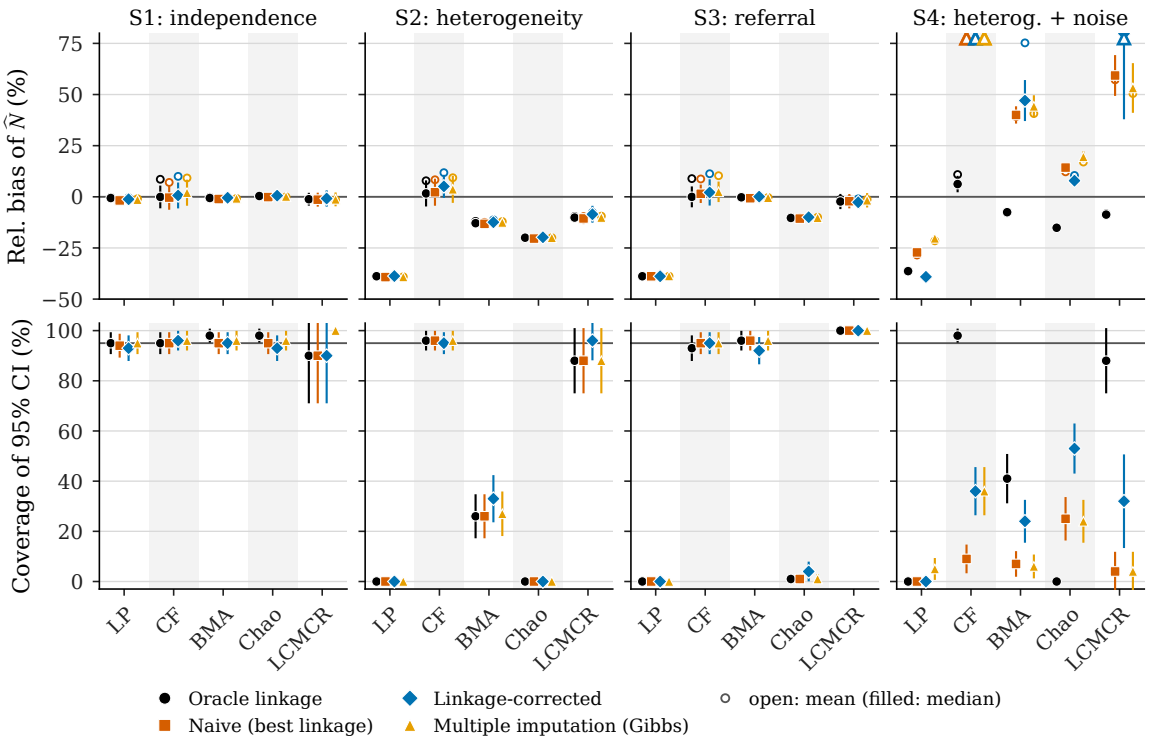


Figure 4. Synthetic data, scenarios S1–S4. Top: median relative bias of $\widehat{N}$ (filled; bars ±2 MCSE) and mean relative bias (open). Bottom: coverage of nominal 95% intervals (bars ±2 MCSE). Triangles on the upper edge mark values beyond the axis. Linkage-corrected: LE correction with audit rates, interval with v_{link}. $R = 100$; LCMCR on 10 (S1, S3) or 25 (S2, S4) replicates.

intervals for the realised N. At a coverage of 0.95 its Monte Carlo standard error (MCSE), $\{c(1-c)/R\}^{1/2}$, is 0.022, 0.015, 0.034, 0.044 and 0.069 for $R = 100$, 200, 40, 25 and 10. Medians carry a distribution-free MCSE (Appendix E.7).

8.2 Results

(Q1) Heterogeneity and referral. Figure 4 and Table 7 summarise S1–S4. In S1 every estimator is close to unbiased: the oracle medians lie within ±1.2% and the coverages between 0.90 and 0.98. Clean records are linked almost perfectly (pairwise precision and recall about 0.99 in S1–S3), so every S1 pipeline lies within ±2.0%. The closed form is unbiased in median (−0.1%) but heavy-tailed (mean +8.6%; median interval width $1.09N$, against $0.10N$ for BMA). Chao's estimator is +0.4%, matching its population limit under independent capture with unequal list probabilities, where Proposition 4.9 makes it an upper bound.

Heterogeneity (S2) separates the estimators as Section 4 predicts. LP is −38.8% (population limit −39.2%; downward by Proposition 4.7(a)), and Chao's estimator is −20.0% (limit −19.8%; a lower bound by Theorem 4.8). BMA is −12.9% with coverage 0.26, because BIC selects pairwise models that do not absorb the heterogeneity. The closed form is +1.7% in median, since the population top interaction is small ($\omega = -0.074$), but it remains heavy-tailed. LCMCR, correctly specified here, is −10.2% with coverage 0.88 (MCSE 0.065, $R = 25$). Its low-visibility class holds half of the incidents, with capture probabilities of 0.06–0.15, and four lists barely identify it (Remark 6.2); its fits also fail our convergence protocol (Q6).

Table 7. Synthetic data, scenarios S1–S4. Median (MAD): median relative bias of $\widehat{N}$ in %, with 1.4826× the median absolute deviation in percentage points. Mean: mean relative bias in %; the superscript counts replicates with $\widehat{N} > 5N$. Cov.: coverage of nominal 95% intervals in %. Corrected: LE correction with audit rates, interval with v_{link}. MI: 20 Gibbs linkage draws with Rubin's rules. $R = 100$; LCMCR on 10 (S1, S3) or 25 (S2, S4) replicates.

		LP			CF			BMA			Chao			LCMCR		
Scen.	Pipeline	Median (MAD)	Mean	Cov.	Median (MAD)	Mean	Cov.	Median (MAD)	Mean	Cov.	Median (MAD)	Mean	Cov.	Median (MAD)	Mean	Cov.
S1	Oracle	−0.6 (5)	−0.3	95	−0.1 (19)	+8.6	95	−0.6 (2)	−0.2	98	+0.4 (3)	+0.2	98	−1.2 (2)	−1.1	90
	Naive	−1.8 (4)	−1.3	94	−0.4 (21)	+7.0	95	−1.1 (2)	−1.1	95	−0.1 (3)	−0.4	95	−1.4 (2)	−1.6	90
	Corrected	−1.2 (5)	−0.4	93	+0.7 (21)	+10.0	96	−0.5 (2)	−0.2	95	+0.6 (3)	+0.6	93	−0.8 (1)	−1.1	90
	MI	−1.4 (4)	−0.9	95	+2.0 (21)	+9.2	96	−0.8 (2)	−0.8	96	+0.1 (3)	−0.0	96	−1.2 (2)	−1.5	100
S2	Oracle	−38.8 (2)	−39.0	0	+1.7 (21)	+7.9	96	−12.9 (4)	−12.0	26	−20.0 (2)	−19.9	0	−10.2 (4)	−9.6	88
	Naive	−39.2 (2)	−39.2	0	+2.3 (25)	+8.3	96	−13.2 (4)	−12.5	26	−20.4 (2)	−20.3	0	−10.6 (4)	−9.7	88
	Corrected	−38.7 (3)	−39.0	0	+5.1 (25)	+11.8	95	−12.4 (4)	−11.3	33	−19.8 (3)	−19.5	0	−8.5 (7)	−7.6	96
	MI	−39.2 (2)	−39.1	0	+3.7 (24)	+9.3	96	−12.5 (4)	−12.1	27	−20.0 (2)	−20.0	0	−10.2 (4)	−9.5	88
S3	Oracle	−38.8 (2)	−38.9	0	−0.0 (21)	+8.9	93	−0.2 (3)	−0.1	96	−10.3 (2)	−10.1	1	−2.3 (2)	−1.3	100
	Naive	−38.8 (2)	−38.9	0	+1.5 (23)	+8.7	95	−0.7 (3)	−0.7	96	−10.7 (2)	−10.4	1	−2.1 (2)	−1.4	100
	Corrected	−38.9 (2)	−38.9	0	+2.1 (23)	+11.3	95	+0.1 (3)	+0.2	92	−10.0 (2)	−9.7	4	−2.7 (2)	−1.0	100
	MI	−38.7 (2)	−38.8	0	+2.4 (23)	+10.3	95	−0.4 (2)	−0.3	96	−10.3 (2)	−10.0	1	−1.7 (3)	−1.3	100
S4	Oracle	−36.3 (2)	−36.2	0	+6.2 (18)	+10.9	98	−7.5 (4)	−7.8	41	−15.1 (2)	−15.2	0	−8.7 (3)	−8.3	88
	Naive	−27.2 (6)	−28.5	0	+168.1 (108)	+204.3[7]	9	+40.0 (19)	+39.2	7	+14.3 (10)	+12.2	25	+59.3 (13)	+57.2	4
	Corrected	−39.1 (5)	−39.3	0	+458.8 (638)	> +10[4][54]	36	+47.1 (41)	+75.3[1]	24	+7.9 (15)	+10.4	53	+79.8 (72)	+344.4[2]	32
	MI	−20.3 (8)	−21.5	5	+87.3 (44)	+97.8	36	+44.2 (17)	+40.7	6	+19.5 (11)	+17.0	24	+53.2 (15)	+50.5	4

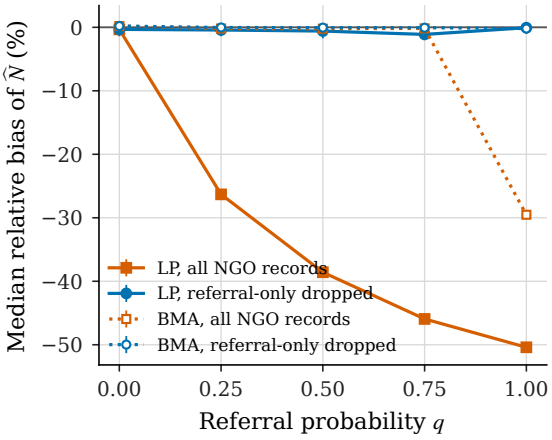


Figure 5. Synthetic data, S7 (independent capture, oracle linkage, $R = 200$ per q). Median relative bias of $\widehat{N}$ against the referral probability q, using all NGO records or after dropping referral-only records by their provenance flag (Proposition 4.10(c)). Bars: ±2 MCSE. Coverage is in Table 20.

Referral at $q = 0.5$ (S3) biases LP by −38.8%, against the limit of −38.7% from (4.8). Dropping the referral-only NGO records by their provenance flag (Proposition 4.10(c)) gives −1.1%, with coverage 0.93. The four-list estimators need no correction (BMA −0.2%, closed form −0.0%), because the other two lists are independent of the referral channel (Proposition 4.10(e)). Chao's estimator is −10.3%, and LCMCR −2.3%. The sweep S7 (Figure 5) confirms the limits. At $q = 0.25$, 0.5, 0.75 and 1, LP on all records is −26.3%, −38.6%, −45.9% and −50.4%, against −26.4%, −38.7%, −45.9% and −50.6%; after recoding it lies between −1.1% and −0.1%, with coverage 0.945–0.970. BMA on all records fails only at $q = 1$ (−29.5%, coverage 0.01), where every police record reaches the NGO, the police-only cells are structurally empty and ω is undefined. Recoding repairs this too (−0.2%; Table 20).

In S4, with oracle linkage, LP is −36.3%, Chao −15.1%, BMA −7.5% (coverage 0.41), LCMCR −8.7% (coverage 0.88) and the closed form +6.2% (coverage 0.98, median width $0.94N$). Q4 takes up the effect of linkage.

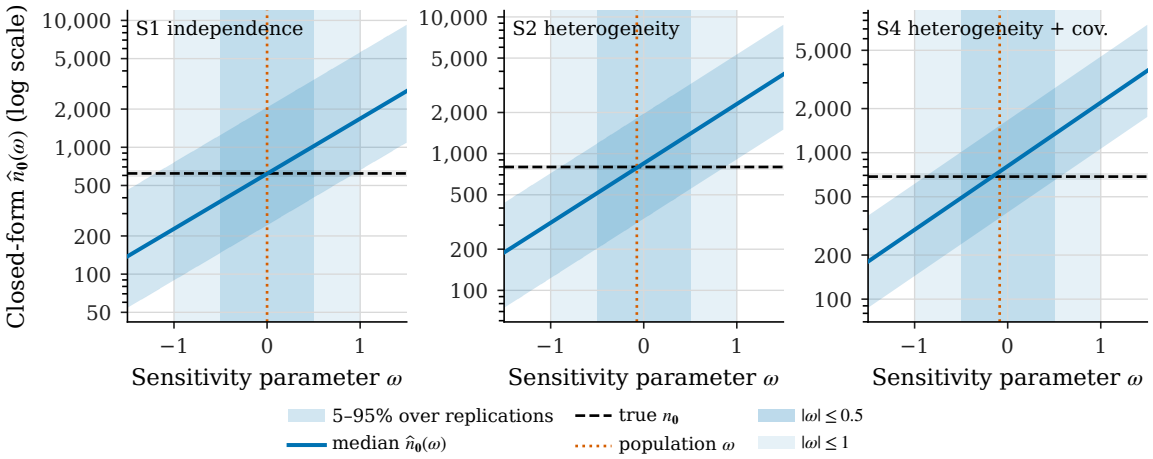


Figure 6. Synthetic data, S6. Closed-form $\widehat{n_0}(\omega)$ on the oracle tables of S1, S2 and S4 ($R = 100$): median curve and 5–95% band over replicates, on a log scale. Shading: $|\omega| \le 0.5$ and $|\omega| \le 1$. Dashed: mean true n_0 (grey band: its 5–95% range). Dotted: mean population ω.

(Q2) Sensitivity bands. On the oracle tables (S6; Figure 6, Table 21), the closed form at $\omega = 0$ has a median error in n_0 of −0.2% (S1), +4.1% (S2), −0.1% (S3) and +18.2% (S4). The population top interaction is 0 in S1 and S3 and negative in S2 (−0.074) and S4 (−0.086). With $K = 4$, $\bar{\omega} = \omega$, so Proposition 4.6 predicts the observed overstatement, although the S4 median exceeds the asymptotic $e^{-\bar{\omega}} - 1 = +9.0\%$ (S2: +7.7%). The estimated bounds of Corollary 4.5(b) contain n_0 in 87%, 93%, 87% and 98% of replicates (S1–S4) for $\Omega = 1$, and in 50–71% for $\Omega = 0.5$. They are point estimates of the identified set, with no allowance for sampling error, so they can miss n_0 even when $|\omega| \le \Omega$ holds (S1, S3). The union of the Wald intervals for n_0 over $|\omega| \le \Omega$ (the construction of Corollary 4.5(c) with $c = z_{0.975}$) covers in 99–100% of replicates for $\Omega = 0.5$ and in all of them for $\Omega = 1$. The price is width. At $\Omega = 1$ the bounds span a factor of $e^2 \approx 7.4$, and the median S1 bounds are $[228, 1682]$, against a mean true n_0 of 622. The bands behave as the theory says, but without a substantive bound on ω they are too wide to inform policy. This is the identification problem of Theorem 4.3, not a feature of the simulation.

(Q3) The LE correction when its model holds. It works (S8; Figure 7; cell-by-cell values in Table 17). Across the grid the naive BMA is biased by +2.4% to +62.7%, more for larger δ. With the true (δ, ε) the corrected estimators are median-unbiased: BMA −0.3% to +0.5%, LP −0.7% to +0.3%, Chao −0.4% to +1.5%, and the heavy-tailed closed form −3.2% to +7.3%. The intervals need v_{link}. With it, BMA covers in 0.945–0.985 of replicates (oracle: 0.950–0.970); without it, as in the first run, BMA covers in 0.725–0.965, worst at $\delta = 0.2$. LP and Chao cover in 0.940–0.980 and 0.930–0.960 with v_{link}, and in 0.790–0.920 and 0.630–0.895 without. The closed form drops to 0.905 at $\delta = 0.2$, $\varepsilon = 0$, where its forward-difference gradient is inaccurate.

With audit-estimated rates the point estimates stay median-unbiased (BMA −1.5% to +1.1%), and the audit's own 95% intervals cover the realised δ in 0.895–0.95 of replicates and the realised ε in 0.91–0.93. But v_{link} alone now undercovers (BMA 0.750–0.835, Chao 0.695–0.810), because it ignores the audit error. Adding the audit variance of Corollary 5.8(iii) restores coverage: BMA 0.925–0.985, LP 0.925–0.980, CF 0.920–0.975 and Chao 0.905–0.970. As Remark C.9 predicts, the sum is conservative in some cells, and it costs width: the median BMA width is $0.17N$–$0.44N$, against $0.12N$–$0.22N$ with known rates.

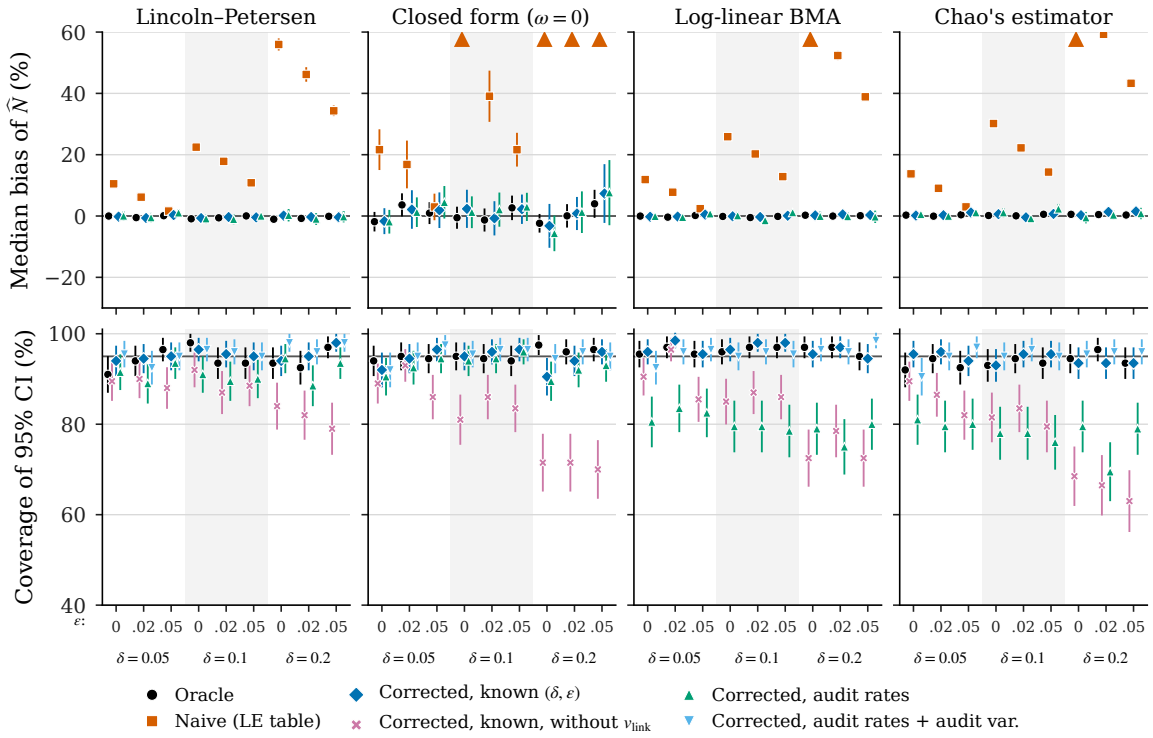


Figure 7. Synthetic data, S8: observed tables generated by the LE model (independent capture, $N = 2000$, $R = 200$ per (δ, ε) cell). Top: median relative bias of $\widehat{N}$ (bars ±2 MCSE; triangles: off the axis). Bottom: coverage of nominal 95% intervals (bars ±2 MCSE). Known: correction with the true (δ, ε). Audit rates: rates from a simulated random audit (200 records, 100 linked edges).

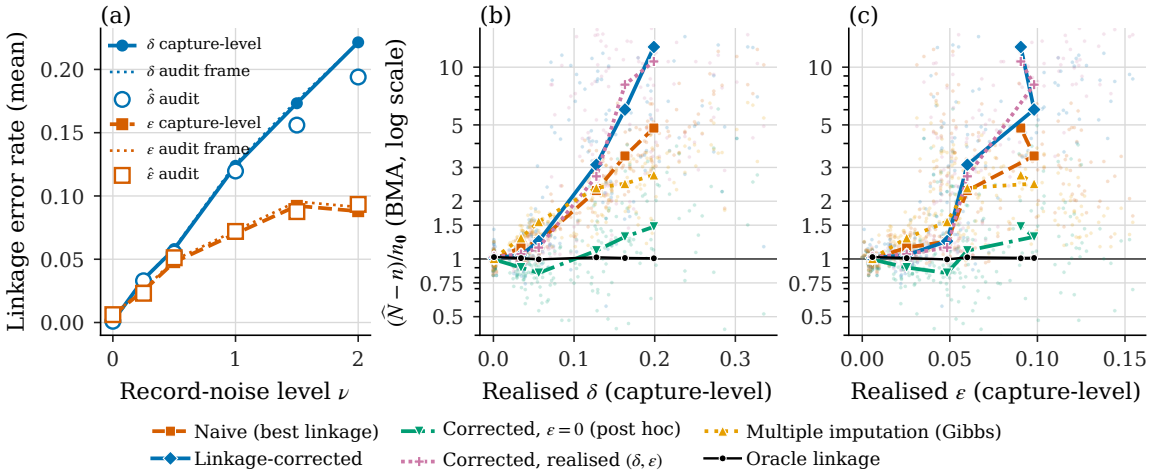


Figure 8. Synthetic data, S5 (independent capture, record-noise sweep, $R = 40$ per level). (a) Means over replicates of the realised capture-level rates δ and ε of the point linkage, the same rates on the audit frame, and the audit estimates. (b, c) Dark-figure ratio $(\widehat{N} - n)/n_0$ for BMA against the realised δ and ε, on a log scale, where n is the true number of recorded incidents. Points: replicates. Lines: medians by noise level. The $\varepsilon = 0$ variant is post hoc.

Table 8. Synthetic data. BMA under realistic linkage: the S5 noise sweep (independent capture, $R = 40$ per level) and S4 ($R = 100$). Rates: means over replicates of the realised capture-level false-match rate ε of the point linkage, its audit estimate $\hat{\varepsilon}$, and the best-fit ε^*. Remaining columns: median relative bias of $\widehat{N}$ in %; the superscript counts replicates with $\widehat{N} > 5N$. LE-corr.: audit rates. Realised: realised rates. † Post hoc variant, added after the first run. ‡ Infeasible diagnostic that uses the oracle table. Generated from `results/sim_results.json`.

	False-match rate			Median relative bias of $\widehat{N}$ (%)						
	ε	$\hat{\varepsilon}$	ε^*	Oracle	Naive	LE-corr.	Realised	ε=0†	Best-fit‡	MI
S5, $\nu = 0$	0.006	0.006	0.010	+0.7	0.0	+0.4	+0.8	−0.2	+0.5	+0.1
S5, $\nu = 0.25$	0.025	0.023	0.011	+0.3	+4.5	+1.5	+1.8	−3.1	−0.3	+8.7
S5, $\nu = 0.5$	0.048	0.051	0.015	−0.2	+7.2	+7.5	+4.5	−5.0	−0.7	+17.1
S5, $\nu = 1$	0.070	0.072	0.008	+0.6	+40.4	+63.4[1]	+52.1	+3.0	+9.2	+41.1
S5, $\nu = 1.5$	0.092	0.088	0.012	+0.4	+75.3	+158.5[7]	+219.7[5]	+9.7	+22.4	+44.2
S5, $\nu = 2$	0.088	0.093	0.002	+0.3	+117.8	+367.2[18]	+306.9[14]	+15.0	+27.0	+53.3
S4	0.077	0.079	0.000	−7.5	+40.0	+47.1[1]	+38.5	−4.5	+4.3	+44.2

(Q4) The LE correction under realistic linkage. It fails (S4, S5; Figure 8, Table 8). In S4 the point linkage has pairwise precision 0.905 and recall 0.787. On average it misses 37.9 of 91.6 within-list duplicates and leaves 126.7 more clusters than there are recorded incidents, so the naive BMA is +40.0% (coverage 0.07). The audit estimates its targets well: $\hat{\delta} = 0.107$ against 0.111 on its frame, and $\hat{\varepsilon} = 0.079$ against 0.081, with interval coverage 0.96 and 0.98. Yet the LE-corrected BMA is +47.1% (mean +75.3%, one blow-up), worse than naive, with coverage 0.24 (0.10 without v_{link}). The audit variance raises coverage to 0.76 only by making the median interval $2.36N$ wide. The corrected closed form is +458.8%, with 54 blow-ups in 100 replicates.

The audit is not to blame: with the realised rates the correction still gives +38.5%. The model is. The best-fitting scalar LE has $\varepsilon^* = 0.000$ in every S4 replicate ($\delta^* = 0.079$ on average), and in S5 its mean ε^* lies between 0.002 and 0.015, although 7–9% of singletons are falsely merged at $\nu \geq 1$ and in S4. Swaps dominate the errors (Remark 5.9). Under the one-record-per-list constraint, a record whose true partner is degraded by noise tends to be linked to a wrong partner from the same list. True clusters $\{a_1, a_2\}$ and $\{b_2\}$ (subscripts denote lists) become $\{a_1, b_2\}$ and $\{a_2\}$, and the table does not change. The audit correctly counts one split true pair and one false edge, so both $\hat{\delta}$ and $\hat{\varepsilon}$ rise. With $K = 4$ the two spurious corrections do not cancel, and here they over-correct. Missed duplicates add singletons that LE does not model either (Remark C.8). In S5 the correction helps at $\nu = 0.25$ (+1.5% against +4.5% naive) and is neutral at $\nu = 0.5$ (+7.5% against +7.2%). From $\nu = 1$ on it is worse than naive: +63.4%, +158.5% and +367.2%, against +40.4%, +75.3% and +117.8%.

Two further pipelines bracket what is achievable. The $\varepsilon = 0$ variant is the best feasible pipeline in the study: −4.5% with coverage 0.69 in S4, and −5.0% to +15.0% across S5. But we chose it after seeing the first run. It ignores false matches that the audit shows to be frequent, and it would be wrong whenever false matches do change the table, as in S8. It is a post hoc observation, not a validated method. The infeasible best-fit LE gives +4.3% with coverage 0.93 in S4, but +9.2%, +22.4% and +27.0% at $\nu = 1$, 1.5 and 2. There its minimised Pearson distance reaches 143 (S4: 35; $\nu = 0$: 0.3), so at high noise no scalar (δ, ε) represents the errors. The LE correction is therefore a stylised forward model, valid when the errors are LE-like (Q3), and a random audit cannot by itself tell whether they are. Fitting "table-effective"

rates on audited clusters, the feasible analogue of $(\delta^*, \varepsilon^*)$, and comparing them with $(\hat{\delta}, \hat{\varepsilon})$ would diagnose swap-dominated errors. We have not implemented this diagnostic.

(Q5) Multiple imputation against the point linkage. In S1–S3, with clean records, MI, naive and oracle agree (Table 7). Under noise MI does not beat the naive table except at the highest noise levels (Table 18). In S4 the BMA is +44.2% against +40.0% naive (coverage 0.06 and 0.07). In S5, MI is worse at $\nu = 0.25$ and 0.5 (+8.7% against +4.5%; +17.1% against +7.2%) and about equal at $\nu = 1$ (+41.1% against +40.4%). It is better only at $\nu = 1.5$ and 2 (+44.2% against +75.3%; +53.3% against +117.8%), and even there its coverage is only 0.15–0.20. The draws split more true clusters than the point linkage: in S4 their pairwise recall is 0.710 against 0.787, and they leave 177.6 surplus clusters against 126.7. MI therefore propagates uncertainty around a worse-centred table. This does not contradict Proposition 5.11(iv), which assumes exact draws from a correct linkage posterior; the plug-in pairwise-composite target is neither. Mixing is not the problem. In S4 the lag-1 autocorrelation of the number of links is −0.033, and about 343 of 2186 active records change cluster in each sweep. The first run's Bernoulli-and-greedy imputation was worse (+64.5%; Table 16). We tried neither a fully Bayesian linkage model nor calibrating π on the audit sample.

(Q6) LCMCR computation. Each fit uses four chains of 7500 iterations (burn-in 2500, thinning 5; 4000 pooled draws) from overdispersed starting values for n_0. Appendix D.5 requires $\widehat{R} < 1.01$ and bulk and tail ESS of at least 400. All S1 oracle fits and 80% of the S1 naive fits meet this protocol; in S3, 10% (oracle) and 40% (naive) do. The S2 and S4 fits do not (Table 19). Only 8–12% reach $\widehat{R} < 1.01$, the median $\widehat{R}$ is 1.024–1.048 (maxima up to 1.341), the median bulk ESS is 95–174, and 0–4% meet the protocol. The LCMCR results for S2–S4 therefore come from a reduced protocol. The corrected and MI fits pool one chain per imputation, so they have no $\widehat{R}$ at all.

Two pieces of evidence suggest that the shortfall does not drive the reported biases. The four chain medians of a fit differ by a median of 2.9% of N in S2 (maximum 13.2%), about 2% in S4 with oracle linkage (maximum 7.4%) and 5.9% with naive linkage (maximum 20.6%). The first two are small next to the oracle biases of −10.2% and −8.7%. The first run, with single chains of 6000 iterations on 40 replicates, also gave similar medians (−11.2% in S2 and −8.3% in S4; Table 16). This is evidence, not proof. LCMCR should not be used on real data until a run meets the protocol.

8.3 What the simulation does and does not show

The simulation shows four things. First, the identification results of Section 4 hold in finite samples. Referral and heterogeneity bias Lincoln–Petersen by close to their population limits, provenance recoding or lists unconnected to the referral channel remove the referral bias, and Chao's estimator falls below N under incident heterogeneity and slightly above it under list heterogeneity. Second, when LE holds, the correction of Corollary 5.8 removes the bias, and its intervals are calibrated once v_{link} and the audit variance are included. Third, under realistic FS-EM linkage the same correction, with well-estimated rates, over-corrects, because swaps dominate the errors and scalar LE cannot represent them. Fourth, multiple imputation from a plug-in pairwise posterior does not repair a biased linkage.

The simulation does not show how the methods perform on real lists. The worlds are not calibrated to any source, and they use one population size ($N = 2000$) and one number of lists ($K = 4$); calibration is part of the analysis plan (Section 9). Nor does it evaluate the fully Bayesian linkage of Algorithm 1, with uncertain m and u and duplicates resolved jointly, the targeted review of Algorithm 3, a feasible estimator of table-effective linkage rates, or the covariate, small-area and funnel models of Sections 6.2, 6.3 and 7. Its LCMCR results for S2–S4 rest on a sampler that does not meet our own convergence protocol, and

LCMCR is never refitted over a grid of ω or q. Finally, the post hoc $\varepsilon = 0$ variant performed best under realistic linkage, but nothing here licenses it as a default.

9 A pre-registered analysis plan

This is the analysis plan we will register, with the commit hash of the analysis code, before any real record is accessed (Nosek et al., 2018); any departure will be reported as a deviation, with its reason. The estimators are those tested on synthetic data in Section 8; the sensitivity devices of Section 6.1 and the linkage diagnostics below will be tested on synthetic worlds with known $\bar{\omega}$, q and linkage errors before registration. The plan contains no real-data results.

9.1 Estimands

Primary estimand. $N_{V,d}$ is the number of *criminal antisemitic incidents*, under the harmonised scope of Section 2.5, that occur in a pilot domain d (a jurisdiction and a period) and reach at least one recording channel: the population U_d^V of Section 7, which is what the lists identify. The candidate pilot is New York City, whose police list is public at complaint level with an anti-Jewish bias field (Section 2.1), for 2024–2025, after the Audit's October 2023 scope change (Section 2.2); both are fixed at registration after a check for interruptions in any list's reporting.

Secondary estimands.

(i) The dark figure $n_{\mathbf{0}} = N_{V,d} - n$ and the ratio $D_1 = N_{V,d}/n_{1+}$.

(ii) The list coverages $p_k = n_{k+}/N_{V,d}$. Because list 1 is the NYPD's, observed before submission, and $N_{V,d}$ conditions on reaching a channel, p_1 corresponds to $\rho_{\text{pol}}\rho_{\text{cls}}$ (Proposition 7.2); comparing NYPD with NIBRS records estimates ρ_{sub}.

(iii) $N_{V,d}$ by conduct class (assault; threat or intimidation; property damage), if each class has enough linked records to pass the checks of Section 9.6.

(iv) Scope variants: $N_{V,d}$ with and without incidents carrying the Israel/Zionism content flag (Section 2.5), reported side by side with no preferred variant.

(v) Exploratory: borough-by-year counts from the small-area model (Equation (6.9)), which has not been evaluated by simulation, released only under the disclosure rules of Section 10.

(vi) Scenarios for all in-scope incidents, $N_d = N_{V,d}/\rho_{\text{vic}}$, over the grid of ρ_{vic} in Table 10. No estimate of ρ_{vic} for antisemitic incidents exists (Section 2.4), so these are sensitivity analyses, not estimates.

9.2 Lists and data-sharing agreements

Table 9 lists the records we will request, numbered as in Section 2 with the NYPD as list 1. The public files lack exact dates, locations and provenance, so they serve only to check the extracts. Every agreement will specify purpose limitation, minimum necessary fields, linkage inside a secure enclave (Section 10), output review for disclosure risk only, so that providers have no veto over findings, and retention and destruction schedules.

9.3 Linkage protocol

1. *Ontology coding.* Two coders independently code every record on the axes of Section 2.5, blind to the other lists; a third resolves disagreements.

Table 9. Lists to be acquired for the candidate pilot (New York City). Fields marked † are held only inside the secure enclave (Section 10).

k	Holder	Fields requested	Agreement
1	NYPD; NIBRS extract for the same agencies	Complaint ID; date and time; geocoded location†; offence; bias code(s); target type; narrative†; date submitted to state/FBI	Research data-use agreement; IRB approval
2	ADL	Incident ID; date; location†; ADL category; narrative†; **provenance** (victim, police, media, named partner); content flag	Data-use agreement; disclosure-only output review
3	SCN	Log ID; date; location†; type; narrative†; whether referred to law enforcement; whether shared with ADL	Data-use agreement
4	Campus or community self-report portal(s)	Report ID; date; location†; conduct; bias category identifying anti-Jewish bias; narrative†	Agreement with each operator
aux	BJS (NCVS); Dept. of Education (Clery)	Published hate-crime reporting estimates; institution-level religion-bias counts	Public; no agreement needed

2. *Within-list deduplication*, so that each list holds at most one record per incident (Section 3).
3. *Standardisation and encoding.* Dates are standardised to the day; locations are geocoded and encoded together with institution names (Section 10). Narratives never leave the enclave.
4. *Blocking and comparison.* Candidate pairs are restricted to the same or adjacent precinct and a date difference of at most 14 days. The comparison vector $\gamma_{rr'}$ has levels for date difference (0, 1, 2–3, 4–14 days), distance band, conduct agreement, target-type agreement and encoded-location similarity.
5. *Model-based linkage.* $m(\gamma)$ and $u(\gamma)$ are estimated by Algorithm 2 (Fellegi and Sunter, 1969), and the posterior over the linkage structure Λ is sampled by multifile Bayesian matching (Section 6.4; Aleshin-Guendel and Sadinle, 2023).
6. *Targeted clerical review* of uncertain links by Algorithm 3 (Theorem C.10), by two reviewers blind to which estimate a pair would move, with Cohen's κ reported; reviewed pairs are fixed. Targeted review is not a probability sample and is *not* used to estimate error rates.
7. *Random audit.* After targeted review, a stratified random audit of at least 300 candidate pairs, allocated as in Proposition C.13 and with a record-level stratum of at least 100 records for errors outside the blocks, estimates the count-based LE parameters $(\hat{\delta}, \hat{\varepsilon})$ and the pairwise precision and recall. A simple random sample of at least 50 connected components of the candidate-pair graph, resolved in full by a reviewer, gives the true and estimated tables $\boldsymbol{t}^{\mathrm{a}}$, $\boldsymbol{n}^{\mathrm{a}}$ of Definition 5.10.
8. *Propagation (primary).* Linkage uncertainty is propagated into $N_{V,d}$ by mixing over $M = 100$ linkage draws (Algorithm 1), with the Monte Carlo error checked as in Appendix D. Proposition 5.11 gives the variance decomposition we report.
9. *LE diagnostic.* A swap leaves the table unchanged, but the audit counts it as missed pairs plus false edges, so a correction at $(\hat{\delta}, \hat{\varepsilon})$ removes errors that are not there (Remark 5.9). We report the

table-effective rates $(\delta_{\text{eff}}, \varepsilon_{\text{eff}})$ of Definition 5.10, the corrected estimates (Corollary 5.8) at both pairs of rates, and the three checks after Definition 5.10 of whether the errors are LE-like: the share of swap-like missed pairs, the bootstrap calibration of X^2_{eff}, and the agreement of count-based and table-effective rates. Corrected estimates are diagnostics, never the estimate.

10. *Freeze.* The linkage is frozen, and its diagnostics archived, before any estimate of $N_{V,d}$ is computed.

9.4 Models and the model-selection rule

The following are fixed in advance.

M1, primary. Latent-class multiple-systems estimation with a Dirichlet-process prior (Equation (6.1); Manrique-Vallier, 2016), with the covariates conduct class, target type and setting, the prior of Section 6 with $p(N) \propto 1/N$ and truncation $J^\star = 10$, and linkage uncertainty propagated by Algorithm 1. The sensitivity parameters Δ and q enter as in Section 6.1; the primary fit has $\Delta = 0$ and $q = 0$, or $q = \hat{q}$ under rule D4.

M2, secondary. Bayesian model averaging over hierarchical log-linear models with all main effects and up to two-way interactions, weighted by BIC, with $\bar{\omega} = \Delta$ in the closed form of Theorem 4.4 (Fienberg, 1972; Madigan and York, 1997).

M3, diagnostic. Chao's estimator (Theorem 4.8; Chao, 1987), reported in every case. It is a lower bound under heterogeneity across incidents (model M_h) but can exceed $N_{V,d}$ when capture probabilities differ across lists (Proposition 4.9), so it is a diagnostic reference, not a floor. The only assumption-free floor is n.

Campus stratum. List 4 operates only on campus, so in M1 and M2 non-campus incidents carry a declared structural zero for list 4: the campus stratum is estimated from all four lists and the rest from lists 1–3 (Example 2.1; Assumption A3), and $N_{V,d}$ is their sum.

Selection rule. M1 is primary if it passes the posterior predictive check of Section 9.6. If M1 fails and M2 passes, M2 becomes primary. If both fail, we report no point estimate of $N_{V,d}$: only n, M3 and the bounds of Corollary 4.5. If the posterior median of the primary model falls below M3, we flag the result: under M_h this indicates negative dependence or misfit, under list heterogeneity the upward bias of M3 (Proposition 4.9). The flag does not change the primary model. No list, stratum or covariate is added or removed after estimates are seen.

9.5 Sensitivity analyses

Table 10 gives the pre-specified grid and states how each parameter enters the primary model.

- *Dependence offset* Δ, on the oriented scale of Proposition 4.6 (positive Δ raises the dark figure). In M1 every posterior draw's odds of being unrecorded are multiplied by e^{Δ} (6.3), which shifts its implied $\bar{\omega}$ by Δ and needs no refit. In M2, $\bar{\omega} = \Delta$ in Theorem 4.4.
- *Referral* q. Where provenance identifies how list 2 learned of every record, the table is recoded as in Proposition 4.10(c) and q is estimated, not varied. Otherwise M1 is refitted with the referral-augmented sampler of Section 6.1 at each grid value, using provenance wherever it is observed; if provenance yields $\hat{q}$, we add $\hat{q}$ and its 90% interval endpoints to the grid. M2 is reported on the recoded table only.

Table 10. Pre-specified sensitivity grid. Δ: departure of the oriented top interaction from the fitted model (Equation (6.3); Theorem 4.4 and Corollary 4.5); q: referral fraction from the police list to the audit (Proposition 4.10); ρ_{vic}: share of incidents that reach a recording channel (Proposition 7.2), used only for the scenarios $N_d = N_{V,d}/\rho_{\text{vic}}$.

Parameter	Enters M1 by	Grid values	Plausible region
Δ	scaling each draw's odds $\pi_{\mathbf{0}}/(1-\pi_{\mathbf{0}})$ by e^{Δ}; no refit	$-1, -0.5, 0, 0.5, 1$	$\lvert\Delta\rvert \le 0.5$
q	provenance recoding, else referral-augmented refit	0, 0.05, 0.10, 0.20, 0.30 (plus $\hat{q}$)	$q \le 0.20$
Linkage	mixing over draws under each blocking rule	strict, primary, loose	primary (all three under D1(b))
ρ_{vic}	shift of Δ by Δ_{vic} (7.2); scenarios only	1, 0.75, 0.5, 0.25	none declared

- *Linkage.* Mixing over linkage draws under the primary, a strict (same precinct, 7 days) and a loose (two precincts, 28 days) blocking rule. The LE-corrected estimates of Section 9.3 are reported beside the grid as diagnostics; there is no (δ, ε) grid.
- *Reaching a channel, ρ_{vic}* (secondary estimand (vi) only). Incidents that reach no channel change only the unobservable cell, so each ρ_{vic} scenario is the Δ grid shifted by Δ_{vic} of (7.2), computed from the same draws.

M1 is refitted only along q and the linkage variants (15 fits); the 75 cells for $N_{V,d}$ follow by post-processing. We also vary, one at a time, the scope-coding rule, the Dirichlet-process concentration prior and $J^{\star} \in \{10, 20\}$. For every cell we report the posterior median and 90% interval of $N_{V,d}$, and we report the *tipping points*: the smallest $|\Delta|$, and separately the smallest q, at which a conclusion under D5 would change. The plausible region enters only the decision rules.

9.6 Validation

Calibration before unblinding. Before fitting real data, we rerun the simulation pipeline of Section 8 on synthetic worlds calibrated to the observed list sizes and to the audit, with records that carry attribute noise and are linked by FS-EM (errors are *not* generated from LE) at noise levels that reproduce the audited precision and recall. The primary pipeline must attain at least 85% coverage for nominal 90% intervals, or its intervals are reported as uncalibrated; its median relative bias there is reported beside the estimate.

Held-out-list prediction. For each $k \in [K]$ we fit the selected model to the other $K-1$ lists, obtaining a posterior for $N_{V,d}$. Let n_{-k} be the number of incidents those lists capture. If list k captures incidents with a common probability, independently of the other lists, then given n_{k+} its captures are a uniformly random subset of U_d^V, and the number m_k of list-k records linked to at least one other list satisfies

$$m_k \mid N_{V,d},\, n_{-k},\, n_{k+} \;\sim\; \text{Hypergeometric}\big(N_{V,d},\, n_{-k},\, n_{k+}\big) \;\approx\; \text{Binomial}\big(n_{k+},\, n_{-k}/N_{V,d}\big)\,. \tag{9.1}$$

We report the posterior predictive p-value of the observed m_k for each k; it tests the $(K-1)$-list model and the independence and homogeneity of list k jointly, a small value (excess overlap) pointing to referral and a large one to specialisation.

Triangulation against NCVS reporting rates. Let r be the most recent BJS estimate of the share of hate-crime victimisations reported to police (Kena and Thompson, 2021), and r^{+} the upper end of its 95% interval. Since reaching the police implies reaching a channel, $r \approx \rho_{\text{vic}}\rho_{\text{pol}}$, and $p_1 \approx \rho_{\text{pol}}\rho_{\text{cls}} \le r/\rho_{\text{vic}}$. With ρ_{vic} unidentified, the check constrains ρ_{vic} rather than p_1: we report the implied upper bound $\rho_{\text{vic}} \le r^{+}/p_1$ and

flag the result if its posterior probability of falling below the smallest grid value 0.25 exceeds 0.5. A flag would suggest scope mismatch, false matches that inflate overlaps, or unmodelled negative dependence. The NCVS excludes institutional victims and is not specific to anti-Jewish bias (Section 2.4), so this check triangulates; it does not identify anything.

Posterior predictive check. Following Gelman et al. (2013), we compare replicated capture tables with the observed table through the Freeman–Tukey discrepancy over observed histories,

$$T(\boldsymbol{n}, \boldsymbol{\mu}) = \sum_{\boldsymbol{h} \in \mathcal{H}_+} \left(\sqrt{n_{\boldsymbol{h}}} - \sqrt{\mu_{\boldsymbol{h}}}\right)^2, \tag{9.2}$$

and through the frequency counts f_j and the list sizes n_{k+}. A model passes if the posterior predictive p-value of T lies in $[0.05, 0.95]$.

9.7 Decision rules

These rules state which results would change our conclusions.

D1. *Linkage gate.* (a) If the random audit (Proposition C.13) estimates δ above 0.20 or ε above 0.05, we do not estimate $N_{V,d}$; we revise blocking and comparisons, re-link and re-audit, and report a deviation. (b) If any LE check fails, or the corrected estimate at the table-effective rates lies outside the 90% interval of the mixed estimate, linkage is reported as a dominant source of uncertainty, and a statement under D5 must then hold under all three linkage variants. The corrected estimate never replaces the mixed one.

D2. *Ontology gate.* If Cohen's κ for the in-scope decision is below 0.6 for any list, on a random sample of at least 200 records (or all of them if fewer), scope-specific estimates are labelled exploratory.

D3. *Model adequacy.* As in Section 9.4; if no model passes, we report only n, M3 and bounds.

D4. *Referral switch.* If list 2's held-out check shows excess overlap ($p < 0.05$), or provenance gives $\hat{q} > 0.10$ with a 90% interval excluding 0.05, the primary fit uses the provenance-recoded table, or the referral-augmented M1 at $q = \hat{q}$ where provenance is incomplete.

D5. *Substantive statements.* We will say that the police list misses *most* in-scope incidents that reach a recording channel only if $\mathbb{P}(p_1 < 0.5 \mid \text{data}) \geq 0.95$ under the primary model *and* the posterior median of p_1 is below 0.5 throughout the plausible region of Table 10; the symmetric condition governs the statement that it captures most of them. A statement that holds under the primary model but fails somewhere in the plausible region is reported as dependent on unverifiable assumptions, with its tipping point. No statement about all incidents, N_d, is made; only the ρ_{vic} scenarios are reported.

D6. *Negative result.* If the held-out-list check fails for two or more lists, or the NCVS triangulation is flagged, we conclude that these lists do not support MSE in the pilot domain, and report that conclusion with the same prominence as an estimate.

9.8 Reporting and replication

The report will contain a flow diagram of records; the capture-history table, subject to Section 10; linkage diagnostics (audit and table-effective rates, corrected versus mixed estimates, review agreement, the decomposition of Proposition 5.11); M1–M3 with the full grid and tipping points; all validation results; and

Table 11. Phased timeline. Durations are planning targets, not commitments of the data holders.

Phase	Months	Activities	Exit criterion
0. Registration	0–3	Register plan and code hash; IRB review; negotiate data-use agreements; convene community advisory group (Section 10)	Plan registered; IRB approval; signed agreements
1. Acquisition	3–8	Receive extracts in enclave; ontology coding; within-list deduplication	Ontology gate D2 assessed
2. Linkage	8–12	Encoding; blocking; FS–EM and Bayesian matching; clerical review (Algorithm 3); random audit	Linkage gate D1 passed; linkage frozen
3. Calibration	12–14	Calibrated simulation; blinded model fitting and diagnostics	Coverage check; model selection completed
4. Unblinding	14–17	Estimates, sensitivity grid, validation; disclosure review	Decision rules D3–D6 applied
5. Release	17–20	Briefings with community advisory group and data holders; manuscript; replication package	Public release

a log of deviations, following Bird and King (2018) and Lum et al. (2013). The replication package will contain the version-tagged `darkfig` code, the ontology codebook, the review and audit protocols, synthetic data with the extracts' schema and every released aggregate table; record-level data cannot be redistributed.

9.9 Timeline

Table 11 gives the phases. Month 0 is the date of registration; estimation begins only after the linkage freeze and the calibration study.

10 Ethics, privacy and responsible reporting

The records these methods are built for describe people at a moment of victimisation. A record reveals or implies a person's religion or perceived religion, the fact that they were targeted, and often where they live, work or worship; some records concern suspects, who may be minors and have not been convicted. This section sets out the safeguards that the plan of Section 9 commits to and the risks that remain when they work.

10.1 Consent, review and agreements

None of the lists was compiled for research, and the people recorded in them did not consent to linkage. Before any data are transferred, we will ask an institutional review board (IRB) whether a waiver of consent is justified, which turns on whether the safeguards below keep risk minimal and whether contacting victims would itself cause harm. The data-use agreements of Section 9.2 restrict use to the registered protocol, prohibit re-identification and any further linkage, require the minimum necessary fields, set retention and destruction dates (linkage keys and encodings are destroyed when the replication package is released), and oblige us to notify data holders of any breach. Victim-service organisations will be consulted on whether research use should be disclosed to people who report in future, for example in a portal's privacy notice; that trade-off against reporting rates belongs to the community and the data holders as well as to us.

10.2 Privacy-preserving record linkage and its limits

Linkage (Section 9.3) separates roles. Each data holder encodes the identifying fields (locations and institution names; names of persons are neither requested nor needed) and sends them, with non-identifying attributes, to a linkage unit inside a secure enclave; analysts receive only capture histories and covariates. Encoding uses Bloom filters (Schnell et al., 2009): the q-grams of each string are hashed into a bit array, and the Dice coefficient of two arrays approximates the q-gram similarity of the strings, allowing approximate matching without exchanging plaintext.

Bloom-filter encodings are not anonymous: frequency-based cryptanalysis recovers encoded values when an attacker holds a reference list with similar value frequencies (Christen et al., 2019; Kuzu et al., 2011; Niedermeyer et al., 2014). Our setting is unfavourable in a specific way. The addresses of synagogues, schools and other Jewish institutions are public, and incidents that target institutions cluster on them, so a reference list exists for exactly the records whose exposure would be most harmful. We will treat encodings as personal data, generate them with keyed hash functions whose key the data holders keep, keep them inside the enclave and destroy them on schedule. Where an agreement permits matching on plaintext inside the enclave, we will prefer that, because it removes the attack surface without exporting anything.

10.3 Enclaves, aggregate release and disclosure control

All record-level processing, including clerical review and the audit, takes place in a secure research environment, and only aggregates leave it, after output review. Released tables are subject to small-cell suppression with a minimum cell size fixed in the agreements; we propose 10 (Hundepool et al., 2012). Capture-history tables need more, because list holders publish their own totals: NYPD open data and the H.E.A.T. Map make the list sizes n_{k+} approximately public, so a single suppressed cell could be recovered by subtraction. Complementary suppression, or release only of coarsened summaries such as the frequency counts f_j, will ensure that no suppressed cell is determined by released and published totals together (Hundepool et al., 2012). Small-area outputs are released only as model-based estimates with intervals, never as maps of point locations.

Formal differential privacy (optional). For any released small-area counts, we may add a formal guarantee. A randomised mechanism $\mathcal{M}$ is $\varepsilon_{\mathrm{DP}}$-differentially private if, for every pair of datasets D, D' that differ by the addition or removal of one incident, and for every measurable set S,

$$\mathbb{P}\{\mathcal{M}(D) \in S\} \le e^{\varepsilon_{\mathrm{DP}}}\, \mathbb{P}\{\mathcal{M}(D') \in S\} \tag{10.1}$$

(Dwork and Roth, 2014; Dwork et al., 2006); $\varepsilon_{\mathrm{DP}}$ is unrelated to the linkage parameter ε. Each incident falls in exactly one cell of $(n_{\boldsymbol{h}})_{\boldsymbol{h}\in\mathcal{H}_+}$, so the table has ℓ_1-sensitivity 1, and adding independent Laplace$(0, 1/\varepsilon_{\mathrm{DP}})$ noise to every cell satisfies (10.1), with noise standard deviation $\sqrt{2}/\varepsilon_{\mathrm{DP}}$ per cell; estimates from the noisy table inherit the guarantee by post-processing (Dwork and Roth, 2014). The guarantee protects incidents, not people: a person targeted in m incidents is protected at level $m\,\varepsilon_{\mathrm{DP}}$.

The cost of this noise is best judged against the posterior uncertainty in N. MSE estimators depend on overlap cells through ratios, so noise is amplified when overlaps are small. For Lincoln–Petersen with n_{11} small relative to the list sizes, the noise adds a relative standard deviation of about $\sqrt{2}/(\varepsilon_{\mathrm{DP}}\, n_{11})$ to first order, against a sampling relative standard deviation of about $\{(1-p_1)(1-p_2)/n_{11}\}^{1/2}$ under independent, homogeneous capture, so

$$\frac{\mathrm{SD}_{\mathrm{DP}}}{\mathrm{SD}_{\mathrm{samp}}} \;\approx\; \frac{\sqrt{2}}{\varepsilon_{\mathrm{DP}}\sqrt{n_{11}(1-p_1)(1-p_2)}}. \tag{10.2}$$

As illustrative arithmetic, with $\varepsilon_{\mathrm{DP}} = 1$ and $p_1 = p_2 = 0.3$, the ratio is about 0.29 for $n_{11} = 50$ (total standard deviation up about 4%) and about 0.90 for $n_{11} = 5$ (up about 35%). Differential privacy is cheap where MSE is informative and expensive where it is already weak. If we use it, inference will treat the noise-free table as latent and include the Laplace likelihood in the model of Section 6, since naive plug-in estimates can fail when a noisy overlap cell is near zero.

10.4 Errors in either direction

Both overestimation and underestimation of the dark figure cause harm, and the methods can err either way. For two lists, or under condition (H) of Theorem 5.5, false non-matches split incidents and push $\widehat{N}$ up, and false matches push it down; with three or more strongly dependent lists the direction can reverse (Remark 5.7). A linkage correction applied under the wrong error model can overshoot (Section 6.4). Referral pushes the Lincoln–Petersen estimate down (Proposition 4.10); more generally the oriented top interaction sets the direction of bias from unmodelled dependence (Proposition 4.6), and scope mismatch can push either way (Example 2.1). An overestimate can cause fear out of proportion to risk, misdirect security resources and, once corrected, be used to discredit monitoring and the people who report. An underestimate erases victims' experience and makes police data look more complete than they are, which can justify inaction.

We therefore commit to three practices. First, we report the observed count n, the only assumption-free floor, next to every estimate, together with Chao's estimator as a diagnostic rather than a floor (Theorem 4.8 and Proposition 4.9). Second, we report the full sensitivity grid rather than a single preferred cell. Third, we state the direction of each known bias source alongside the result.

10.5 Communicating uncertainty

Estimates of unreported hate crime will be quoted without their caveats, so our release materials will follow six rules: (i) lead with an interval, not a point, and with the recorded count as the only firm floor ("n incidents recorded; most plausibly between A and B in all"); (ii) state the scope every time (criminal incidents that reach a recording channel, the jurisdiction, the period), and never quote a "percentage unreported" without its interval; (iii) do not compare one list's published total with another's as if they counted the same thing (Section 2.5); (iv) do not describe changes across definitional or reporting-system breaks, such as 2021 and October 2023, as trends (Sections 2.1 and 2.2); (v) show sensitivity visually, as $\widehat{N}$ against Δ with the plausible region shaded; and (vi) provide a plain-language summary and briefings that cover what the estimate does *not* show.

10.6 Engagement with Jewish communities

The work concerns a community that bears the harm being measured. Before data acquisition we will convene an advisory group of Jewish communal and security organisations, synagogue and campus organisations, victim-service providers, privacy and civil-liberties experts, and researchers who study hate crime against other groups. It will be consulted on the ontology and scope variants (Section 2.5), the disclosure rules and the communication of results, with no veto over findings. Funding sources and any relationship between the research team and the data holders will be disclosed in every output. Because communities differ in how they relate to the police and to communal organisations, which is itself a source of capture heterogeneity, we will seek input across denominations, languages and neighbourhoods.

10.7 Misuse

We foresee four kinds of misuse. *Targeting*: fine-grained maps of incidents or of "under-reporting" could point offenders to vulnerable institutions or to places where offences are least likely to be recorded, so we release no point locations and only coarse, model-based small-area outputs. *Selective quotation* of one end of an interval or one cell of the grid is checkable because the full grid and the decision rules (Section 9.7) are published. *Definitional proxy wars*: the scope variants with and without Israel/Zionism-related content are reported side by side and are not a ruling on any definition of antisemitism (Section 2.5). *Function creep*: a linkage infrastructure built to count victims could be repurposed to monitor people, including lawful protesters; the agreements confine linkage to the registered protocol, require keys and encodings to be destroyed, and no offender-level or protester-level dataset will be created.

11 Discussion

What the framework makes possible. The usual argument about antisemitic incidents asks whose count is right. The framework asks instead what population every list samples, and how each list samples it. Once the target is defined (Section 3) as the in-scope incidents that reach a recording channel, the police count, an NGO audit and a community-security log become *data about* the same unknown total, not competitors for it. Their overlaps are informative, their dependence is modelled rather than assumed away (Proposition 4.10), and the part of the answer that the data cannot deliver is isolated in a single interpretable parameter ω (Theorem 4.4 and Corollary 4.5). The sensitivity frontier tells a reader how much unmeasured dependence a stated conclusion can withstand (Theorem 4.11). Linkage error, which in practice is often handled silently, becomes an explicit operator whose first-order effect can be computed (Theorem 5.5). The correction is valid only under LE-type errors (Corollary 5.8); otherwise we mix over linkage draws (Algorithm 1) and use the correction as a diagnostic.

The funnel decomposition (Proposition 7.2 and Theorem 7.3) turns an estimate of the dark figure into something a policymaker can act on. A gap between two jurisdictions' recorded rates can arise because victims there are less willing to come forward, because the police are less likely to recognise a bias motivation, or because the agency does not submit its data. Each of these calls for a different response: community outreach, officer training or reporting infrastructure. Raw counts cannot tell these explanations apart. The decomposition can, to the extent that the stage-specific rates are identified by the auxiliary data of Table 4. Incidents that never reach a recording channel remain beyond the reach of every list, so that stage rests on victim surveys.

Limitations. We stress six.

Untestable assumptions. No method escapes the basic non-identifiability of the unobserved cell (Theorem 4.3). Our approach makes the identifying assumption explicit and adjustable, but it does not remove it. A sensitivity interval is only as credible as the range chosen for ω, which must be argued from how the lists are generated and must in general be two-sided (Proposition 4.6, Example A.7). A one-sided range is justified only by a mechanism of known sign, such as referral between two lists (Remark A.9); with three or more lists, referral itself can push the bias either way. Latent-class models replace the assumption of no highest-order interaction with conditional independence within classes. That is weaker in some directions and not in others. The Chao bound is a lower bound only when heterogeneity runs across incidents. Heterogeneity across lists can turn it into an upper bound (Proposition 4.9), so it is a diagnostic, not a floor; the only assumption-free floor is the observed count n.

Scope harmonisation. Harmonisation rules (Sections 2.5 and 9) can align police and NGO lists only as far as their records contain the information needed to apply a common definition; residual mismatch behaves like coverage error and cannot be fully separated from it.

Data access. The methods need record-level access to sensitive records of victims who did not consent to linkage. Agreements, privacy-preserving linkage (Schnell et al., 2009), disclosure control (Hundepool et al., 2012) and formal privacy (Dwork and Roth, 2014) each degrade linkage quality or precision, encodings can be attacked (Section 10.2), and the ethics protocol may rightly rule out some linkages altogether.

The closed-population idealisation. Incidents occur in continuous time, lists have different reporting windows and lags, and records are added or revised after the reference period ends. Treating each state–year domain as a closed population is a convenient approximation. Its error is concentrated at period boundaries and in lists with long reporting delays.

Approximate linkage-error models. The operator $\mathcal{T}(\delta, \varepsilon)$ assumes that detachments and false merges occur independently across records, at rates that are at most list-specific (Remark C.7). It has no move for a *swap*, in which a record is linked to the wrong partner. A swap leaves the table unchanged, but an audit counts it as missed pairs plus false edges, so a correction at audit-based rates removes errors that are not there (Remark 5.9). In our simulation, where records carry realistic attribute noise and are linked by FS-EM, swaps dominate and the correction does over-correct (Section 8). Real errors also cluster in dense places, in generic incident descriptions and in records with missing fields, and may be correlated with capture history itself. The correction is therefore a diagnostic. Mixing over linkage draws (Algorithm 1) is our primary approach, but it propagates only the uncertainty that the linkage posterior expresses and cannot remove a bias shared by every draw (in our simulation, approximate draws from a plug-in pairwise posterior were centred no better than the point linkage); the random audit and the calibration study of Section 9.6 are the guard. The modular posterior (Equation (6.11)) also ignores the information that the capture model carries about which records match, trading some efficiency for robustness.

Unevaluated components. The small-area model of Section 6.3 is specified but has not been evaluated by simulation, and the sensitivity devices and linkage diagnostics of the analysis plan are to be tested on synthetic data before registration. Until then their operating characteristics are unknown.

Extensions. Three extensions seem most valuable. First, *open populations and continuous time*: modelling incident occurrence as a point process and each list as a time-varying, delayed thinning of it would remove the closed-population idealisation and connect dark-figure estimation to nowcasting of reporting delays. Second, *online incidents*: here the unit of analysis is itself unsettled, and platform moderation and automated detection act as lists with their own selection and classification error, a direct analogue of the funnel's classification stage. Third, *international comparison*. Community lists such as the UK's Community Security Trust, which documents its police referrals (Community Security Trust, 2026), and surveys of Jewish people's experiences of non-reporting (European Union Agency for Fundamental Rights, 2018, 2024) exist outside the United States (Sections 2.3 and 2.4). Combined country by country, with the funnel as a common accounting framework, they would allow cross-national comparisons of reporting, not only of recorded counts.

Why measure what is missing. A count that omits its dark figure can mislead in both directions. Rising counts may reflect greater willingness to report, better classification or wider agency participation as much as rising incidence. Falling counts may reflect lost trust or lapsed reporting. Resources allocated in proportion to recorded incidents reward the communities and agencies that report, which are not necessarily those that are most affected. Nor is there reason to expect the dark figure to be random noise. If it is concentrated where victims expect least from reporting, its size is itself a measure of the distance between

a community and the institutions meant to protect it. Estimating it honestly, with explicit assumptions, propagated uncertainty and a plan fixed before the data are seen, is how a measurement programme takes seriously the incidents that nobody wrote down.

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Code and data availability. The code for the simulation study (the `darkfig` Python package, the experiment driver with its deterministic seeds, and the scripts that produce every figure and table of Section 8) is in the paper's repository, together with the simulation output. All data in this paper are synthetic and are regenerated by that code. No real incident data were used.

References

Johan Albrecht, Delphine François, and Koen Schoors. A Shapley decomposition of carbon emissions without residuals. *Energy Policy*, 30(9):727–736, 2002. doi: 10.1016/S0301-4215(01)00131-8.

Serge Aleshin-Guendel and Mauricio Sadinle. Multifile partitioning for record linkage and duplicate detection. *Journal of the American Statistical Association*, 118(543):1786–1795, 2023. doi: 10.1080/01621459.2021.2013242.

Serge Aleshin-Guendel, Mauricio Sadinle, and Jon Wakefield. The central role of the identifying assumption in population size estimation. *Biometrics*, 80(1):ujad028, 2024. doi: 10.1093/biomtc/ujad028.

Juha M. Alho. Logistic regression in capture-recapture models. *Biometrics*, 46(3):623–635, 1990. doi: 10.2307/2532083.

Elizabeth S. Allman, Catherine Matias, and John A. Rhodes. Identifiability of parameters in latent structure models with many observed variables. *The Annals of Statistics*, 37(6A):3099–3132, 2009. doi: 10.1214/09-AOS689.

B. W. Ang. Decomposition analysis for policymaking in energy: which is the preferred method? *Energy Policy*, 32(9):1131–1139, 2004. doi: 10.1016/S0301-4215(03)00076-4.

B. W. Ang. LMDI decomposition approach: A guide for implementation. *Energy Policy*, 86:233–238, 2015. doi: 10.1016/j.enpol.2015.07.007.

Anti-Defamation League. Audit of Antisemitic Incidents 2021. Technical report, ADL Center on Extremism, New York, 2022. URL https://www.adl.org/resources/report/audit-antisemitic-incidents-2021. Accessed November 2026.

Anti-Defamation League. Audit of Antisemitic Incidents 2024. Technical report, ADL Center on Extremism, New York, April 2025. URL https://www.adl.org/resources/report/audit-antisemitic-incidents-2024. Accessed November 2026.

Anti-Defamation League. Audit of Antisemitic Incidents 2025. Technical report, ADL Center on Extremism, New York, May 2026a. URL https://www.adl.org/resources/report/audit-antisemitic-incidents-2025. Accessed November 2026.

Anti-Defamation League. 2025 marked third-highest year on record for Anti-Jewish hate crimes, FBI reports. Press release, August 2026b. URL https://www.adl.org/resources/press-release/2025-marked-third-highest-year-record-anti-jewish-hate-crimes-fbi-reports. Accessed November 2026.

Anti-Defamation League. H.E.A.T. Map: Hate, Extremism, Antisemitism, Terrorism. Interactive map and downloadable incident data, ADL Center on Extremism, n.d. URL https://www.adl.org/resources/tools-to-track-hate/heat-map. Accessed November 2026.

Patrick Ball and Megan Price. The statistics of genocide. *CHANCE*, 31(1), 2018. URL https://chance.amstat.org/?p=11896.

M. J. Bayarri, James O. Berger, and Fei Liu. Modularization in Bayesian analysis, with emphasis on analysis of computer models. *Bayesian Analysis*, 4(1):119–150, 2009. doi: 10.1214/09-BA404.

Thomas R. Belin and Donald B. Rubin. A method for calibrating false-match rates in record linkage. *Journal of the American Statistical Association*, 90(430):694–707, 1995.

Julian Besag, Jeremy York, and Annie Mollié. Bayesian image restoration, with two applications in spatial statistics. *Annals of the Institute of Statistical Mathematics*, 43(1):1–20, 1991. doi: 10.1007/BF00116466.

Michael Betancourt. A conceptual introduction to Hamiltonian Monte Carlo. arXiv:1701.02434, 2017.

Olivier Binette and Rebecca C. Steorts. (Almost) all of entity resolution. *Science Advances*, 8(12):eabi8021, 2022. doi: 10.1126/sciadv.abi8021.

Sheila M. Bird and Ruth King. Multiple systems estimation (or capture-recapture estimation) to inform public policy. *Annual Review of Statistics and Its Application*, 5:95–118, 2018. doi: 10.1146/annurev-statistics-031017-100641.

Yvonne M. M. Bishop, Stephen E. Fienberg, and Paul W. Holland. *Discrete Multivariate Analysis: Theory and Practice*. MIT Press, Cambridge, MA, 1975.

California Department of Justice. Hate Crime in California 2024. Technical report, California Department of Justice, Sacramento, CA, 2025. URL https://data-openjustice.doj.ca.gov/sites/default/files/2025-06/Hate%20Crime%20In%20CA%202024.pdf. Accessed November 2026.

California Department of Justice. Hate Crime in California 2025. Technical report, California Department of Justice, Sacramento, CA, 2026. URL https://data-openjustice.doj.ca.gov/sites/default/files/2026-07/Hate%20Crime%20In%20CA%202025f.pdf. Accessed November 2026.

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017. doi: 10.18637/jss.v076.i01.

Anne Chao. Estimating the population size for capture-recapture data with unequal catchability. *Biometrics*, 43(4):783–791, 1987. doi: 10.2307/2531532.

Anne Chao, P. K. Tsay, Sheng-Hsiang Lin, Wen-Yi Shau, and Day-Yu Chao. The applications of capture-recapture models to epidemiological data. *Statistics in Medicine*, 20(20):3123–3157, 2001. doi: 10.1002/sim.996.

Douglas G. Chapman. Some properties of the hypergeometric distribution with applications to zoological sample censuses. *University of California Publications in Statistics*, 1(7):131–160, 1951.

Peter Christen. *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Data-Centric Systems and Applications. Springer, Berlin, Heidelberg, 2012. doi: 10.1007/978-3-642-31164-2.

Peter Christen, Thilina Ranbaduge, Dinusha Vatsalan, and Rainer Schnell. Precise and fast cryptanalysis for Bloom filter based privacy-preserving record linkage. *IEEE Transactions on Knowledge and Data Engineering*, 31(11):2164–2177, 2019. doi: 10.1109/TKDE.2018.2874004.

William G. Cochran. *Sampling Techniques*. Wiley, New York, third edition, 1977.

Community Security Trust. Antisemitic Incidents Report 2025. Technical report, Community Security Trust, London, February 2026. URL https://cst.org.uk/research/cst-publications/antisemitic-incidents-report-2025. Accessed November 2026.

Congressional Research Service. Federal data on hate crimes in the United States. CRS Report R46318, Congressional Research Service, 2025. URL https://www.congress.gov/crs-product/R46318. Updated December 16, 2025. Accessed November 2026.

Brent A. Coull and Alan Agresti. The use of mixed logit models to reflect heterogeneity in capture-recapture studies. *Biometrics*, 55(1):294–301, 1999. doi: 10.1111/j.0006-341X.1999.00294.x.

George B. Dantzig. Discrete-variable extremum problems. *Operations Research*, 5(2):266–288, 1957.

J. N. Darroch. The multiple-recapture census: I. Estimation of a closed population. *Biometrika*, 45(3/4):343–359, 1958. doi: 10.1093/biomet/45.3-4.343.

Loredana Di Consiglio and Tiziana Tuoto. Coverage evaluation on probabilistically linked data. *Journal of Official Statistics*, 31(3):415–429, 2015. doi: 10.1515/jos-2015-0025.

Loredana Di Consiglio and Tiziana Tuoto. Population size estimation and linkage errors: the multiple lists case. *Journal of Official Statistics*, 34(4):889–908, 2018. doi: 10.2478/jos-2018-0044.

Ye Ding and Stephen E. Fienberg. Dual system estimation of census undercount in the presence of matching error. *Survey Methodology*, 20(2):149–158, 1994.

Ye Ding and Stephen E. Fienberg. Multiple sample estimation of population and census undercount in the presence of matching errors. *Survey Methodology*, 22(1), 1996. URL https://www150.statcan.gc.ca/n1/en/catalogue/12-001-X199600114385.

Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–407, 2014. doi: 10.1561/0400000042.

Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography Conference (TCC 2006)*, volume 3876 of *Lecture Notes in Computer Science*, pages 265–284, Berlin, Heidelberg, 2006. Springer. doi: 10.1007/11681878_14.

Bradley Efron and Charles Stein. The jackknife estimate of variance. *The Annals of Statistics*, 9(3):586–596, 1981.

Ted Enamorado, Benjamin Fifield, and Kosuke Imai. Using a probabilistic model to assist merging of large-scale administrative records. *American Political Science Review*, 113(2):353–371, 2019. doi: 10.1017/S0003055418000783.

European Union Agency for Fundamental Rights. Experiences and perceptions of antisemitism: Second survey on discrimination and hate crime against Jews in the EU. Technical report, Publications Office of the European Union, Luxembourg, 2018. URL https://fra.europa.eu/en/publication/2018/experiences-and-perceptions-antisemitism-second-survey-discrimination-and-hate. Accessed November 2026.

European Union Agency for Fundamental Rights. Jewish people's experiences and perceptions of antisemitism. Third FRA survey on discrimination and hate crime against Jews, Publications Office of the European Union, Luxembourg, July 2024. URL https://fra.europa.eu/en/publication/2024/experiences-and-perceptions-antisemitism-third-survey. Accessed November 2026.

Federal Bureau of Investigation. About Hate Crime Statistics, 2018. Hate Crime Statistics, 2018, Criminal Justice Information Services Division, U.S. Department of Justice, 2019. URL https://ucr.fbi.gov/hate-crime/2018/resource-pages/about-hate-crime.pdf. Accessed November 2026.

Federal Bureau of Investigation. Supplemental Hate Crime Statistics, 2021. Technical report, Uniform Crime Reporting Program, Criminal Justice Information Services Division, March 2023. URL https://le.fbi.gov/file-repository/supplemental-hate-crime-statistics-2021.pdf. Accessed November 2026.

Federal Bureau of Investigation. FBI releases 2023 Crime in the Nation statistics. Press release, U.S. Department of Justice, September 2024. URL https://www.fbi.gov/news/press-releases/fbi-releases-2023-crime-in-the-nation-statistics.

Federal Bureau of Investigation. UCR Summary of Reported Crimes in the Nation, 2024. Technical report, Uniform Crime Reporting Program, Criminal Justice Information Services Division, 2025. URL https://cde.ucr.cjis.gov/LATEST/resources/reports/UCR%20Summary%20of%20Reported%20Crimes%20in%20the%20Nation%202024.pdf. Accessed November 2026.

Federal Bureau of Investigation. Hate Crime Statistics, 2025. Crime Data Explorer, Uniform Crime Reporting Program, 2026. URL https://cde.ucr.cjis.gov/. Accessed November 2026.

Federal Bureau of Investigation. Hate Crime Data Collection Guidelines and Training Manual. Technical report, Uniform Crime Reporting Program, Criminal Justice Information Services Division, n.d. URL https://le.fbi.gov/file-repository/hate-crime-data-collection-guidelines-and-training-manual.pdf. Undated current edition. Accessed November 2026.

Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64 (328):1183–1210, 1969. doi: 10.2307/2286061.

William Feller. *An Introduction to Probability Theory and Its Applications*, volume I. Wiley, New York, third edition, 1968.

Stephen E. Fienberg. The multiple recapture census for closed populations and incomplete 2^k contingency tables. *Biometrika*, 59(3):591–603, 1972. doi: 10.1093/biomet/59.3.591.

Stephen E. Fienberg and Alessandro Rinaldo. Maximum likelihood estimation in log-linear models. *The Annals of Statistics*, 40(2):996–1023, 2012. doi: 10.1214/12-AOS986.

Anna Freni-Sterrantino, Massimo Ventrucci, and Håvard Rue. A note on intrinsic conditional autoregressive models for disconnected graphs. *Spatial and Spatio-temporal Epidemiology*, 26:25–34, 2018. doi: 10.1016/j.sste.2018.04.002.

John J. Gart and James R. Zweifel. On the bias of various estimators of the logit and its variance with application to quantal bioassay. *Biometrika*, 54(1/2):181–187, 1967. doi: 10.1093/biomet/54.1-2.181.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. Texts in Statistical Science. Chapman and Hall/CRC, Boca Raton, FL, third edition, 2013. ISBN 978-1-4398-4095-5.

Susanna C. Gerritse, Peter G. M. van der Heijden, and Bart F. M. Bakker. Sensitivity of population size estimation for violating parametric assumptions in log-linear models. *Journal of Official Statistics*, 31(3):357–379, 2015. doi: 10.1515/jos-2015-0022.

John Geweke. Getting it right: Joint distribution tests of posterior simulators. *Journal of the American Statistical Association*, 99(467):799–804, 2004. doi: 10.1198/016214504000001132.

Thomas N. Herzog, Fritz J. Scheuren, and William E. Winkler. *Data Quality and Record Linkage Techniques*. Springer, New York, 2007. doi: 10.1007/0-387-69505-2.

Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics*, 19(3):293–325, 1948.

Matthew D. Hoffman and Andrew Gelman. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593–1623, 2014.

R. M. Huggins. On the statistical analysis of capture experiments. *Biometrika*, 76(1):133–140, 1989. doi: 10.1093/biomet/76.1.133.

Anco Hundepool, Josep Domingo-Ferrer, Luisa Franconi, Sarah Giessing, Eric Schulte Nordholt, Keith Spicer, and Peter-Paul de Wolf. *Statistical Disclosure Control*. Wiley Series in Survey Methodology. Wiley, Chichester, 2012. doi: 10.1002/9781118348239.

Guido W. Imbens and Charles F. Manski. Confidence intervals for partially identified parameters. *Econometrica*, 72 (6):1845–1857, 2004. doi: 10.1111/j.1468-0262.2004.00555.x.

International Holocaust Remembrance Alliance. Working Definition of Antisemitism. Non-legally binding working definition adopted by the IHRA Plenary, 2016. URL https://holocaustremembrance.com/resources/working-definition-antisemitism. Accessed November 2026.

International Working Group for Disease Monitoring and Forecasting. Capture-recapture and multiple-record systems estimation I: History and theoretical development. *American Journal of Epidemiology*, 142(10):1047–1058, 1995.

Hemant Ishwaran and Lancelot F. James. Gibbs sampling methods for stick-breaking priors. *Journal of the American Statistical Association*, 96(453):161–173, 2001. doi: 10.1198/016214501750332758.

Matthew A. Jaro. Advances in record-linkage methodology as applied to matching the 1985 Census of Tampa, Florida. *Journal of the American Statistical Association*, 84(406):414–420, 1989.

Jerusalem Declaration on Antisemitism. The Jerusalem Declaration on Antisemitism. Declaration and signatory list, 2021. URL https://jerusalemdeclaration.org/. Accessed November 2026.

Noa Kallioinen, Topi Paananen, Paul-Christian Bürkner, and Aki Vehtari. Detecting and diagnosing prior and likelihood sensitivity with power-scaling. *Statistics and Computing*, 34:57, 2024. doi: 10.1007/s11222-023-10366-5.

Hans Kellerer, Ulrich Pferschy, and David Pisinger. *Knapsack Problems*. Springer, Berlin, 2004.

Grace Kena and Alexandra Thompson. Hate crime victimization, 2005–2019. BJS Report NCJ 300954, Bureau of Justice Statistics, U.S. Department of Justice, September 2021. URL https://bjs.ojp.gov/library/publications/hate-crime-victimization-2005-2019. Accessed November 2026.

Leonhard Knorr-Held. Bayesian modelling of inseparable space-time variation in disease risk. *Statistics in Medicine*, 19(17–18):2555–2567, 2000.

Mehmet Kuzu, Murat Kantarcioglu, Elizabeth Durham, and Bradley Malin. A constraint satisfaction cryptanalysis of Bloom filters in private record linkage. In *Privacy Enhancing Technologies (PETS 2011)*, volume 6794 of *Lecture Notes in Computer Science*, pages 226–245, Berlin, Heidelberg, 2011. Springer. doi: 10.1007/978-3-642-22263-4_13.

P. Lahiri and Michael D. Larsen. Regression analysis with linked data. *Journal of the American Statistical Association*, 100(469):222–230, 2005. doi: 10.1198/016214504000001277.

Brendan Lantz and Marin R. Wenger. Are Asian victims less likely to report hate crime victimization to the police? Implications for research and policy in the wake of the COVID-19 pandemic. *Crime & Delinquency*, 68(8): 1292–1319, 2022. doi: 10.1177/00111287211041521.

Michael D. Larsen and Donald B. Rubin. Iterative automated record linkage using mixture models. *Journal of the American Statistical Association*, 96(453):32–41, 2001.

Erich L. Lehmann and Joseph P. Romano. *Testing Statistical Hypotheses*. Springer, New York, third edition, 2005.

Daniel Lewandowski, Dorota Kurowicka, and Harry Joe. Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis*, 100(9):1989–2001, 2009. doi: 10.1016/j.jmva.2009.04.008.

Frederick C. Lincoln. Calculating waterfowl abundance on the basis of banding returns. Circular 118, U.S. Department of Agriculture, Washington, DC, May 1930.

Torgny Lindvall. *Lectures on the Coupling Method*. Wiley, New York, 1992.

William A. Link. Nonidentifiability of population size from capture-recapture data with heterogeneous detection probabilities. *Biometrics*, 59(4):1123–1130, 2003. doi: 10.1111/j.0006-341X.2003.00129.x.

Thomas A. Louis. Estimating a population of parameter values using Bayes and empirical Bayes methods. *Journal of the American Statistical Association*, 79(386):393–398, 1984.

Kristian Lum, Megan Emily Price, and David Banks. Applications of multiple systems estimation in human rights research. *The American Statistician*, 67(4):191–200, 2013. doi: 10.1080/00031305.2013.821093.

David Madigan and Jeremy C. York. Bayesian methods for estimation of the size of a closed population. *Biometrika*, 84(1):19–31, 1997. doi: 10.1093/biomet/84.1.19.

Daniel Manrique-Vallier. Bayesian population size estimation using Dirichlet process mixtures. *Biometrics*, 72(4): 1246–1254, 2016. doi: 10.1111/biom.12502.

Charles F. Manski. *Partial Identification of Probability Distributions*. Springer Series in Statistics. Springer, New York, 2003. doi: 10.1007/b97478.

Neil G. Marchant, Andee Kaplan, Daniel N. Elazar, Benjamin I. P. Rubinstein, and Rebecca C. Steorts. d-blink: Distributed end-to-end Bayesian entity resolution. *Journal of Computational and Graphical Statistics*, 30(2): 406–421, 2021. doi: 10.1080/10618600.2020.1825451.

Matthew A. Masten and Alexandre Poirier. Inference on breakdown frontiers. *Quantitative Economics*, 11(1):41–111, 2020. doi: 10.3982/QE1288.

Madeline Masucci and Lynn Langton. Hate crime victimization, 2004–2015. Special Report NCJ 250653, Bureau of Justice Statistics, U.S. Department of Justice, 2017. URL https://bjs.ojp.gov/library/publications/hate-crime-victimization-2004-2015.

Jack McDevitt, Jennifer M. Balboni, Susan Bennett, Joan C. Weiss, Stan Orchowsky, and Lisa Walbolt. Improving the quality and accuracy of bias crime statistics nationally: An assessment of the first ten years of bias crime data collection. In Barbara Perry, editor, *Hate and Bias Crime: A Reader*, pages 77–90. Routledge, New York, 2003.

Xiao-Li Meng. Multiple-imputation inferences with uncongenial sources of input. *Statistical Science*, 9(4):538–558, 1994.

Mitzi Morris, Katherine Wheeler-Martin, Dan Simpson, Stephen J. Mooney, Andrew Gelman, and Charles DiMaggio. Bayesian hierarchical spatial models: Implementing the Besag York Mollié model in Stan. *Spatial and Spatio-temporal Epidemiology*, 31:100301, 2019. doi: 10.1016/j.sste.2019.100301.

New York City Police Department. NYPD Hate Crimes. NYC Open Data, dataset bqiq-cu78, 2026. URL https://data.cityofnewyork.us/Public-Safety/NYPD-Hate-Crimes/bqiq-cu78. Accessed November 2026.

Jerzy Neyman. On the two different aspects of the representative method: the method of stratified sampling and the method of purposive selection. *Journal of the Royal Statistical Society*, 97(4):558–625, 1934. doi: 10.2307/2342192.

Frank Niedermeyer, Simone Steinmetzer, Martin Kroll, and Rainer Schnell. Cryptanalysis of basic Bloom filters used for privacy preserving record linkage. *Journal of Privacy and Confidentiality*, 6(2):59–79, 2014. doi: 10.29012/jpc.v6i2.640.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Office of the New York State Comptroller. The concerning growth of hate crime in New York State. Technical report, Office of Budget Policy and Analysis, Office of the New York State Comptroller, Albany, NY, August 2024. URL https://www.osc.ny.gov/reports/concerning-growth-hate-crime-new-york-state. Accessed November 2026.

David L. Otis, Kenneth P. Burnham, Gary C. White, and David R. Anderson. Statistical inference from capture data on closed animal populations. *Wildlife Monographs*, 62:3–135, 1978.

Guillermo Owen. Multilinear extensions of games. *Management Science*, 18(5, Part 2):P64–P79, 1972. doi: 10.1287/mnsc.18.5.64.

Omiros Papaspiliopoulos, Gareth O. Roberts, and Martin Sköld. A general framework for the parametrization of hierarchical models. *Statistical Science*, 22(1):59–73, 2007. doi: 10.1214/088342307000000014.

C. G. J. Petersen. The yearly immigration of young plaice into the Limfjord from the German Sea. *Report of the Danish Biological Station to the Ministry of Fisheries*, 6, 1896.

Frank S. Pezzella, Matthew D. Fetzer, and Tyler Keller. The dark figure of hate crime underreporting. *American Behavioral Scientist*, 2019. doi: 10.1177/0002764218823844. Published online 28 January 2019.

Martyn Plummer. Cuts in Bayesian graphical models. *Statistics and Computing*, 25(1):37–43, 2015. doi: 10.1007/s11222-014-9503-z.

Andrea Riebler, Sigrunn H. Sørbye, Daniel Simpson, and Håvard Rue. An intuitive Bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4):1145–1165, 2016. doi: 10.1177/0962280216660421.

Paul R. Rosenbaum. *Observational Studies*. Springer, New York, second edition, 2002. doi: 10.1007/978-1-4757-3692-2.

Gian-Carlo Rota. On the foundations of combinatorial theory I. Theory of Möbius functions. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 2(4):340–368, 1964. doi: 10.1007/BF00531932.

Judith Rousseau and Kerrie Mengersen. Asymptotic behaviour of the posterior distribution in overfitted mixture models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(5):689–710, 2011. doi: 10.1111/j.1467-9868.2011.00781.x.

Donald B. Rubin. *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, New York, 1987. doi: 10.1002/9780470316696.

Mauricio Sadinle. Detecting duplicates in a homicide registry using a Bayesian partitioning approach. *The Annals of Applied Statistics*, 8(4):2404–2434, 2014. doi: 10.1214/14-AOAS779.

Mauricio Sadinle. Bayesian estimation of bipartite matchings for record linkage. *Journal of the American Statistical Association*, 112(518):600–612, 2017. doi: 10.1080/01621459.2016.1148612.

Mauricio Sadinle. Bayesian propagation of record linkage uncertainty into population size estimation of human rights violations. *The Annals of Applied Statistics*, 12(2):1013–1038, 2018. doi: 10.1214/18-AOAS1178.

Mauricio Sadinle and Stephen E. Fienberg. A generalized Fellegi–Sunter framework for multiple record linkage with application to homicide record systems. *Journal of the American Statistical Association*, 108(502):385–397, 2013.

Lalitha Sanathanan. Estimating the size of a multinomial population. *The Annals of Mathematical Statistics*, 43(1): 142–152, 1972. doi: 10.1214/aoms/1177692709.

Rainer Schnell, Tobias Bachteler, and Jörg Reiher. Privacy-preserving record linkage using Bloom filters. *BMC Medical Informatics and Decision Making*, 9:41, 2009. doi: 10.1186/1472-6947-9-41.

Secure Community Network. September monthly update. Periodic threat summary, 2024. URL https://www.securecommunitynetwork.org/articles/september-monthly-update/. Accessed November 2026.

Secure Community Network. Monitoring and acting on threats. Organisational web page, n.d.a. URL https://www.securecommunitynetwork.org/monitoring-the-threat-landscape/. Accessed November 2026.

Secure Community Network. Who we are. Organisational web page, n.d.b. URL https://www.securecommunitynetwork.org/about. Accessed November 2026.

Jayaram Sethuraman. A constructive definition of Dirichlet priors. *Statistica Sinica*, 4(2):639–650, 1994.

Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, number 28 in Annals of Mathematics Studies, pages 307–317. Princeton University Press, Princeton, NJ, 1953. doi: 10.1515/9781400881970-018.

Anthony F. Shorrocks. Inequality decomposition by factor components. *Econometrica*, 50(1):193–211, 1982. doi: 10.2307/1912537.

Anthony F. Shorrocks. Decomposition procedures for distributional analysis: a unified framework based on the Shapley value. *The Journal of Economic Inequality*, 11(1):99–126, 2013. doi: 10.1007/s10888-011-9214-z.

Bernard W. Silverman. Multiple-systems analysis for the quantification of modern slavery: classical and Bayesian approaches. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 183(3):691–736, 2020. doi: 10.1111/rssa.12505.

Daniel Simpson, Håvard Rue, Andrea Riebler, Thiago G. Martins, and Sigrunn H. Sørbye. Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science*, 32(1):1–28, 2017. doi: 10.1214/16-STS576.

Sigrunn Holbek Sørbye and Håvard Rue. Scaling intrinsic Gaussian Markov random field priors in spatial modelling. *Spatial Statistics*, 8:39–51, 2014. doi: 10.1016/j.spasta.2013.06.004.

Rebecca C. Steorts, Rob Hall, and Stephen E. Fienberg. A Bayesian approach to graphical record linkage and deduplication. *Journal of the American Statistical Association*, 111(516):1660–1672, 2016. doi: 10.1080/01621459.2015.1105807.

Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration. arXiv:1804.06788, 2018.

Andrea Tancredi and Brunero Liseo. A hierarchical Bayesian approach to record linkage and population size problems. *The Annals of Applied Statistics*, 5(2B):1553–1585, 2011. doi: 10.1214/10-AOAS447.

Andrea Tancredi, Rebecca Steorts, and Brunero Liseo. A unified framework for de-duplication and population size estimation (with discussion). *Bayesian Analysis*, 15(2):633–682, 2020. doi: 10.1214/19-BA1146.

Daniel Turek and David Fletcher. Model-averaged Wald confidence intervals. *Computational Statistics & Data Analysis*, 56(9):2809–2815, 2012. doi: 10.1016/j.csda.2012.03.002.

United States Congress. Jeanne Clery Disclosure of Campus Security Policy and Campus Crime Statistics Act. 20 U.S.C. § 1092(f), as amended, 1990a. URL https://www.law.cornell.edu/uscode/text/20/1092. Accessed November 2026.

United States Congress. Hate Crime Statistics Act. Pub. L. No. 101-275, 104 Stat. 140 (April 23, 1990), codified as amended at 34 U.S.C. § 41305 (formerly 28 U.S.C. § 534 note), 1990b. URL https://www.govinfo.gov/content/pkg/STATUTE-104/pdf/STATUTE-104-Pg140.pdf. Accessed November 2026.

U.S. Department of Education. Voluntary disclosure of hate crime sub-categories. Dear Colleague Letter GEN-24-09, Federal Student Aid, July 2024. URL https://fsapartners.ed.gov/knowledge-center/library/dear-colleague-letters/2024-07-29/voluntary-disclosure-hate-crime-sub-categories. Dated July 29, 2024. Accessed November 2026.

U.S. Department of Education. Institutional Security Policies and Crime Statistics. 34 C.F.R. § 668.46, Electronic Code of Federal Regulations, 2026. URL https://www.ecfr.gov/current/title-34/subtitle-B/chapter-VI/part-668/subpart-D/section-668.46. Accessed November 2026.

U.S. Department of Education, Office of Postsecondary Education. Campus Safety and Security Data Analysis Cutting Tool. Online database of Campus Safety and Security Survey data, n.d. URL https://ope.ed.gov/campussafety/. Accessed November 2026.

U.S. Department of Justice. 2021 Hate Crime Statistics. Hate Crimes web portal, 2023. URL https://www.justice.gov/hatecrimes/2021-hate-crime-statistics. Accessed November 2026.

U.S. Department of Justice, Community Relations Service. 2023 Hate Crime Statistics. Summary of the FBI's 2023 hate crime data, 2024. URL https://www.justice.gov/crs/news/2023-hate-crime-statistics.

Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.

Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718, 2021. doi: 10.1214/20-BA1221.

William E. Winkler. Using the EM algorithm for weight computation in the Fellegi–Sunter model of record linkage. In *Proceedings of the Section on Survey Research Methods*, pages 667–671. American Statistical Association, 1988.

William E. Winkler. String comparator metrics and enhanced decision rules in the Fellegi–Sunter model of record linkage. In *Proceedings of the Section on Survey Research Methods*, pages 354–359. American Statistical Association, 1990.

Kirk M. Wolter. Some coverage error models for census data. *Journal of the American Statistical Association*, 81 (394):338–346, 1986. doi: 10.1080/01621459.1986.10478277.

Heather Zaykowski. Racial disparities in hate crime reporting. *Violence and Victims*, 25(3):378–394, 2010. doi: 10.1891/0886-6708.25.3.378.

Appendix A Proofs for Section 4

Throughout, $\sum_{\boldsymbol{h}}$ and $\prod_{\boldsymbol{h}}$ without qualification run over $\mathcal{H}_+$, and $\bar{\omega} = (-1)^K \omega$. We use the identities (3.3) and the key identity (3.10) repeatedly. For completeness, (3.10) follows from (3.8) by isolating the term $T = \emptyset$:

$$\omega = (-1)^K \log \mu_{\mathbf{0}} + \sum_{\boldsymbol{h}} (-1)^{K-|\boldsymbol{h}|} \log \mu_{\boldsymbol{h}} \implies \log \mu_{\mathbf{0}} = (-1)^K \omega - \sum_{\boldsymbol{h}} (-1)^{|\boldsymbol{h}|} \log \mu_{\boldsymbol{h}} = \bar{\omega} + \sum_{\boldsymbol{h}} s_{\boldsymbol{h}} \log \mu_{\boldsymbol{h}}.$$

The Möbius sums (3.7) for $S \neq \emptyset$ annihilate additive constants, because $\sum_{T \subseteq S} (-1)^{|S|-|T|} = 0$. Hence ω computed from $\boldsymbol{\pi}$ equals ω computed from $\boldsymbol{\mu} = N\boldsymbol{\pi}$, and (3.10) applies to $\boldsymbol{\pi}$. Writing $\boldsymbol{\pi}_+ = (1 - \pi_{\mathbf{0}})\boldsymbol{\pi}^c$ and using the degree-one homogeneity of $\mathcal{A}$ gives (3.11). Finally, (3.9) follows by splitting the sum (3.8) into the sets $T \not\ni k$, for which $(-1)^{K-|T|} = -(-1)^{K-1-|T|}$, and $T = T' \cup \{k\}$ with $T' \subseteq [K] \setminus \{k\}$, for which $(-1)^{K-|T|} = (-1)^{K-1-|T'|}$.

A.1 Preliminary lemmas

Proof of Lemma 4.2. Fix ω. *Injectivity.* Let $\boldsymbol{u}, \boldsymbol{u}' \in \Theta_\omega$ with $\boldsymbol{\mu}_+(\boldsymbol{u}) = \boldsymbol{\mu}_+(\boldsymbol{u}')$, and let $\boldsymbol{\mu}, \boldsymbol{\mu}'$ be the full mean vectors defined by (3.6). Both satisfy (3.10) with the same ω, so $\mu_{\mathbf{0}} = e^{\bar{\omega}} \mathcal{A}(\boldsymbol{\mu}_+) = \mu'_{\mathbf{0}}$. Hence $\boldsymbol{\mu} = \boldsymbol{\mu}'$ on all of $\mathcal{H}$, and $\boldsymbol{u} = \boldsymbol{u}'$ by Lemma 3.3. *Surjectivity.* Given $\boldsymbol{\mu}_+ \in (0, \infty)^{\mathcal{H}_+}$, set $\mu_{\mathbf{0}} = e^{\bar{\omega}} \mathcal{A}(\boldsymbol{\mu}_+)$ and define $\boldsymbol{u}$ from the full vector $\boldsymbol{\mu}$ by (3.7). By Lemma 3.3, $\boldsymbol{u}$ reproduces $\boldsymbol{\mu}$ through (3.6), so $\boldsymbol{\mu}_+(\boldsymbol{u}) = \boldsymbol{\mu}_+$. Moreover,

$$u_{[K]} = (-1)^K \log \mu_{\mathbf{0}} + \sum_{\boldsymbol{h}} (-1)^{K-|\boldsymbol{h}|} \log \mu_{\boldsymbol{h}} = \omega + \sum_{\boldsymbol{h}} \left\{ (-1)^K s_{\boldsymbol{h}} + (-1)^{K-|\boldsymbol{h}|} \right\} \log \mu_{\boldsymbol{h}} = \omega,$$

because $(-1)^K s_{\boldsymbol{h}} = (-1)^{K+|\boldsymbol{h}|+1} = -(-1)^{K-|\boldsymbol{h}|}$. Thus $\boldsymbol{u} \in \Theta_\omega$, and the inverse is as stated. □

Lemma A.1 (Block independence). *Suppose that under (M) the history splits into independent blocks: $[K] = A \sqcup B$ with $A, B \neq \emptyset$ and $\pi_{\boldsymbol{h}} = \pi^A_{\boldsymbol{h}_A} \pi^B_{\boldsymbol{h}_B}$, where $\boldsymbol{h}_A$ and $\boldsymbol{h}_B$ are the sub-histories on A and B. Then $u_S = 0$ for every S that meets both A and B; in particular $\omega = 0$.*

Proof. Write $\log \pi_{\boldsymbol{h}} = a(\boldsymbol{h}_A) + b(\boldsymbol{h}_B)$, and for $T \subseteq [K]$ write $T_A = T \cap A$ and $T_B = T \cap B$. Let S meet both blocks, with $S_A = S \cap A \neq \emptyset$ and $S_B = S \cap B \neq \emptyset$. By (3.7),

$$\sum_{T \subseteq S} (-1)^{|S|-|T|} a(T_A) = \Big\{ \sum_{T_A \subseteq S_A} (-1)^{|S_A|-|T_A|} a(T_A) \Big\} \Big\{ \sum_{T_B \subseteq S_B} (-1)^{|S_B|-|T_B|} \Big\} = 0,$$

because the second factor equals $(1-1)^{|S_B|} = 0$. The same argument with the roles of A and B exchanged disposes of b, so $u_S = 0$. Since $[K]$ meets both blocks, $\omega = u_{[K]} = 0$. □

Lemma A.2 (Total variation under (M)). *Under (M), let P and P' be the laws of the observed table $\boldsymbol{n}_+$ under $(N, \boldsymbol{\pi})$ and $(N', \boldsymbol{\pi}')$, where $\boldsymbol{\pi}^c = \boldsymbol{\pi}'^c$. Then* $\mathrm{TV}(P, P') = \mathrm{TV}\big(\mathrm{Binomial}(N, 1-\pi_{\mathbf{0}}), \mathrm{Binomial}(N', 1-\pi'_{\mathbf{0}})\big)$.

Proof. By Lemma 3.2, $P(\boldsymbol{n}_+) = b(n)\, \mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c)$ and $P'(\boldsymbol{n}_+) = b'(n)\, \mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c)$, where b and b' are the two binomial probability mass functions and $n = \sum_{\boldsymbol{h}} n_{\boldsymbol{h}}$. Hence $2\,\mathrm{TV}(P, P') = \sum_{n \geq 0} |b(n) - b'(n)| \sum_{\boldsymbol{n}_+ : \sum_{\boldsymbol{h}} n_{\boldsymbol{h}} = n} \mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c) = \sum_{n \geq 0} |b(n) - b'(n)|$. □

A.2 Proof of Theorem 4.3

Part (a). Let $\boldsymbol{\mu}_+ = \boldsymbol{\mu}_+(\boldsymbol{u})$. By Lemma 4.2, for each ω' there is exactly one $\boldsymbol{u}(\omega') \in \Theta_{\omega'}$ with $\boldsymbol{\mu}_+(\boldsymbol{u}(\omega')) = \boldsymbol{\mu}_+$. By (3.10), $\mu_{\mathbf{0}}(\boldsymbol{u}(\omega')) = e^{(-1)^K\omega'}\mathcal{A}(\boldsymbol{\mu}_+) = e^{(-1)^K(\omega'-\omega)}\mu_{\mathbf{0}}(\boldsymbol{u})$. In (3.7) only the $T = \emptyset$ term changes along the curve, so $u_S(\omega') - u_S = (-1)^{|S|}\{\log\mu_{\mathbf{0}}(\boldsymbol{u}(\omega')) - \log\mu_{\mathbf{0}}(\boldsymbol{u})\} = (-1)^{K-|S|}(\omega'-\omega)$. Under (P), the observed table has law $\bigotimes_{\boldsymbol{h}}\text{Poisson}(\mu_{\boldsymbol{h}})$. This depends on $\boldsymbol{u}$ only through $\boldsymbol{\mu}_+$, so the law and the likelihood (3.5) are constant along the curve. The map $\omega' \mapsto e^{(-1)^K(\omega'-\omega)}$ is a bijection from $\mathbb{R}$ onto $(0,\infty)$, which gives the stated ranges of $\mu_{\mathbf{0}}$ and $\mathbb{E}N = \mu_{\text{obs}} + \mu_{\mathbf{0}}$. Every u_S moves with slope ± 1, so none is identified.

Part (b), first claims. The conditional law of $\boldsymbol{n}_+$ given n is $\mathcal{M}(\cdot\,; n, \boldsymbol{\pi}^c)$ by Lemma 3.2. Fix ω and $\boldsymbol{\pi}^c$ in the interior of the simplex on $\mathcal{H}_+$. Equation (3.11) determines a unique $\pi_{\mathbf{0}} \in (0,1)$, namely $\pi_{\mathbf{0}} = e^{\bar{\omega}}\mathcal{A}(\boldsymbol{\pi}^c)/\{1 + e^{\bar{\omega}}\mathcal{A}(\boldsymbol{\pi}^c)\}$. Setting $\boldsymbol{\pi}_+ = (1-\pi_{\mathbf{0}})\boldsymbol{\pi}^c$ then determines $\boldsymbol{\pi}$. The vector so defined satisfies $\pi_{\mathbf{0}} = e^{\bar{\omega}}\mathcal{A}(\boldsymbol{\pi}_+)$, and it also satisfies (3.10) with its own top interaction $\omega(\boldsymbol{\pi})$; comparing the two gives $\omega(\boldsymbol{\pi}) = \omega$. Conversely, $\boldsymbol{\pi}^c$ and ω determine $\pi_{\mathbf{0}}$, and hence $\boldsymbol{\pi}$. This proves the bijection.

Part (b), no ratio-consistent estimator. Fix $\boldsymbol{\pi}^c$ in the interior and $\omega \neq \omega'$. Let $a, a' \in (0,1)$ be the corresponding values of $\pi_{\mathbf{0}}$, and $\boldsymbol{\pi}, \boldsymbol{\pi}'$ the corresponding probability vectors. By (3.11), $\operatorname{logit} a - \operatorname{logit} a' = (-1)^K(\omega - \omega') \neq 0$, so $c := (1-a)/(1-a') \neq 1$. Take integers $N_\nu \to \infty$ and set $N'_\nu = \lceil cN_\nu \rceil$. Then the binomial means $m_\nu = N_\nu(1-a)$ and $m'_\nu = N'_\nu(1-a')$ satisfy $0 \le m'_\nu - m_\nu < 1$, and $N'_\nu/N_\nu \to c$. Let P_ν and P'_ν be the laws of the observed table under $(N_\nu, \boldsymbol{\pi})$ and $(N'_\nu, \boldsymbol{\pi}')$.

Step 1: $\liminf_\nu\{1 - \text{TV}(P_\nu, P'_\nu)\} > 0$. By Lemma A.2 it suffices to compare $b_\nu = \text{Binomial}(N_\nu, 1-a)$ and $b'_\nu = \text{Binomial}(N'_\nu, 1-a')$. Their variances are $\sigma_\nu^2 = m_\nu a$ and $\sigma_\nu'^2 = m'_\nu a'$. Fix $C > 0$. The de Moivre–Laplace local limit theorem (Feller, 1968, Ch. VII) gives, uniformly over integers x with $|x - m_\nu| \le C\sigma_\nu$,

$$b_\nu(x) = \{1 + o(1)\}\,\sigma_\nu^{-1}\phi(z_x), \qquad z_x = (x - m_\nu)/\sigma_\nu,$$

where ϕ is the standard normal density. Because $(m'_\nu - m_\nu)/\sigma_\nu \to 0$ and $\sigma'_\nu/\sigma_\nu \to r^{1/2}$ with $r = a'/a$, the same theorem applied to b'_ν gives, uniformly over the same range, $b'_\nu(x) = \{1 + o(1)\}\,\sigma_\nu^{-1}\phi_r(z_x)$, where $\phi_r(z) = r^{-1/2}\phi(zr^{-1/2})$ is the $\text{N}(0, r)$ density. Therefore

$$\begin{aligned} 1 - \text{TV}(b_\nu, b'_\nu) = \sum_x \min\{b_\nu(x), b'_\nu(x)\} &\ge \{1 + o(1)\} \sum_{|z_x| \le C} \sigma_\nu^{-1} \min\{\phi(z_x), \phi_r(z_x)\} \\ &\longrightarrow \int_{-C}^{C} \min\{\phi(z), \phi_r(z)\}\, dz > 0, \end{aligned}$$

since the sum on the right of the inequality is a Riemann sum with mesh $\sigma_\nu^{-1} \to 0$.

Step 2. Suppose some estimator satisfies $\widehat{N}/N \to 1$ in probability as $N \to \infty$, for every fixed $\boldsymbol{\pi}$. Assume $c > 1$; otherwise exchange the roles of the two sequences. Consider the test $\varphi_\nu = \mathbf{1}\{\widehat{N} > (1+c)N_\nu/2\}$. Under P_ν, $\widehat{N}/N_\nu \to 1 < (1+c)/2$, so $\mathbb{E}_{P_\nu}\varphi_\nu \to 0$. Under P'_ν, $\widehat{N}/N_\nu = (\widehat{N}/N'_\nu)(N'_\nu/N_\nu) \to c > (1+c)/2$, so $\mathbb{E}_{P'_\nu}(1 - \varphi_\nu) \to 0$. But every test satisfies $\mathbb{E}_P\varphi + \mathbb{E}_{P'}(1-\varphi) \ge 1 - \text{TV}(P, P')$, and by Step 1 the right-hand side has a positive lim inf. This contradiction completes the proof. □

A.3 Proof of Theorem 4.4

Part (a). With ω fixed, the parameter space is Θ_ω. By (3.5), the log-likelihood depends on $\boldsymbol{u}$ only through $\boldsymbol{\mu}_+(\boldsymbol{u})$: $\ell(\boldsymbol{u}) = \sum_{\boldsymbol{h}}\{n_{\boldsymbol{h}}\log\mu_{\boldsymbol{h}}(\boldsymbol{u}) - \mu_{\boldsymbol{h}}(\boldsymbol{u})\} + \text{const}$. By Lemma 4.2, $\boldsymbol{u} \mapsto \boldsymbol{\mu}_+(\boldsymbol{u})$ is a bijection from Θ_ω onto $(0,\infty)^{\mathcal{H}_+}$. Maximising ℓ over Θ_ω is therefore equivalent to maximising $g(\boldsymbol{\mu}_+) = \sum_{\boldsymbol{h}}(n_{\boldsymbol{h}}\log\mu_{\boldsymbol{h}} - \mu_{\boldsymbol{h}})$ over the open orthant, and g is a sum of functions of single coordinates. If $n_{\boldsymbol{h}} > 0$, the map $\mu \mapsto n_{\boldsymbol{h}}\log\mu - \mu$

is strictly concave with unique maximiser $n_{\boldsymbol{h}}$. If $n_{\boldsymbol{h}} = 0$, the map $\mu \mapsto -\mu$ has no maximiser on $(0, \infty)$. Hence a maximiser exists if and only if every $n_{\boldsymbol{h}} > 0$, and it is then unique, with $\widehat{\boldsymbol{\mu}}_+ = \boldsymbol{n}_+$. The maximum likelihood estimator $\widehat{\boldsymbol{u}}$ is the preimage of $\boldsymbol{n}_+$, so $\widehat{\mu}_{\boldsymbol{0}} = \exp(\widehat{u}_\emptyset) = e^{\bar{\omega}}\mathcal{A}(\boldsymbol{n}_+)$ by Lemma 4.2.

Part (b). By Theorem 4.3(b), with ω fixed, $\boldsymbol{\pi}^c$ ranges over the interior of the simplex. The conditional likelihood $\mathcal{M}(\boldsymbol{n}_+; n, \boldsymbol{\pi}^c) \propto \prod_{\boldsymbol{h}} (\pi^c_{\boldsymbol{h}})^{n_{\boldsymbol{h}}}$ has an interior maximiser if and only if every $n_{\boldsymbol{h}} > 0$, and it is then unique and equal to $\boldsymbol{n}_+/n$. By (3.11) and homogeneity, $\widehat{\pi}_{\boldsymbol{0}}/(1-\widehat{\pi}_{\boldsymbol{0}}) = e^{\bar{\omega}}\mathcal{A}(\boldsymbol{n}_+/n) = e^{\bar{\omega}}\mathcal{A}(\boldsymbol{n}_+)/n$. Hence $n/(1-\widehat{\pi}_{\boldsymbol{0}}) = n\{1+\widehat{\pi}_{\boldsymbol{0}}/(1-\widehat{\pi}_{\boldsymbol{0}})\} = n+e^{\bar{\omega}}\mathcal{A}(\boldsymbol{n}_+) = \widehat{N}(\omega)$. For the binomial factor $B(N) = \binom{N}{n}\widehat{\pi}_{\boldsymbol{0}}^{N-n}(1-\widehat{\pi}_{\boldsymbol{0}})^n$ and $N > n$, we have $B(N)/B(N-1) = N\widehat{\pi}_{\boldsymbol{0}}/(N-n)$. This is at least one if and only if $N \le n/(1-\widehat{\pi}_{\boldsymbol{0}}) = \widehat{N}(\omega)$, with equality only when $N = \widehat{N}(\omega)$. So B is non-decreasing up to $\lfloor \widehat{N}(\omega) \rfloor$ and strictly decreasing afterwards. Its maximum is attained at $\lfloor \widehat{N}(\omega) \rfloor$, and also at $\widehat{N}(\omega) - 1$ if $\widehat{N}(\omega)$ is an integer.

Part (c). For $K = 2$, $\mathcal{H}_+ = \{10, 01, 11\}$ with signs $(+1, +1, -1)$, so $\widehat{n}_{\boldsymbol{0}}(\omega) = e^{\omega} n_{10} n_{01}/n_{11}$, and $\omega = \log\{\mu_{11}\mu_{00}/(\mu_{10}\mu_{01})\}$ by (3.8). At $\omega = 0$, $\widehat{N}(0) = n_{10} + n_{01} + n_{11} + n_{10}n_{01}/n_{11} = (n_{10} + n_{11})(n_{01} + n_{11})/n_{11} = n_{1+}n_{2+}/n_{11}$.

Part (d). In the regime of Definition 4.1, $n_{\boldsymbol{h}}/\mu_{\boldsymbol{h}} \to 1$ in probability for every $\boldsymbol{h} \in \mathcal{H}$, so $\mathbb{P}(n_{\boldsymbol{h}} > 0 \ \forall \boldsymbol{h}) \to 1$. On this event, a Taylor expansion of $\log(1+x)$ at $x = (n_{\boldsymbol{h}} - \mu_{\boldsymbol{h}})/\mu_{\boldsymbol{h}} = O_p(\tau^{-1/2})$ gives

$$\log n_{\boldsymbol{h}} - \log \mu_{\boldsymbol{h}} = \frac{n_{\boldsymbol{h}} - \mu_{\boldsymbol{h}}}{\mu_{\boldsymbol{h}}} + O_p(\tau^{-1}). \tag{A.1}$$

Since v_+ and v are of exact order τ^{-1}, remainders of order $O_p(\tau^{-1})$ are negligible after division by $v_+^{1/2}$ or $v^{1/2}$.

(i) By (3.10), $\log \widehat{n}_{\boldsymbol{0}}(\omega) - \log \mu_{\boldsymbol{0}} = \sum_{\boldsymbol{h}} s_{\boldsymbol{h}}(\log n_{\boldsymbol{h}} - \log \mu_{\boldsymbol{h}})$, whose linear part is $L_1 = \sum_{\boldsymbol{h}} s_{\boldsymbol{h}}(n_{\boldsymbol{h}} - \mu_{\boldsymbol{h}})/\mu_{\boldsymbol{h}}$. Under (P) the summands are independent with variances $s_{\boldsymbol{h}}^2/\mu_{\boldsymbol{h}} = 1/\mu_{\boldsymbol{h}}$, so $\operatorname{Var} L_1 = v_+$. Each standardised Poisson count is asymptotically normal, so $L_1/v_+^{1/2} \Rightarrow \mathrm{N}(0,1)$. Under (M), the multivariate central limit theorem gives $N^{-1/2}(\boldsymbol{n} - N\boldsymbol{\pi}) \Rightarrow \mathrm{N}(0, \operatorname{diag}\boldsymbol{\pi} - \boldsymbol{\pi}\boldsymbol{\pi}^\top)$. Since $N^{1/2}L_1 = \sum_{\boldsymbol{h}}(s_{\boldsymbol{h}}/\pi_{\boldsymbol{h}})N^{-1/2}(n_{\boldsymbol{h}} - N\pi_{\boldsymbol{h}})$, its limit variance is $\sum_{\boldsymbol{h}} s_{\boldsymbol{h}}^2/\pi_{\boldsymbol{h}} - (\sum_{\boldsymbol{h}} s_{\boldsymbol{h}})^2 = \sum_{\boldsymbol{h}} 1/\pi_{\boldsymbol{h}} - 1 = N(v_+ - N^{-1})$, by (3.3). This quantity is positive because each $\pi_{\boldsymbol{h}} < 1$.

(ii) Now $\log \widehat{n}_{\boldsymbol{0}}(\omega) - \log n_{\boldsymbol{0}}$ equals $\sum_{\boldsymbol{h}} s_{\boldsymbol{h}}(\log n_{\boldsymbol{h}} - \log \mu_{\boldsymbol{h}}) - (\log n_{\boldsymbol{0}} - \log \mu_{\boldsymbol{0}})$, whose linear part is $L_2 = \sum_{\boldsymbol{h} \in \mathcal{H}} c_{\boldsymbol{h}}(n_{\boldsymbol{h}} - \mu_{\boldsymbol{h}})$ with $c_{\boldsymbol{h}} = s_{\boldsymbol{h}}/\mu_{\boldsymbol{h}}$ for $\boldsymbol{h} \in \mathcal{H}_+$ and $c_{\boldsymbol{0}} = -1/\mu_{\boldsymbol{0}}$. Under (P), $\operatorname{Var} L_2 = \sum_{\boldsymbol{h} \in \mathcal{H}} c_{\boldsymbol{h}}^2 \mu_{\boldsymbol{h}} = v$. Under (M), the limit variance of $N^{1/2}L_2$ is

$$\sum_{\boldsymbol{h} \in \mathcal{H}} (Nc_{\boldsymbol{h}})^2 \pi_{\boldsymbol{h}} - \Big(\sum_{\boldsymbol{h} \in \mathcal{H}} Nc_{\boldsymbol{h}}\pi_{\boldsymbol{h}}\Big)^2 = \sum_{\boldsymbol{h} \in \mathcal{H}} \frac{1}{\pi_{\boldsymbol{h}}} - \Big(\sum_{\boldsymbol{h} \in \mathcal{H}_+} s_{\boldsymbol{h}} - 1\Big)^2 = \sum_{\boldsymbol{h} \in \mathcal{H}} \frac{1}{\pi_{\boldsymbol{h}}} = Nv,$$

again by (3.3). So $L_2/v^{1/2} \Rightarrow \mathrm{N}(0,1)$ under both models. Finally, $\widehat{N}(\omega) - N = \widehat{n}_{\boldsymbol{0}}(\omega) - n_{\boldsymbol{0}} = \mu_{\boldsymbol{0}}\{L_2 + o_p(v^{1/2})\}$, so $(\widehat{N}(\omega) - N)/(\mu_{\boldsymbol{0}} v^{1/2}) \Rightarrow \mathrm{N}(0,1)$, and $\mu_{\boldsymbol{0}}^2 v = \mu_{\boldsymbol{0}}^2 v_+ + \mu_{\boldsymbol{0}}$.

(iii) The ratio $\widehat{v}_+/v_+$ is a weighted average of the ratios $\mu_{\boldsymbol{h}}/n_{\boldsymbol{h}}$, with weights $\mu_{\boldsymbol{h}}^{-1}/v_+$, and each ratio tends to one in probability. The same holds for $\widehat{v}/v$, because $\widehat{n}_{\boldsymbol{0}}(\omega)/\mu_{\boldsymbol{0}} \to 1$ by (i).

The $K = 2$ variance in Remark A.4. With $\omega = 0$, $\mu_{00} = \mu_{10}\mu_{01}/\mu_{11}$, and

$$\mu_{00}^2\Big(\frac{1}{\mu_{10}} + \frac{1}{\mu_{01}} + \frac{1}{\mu_{11}}\Big) + \mu_{00} = \frac{\mu_{10}\mu_{01}}{\mu_{11}^3}\big(\mu_{01}\mu_{11} + \mu_{10}\mu_{11} + \mu_{10}\mu_{01} + \mu_{11}^2\big) = \frac{\mu_{10}\mu_{01}\,\mu_{1+}\mu_{2+}}{\mu_{11}^3}. \qquad \square$$

A.4 Proof of Corollary 4.5

Part (a). Under (P), the law $\bigotimes_{\boldsymbol{h}} \mathrm{Poisson}(\mu_{\boldsymbol{h}})$ of the observed table identifies $\boldsymbol{\mu}_+$, and conversely. By Lemma 4.2, the parameter values compatible with $\boldsymbol{\mu}_+$ and with $|\omega'| \le \Omega$ are exactly $\{\boldsymbol{u}(\omega') : |\omega'| \le \Omega\}$,

one for each ω', with $\mu_{\mathbf{0}} = e^{(-1)^K\omega'}\mathcal{A}(\boldsymbol{\mu}_+)$. As ω' runs over $[-\Omega,\Omega]$, so does $(-1)^K\omega'$, and the increasing map $t \mapsto e^t\mathcal{A}(\boldsymbol{\mu}_+)$ carries $[-\Omega,\Omega]$ onto the stated interval, with each point attained exactly once. Under (M), the conditional law of $\boldsymbol{n}_+$ given any $n \geq 1$ identifies $\boldsymbol{\pi}^c$. By Theorem 4.3(b), each ω' then corresponds to exactly one $\boldsymbol{\pi}$, and its odds are $\pi_{\mathbf{0}}/(1-\pi_{\mathbf{0}}) = e^{(-1)^K\omega'}\mathcal{A}(\boldsymbol{\pi}^c)$ by (3.11). The rest of the argument is as under (P).

Part (b). By (4.2), $\widehat{n}_{\mathbf{0}}(\omega) = e^{(-1)^K\omega}\widehat{n}_{\mathbf{0}}(0)$, and $(-1)^K\omega$ ranges over $[-\Omega,\Omega]$.

Part (c). Let $\alpha \in (0,\frac{1}{2})$, $\widehat{\theta} = \log\widehat{n}_{\mathbf{0}}(0)$, $\theta_0 = \log\mathcal{A}(\boldsymbol{\mu}_+)$ and $t = \bar{\omega} \in [-\Omega,\Omega]$, so that $\log\mu_{\mathbf{0}} = \theta_0 + t$ and $\log\widehat{n}_{\mathbf{0}}(\omega) = \widehat{\theta} + t$. By Theorem 4.4(d)(i), $Z = (\widehat{\theta}-\theta_0)/v_+^{1/2} \Rightarrow \mathrm{N}(0,1)$, and $\rho = (\widehat{v}_+/v_+)^{1/2} \to 1$. The coverage event is

$$\Big\{\frac{t-\Omega}{v_+^{1/2}} - c\rho \leq Z \leq \frac{t+\Omega}{v_+^{1/2}} + c\rho\Big\}.$$

For $d \geq 0$, the equation $\Phi(c+d)-\Phi(-c) = 1-\alpha$ has a unique root $c(d) \in [z_{1-\alpha}, z_{1-\alpha/2}]$, because the left side increases in c, is at most $1-\alpha$ at $c = z_{1-\alpha}$, and is at least $1-\alpha$ at $c = z_{1-\alpha/2}$. Moreover, $c(d) \to z_{1-\alpha}$ as $d \to \infty$. If $\Omega = 0$, then $c = z_{1-\alpha/2}$ and the coverage tends to $1-\alpha$. If $\Omega > 0$, then $d = 2\Omega\widehat{v}_+^{-1/2} \to \infty$ in probability, so $c \to z_{1-\alpha}$. If $|t| < \Omega$, both ends of the event diverge and the coverage tends to one. If $t = \Omega$, the upper end diverges and the lower end tends to $-z_{1-\alpha}$, so the coverage tends to $1-\alpha$; the case $t = -\Omega$ is symmetric. Replacing c by $z_{1-\alpha/2} \geq c$ only enlarges the interval. □

A.5 Proof of Proposition 4.6

By (4.2), $\widehat{n}_{\mathbf{0}}(0) = e^{-\bar{\omega}}\widehat{n}_{\mathbf{0}}(\omega)$, where ω is the true value. By Theorem 4.4(d)(ii), $\log\widehat{n}_{\mathbf{0}}(\omega) - \log n_{\mathbf{0}} = O_p(\tau^{-1/2})$, so $\widehat{n}_{\mathbf{0}}(0)/n_{\mathbf{0}} \to e^{-\bar{\omega}}$. Since $n/N \to 1-\pi_{\mathbf{0}}$ and $n_{\mathbf{0}}/N \to \pi_{\mathbf{0}}$ in probability under either model, $\widehat{N}(0)/N = n/N + \{\widehat{n}_{\mathbf{0}}(0)/n_{\mathbf{0}}\}(n_{\mathbf{0}}/N) \to 1-\pi_{\mathbf{0}}+e^{-\bar{\omega}}\pi_{\mathbf{0}}$. Multiplying (3.9) by $(-1)^K$ gives $\bar{\omega} = (-1)^{K-1}\{\psi_k(0)-\psi_k(1)\}$. Imposing $\omega = 0$ in (3.9) sets $\psi_k(0) = \psi_k(1)$. Part (a): for $K = 2$, $\bar{\omega} = \omega = \log\mathrm{OR}_{12}$ by (3.8). Part (b): for $K = 3$ and $k = 3$, (3.9) gives $\psi_3(1) = \log\{\mu_{111}\mu_{001}/(\mu_{101}\mu_{011})\} = \log\mathrm{OR}_{12|h_3=1}$ and $\psi_3(0) = \log\{\mu_{110}\mu_{000}/(\mu_{100}\mu_{010})\} = \log\mathrm{OR}_{12|h_3=0}$, so $\bar{\omega} = \psi_3(0)-\psi_3(1)$. By Lemma 4.2, the pairwise terms are unrestricted in the $\omega = 0$ model, so any pattern of pairwise dependence is compatible with any value of ω. Part (c) restates the sign rule. □

A.6 Proof of Proposition 4.7

Under the assumptions, the histories are i.i.d. with $\pi_{\boldsymbol{h}} = \vartheta_{|\boldsymbol{h}|}$, where $\vartheta_j = \int p^j(1-p)^{K-j}\,dF(p)$. By (3.8) applied to $\boldsymbol{\pi}$,

$$\omega = \sum_{j=0}^{K}(-1)^{K-j}\binom{K}{j}\log\vartheta_j = \Delta^K\mathcal{L}(0), \qquad \mathcal{L}(t) = \log\int p^t(1-p)^{K-t}\,dF(p),$$

where $\Delta f(t) = f(t+1)-f(t)$. The second equality is the binomial expansion of $\Delta^K = (E-I)^K$, where E is the unit shift and I the identity; the two operators commute. Since F is supported in $[\eta,1-\eta]$, $X = \mathrm{logit}\,p$ is bounded. Hence $\mathcal{L}(t) = \log\int(1-p)^K e^{tX}dF$ is finite and infinitely differentiable on $\mathbb{R}$, and differentiation under the integral is justified by bounded convergence on compact t-sets. The family G_t, with $dG_t/dF \propto (1-p)^K e^{tX} = p^t(1-p)^{K-t}$, is the exponential family generated by X under G_0, and $\mathcal{L}-\mathcal{L}(0)$ is the cumulant generating function of X under G_0. Hence $\log\mathbb{E}_{G_t}e^{sX} = \mathcal{L}(t+s)-\mathcal{L}(t)$, and $\mathcal{L}^{(r)}(t) = \kappa_r(t)$ for every $r \geq 1$. For smooth f, $\Delta f(t) = \int_0^1 f'(t+u)\,du$. By induction on K,

$$\Delta^K f(t) = \int_{[0,1]^{K-1}}\big\{f^{(K-1)}(t+1+\textstyle\sum_{j<K}u_j) - f^{(K-1)}(t+\sum_{j<K}u_j)\big\}\,du = \int_{[0,1]^K} f^{(K)}\big(t+\textstyle\sum_{j\leq K}u_j\big)\,du.$$

With $f = \mathcal{L}$ and $t = 0$, this gives $\omega = \mathbb{E}\{\kappa_K(S_K)\}$, which is (4.5).

Part (a). The density of G_t with respect to F is bounded above and away from zero on $[\eta, 1-\eta]$. Since F is non-degenerate, X is non-degenerate under every G_t, so $\kappa_2(t) = \mathrm{Var}_{G_t}(X) > 0$ and $\omega > 0$. Proposition 4.6(a) then gives underestimation.

Part (b): both signs occur for every $K \geq 3$. Let $\beta_r(s)$ denote the rth cumulant of a Bernoulli(s) variable. Then $\beta_2(s) = s(1-s)$ and $\beta_{r+1}(s) = s(1-s)\beta_r'(s)$; the recursion follows by differentiating the cumulant generating function $\log(1 - s + se^{\lambda})$ with respect to its natural parameter. By induction, β_r is a polynomial of degree r with leading coefficient $(-1)^{r-1}(r-1)! \neq 0$, and it vanishes at $s = 0$ and $s = 1$ for $r \geq 2$. For $r \geq 3$, β_{r-1} is not identically zero and vanishes at both endpoints, so β_{r-1}' takes both signs on $(0,1)$: if it had one sign, β_{r-1} would be monotone between two zeros and hence identically zero. Therefore $\beta_r = s(1-s)\beta_{r-1}'$ takes both signs on $(0,1)$.

Now take a two-point F with atoms $p^{(1)} < p^{(2)}$, where $p^{(1)} \in (\eta, 1-\eta)$ is fixed and $p^{(2)} \in [\eta, 1-\eta]$, where $\mathrm{logit}\, p^{(2)} - \mathrm{logit}\, p^{(1)} = \delta$, and weights $1-w$ and w. Under G_t, X is a shifted and scaled Bernoulli variable, with jump δ and success probability

$$\varsigma(t) = \frac{w\,(p^{(2)})^t(1-p^{(2)})^{K-t}}{(1-w)(p^{(1)})^t(1-p^{(1)})^{K-t} + w\,(p^{(2)})^t(1-p^{(2)})^{K-t}},$$

so that $\kappa_K(t) = \delta^K \beta_K(\varsigma(t))$, since a shift changes only the first cumulant. Let $\delta \downarrow 0$, so that $p^{(2)} \downarrow p^{(1)}$. Then $\varsigma(t) \to w$ uniformly on $[0, K]$, so $\omega/\delta^K = \mathbb{E}\{\beta_K(\varsigma(S_K))\} \to \beta_K(w)$. Choosing w with $\beta_K(w) > 0$, or with $\beta_K(w) < 0$, and δ small enough gives ω of either sign. The final claim of (b) follows from Proposition 4.6. □

A.7 Proofs of Theorem 4.8 and Proposition 4.9

Proof of Theorem 4.8 (Chao, 1987). Conditionally on $\boldsymbol{p}$, $|\boldsymbol{h}_i| \sim \mathrm{Binomial}(K, p_i)$ independently over i, so $\mathbb{E}f_j = \binom{K}{j}\sum_i p_i^j(1-p_i)^{K-j}$. Let $x_i = (1-p_i)^{K/2}$ and $y_i = p_i(1-p_i)^{K/2-1}$; the exponents are non-negative because $K \geq 2$, with $0^0 = 1$. Then $\sum_i x_i^2 = \mathbb{E}f_0$, $K\sum_i x_i y_i = \mathbb{E}f_1$ and $\binom{K}{2}\sum_i y_i^2 = \mathbb{E}f_2$. By the Cauchy–Schwarz inequality,

$$\frac{(\mathbb{E}f_1)^2}{K^2} = \Big(\sum_i x_i y_i\Big)^2 \leq \sum_i x_i^2 \sum_i y_i^2 = \mathbb{E}f_0\,\frac{\mathbb{E}f_2}{\binom{K}{2}},$$

which rearranges to (4.6), since $\binom{K}{2}/K^2 = (K-1)/(2K)$. The second form follows from $N = \sum_j \mathbb{E}f_j = \mathbb{E}n + \mathbb{E}f_0$. Equality holds if and only if $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. When every $p_i \in (0,1)$, all $x_i > 0$ and $y_i/x_i = p_i/(1-p_i)$, so equality holds if and only if the p_i are all equal. If the p_i are i.i.d. from F, the histories are i.i.d., and by the strong law $f_j/N \to \varphi_j = \binom{K}{j}\int p^j(1-p)^{K-j}dF$ almost surely, with $\varphi_2 > 0$ because F is concentrated on $(0,1)$. Therefore $\widehat{N}_{\mathrm{C}}/N \to 1 - \varphi_0 + \frac{K-1}{K}\varphi_1^2/(2\varphi_2) = \lambda_{\mathrm{C}}$. The Cauchy–Schwarz inequality in $L^2(F)$, applied to $x(p) = (1-p)^{K/2}$ and $y(p) = p(1-p)^{K/2-1}$, gives $\lambda_{\mathrm{C}} \leq 1$. Equality holds if and only if $p/(1-p)$ is F-almost surely constant, that is, if and only if F is a point mass. □

Proof of Proposition 4.9. Under M_t, every incident has $\mathbb{P}(\boldsymbol{h}_i = \boldsymbol{h}_T) = \prod_{k\in T} a_k \prod_{k\notin T}(1-a_k) = P\prod_{k\in T} r_k$, where $P = \prod_k(1-a_k)$ and $r_k = a_k/(1-a_k) > 0$. Hence $\mathbb{E}f_j = NPe_j(\boldsymbol{r})$, where e_j is the elementary symmetric polynomial of degree j. Using $e_1^2 = \sum_k r_k^2 + 2e_2$, the claimed inequality $\frac{K-1}{K}e_1^2/(2e_2) \geq 1$ is equivalent to $(K-1)e_1^2 \geq K(e_1^2 - \sum_k r_k^2)$, that is, to $K\sum_k r_k^2 \geq (\sum_k r_k)^2$. This is the Cauchy–Schwarz inequality, with equality if and only if all r_k, equivalently all a_k, are equal. By the strong law, $f_j/N \to Pe_j(\boldsymbol{r})$ almost surely. Hence $\widehat{N}_{\mathrm{C}}/N \to 1 - P + P\,\frac{K-1}{K}e_1^2/(2e_2) \geq 1$. □

A.8 Proof of Proposition 4.10

Part (a). We have $\mathbb{P}(Y_1 = 1, Y_2 = 1) = p_1\mathbb{P}(\max(D, R) = 1) = p_1\{1-(1-p_2)(1-q)\} = p_1\{p_2+q(1-p_2)\}$. Also $\mathbb{P}(Y_1 = 1, Y_2 = 0) = p_1(1 - p_2)(1 - q)$, and when $Y_1 = 0$ we have $Y_2 = D$, which gives π_{01} and π_{00}. Then

$$\mathrm{OR}_{12} = \frac{p_1\{p_2 + q(1 - p_2)\}(1 - p_1)(1 - p_2)}{p_1(1 - p_2)(1 - q)(1 - p_1)p_2} = \frac{p_2(1 - q) + q}{p_2(1 - q)} = 1 + \frac{q}{p_2(1 - q)},$$

and $u_{12} = \omega = \log \mathrm{OR}_{12}$ by (3.8) with $K = 2$. The monotonicity claims are immediate.

Part (b). By the strong law, $n_{1+}/N \to p_1, n_{2+}/N \to \pi_{11}+\pi_{01} = p_2+p_1q(1-p_2)$ and $n_{11}/N \to \pi_{11} > 0$. Hence $\widehat{N}_{\mathrm{LP}}/N \to p_1\{p_2 + p_1q(1 - p_2)\}/[p_1\{p_2 + q(1 - p_2)\}]$. Subtracting this limit from one gives $q(1 - p_2)(1 - p_1)/\{p_2 + q(1 - p_2)\}$. Also $\widehat{n}^{\mathrm{LP}}_{\mathbf{0}} = n_{10}n_{01}/n_{11}$, so $\widehat{n}^{\mathrm{LP}}_{\mathbf{0}}/n_{\mathbf{0}} \to \pi_{10}\pi_{01}/(\pi_{11}\pi_{00}) = 1/\mathrm{OR}_{12}$.

Part (c). The recoded histories $(Y_1, Y_2^\circ) = (Y_1, D)$ are i.i.d. with independent components. Their odds ratio is therefore one, so $\omega = 0$. By the strong law, $\widehat{N}^\circ/N \to p_1p_2/(p_1p_2) = 1$. For general K, the recoded histories *are* the direct-capture histories $(Y_1, D, Y_3, \ldots, Y_K)$, so any restriction those satisfy carries over unchanged.

Part (d). Let $Z_i = R_iY_{i1}$ and $U^\circ = \{i : Z_i = 0\}$. Membership in U° is decided independently for each incident, and every flagged incident is observed, so $n_Z = N - |U^\circ|$. On $\{Z = 0\}$, either $Y_1 = 0$ or $R = 0$, and in both cases $Y_2 = D$. Hence

$$\begin{aligned}
&\mathbb{P}(Y_1 = 1, Y_2 = 1, Z = 0) = p_1(1 - q)p_2, && \mathbb{P}(Y_1 = 1, Y_2 = 0, Z = 0) = p_1(1 - q)(1 - p_2),\\
&\mathbb{P}(Y_1 = 0, Y_2 = 1, Z = 0) = (1 - p_1)p_2, && \mathbb{P}(Y_1 = 0, Y_2 = 0, Z = 0) = (1 - p_1)(1 - p_2),
\end{aligned}$$

whose odds ratio is one. In the reduced table, the police total is $n_{1+} - n_Z$, the both-lists cell is m_{11}, and the NGO-only cell is n_{01}, since every incident with $Y_1 = 0$ is unflagged. By the strong law,

$$\frac{(n_{1+} - n_Z)(m_{11} + n_{01})}{N\, m_{11}} \to \frac{p_1(1 - q)\{p_1(1 - q)p_2 + (1 - p_1)p_2\}}{p_1(1 - q)p_2} = 1 - p_1q,$$

and $n_Z/N \to p_1q$. Adding the two limits gives $\widehat{N}^\ddagger/N \to 1$.

Part (e). The history law factorises as $\pi_{\boldsymbol{h}} = \pi^{\{1,2\}}_{(h_1,h_2)}\, \pi^{B}_{\boldsymbol{h}_B}$ with $B = \{3, \ldots, K\}$. By Lemma A.1, $u_S = 0$ whenever S meets both $\{1, 2\}$ and B; in particular $\omega = 0$. The model of Theorem 4.4 with $\omega = 0$ is therefore correctly specified, and $\widehat{n}_{\mathbf{0}}(0)/n_{\mathbf{0}} \to 1$ by Theorem 4.4(d)(ii). □

A.9 Proof of Theorem 4.11

Part (a). By Corollary 4.5(a), the parameter values observationally equivalent to the truth with $|\omega'| \le \Omega$ all share $\boldsymbol{\mu}_+$, and hence $\mu_{1+} = \sum_{\boldsymbol{h}:h_1=1} \mu_{\boldsymbol{h}}$. Their dark figures fill $[e^{-\Omega}\mathcal{A}(\boldsymbol{\mu}_+), e^{\Omega}\mathcal{A}(\boldsymbol{\mu}_+)]$. So C_κ holds at all of them if and only if $e^{-\Omega}\mathcal{A}(\boldsymbol{\mu}_+) > \kappa\mu_{1+}$, that is, if and only if $\Omega < \Omega^*_\kappa$.

Part (b). Let $t = \bar{\omega} \ge -\Omega$. Since $\widehat{n}_{\mathbf{0}}(\omega) = e^t\widehat{n}_{\mathbf{0}}(0)$, we have $\log \widehat{n}_{\mathbf{0}}(0) - \Omega \le \log \widehat{n}_{\mathbf{0}}(\omega)$. Define the pivot

$$W = \log \frac{\widehat{n}_{\mathbf{0}}(\omega)}{n_{1+}} - \log \frac{\mu_{\mathbf{0}}}{\mu_{1+}} = \sum_{\boldsymbol{h}} s_{\boldsymbol{h}} \log \frac{n_{\boldsymbol{h}}}{\mu_{\boldsymbol{h}}} - \log \frac{n_{1+}}{\mu_{1+}},$$

where the second expression uses (3.10) and does not involve ω. When C_κ fails, $\log\{\mu_{\mathbf{0}}/(\kappa\mu_{1+})\} \le 0$, and therefore

$$\{\widehat{\Omega}_{\kappa,\alpha} > \Omega\} = \left\{\log \frac{\widehat{n}_{\mathbf{0}}(0)}{\kappa n_{1+}} - \Omega > z_{1-\alpha}\widehat{s}_1\right\} \subseteq \left\{\log \frac{\widehat{n}_{\mathbf{0}}(\omega)}{\kappa n_{1+}} > z_{1-\alpha}\widehat{s}_1\right\} \subseteq \{W > z_{1-\alpha}\widehat{s}_1\}.$$

By (A.1) and $\log(n_{1+}/\mu_{1+}) = (n_{1+} - \mu_{1+})/\mu_{1+} + O_p(\tau^{-1})$, the linear part of W is $\sum_{\boldsymbol{h}} c_{\boldsymbol{h}}(n_{\boldsymbol{h}} - \mu_{\boldsymbol{h}})$ with $c_{\boldsymbol{h}} = s_{\boldsymbol{h}}/\mu_{\boldsymbol{h}} - h_1/\mu_{1+}$. Under (P), its variance is

$$\sum_{\boldsymbol{h}} c_{\boldsymbol{h}}^2 \mu_{\boldsymbol{h}} = \sum_{\boldsymbol{h}} \frac{1}{\mu_{\boldsymbol{h}}} - \frac{2}{\mu_{1+}} \sum_{\boldsymbol{h}:h_1=1} s_{\boldsymbol{h}} + \frac{1}{\mu_{1+}} = v_+ + \frac{1}{\mu_{1+}} =: s_1^2,$$

by the second identity in (3.3). Under (M), the multinomial covariance subtracts $(\sum_{\boldsymbol{h}} c_{\boldsymbol{h}}\mu_{\boldsymbol{h}})^2/N = (\sum_{\boldsymbol{h}} s_{\boldsymbol{h}} - 1)^2/N = 0$, by the first identity. So $W/s_1 \Rightarrow \mathrm{N}(0,1)$ under both models, and $\widehat{s}_1/s_1 \to 1$ as in Theorem 4.4(d)(iii). Hence $\limsup \mathbb{P}(\widehat{\Omega}_{\kappa,\alpha} > \Omega) \le \lim \mathbb{P}(W > z_{1-\alpha}\widehat{s}_1) = \alpha$. If $t = -\Omega$ and $\mu_{\boldsymbol{0}} = \kappa\mu_{1+}$, both inclusions are equalities, and the limit equals α.

Part (c). In the regime of Definition 4.1, $\log \widehat{n}_{\boldsymbol{0}}(0) - \log \mathcal{A}(\boldsymbol{\mu}_+) \to 0$ and $\log n_{1+} - \log \mu_{1+} \to 0$ in probability. By homogeneity, $\mathcal{A}(\boldsymbol{\mu}_+)/\mu_{1+} = \mathcal{A}(\boldsymbol{m}_+)/m_{1+}$ does not depend on τ, and $\widehat{s}_1 \to 0$. Hence $\widehat{\Omega}_{\kappa,\alpha} \to \Omega_\kappa^*$ in probability, and $\mathbb{P}(\widehat{\Omega}_{\kappa,\alpha} > \Omega) \to 1$ whenever $\Omega < \Omega_\kappa^*$. □

A.10 Further remarks on Section 4

Remark A.3 (The restriction on ω cannot be tested). Every goodness-of-fit statistic, information criterion and posterior predictive check is a function of the observed table. Its law is constant along the curve in Theorem 4.3(a). Model search over log-linear models (Section 6) therefore chooses among restrictions on lower-order terms *given* a restriction on ω, never the restriction on ω itself. A non-hierarchical model that sets some lower-order terms to zero and leaves ω free can identify ω, because the observed table then over-determines the remaining parameters. Each such restriction, however, is itself a statement about the unobservable cell, since every u_S involves $\log \mu_{\boldsymbol{0}}$ through the $T = \emptyset$ term of (3.7). That is identification by functional form, not by data (compare Fienberg, 1972; International Working Group for Disease Monitoring and Forecasting, 1995).

Remark A.4 (Intervals). Theorem 4.4(d) gives two Wald intervals on the log scale. For the expected dark figure under (P), $\widehat{n}_{\boldsymbol{0}} \exp(\pm z_{1-\alpha/2}\widehat{v}_+^{1/2})$ has asymptotic coverage $1 - \alpha$. For the realised total under (P) or (M), the interval

$$\left[\, n + \widehat{n}_{\boldsymbol{0}}\, e^{-z_{1-\alpha/2}\widehat{v}^{1/2}},\ n + \widehat{n}_{\boldsymbol{0}}\, e^{z_{1-\alpha/2}\widehat{v}^{1/2}} \,\right]$$

has asymptotic coverage $1-\alpha$, and its lower end never falls below n. This is in the spirit of the log-transformed interval of Chao (1987). For $K = 2$ and $\omega = 0$, a direct computation gives $\mu_{\boldsymbol{0}}^2 v_+ + \mu_{\boldsymbol{0}} = \mu_{1+}\mu_{2+}\mu_{10}\mu_{01}/\mu_{11}^3$, the familiar large-sample variance of the Lincoln–Petersen estimator (Bishop et al., 1975). For the illustrative Table 3, $\widehat{v}_+ = 0.104$, the 95% interval for $\mu_{\boldsymbol{0}}$ is $[382, 1355]$, and the interval for N is $[1141, 2121]$.

Remark A.5 (Zero cells). If $n_{\boldsymbol{h}} = 0$ for some $\boldsymbol{h} \in \mathcal{H}_+$, the likelihood has no maximiser in the interior. If all empty cells have odd (respectively even) $|\boldsymbol{h}|$, the formal limit of (4.2) is 0 (respectively $+\infty$). With empty cells of both parities it is undefined. We use three remedies, in order of preference. (1) Add $\frac{1}{2}$ to every observed cell. Under (P), $\mathbb{E}\log(n_{\boldsymbol{h}} + \frac{1}{2}) = \log \mu_{\boldsymbol{h}} + O(\mu_{\boldsymbol{h}}^{-2})$, whereas the bias of $\log n_{\boldsymbol{h}}$, conditional on $n_{\boldsymbol{h}} > 0$, is of order $\mu_{\boldsymbol{h}}^{-1}$. The corrected estimator $e^{(-1)^K\omega}\mathcal{A}(\boldsymbol{n}_+ + \frac{1}{2})$ therefore removes the leading bias of $\log \widehat{n}_{\boldsymbol{0}}$ (compare Gart and Zweifel, 1967). For $K = 2$, Chapman (1951) gives the analogous correction for Lincoln–Petersen. (2) Fit a hierarchical submodel without the interactions that the empty cells cannot support. Existence of its maximum likelihood estimator is a polyhedral condition on the support of the table (Fienberg and Rinaldo, 2012). (3) Place proper priors on the u-terms (Madigan and York, 1997); see Section 6. Sparse tables are the rule in small domains, which is one motivation for the hierarchical models of Section 6.

Remark A.6 (Every log-linear estimator is an alternating product). For any hierarchical generating class $\mathcal{S}$ with $[K] \notin \mathcal{S}$, every parameter value satisfies $\omega = 0$, so $\mu_{\mathbf{0}} = \mathcal{A}(\boldsymbol{\mu}_+)$ by (3.10). By equivariance, the maximum likelihood estimator under $\mathcal{S}$ is therefore $\widehat{\mu}_{\mathbf{0}}^{\mathcal{S}} = \mathcal{A}(\widehat{\boldsymbol{\mu}}_+^{\mathcal{S}})$, the alternating product of the *fitted* observed cells. In the same way, each posterior draw of $\boldsymbol{\mu}_+$ maps to a draw $\mathcal{A}(\boldsymbol{\mu}_+)$ of $\mu_{\mathbf{0}}$. Log-linear multiple-systems estimators thus differ only in how they smooth the observed table before applying $\mathcal{A}$; the model of Theorem 4.4 smooths nothing. With a non-zero offset ω_0 in a proper submodel, the fitted cells $\widehat{\boldsymbol{\mu}}_+^{\mathcal{S}}$ depend on ω_0. The exact multiplicative dependence on ω in (4.2) is therefore specific to the model without a K-way term.

Example A.7 (Visibility heterogeneity can reverse the bias). Suppose 95% of incidents have $p = 0.1$ and 5% are highly visible with $p = 0.9$. Computing ω exactly from $\pi_{\boldsymbol{h}} = \int p^{|\boldsymbol{h}|}(1-p)^{K-|\boldsymbol{h}|}\,dF$ gives $\omega = 1.56$, 2.53 and -0.49 for $K = 2, 3, 4$, so by (4.4) the limit of $\widehat{n}_{\mathbf{0}}(0)/n_{\mathbf{0}}$ is 0.21, 12.5 and 1.63. Lincoln–Petersen recovers a fifth of the dark figure. For $K = 3$ the $\omega = 0$ model, with all two-way interactions, overstates the dark figure 12.5-fold and N about 9-fold, because the few highly visible incidents pile up in n_{111}, which enters $\mathcal{A}$ in the numerator. Simpler three-list models understate it instead, and over a grid of two-class mixtures the all-two-way limit ranges from 0.37 to 25 (Remark A.8). These are exact population limits for stylised models. The lesson is not that three lists are worse than two, but that "heterogeneity" fixes neither the sign nor the size of the bias, which is why Section 6 models heterogeneity directly through latent classes.

Remark A.8 (More on Example A.7). In the population of Example A.7 with $K = 3$, simpler log-linear models understate the dark figure: the limit of $\widehat{n}_{\mathbf{0}}/n_{\mathbf{0}}$ is 0.24 under independence, 0.30 with one two-way interaction and 0.69 with two. The magnitude of the bias of the all-two-way model is also fragile. Over two-class mixtures with low visibility $p \in \{0.05, 0.1, 0.2\}$, high visibility $p \in \{0.7, 0.8, 0.9\}$ and a highly visible share in $\{0.02, 0.05, 0.1, 0.2\}$, its limit ranges from 0.37 to 25.

Remark A.9 (Calibrating Ω from referral). For $K = 2$, referral produces only $\omega \geq 0$. Suppose the analyst is willing to bound the ingestion rate, $q \leq \bar{q}$, and the direct coverage, $p_2 \geq \underline{p}_2$. Then (4.7) gives the one-sided restriction $0 \leq \omega \leq \log\{1 + \bar{q}/(\underline{p}_2(1-\bar{q}))\}$. For example, $\bar{q} = \underline{p}_2 = 0.2$ gives $e^{\Omega} = 2.25$. The bounds of Corollary 4.5 then become $[\widehat{n}_{\mathbf{0}}(0),\ e^{\Omega}\widehat{n}_{\mathbf{0}}(0)]$. This translates the abstract ω into two quantities that list holders can speak to.

For $K \geq 3$ the one-sided conclusion fails, because (4.7) is a two-list result and referral involving several lists can give $\bar{\omega}$ of either sign (Example A.10). With three or more lists the analyst should therefore calibrate $\bar{\omega}$, which governs the direction of bias (Proposition 4.6), from an explicit model of the referral channels, not assume a sign for ω.

Example A.10 (Referral among three lists can bias either way). Keep the police-to-NGO referral of Proposition 4.10 with $p_1 = 0.3$, $p_2 = 0.2$, $q = 0.3$. Add a third list with direct coverage $p_3 = 0.2$ that also ingests, with probability 0.3, incidents recorded by both police *and* the NGO. Then $\omega = +1.15$ and $\bar{\omega} = -1.15$, so the $\omega = 0$ estimator overstates the dark figure by a factor of about 3.1. If the third list instead ingests incidents recorded by police *or* the NGO, then $\omega = -1.15$, and the dark figure is understated by the same factor. This is why Remark A.9 confines the one-sided calibration of Ω to $K = 2$.

Example A.11 (Illustrative numbers, continued). For Table 3, the claim "the dark figure exceeds the police count" ($\kappa = 1$) has plug-in frontier $\log(720/400) = 0.59$, which estimates Ω_1^*. Ignoring sampling error, ω would have to move the dark figure by a factor of 1.8 to overturn it. Once sampling error is included ($\widehat{s}_1 = 0.33$, $\alpha = 0.05$), $\widehat{\Omega}_{1,0.05} = 0.05$, so the claim is barely robust at all. The weaker claim with $\kappa = \frac{1}{2}$ has

$\widehat{\Omega}_{1/2,0.05} = 0.74$ and survives any departure up to a factor of $e^{0.74} \approx 2.1$. With counts of this size, only the weaker claim could responsibly be reported.

Appendix B Proofs for Section 5

We use the notation of Section 5. Cells are indexed by non-empty supports $g, h \subseteq [K]$. Sums over g run over non-empty g. For a fixed list k we use the alternating-sum identity

$$\sum_{\emptyset \neq g \subseteq J} (-1)^{|g|+1} \prod_{j \in g} x_j = 1 - \prod_{j \in J} (1 - x_j), \qquad \text{in particular} \qquad \sum_{\emptyset \neq g \subseteq J} (-1)^{|g|+1} = 1 \quad (J \neq \emptyset), \tag{B.1}$$

which follows by expanding the product. Norms on $\mathbb{R}^{\mathcal{H}_+}$ are ℓ_1 unless stated otherwise.

B.1 Proof of Proposition C.1

We give the standard Neyman–Pearson argument (Lehmann and Romano, 2005, Ch. 3); the result is Theorem 1 of Fellegi and Sunter (1969). Use the conventions $w = +\infty$ if $u(\gamma) = 0 < m(\gamma)$ and $w = -\infty$ if $m(\gamma) = 0 < u(\gamma)$, and discard γ with $m(\gamma) = u(\gamma) = 0$. Because $T_N < T_L$, the link and non-link regions of $\phi^\star$ are disjoint, so $\phi^\star$ is a decision rule. Let ϕ satisfy $P_{FL}(\phi) \le P^\star_{FL}$ and $P_{ML}(\phi) \le P^\star_{ML}$.

Put $c = e^{T_L}$. By construction, $\phi^\star_L(\gamma) = 1$ if $m(\gamma) > c\,u(\gamma)$ and $\phi^\star_L(\gamma) = 0$ if $m(\gamma) < c\,u(\gamma)$. Hence $\{\phi^\star_L(\gamma) - \phi_L(\gamma)\}\{m(\gamma) - c\,u(\gamma)\} \ge 0$ for every γ. Summing over γ,

$$\sum_\gamma m(\gamma)\{\phi^\star_L(\gamma) - \phi_L(\gamma)\} \;\ge\; c\,\{P^\star_{FL} - P_{FL}(\phi)\} \;\ge\; 0. \tag{B.2}$$

Similarly, with $c' = e^{T_N}$, $\phi^\star_N(\gamma) = 1$ if $m(\gamma) < c'u(\gamma)$ and $\phi^\star_N(\gamma) = 0$ if $m(\gamma) > c'u(\gamma)$. Hence $\{\phi^\star_N - \phi_N\}\{c'u - m\} \ge 0$ pointwise, and

$$c' \sum_\gamma u(\gamma)\{\phi^\star_N(\gamma) - \phi_N(\gamma)\} \;\ge\; P^\star_{ML} - P_{ML}(\phi) \;\ge\; 0. \tag{B.3}$$

Since $\phi_C = 1 - \phi_L - \phi_N$, (B.2) and $P_{ML}(\phi) \le P^\star_{ML}$ give

$$\textstyle\sum_\gamma m\phi_C = 1 - \sum_\gamma m\phi_L - P_{ML}(\phi) \;\ge\; 1 - \sum_\gamma m\phi^\star_L - P^\star_{ML} = \sum_\gamma m\phi^\star_C.$$

Likewise (B.3) and $P_{FL}(\phi) \le P^\star_{FL}$ give $\sum_\gamma u\phi_C = 1 - P_{FL}(\phi) - \sum_\gamma u\phi_N \ge 1 - P^\star_{FL} - \sum_\gamma u\phi^\star_N = \sum_\gamma u\phi^\star_C$. For any proportion p of matches, the review probability is $p\sum m\phi_C + (1-p)\sum u\phi_C$, which $\phi^\star$ therefore minimises. □

B.2 Proof of Proposition 5.3

Consider one true incident with support h, and let $S \subseteq h$ be the set of lists whose records do not detach. Then $\mathbb{P}(S = g) = (1-\delta)^{|g|}\delta^{|h|-|g|}$ for $g \subseteq h$. The incident produces the cluster vector

$$\boldsymbol{Z}_h = \boldsymbol{e}_S \mathbf{1}\{S \neq \emptyset\} + \sum_{k \in h \setminus S} \boldsymbol{e}_{\{k\}}.$$

If $S = \{j\}$, then $\boldsymbol{e}_S = \boldsymbol{e}_{\{j\}}$ and the remaining singletons lie in other cells. In every case $\boldsymbol{Z}_h$ is a 0/1 vector with at most $|h|$ non-zero entries. For $|g| \ge 2$, $\mathbb{E}(\boldsymbol{Z}_h)_g = \mathbb{P}(S = g) = \mathbf{1}\{g \subseteq h\}\,(1-\delta)^{|g|}\delta^{|h|-|g|}$. For $g = \{k\}$, $\mathbb{E}(\boldsymbol{Z}_h)_{\{k\}} = \mathbf{1}\{k \in h\}\,\{\mathbb{P}(S = \{k\}) + \mathbb{P}(k \notin S)\} = \mathbf{1}\{k \in h\}\,\{(1-\delta)\delta^{|h|-1} + \delta\}$. This is (5.2).

Detachments are independent across records, hence across incidents, and $\tilde{\boldsymbol{n}}$ is the sum of the independent vectors $\boldsymbol{Z}_{h_b}$ over the true incidents b. So $\mathbb{E}\tilde{\boldsymbol{n}} = \mathcal{D}_\delta \boldsymbol{t}$.

(i) Every term of (5.2) vanishes unless $g \subseteq h$. For $g = h$ with $|h| \geq 2$ the singleton term is absent and the entry is $(1-\delta)^{|h|}$. For $g = h = \{k\}$ it is $(1-\delta) + \delta = 1$. A matrix that is triangular with respect to a linear extension of inclusion has determinant equal to the product of its diagonal entries, which is positive for $\delta < 1$.

(ii) The clusters of $\boldsymbol{Z}_h$ partition the records of the incident, so $(\boldsymbol{Z}_h)_{k+} = \mathbf{1}\{k \in h\} = (\boldsymbol{e}_h)_{k+}$ pathwise. Taking expectations, $(\mathcal{D}_\delta \boldsymbol{e}_h)_{k+} = (\boldsymbol{e}_h)_{k+}$, and linearity gives the claim. The column sum is $\mathbb{E}\sum_g (\boldsymbol{Z}_h)_g = \mathbb{P}(S \neq \emptyset) + \mathbb{E}|h \setminus S| = 1 - \delta^{|h|} + |h|\delta$.

(iii) Let $\mathcal{U} = \{h : |h| \geq 2\}$ and let $\mathcal{S}$ be the set of singletons. Since $g \subseteq h$ is impossible for $g \in \mathcal{U}$ and $h \in \mathcal{S}$, we have $\boldsymbol{y}_{\mathcal{U}} = (\mathcal{D}_\delta)_{\mathcal{U}\mathcal{U}} \boldsymbol{x}_{\mathcal{U}}$. On $\mathcal{U} \times \mathcal{U}$ the singleton term vanishes, so $(\mathcal{D}_\delta)_{gf} = (1-\delta)^{|g|}\delta^{|f|-|g|}\mathbf{1}\{g \subseteq f\}$. Let $E_{hg} = (-\delta)^{|g|-|h|}(1-\delta)^{-|g|}\mathbf{1}\{h \subseteq g\}$ for $h, g \in \mathcal{U}$. For $h, f \in \mathcal{U}$, every g with $h \subseteq g \subseteq f$ lies in $\mathcal{U}$. Hence

$$(E\,\mathcal{D}_\delta)_{hf} = \sum_{h \subseteq g \subseteq f} (-\delta)^{|g|-|h|}(1-\delta)^{-|g|}(1-\delta)^{|g|}\delta^{|f|-|g|} = \delta^{|f|-|h|} \sum_{h \subseteq g \subseteq f} (-1)^{|g|-|h|} = \mathbf{1}\{h = f\},$$

by the binomial theorem. Thus $\boldsymbol{x}_{\mathcal{U}} = E\boldsymbol{y}_{\mathcal{U}}$, which is (5.3). By (ii), $y_{k+} = x_{k+} = x_{\{k\}} + \sum_{h \ni k, |h| \geq 2} x_h$, which is the first equality in (5.4). For the second, substitute (5.3) and exchange sums:

$$\sum_{h \ni k,\, |h| \geq 2} x_h = \sum_{g \ni k,\, |g| \geq 2} \frac{y_g}{(1-\delta)^{|g|}} \sum_{h:\, k \in h \subseteq g,\ |h| \geq 2} (-\delta)^{|g|-|h|}.$$

Writing $h = \{k\} \cup h'$ with $\emptyset \neq h' \subseteq g \setminus k$, the inner sum is $(1-\delta)^{|g|-1} - (-\delta)^{|g|-1}$. Since $y_{k+} = y_{\{k\}} + \sum_{g \ni k, |g| \geq 2} y_g$, the coefficient of y_g in $x_{\{k\}}$ is $1 - (1-\delta)^{-1} + (-\delta)^{|g|-1}(1-\delta)^{-|g|}$, as claimed.

(iv) The records of lists j and k of an incident with $h \supseteq \{j, k\}$ lie in a common cluster if and only if $j, k \in S$. If either detaches, it is a singleton. No cluster contains records from two different true incidents, because (LE-D) only splits. Hence $\tilde{n}_{jk}$ is a sum of t_{jk} independent Bernoulli$((1-\delta)^2)$ variables.

(v) $\tilde{\boldsymbol{n}} - \mathcal{D}_\delta \boldsymbol{t} = \sum_b (\boldsymbol{Z}_{h_b} - \mathbb{E}\boldsymbol{Z}_{h_b})$ with independent summands, and $\mathbb{E}\|\boldsymbol{Z}_h - \mathbb{E}\boldsymbol{Z}_h\|_2^2 \leq \mathbb{E}\|\boldsymbol{Z}_h\|_2^2 \leq |h| \leq K$. □

B.3 Proof of Lemma C.2

Condition on $\tilde{\Lambda}$ throughout. Let C be its set of clusters, $\mathrm{type}(c) \in \mathcal{H}_+$ the support of $c \in C$, $\mathcal{S}_k$ the list-k singletons, $\mathcal{S} = \bigcup_k \mathcal{S}_k$ (so $|\mathcal{S}| = A$), and $P_k = \{c \in C : k \notin \mathrm{type}(c)\}$, so $|P_k| = C_k := C_k(\tilde{\boldsymbol{n}})$. Let $\chi_r \sim \mathrm{Bernoulli}(\varepsilon)$, $r \in \mathcal{S}$, be the selection indicators, $\mathcal{F} = \{r : \chi_r = 1\}$, and process $\mathcal{F}$ in uniformly random order.

Idealised process. For each $r \in \mathcal{S}_k$ let B_r be uniform on P_k, with the B_r independent of each other and of (χ, order). Note that $r \notin P_k$. Put $\boldsymbol{m}_r = -\boldsymbol{d}_{\mathrm{type}(B_r) \cup k, k}$ and $\boldsymbol{X} = \sum_{r \in \mathcal{F}} \boldsymbol{m}_r$. Then

$$\mathbb{E}\boldsymbol{X} = \sum_k \varepsilon A_k \sum_{g \not\ni k} \frac{\tilde{n}_g}{C_k}\left(-\boldsymbol{d}_{g \cup k, k}\right) = \varepsilon\, \boldsymbol{q}(\tilde{\boldsymbol{n}}), \qquad A_k = \tilde{n}_{\{k\}}.$$

The three cells $g \cup k$, g and $\{k\}$ of $\boldsymbol{d}_{g \cup k, k}$ are distinct, so $\|\boldsymbol{m}_r\|_2^2 = 3$ and $\|\boldsymbol{m}_r\|_1 = 3$. Set $\boldsymbol{\xi} = \boldsymbol{X} - \varepsilon \boldsymbol{q}(\tilde{\boldsymbol{n}}) = \sum_{r \in \mathcal{S}} (\chi_r \boldsymbol{m}_r - \mathbb{E}\chi_r \boldsymbol{m}_r)$. This is a sum of independent mean-zero vectors with $\mathbb{E}\|\boldsymbol{\xi}\|_2^2 \leq \sum_r \mathbb{E}\|\chi_r \boldsymbol{m}_r\|_2^2 = 3\varepsilon A$.

Coupling. Run (LE-M) with the same χ and order. Let $r \in \mathcal{S}_k$ be the j-th element of $\mathcal{F}$ to be processed, and let $m_j \leq j - 1$ be the number of merges performed before it. Call a cluster *touched* if it took part in

an earlier merge, either as the merging singleton or as the target. An untouched cluster is alive and has its original type. A merging singleton dies. A target never dies: it then has at least two records, so it is not a singleton and never merges. Let P_k° be the untouched members of P_k, and let $\mathcal{P}_j$ be the set of current clusters whose support excludes k. Then $P_k^\circ \subseteq \mathcal{P}_j$, $|P_k \setminus P_k^\circ| \le 2m_j$, and $|\mathcal{P}_j \setminus P_k^\circ| \le m_j$, because the touched clusters that are alive are earlier targets. If r is still a singleton, (LE-M) draws V_r uniformly from $\mathcal{P}_j$. (If $\mathcal{P}_j = \emptyset$, the step does nothing and we count it as bad. Then $P_k^\circ = \emptyset$, so $C_k \le 2m_j$ and the bound below is trivial.) We draw (B_r, V_r) from a maximal coupling of the uniform laws on P_k and on $\mathcal{P}_j$, given the past (Lindvall, 1992). For $r \in \mathcal{S} \setminus \mathcal{F}$, for $r \in \mathcal{F}$ that is no longer a singleton at its turn, and when $\mathcal{P}_j = \emptyset$, we draw B_r independently. Conditionally on the past, B_r is uniform on P_k. The B_r are therefore independent and uniform, and independent of (χ, order), so the idealised process has the law described above. A maximal coupling satisfies $\mathbb{P}(B = V = c) = \min\{\mathbb{P}(B = c), \mathbb{P}(V = c)\}$, so

$$\mathbb{P}\big(B_r = V_r \in P_k^\circ \mid \text{past}\big) = \frac{|P_k^\circ|}{\max(C_k, |\mathcal{P}_j|)} \ge \frac{C_k - 2m_j}{C_k + m_j} \ge 1 - \frac{3m_j}{C_k}.$$

Call $r \in \mathcal{F}$ *good* if it is still a singleton at its turn and $B_r = V_r \in P_k^\circ$. Then the actual change in the table is $\boldsymbol{m}_r$, because V_r is untouched and has type $\text{type}(B_r)$. Otherwise r is *bad*. Either (a) it was the target of an earlier merge, so that it is no longer a singleton at its turn, or (b) it is still a singleton but the coupling fails. Let $\boldsymbol{a}_r$ be the actual change caused by $r \in \mathcal{F}$: zero if r is skipped, and otherwise $-\boldsymbol{d}_{\text{type}(V_r)\cup k,k}$ with the current type of V_r. Both $\|\boldsymbol{a}_r\|_1$ and $\|\boldsymbol{m}_r\|_1$ are at most 3. Put $\boldsymbol{\rho}_{\text{int}} = (\boldsymbol{n} - \tilde{\boldsymbol{n}}) - \boldsymbol{X} = \sum_r {}_{\text{bad}}(\boldsymbol{a}_r - \boldsymbol{m}_r)$. Then $\|\boldsymbol{\rho}_{\text{int}}\|_1 \le 6\,\#\{\text{bad}\}$, and $\boldsymbol{n} = \tilde{\boldsymbol{n}} + \varepsilon \boldsymbol{q}(\tilde{\boldsymbol{n}}) + \boldsymbol{\xi} + \boldsymbol{\rho}_{\text{int}}$.

Counting bad steps. For (b), $|\mathcal{F}| \sim \text{Binomial}(A, \varepsilon)$ gives

$$\mathbb{E}\#(\text{b}) \le \mathbb{E}\sum_{j=1}^{|\mathcal{F}|} \frac{3(j-1)}{C_{\min}} = \frac{3}{2C_{\min}}\mathbb{E}\big[|\mathcal{F}|(|\mathcal{F}|-1)\big] = \frac{3\varepsilon^2 A(A-1)}{2C_{\min}}.$$

For (a), map each such $r \in \mathcal{F}$ to the first earlier step r' whose target was r. This map is injective, because each step has one target. The step r' merged, so it is not of type (a). If r' is good, then $B_{r'} = V_{r'} = r \in \mathcal{F}$. Hence $\#(\text{a}) \le \#(\text{b}) + \#\{r' \in \mathcal{F} : B_{r'} \in \mathcal{F}\}$. Since the B are independent of χ, and $c \ne r'$ for $c \in P_k$,

$$\mathbb{E}\#\{r' \in \mathcal{F} : B_{r'} \in \mathcal{F}\} = \sum_k \varepsilon A_k \sum_{c \in P_k \cap \mathcal{S}} \frac{\varepsilon}{C_k} \le \frac{\varepsilon^2 A^2}{C_{\min}}.$$

Therefore $\mathbb{E}\#\{\text{bad}\} \le 2\mathbb{E}\#(\text{b}) + \varepsilon^2 A^2/C_{\min} \le 4\varepsilon^2 A^2/C_{\min}$, and $\mathbb{E}\|\boldsymbol{\rho}_{\text{int}}\|_1 \le 24\varepsilon^2 A^2/C_{\min}$. Finally $\mathbb{E}[\boldsymbol{n} \mid \tilde{\Lambda}] - \tilde{\boldsymbol{n}} - \varepsilon \boldsymbol{q}(\tilde{\boldsymbol{n}}) = \mathbb{E}\boldsymbol{\rho}_{\text{int}}$. □

B.4 Proof of Lemma C.3

Step 1 (a Lipschitz bound for the drift). For $c > 0$ let $\mathcal{K}_c = \{\boldsymbol{x} \ge \boldsymbol{0} : C_k(\boldsymbol{x}) \ge c\|\boldsymbol{x}\|_1 \ \forall k\}$. It is a convex cone, being the non-negative orthant intersected with half-spaces. On $\mathcal{K}_c$,

$$\frac{\partial \boldsymbol{q}}{\partial x_f} = -\sum_k \Big[\mathbf{1}\{f = \{k\}\} \sum_{g \not\ni k} \frac{x_g}{C_k} \boldsymbol{d}_{g\cup k,k} + x_{\{k\}} \sum_{g \not\ni k} \Big(\frac{\mathbf{1}\{f = g\}}{C_k} - \frac{x_g \mathbf{1}\{k \notin f\}}{C_k^2}\Big) \boldsymbol{d}_{g\cup k,k}\Big].$$

Its ℓ_1 norm is at most $3 + 6\sum_k x_{\{k\}}/C_k \le 3 + 6/c$. Hence $\boldsymbol{q}$ is Lipschitz on $\mathcal{K}_c$, with constant $L_c = 3 + 6/c$ in ℓ_1.

Step 2 (location of the tables). Every true incident whose support excludes k leaves at least one cluster whose support excludes k, and distinct incidents leave distinct clusters. So $C_k(\tilde{\boldsymbol{n}}) \ge C_k(\boldsymbol{t})$, and also

$C_k(\mathcal{D}_\delta \boldsymbol{t}) \geq C_k(\boldsymbol{t})$ on taking expectations. The number of clusters never exceeds the number of records, so $\|\tilde{\boldsymbol{n}}\|_1 \leq K\|\boldsymbol{t}\|_1$, and likewise $\|\mathcal{D}_\delta \boldsymbol{t}\|_1 \leq K\|\boldsymbol{t}\|_1$. Under Assumption A5, therefore, $\tilde{\boldsymbol{n}}$ and $\mathcal{D}_\delta \boldsymbol{t}$ lie in $\mathcal{K}_c$ for every large N, with $c = \frac{1}{2}\min_k C_k(\boldsymbol{\tau})/(K\|\boldsymbol{\tau}\|_1) > 0$. Moreover $A \leq K\|\boldsymbol{t}\|_1$ and $C_{\min}(\tilde{\boldsymbol{n}}) \geq \min_k C_k(\boldsymbol{t})$.

Step 3 (the observed table). By Proposition 5.3(v), $\tilde{\boldsymbol{n}} = \mathcal{D}_\delta \boldsymbol{t} + \boldsymbol{\rho}_1$ with $\mathbb{E}\|\boldsymbol{\rho}_1\|_2^2 \leq K\|\boldsymbol{t}\|_1$. By Lemma C.2 and Step 1,

$$\boldsymbol{n} = \mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t}) + \boldsymbol{\rho}_1 + \varepsilon\{\boldsymbol{q}(\tilde{\boldsymbol{n}}) - \boldsymbol{q}(\mathcal{D}_\delta \boldsymbol{t})\} + \boldsymbol{\xi} + \boldsymbol{\rho}_{\text{int}},$$

where $\|\boldsymbol{q}(\tilde{\boldsymbol{n}}) - \boldsymbol{q}(\mathcal{D}_\delta \boldsymbol{t})\|_1 \leq L_c\|\boldsymbol{\rho}_1\|_1$, $\mathbb{E}\|\boldsymbol{\xi}\|_2^2 \leq 3K\|\boldsymbol{t}\|_1$, and $\mathbb{E}\|\boldsymbol{\rho}_{\text{int}}\|_1 \leq 24\varepsilon^2 K^2\|\boldsymbol{t}\|_1^2/\min_k C_k(\boldsymbol{t})$. All constants are free of δ and ε. Chebyshev's inequality, applied in the $(2^K - 1)$-dimensional space, and Markov's inequality give

$$\left\|\boldsymbol{n}/N - \mathcal{T}_{\delta,\varepsilon}(\boldsymbol{\tau}_N)\right\|_1 = O_p^{\text{u}}\big(\varepsilon^2 + N^{-1/2}\big), \tag{B.4}$$

using the homogeneity $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t})/N = \mathcal{T}_{\delta,\varepsilon}(\boldsymbol{\tau}_N)$.

Step 4 (expansion in the rates). On $[0,1)^2 \times \{\boldsymbol{x} : C_k(\boldsymbol{x}) > 0\ \forall k\}$ the map $(\delta, \varepsilon, \boldsymbol{x}) \mapsto \mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x})$ is infinitely differentiable, because $\mathcal{D}_\delta$ is polynomial in δ and $\boldsymbol{q}$ is rational with non-vanishing denominators. It equals $\boldsymbol{x}$ at $\delta = \varepsilon = 0$. Hence, for a compact neighbourhood $\mathcal{V}$ of $\boldsymbol{\tau}$ and r_0 small, $\mathcal{T}_{\delta,\varepsilon}(\mathcal{V})$ lies in the region where F is C^2 whenever $(\delta, \varepsilon) \in [0, r_0]^2$. Taylor's theorem in (δ, ε) then gives, uniformly in $\boldsymbol{x} \in \mathcal{V}$,

$$F(\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x})) = F(\boldsymbol{x}) + \delta\, \nabla F(\boldsymbol{x})^\top \partial_\delta \mathcal{D}_\delta \boldsymbol{x}\big|_{\delta=0} + \varepsilon\, \nabla F(\boldsymbol{x})^\top \boldsymbol{q}(\boldsymbol{x}) + O\big((\delta + \varepsilon)^2\big).$$

Differentiating (5.2) at $\delta = 0$ gives $\partial_\delta(\mathcal{D}_\delta)_{gh}|_0 = -|h|\mathbf{1}\{g = h\} + \mathbf{1}\{g \subseteq h, |h \setminus g| = 1\} + \mathbf{1}\{g = \{k\}, k \in h\}$. For $|h| = 1$ the column vanishes. For $|h| \geq 2$ it equals $\sum_{k\in h}(\boldsymbol{e}_{h\setminus k} + \boldsymbol{e}_{\{k\}} - \boldsymbol{e}_h) = \sum_{k\in h} \boldsymbol{d}_{h,k}$. So $\partial_\delta \mathcal{D}_\delta \boldsymbol{x}|_0 = \sum_{|h|\geq 2} x_h \sum_{k\in h} \boldsymbol{d}_{h,k}$.

Step 5 (assembly). F is Lipschitz on a neighbourhood $\mathcal{W}$ of $\boldsymbol{\tau}$. By (B.4), for every $\nu > 0$ there are r_0 and N_0 such that $\boldsymbol{n}/N \in \mathcal{W}$ and $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{\tau}_N) \in \mathcal{W}$ with probability at least $1 - \nu$ whenever $N \geq N_0$ and $(\delta, \varepsilon) \in [0, r_0]^2$. On this event, $|F(\boldsymbol{n}/N) - F(\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{\tau}_N))|$ is bounded by a constant times the left side of (B.4). Combining with Step 4 at $\boldsymbol{x} = \boldsymbol{\tau}_N \in \mathcal{V}$, and using homogeneity of degree one of F, $\mathcal{T}$ and $\boldsymbol{q}$ and of degree zero of ∇F,

$$\frac{F(\boldsymbol{n}) - F(\boldsymbol{t})}{N} = F(\boldsymbol{n}/N) - F(\boldsymbol{\tau}_N) = \frac{\delta}{N}\nabla F(\boldsymbol{t})^\top \sum_{|h|\geq 2} t_h \sum_{k\in h} \boldsymbol{d}_{h,k} + \frac{\varepsilon}{N}\nabla F(\boldsymbol{t})^\top \boldsymbol{q}(\boldsymbol{t}) + O_p^{\text{u}}\big((\delta + \varepsilon)^2 + N^{-1/2}\big).$$

Inserting the definition (5.5) of $\boldsymbol{q}$ gives (C.2). □

B.5 Proof of Theorem 5.5

(i) By (5.1), $n_{k+} = t_{k+}$. For $K = 2$, $\widehat{N}_{\text{LP}}(\boldsymbol{x}) = x_{1+}x_{2+}/x_{11}$, which gives the first identity. By Theorem 4.4(c), $\widehat{n}_{\boldsymbol{0}}(\boldsymbol{x}) = x_{10}x_{01}/x_{11}$ with $x_{10} = x_{1+} - x_{11}$ and $x_{01} = x_{2+} - x_{11}$. On $0 < y < \min(a, b)$ the function $f(y) = (a - y)(b - y)/y$ has

$$f'(y) = -\frac{(a - y) + (b - y)}{y} - \frac{(a - y)(b - y)}{y^2} < 0.$$

Under (LE-D), a cluster that contains records of both lists consists of records of one true incident with support $\{1, 2\}$, and each such incident yields at most one such cluster. So $\tilde{n}_{11} \leq t_{11}$. Under (LE-M) with $K = 2$, the clusters into which a list-1 singleton can merge are exactly the list-2 singletons, and vice versa. Every merge therefore turns one $\{1\}$-cluster and one $\{2\}$-cluster into a $\{1, 2\}$-cluster, and no merge destroys a $\{1, 2\}$-cluster. Hence $n_{11} \geq \tilde{n}_{11}$. If $\varepsilon = 0$, then $n_{11} \leq t_{11}$ and both estimators weakly increase. If $\delta = 0$, then $n_{11} \geq t_{11}$ and both decrease.

(ii) For $K = 2$, $C_1(\boldsymbol{x}) = x_{\{2\}}$ and $C_2(\boldsymbol{x}) = x_{\{1\}}$. By (5.5), $q(\boldsymbol{x})_{11} = x_{\{1\}}x_{\{2\}}/C_1 + x_{\{2\}}x_{\{1\}}/C_2 = x_{\{1\}} + x_{\{2\}}$. With $\tilde{\boldsymbol{x}} = \mathcal{D}_\delta \boldsymbol{t}$, we have $\tilde{x}_{11} = (1-\delta)^2 t_{11}$ and, by conservation of margins, $\tilde{x}_{\{k\}} = t_{k+} - (1-\delta)^2 t_{11}$. So

$$\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t})_{11} = (1-\delta)^2(1-2\varepsilon)\, t_{11} + \varepsilon(t_{1+} + t_{2+}).$$

By (B.4) and $\tau_{11} > 0$, $n_{11}/t_{11} = (1-\delta)^2(1-2\varepsilon) + \varepsilon(t_{1+} + t_{2+})/t_{11} + O_p^{\mathrm{u}}(\varepsilon^2 + N^{-1/2})$. Moreover $(t_{1+} + t_{2+})/t_{11} = \hat{p}_2^{-1} + \hat{p}_1^{-1}$. The bracket in (5.6) is bounded away from zero for small rates, so inverting and expanding to first order gives (5.6). For $\varepsilon = 0$, $n_{11} = \tilde{n}_{11} \sim \mathrm{Binomial}(t_{11}, (1-\delta)^2)$ by Proposition 5.3(iv). For two lists $j \neq k$ of a K-list system the same argument applies to $\widehat{N}_{\mathrm{LP}}^{(jk)}(\boldsymbol{x}) = x_{j+}x_{k+}/x_{jk}$.

(iii) $F = \widehat{n}_{\mathbf{0}}(\omega;\cdot) = e^{(-1)^K \omega}\mathcal{A}$ is positively homogeneous of degree one (Section 3.1) and smooth on $(0,\infty)^{\mathcal{H}_+}$. Apply Lemma C.3 and divide by $F(\boldsymbol{t})/N = F(\boldsymbol{\tau}_N)$, which is bounded away from zero. Since $\nabla F/F = \nabla \log \mathcal{A}$, the factor $e^{(-1)^K\omega}$ cancels, and the relative error is $\delta\Sigma_\delta - \varepsilon\Sigma_\varepsilon + O_p^{\mathrm{u}}((\delta+\varepsilon)^2 + N^{-1/2})$, where by (C.3)

$$\Sigma_\delta = \sum_{|h|\geq 2} t_h \sum_{k\in h} \partial_{h,k} \log \mathcal{A}(\boldsymbol{t}) = \sum_k \sum_{g\not\ni k} t_{g\cup k}\Big[\frac{1}{t_{\{k\}}} + (-1)^{|g|+1}\Big(\frac{1}{t_{g\cup k}} + \frac{1}{t_g}\Big)\Big],$$
$$\Sigma_\varepsilon = \sum_k \frac{t_{\{k\}}}{C_k(\boldsymbol{t})} \sum_{g\not\ni k} t_g\Big[\frac{1}{t_{\{k\}}} + (-1)^{|g|+1}\Big(\frac{1}{t_{g\cup k}} + \frac{1}{t_g}\Big)\Big].$$

Here we wrote $h = g \cup k$ with $g = h \setminus k \neq \emptyset$, so that $(-1)^{|h|} = (-1)^{|g|+1}$. By (B.1), $\sum_{g\not\ni k}(-1)^{|g|+1} = 1$. Also $\sum_{g\not\ni k} t_{g\cup k} = t_{k+} - t_{\{k\}}$ and $\sum_{g\not\ni k} t_g = C_k(\boldsymbol{t})$. Hence $\Sigma_\delta = \sum_k\{b_k + 1 + a_k\}$ and $\Sigma_\varepsilon = \sum_k\{1 + (t_{\{k\}}/C_k)(1+\bar{a}_k)\}$, which is (5.8). The coefficients are ratios of cells, so they are scale-free.

(iv) $b_k \geq 0$ because $t_{k+} \geq t_{\{k\}}$. If $\boldsymbol{t}$ satisfies (H), then (5.8) gives $\beta_\delta \geq K$ and $\beta_\varepsilon \leq -K$. The functions a_k and $\bar{a}_k$ are continuous on $(0,\infty)^{\mathcal{H}_+}$ and scale-free, so $a_k(\boldsymbol{t}) = a_k(\boldsymbol{\tau}_N) \to a_k(\boldsymbol{\tau})$, and likewise for $\bar{a}_k$. If $\boldsymbol{\tau}$ satisfies (H) strictly, then so does $\boldsymbol{t}$ for all large N. For $K = 2$, $a_1(\boldsymbol{x}) = x_{11}/x_{\{2\}} > 0$ and $\bar{a}_1(\boldsymbol{x}) = x_{\{2\}}/x_{11} > 0$ for every positive $\boldsymbol{x}$, and symmetrically for $k = 2$. The factorised case is Lemma B.1 below, applied to $\boldsymbol{x} = \boldsymbol{\tau}$. □

Lemma B.1 (A sufficient condition for (H)). *Let $\boldsymbol{x}$ be a positive table and fix k. Suppose there are $c_k > 0$ and $\phi_{jk} \in [\frac{1}{2}, 2]$ $(j \neq k)$ such that $x_{g\cup k}/x_g = c_k \prod_{j\in g}\phi_{jk}$ for every non-empty $g \not\ni k$. Then*

$$a_k(\boldsymbol{x}) = c_k\Big\{1 - \prod_{j\neq k}(1-\phi_{jk})\Big\} \geq 0, \qquad \bar{a}_k(\boldsymbol{x}) = c_k^{-1}\Big\{1 - \prod_{j\neq k}\big(1-\phi_{jk}^{-1}\big)\Big\} \geq 0.$$

If every $\phi_{jk} \in (\frac{1}{2}, 2)$, both inequalities are strict, so $\boldsymbol{x}$ satisfies (H) strictly.

Proof. Both identities are (B.1) with $x_j = \phi_{jk}$ and with $x_j = \phi_{jk}^{-1}$. For $\phi \in [\frac{1}{2}, 2]$, both $|1-\phi| \leq 1$ and $|1-\phi^{-1}| \leq 1$, so the products lie in $[-1, 1]$. For $\phi \in (\frac{1}{2}, 2)$ the inequalities are strict, and the products lie in $(-1, 1)$. □

Under independent capture, $\tau_{g\cup k}/\tau_g = p_k/(1-p_k)$ for all g, so $\phi_{jk} \equiv 1$ and $\boldsymbol{\tau}$ satisfies (H) strictly. For $K = 3$ and $k = 3$, the three equations $t_{\{1,3\}}/t_{\{1\}} = c_3\phi_{13}$, $t_{\{2,3\}}/t_{\{2\}} = c_3\phi_{23}$ and $t_{\{1,2,3\}}/t_{\{1,2\}} = c_3\phi_{13}\phi_{23}$ have the unique positive solution

$$c_3 = \frac{t_{\{1,3\}}t_{\{2,3\}}t_{\{1,2\}}}{t_{\{1\}}t_{\{2\}}t_{\{1,2,3\}}}, \qquad \phi_{13} = \frac{t_{\{1,2,3\}}\, t_{\{2\}}}{t_{\{1,2\}}\, t_{\{2,3\}}}, \qquad \phi_{23} = \frac{t_{\{1,2,3\}}\, t_{\{1\}}}{t_{\{1,2\}}\, t_{\{1,3\}}},$$

and similarly for $k = 1, 2$. Each ϕ_{jk} is the odds ratio between lists j and k among the incidents recorded by the third list. This proves the claim of Remark C.4.

B.6 Proof of Corollary 5.6

All quantities below are evaluated at $\boldsymbol{\tau}$, and $\beta(\boldsymbol{t}) = \beta(\boldsymbol{\tau}_N) \to \beta(\boldsymbol{\tau})$ by continuity. We drop $\boldsymbol{\tau}$ from the notation and write t for τ. Let $\theta_k = p_k/(1-p_k)$. Then $t_{g\cup k}/t_g = \theta_k$ for every $g \not\ni k$, so $c_k = \theta_k$ and $\phi_{jk} \equiv 1$ in Lemma B.1, giving $a_k = \theta_k$ and $\bar{a}_k = \theta_k^{-1}$. Next, $t_{k+}/t_{\{k\}} = \sum_{g' \subseteq [K]\setminus k} \prod_{j\in g'} \theta_j = \prod_{j\neq k}(1+\theta_j) = 1/q_{-k}$, so $b_k = q_{-k}^{-1} - 1$ and $\beta_\delta = \sum_k \theta_k + \sum_k q_{-k}^{-1}$. Up to the same proportionality constant, $t_{\{k\}} = p_k q_{-k}$ and $C_k = (1-p_k)(1-q_{-k})$, the latter being the probability of being missed by list k and recorded by another list. Hence $t_{\{k\}}/C_k = \theta_k q_{-k}/(1-q_{-k})$, and $(t_{\{k\}}/C_k)(1+\bar{a}_k) = (1+\theta_k)q_{-k}/(1-q_{-k}) = q_{-k}/\{(1-p_k)(1-q_{-k})\}$, because $1+\theta_k = (1-p_k)^{-1}$. As $\max_k p_k \to 0$, $\theta_k \to 0$ and $q_{-k} \to 1$, so $\beta_\delta \to K$. Also $1 - q_{-k} = \{1+o(1)\}\sum_{j\neq k} p_j$, which gives the stated rate for β_ε. For the sum, write $q = q_{-k}$. The k-th terms of β_δ and β_ε add up to

$$\theta_k + \frac{1}{q} - 1 - \frac{(1+\theta_k)q}{1-q} = \theta_k \frac{1-2q}{1-q} + \frac{(1-q)^2 - q^2}{q(1-q)} = \frac{(1-2q)(\theta_k + q^{-1})}{1-q},$$

which has the sign of $1 - 2q_{-k}$. With equal p, every $q_{-k} = (1-p)^{K-1}$, so $\beta_\delta + \beta_\varepsilon = 0$ exactly when $(1-p)^{K-1} = \frac{1}{2}$. □

B.7 Proof of Corollary 5.8

(i) With $\varepsilon = 0$, $\boldsymbol{n} = \tilde{\boldsymbol{n}}$ and $\mathbb{E}[\hat{\boldsymbol{t}} \mid \Lambda^\star] = \mathcal{D}_\delta^{-1}\mathcal{D}_\delta \boldsymbol{t} = \boldsymbol{t}$. Moreover $\hat{\boldsymbol{t}} - \boldsymbol{t} = \sum_b \mathcal{D}_\delta^{-1}(\boldsymbol{Z}_{h_b} - \mathbb{E}\boldsymbol{Z}_{h_b})$ is a sum of independent, bounded, mean-zero vectors. There are $t_h = N\tau_{N,h}$ incidents of type h, so the Lindeberg–Feller theorem gives $N^{-1/2}(\hat{\boldsymbol{t}} - \boldsymbol{t}) \Rightarrow \mathrm{N}(\boldsymbol{0}, \Sigma)$ with $\Sigma = \sum_h \tau_h \mathcal{D}_\delta^{-1}\operatorname{Cov}(\boldsymbol{Z}_h)\mathcal{D}_\delta^{-\top}$. By homogeneity, $F(\hat{\boldsymbol{t}}) - F(\boldsymbol{t}) = N\{F(\hat{\boldsymbol{t}}/N) - F(\boldsymbol{\tau}_N)\} = \nabla F(\boldsymbol{\tau}_N)^\top(\hat{\boldsymbol{t}} - \boldsymbol{t}) + N\,O_p(\|\hat{\boldsymbol{t}}/N - \boldsymbol{\tau}_N\|^2)$, and the remainder is $O_p(1)$. The delta method gives the conditional limit with $v_{\mathrm{link}} = \nabla F(\boldsymbol{\tau})^\top \Sigma \nabla F(\boldsymbol{\tau})$, which is the stated sum.

Now let the histories be independent and identically distributed with cell probabilities $\boldsymbol{\pi}$. Write $U_N = N^{-1/2}\{F(\boldsymbol{t}) - NF(\boldsymbol{\pi}_+)\}$ and $W_N = N^{-1/2}\{F(\hat{\boldsymbol{t}}) - F(\boldsymbol{t})\}$. Then $U_N \Rightarrow \mathrm{N}(0, v_{\mathrm{cap}})$ by the delta method. Conditionally on $\boldsymbol{t}$, $W_N \Rightarrow \mathrm{N}(0, v_{\mathrm{link}})$ along almost every sequence, because $\boldsymbol{\tau}_N \to \boldsymbol{\pi}_+$ almost surely. Hence $\mathbb{E}[e^{\mathrm{i}(aU_N + bW_N)}] = \mathbb{E}[e^{\mathrm{i}aU_N}\mathbb{E}(e^{\mathrm{i}bW_N} \mid \boldsymbol{t})] \to e^{-a^2 v_{\mathrm{cap}}/2} e^{-b^2 v_{\mathrm{link}}/2}$ by dominated convergence, and $U_N + W_N \Rightarrow \mathrm{N}(0, v_{\mathrm{cap}} + v_{\mathrm{link}})$.

For $K = 2$ only incidents with $h = \{1,2\}$ contribute to v_{link}. With $\varrho = (1-\delta)^2$, $\boldsymbol{Z}_{\{1,2\}}$ equals $\boldsymbol{e}_{11}$ with probability ϱ and $\boldsymbol{e}_{10} + \boldsymbol{e}_{01}$ otherwise. So $\operatorname{Cov}(\boldsymbol{Z}_{\{1,2\}}) = \varrho(1-\varrho)\boldsymbol{v}\boldsymbol{v}^\top$ with $\boldsymbol{v} = \boldsymbol{e}_{11} - \boldsymbol{e}_{10} - \boldsymbol{e}_{01}$. The $K = 2$ case of (5.3)–(5.4) gives $\mathcal{D}_\delta^{-1}\boldsymbol{v} = \boldsymbol{v}/\varrho$, because $\delta(2-\delta) = 1 - \varrho$. For $F = \log \widehat{N}_{\mathrm{LP}}$, $\nabla F(\boldsymbol{\tau})^\top \boldsymbol{v} = -1/\tau_{11}$. Hence $v_{\mathrm{link}} = \tau_{11}(1-\varrho)/(\varrho\,\tau_{11}^2) = (\varrho^{-1} - 1)/\tau_{11}$. Under independent capture $\tau_{11} = p_1 p_2$. The perfect-linkage variance of $\log \widehat{N}_{\mathrm{LP}}$ is $v_{\mathrm{cap}} = (1-p_1)(1-p_2)/(p_1 p_2)$, the variance of the influence function $Y_1/p_1 + Y_2/p_2 - Y_1 Y_2/(p_1 p_2)$. The ratio is (5.9), and it is the same for $\widehat{N}_{\mathrm{LP}}$ as for its logarithm. Finally, $\hat{t}_{11} = n_{11}/\varrho$ and the margins are unchanged, so $\widehat{N}_{\mathrm{LP}}(\hat{\boldsymbol{t}}) = \varrho\, n_{1+} n_{2+}/n_{11}$.

(ii) Write $\Psi = \mathcal{T}_{\delta,\varepsilon}$. Its Jacobian is $\nabla\Psi(\boldsymbol{x}) = \{I + \varepsilon \nabla \boldsymbol{q}(\mathcal{D}_\delta \boldsymbol{x})\}\mathcal{D}_\delta$. This is continuous in $(\delta, \varepsilon, \boldsymbol{x})$ and equals I at $\delta = \varepsilon = 0$. Choose $r_0, r > 0$ such that $\|\nabla\Psi(\boldsymbol{x}) - I\|_{1\to 1} \le \frac{1}{2}$ for $(\delta,\varepsilon) \in [0, r_0]^2$ and $\|\boldsymbol{x} - \boldsymbol{x}_0\|_1 \le 2r$, for every $\boldsymbol{x}_0$ in a neighbourhood $\mathcal{V}$ of $\boldsymbol{\tau}$. Fix such $\boldsymbol{x}_0$ and $\boldsymbol{y}$ with $\|\boldsymbol{y} - \Psi(\boldsymbol{x}_0)\|_1 \le r$. The map $\boldsymbol{x} \mapsto \boldsymbol{x} - \Psi(\boldsymbol{x}) + \boldsymbol{y}$ is a $\frac{1}{2}$-contraction of the ball $\{\|\boldsymbol{x} - \boldsymbol{x}_0\|_1 \le 2r\}$ into itself, since

$$\|\boldsymbol{x} - \Psi(\boldsymbol{x}) + \boldsymbol{y} - \boldsymbol{x}_0\|_1 \le \|(\boldsymbol{x} - \boldsymbol{x}_0) - (\Psi(\boldsymbol{x}) - \Psi(\boldsymbol{x}_0))\|_1 + \|\boldsymbol{y} - \Psi(\boldsymbol{x}_0)\|_1 \le r + r.$$

By Banach's theorem it has a unique fixed point $\hat{\boldsymbol{x}}$, which solves $\Psi(\hat{\boldsymbol{x}}) = \boldsymbol{y}$, and $\|\hat{\boldsymbol{x}} - \boldsymbol{x}_0\|_1 \le 2\|\boldsymbol{y} - \Psi(\boldsymbol{x}_0)\|_1$. Take $\boldsymbol{x}_0 = \boldsymbol{\tau}_N$ and $\boldsymbol{y} = \boldsymbol{n}/N$. By (B.4), $\|\boldsymbol{y} - \Psi(\boldsymbol{\tau}_N)\|_1 \le r$ with probability tending to one, for r_0 small. Hence $\hat{\boldsymbol{t}} = N\hat{\boldsymbol{x}}$ is the unique solution in the ball $\|\boldsymbol{x} - \boldsymbol{t}\|_1 \le 2rN$, and $\|\hat{\boldsymbol{t}}/N - \boldsymbol{\tau}_N\|_1 = O_p^{\mathrm{u}}(\varepsilon^2 + N^{-1/2})$. Lipschitz continuity of F near $\boldsymbol{\tau}$ and homogeneity complete the proof. The contraction is the unprojected

iteration $\boldsymbol{x} \leftarrow \boldsymbol{x} + (\boldsymbol{n} - \Psi(\boldsymbol{x}))$ with step length one. This is the simulation code's iteration while its projection $\max\{\mathbf{0}, \cdot\}$ is inactive, so for small rates that iteration converges geometrically.

(iii) By the implicit function theorem, $\theta \mapsto \hat{\boldsymbol{t}}(\theta)/N$ is continuously differentiable near θ_0, with derivative $-\{\nabla_{\boldsymbol{x}} \mathcal{T}_\theta\}^{-1} \partial_\theta \mathcal{T}_\theta$ evaluated at $\hat{\boldsymbol{t}}(\theta_0)/N$. That point converges to $\boldsymbol{\tau}$ up to $O(\varepsilon^2)$ by (ii). Hence $N^{-1/2}\{F(\hat{\boldsymbol{t}}(\hat{\theta})) - F(\hat{\boldsymbol{t}}(\theta_0))\} = \sqrt{N/m}\, \boldsymbol{c}_N^\top \sqrt{m}(\hat{\theta} - \theta_0) + o_p(1)$ with $\boldsymbol{c}_N \to \boldsymbol{c}$. Since $\hat{\theta}$ is independent of $\boldsymbol{n}$ given $\Lambda^\star$, the two limits are independent, and Slutsky's lemma gives the claim.

The claim in Remark C.9. Let $K = 2$ and $\varepsilon = 0$, and let $\mathsf{R} = n_{11}/t_{11}$ be the realised retention, so that $t_{11}\mathsf{R} \sim \text{Binomial}(t_{11}, \varrho)$. The audit draws $f t_{11}$ of the t_{11} true pairs by simple random sampling and estimates ϱ by the sample proportion $\hat{\mathsf{R}}$ of retained pairs. The margins are unchanged, so $\hat{t}_{11} = n_{11}/\hat{\mathsf{R}}$ and $\log \hat{t}_{11} - \log t_{11} = \log \mathsf{R} - \log \hat{\mathsf{R}}$. Given R, $\mathbb{E}\hat{\mathsf{R}} = \mathsf{R}$ and $\text{Var}(\hat{\mathsf{R}} \mid \mathsf{R}) = \mathsf{R}(1-\mathsf{R})(1-f)t_{11}/\{f t_{11}(t_{11}-1)\}$. Since $\mathsf{R} \to \varrho$ almost surely, the delta method gives $\text{Var}(\log \hat{t}_{11} - \log t_{11}) = \{1+o(1)\}(1-\varrho)(f^{-1}-1)/(\varrho t_{11})$. For $f = 1$, $\hat{\mathsf{R}} = \mathsf{R}$ and $\hat{t}_{11} = t_{11}$. An independent audit of $f t_{11}$ pairs instead has $\text{Var}(\log \hat{\mathsf{R}}) = \{1+o(1)\}(1-\varrho)/(\varrho f t_{11})$, which adds to the linkage variance $(1-\varrho)/(\varrho t_{11})$ of $\log \hat{t}_{11}$ in (i). □

B.8 Proof of Proposition 5.11

(i) is the law of total variance under $p(N, \Lambda \mid D)$. For (ii), $\widehat{\Lambda}$ is fixed given D, so $G(\widehat{\Lambda})$ is a constant, and $\mathbb{E}[(N - G(\widehat{\Lambda}))^2 \mid D] = \text{Var}(N \mid D) + \{\mathbb{E}[N \mid D] - G(\widehat{\Lambda})\}^2$. If $V(\Lambda) \equiv V_w$ on the support, the reported $V(\widehat{\Lambda}) = V_w$ falls short of this by $V_b + \{\mathbb{E}[N \mid D] - G(\widehat{\Lambda})\}^2 \geq V_b$. For (iii), let $N \mid D \sim \text{N}(\bar{N}, s^2)$ with $s^2 = V_w + V_b$, let $b = G(\widehat{\Lambda}) - \bar{N}$, and let $w = z_{\alpha/2} V_w^{1/2}$. The coverage is $\Phi\{(w-b)/s\} + \Phi\{(w+b)/s\} - 1 =: \kappa(b)$. Its derivative is $\kappa'(b) = s^{-1}[\varphi\{(w+b)/s\} - \varphi\{(w-b)/s\}]$, which is ≤ 0 for $b \geq 0$ and ≥ 0 for $b \leq 0$, because $|w+b| \geq |w-b|$ exactly when $b \geq 0$. So κ is maximised at $b = 0$, where it equals $2\Phi(w/s) - 1 = 2\Phi\{z_{\alpha/2}(1 + V_b/V_w)^{-1/2}\} - 1$. For (iv), $N - \bar{Q} = (N - \mathbb{E}[N \mid D]) - (\bar{Q} - \mathbb{E}[N \mid D])$, and the two terms are independent given D, so $\mathbb{E}[(N-\bar{Q})^2 \mid D] = \text{Var}(N \mid D) + \text{Var}(\bar{Q} \mid D) = V_w + V_b + V_b/M$. Finally $\mathbb{E}[\bar{U} \mid D] = V_w$, and $\mathbb{E}[B_M \mid D] = V_b$ because B_M is the unbiased sample variance of M independent draws. □

B.9 Proofs of Theorem C.10 and Propositions C.11 and C.12

(i) For independent $O_1, \ldots, O_L$ with finite ranges, the Hoeffding decomposition (Efron and Stein, 1981; Hoeffding, 1948) is

$$G_T(O_T) = \sum_{T' \subseteq T} (-1)^{|T| - |T'|} \mathbb{E}[G \mid O_{T'}], \qquad T \neq \emptyset.$$

It has $\mathbb{E}[G_T \mid O_{T'}] = 0$ unless $T \subseteq T'$, the terms are mutually uncorrelated, and $\mathbb{E}[G \mid O_S] - \mathbb{E}G = \sum_{\emptyset \neq T \subseteq S} G_T$. Hence $\text{Var}(\mathbb{E}[G \mid O_S]) = \sum_{\emptyset \neq T \subseteq S} \sigma_T^2$, and

$$V(S) = \text{Var}\, G - \text{Var}(\mathbb{E}[G \mid O_S]) = \sum_{T \not\subseteq S} \sigma_T^2.$$

For $T = \{u\}$, $G_{\{u\}} = \mathbb{E}[G \mid O_u] - \mathbb{E}G$. For a binary I_ℓ, $\mathbb{E}[G \mid I_\ell] = \mathbb{E}[G \mid I_\ell = 0] + \Delta_\ell I_\ell$, so $\sigma_\ell^2 = \Delta_\ell^2 \pi_\ell (1 - \pi_\ell)$ and $\text{Cov}(G, I_\ell) = \text{Cov}(\mathbb{E}[G \mid I_\ell], I_\ell) = \Delta_\ell \pi_\ell (1 - \pi_\ell)$. This gives (C.4). The identity $\text{Var}(\mathbb{E}[G \mid O_u]) = \text{Cov}(G, I_\ell)^2 / \{\pi_\ell(1-\pi_\ell)\}$ uses no independence.

(ii) $\sum_{u \notin S} \sigma_u^2 = \sum_u \sigma_u^2 - \sum_{u \in S} \sigma_u^2$. The B largest terms maximise $\sum_{u \in S} \sigma_u^2$ over $|S| \leq B$; exchanging any chosen term for a larger unchosen one can only increase the sum. Let $S^\star$ minimise V over $|S| \leq B$, and write $V(S) = V_1(S) + V_{\text{int}}(S)$, where V_1 is the main-effect part and $0 \leq V_{\text{int}}(S) \leq I_{\text{int}}$. Then $V(S_{\text{g}}) \leq V_1(S^\star) + I_{\text{int}} \leq V(S^\star) + I_{\text{int}}$.

Proof of Proposition C.11. Let $\xi_\ell = I_\ell - \pi_\ell$, $\boldsymbol{X} = \sum_\ell \xi_\ell \boldsymbol{v}_\ell$ and $\boldsymbol{X}_{-\ell} = \boldsymbol{X} - \xi_\ell \boldsymbol{v}_\ell$. Then $\boldsymbol{n}(I) = \bar{\boldsymbol{n}} + \boldsymbol{X}$, and $\bar{\boldsymbol{n}} = \mathbb{E}\boldsymbol{n}(I)$ lies in the convex hull of the attainable tables. By the mean-value theorem,

$$\Delta_\ell = \mathbb{E}\int_0^1 \nabla F\big(\bar{\boldsymbol{n}} + \boldsymbol{X}_{-\ell} + (s - \pi_\ell)\boldsymbol{v}_\ell\big)^\top \boldsymbol{v}_\ell \, ds,$$

and $|\nabla F(\boldsymbol{y})^\top \boldsymbol{v}_\ell - g_\ell| \le H\|\boldsymbol{y} - \bar{\boldsymbol{n}}\|_2 \, \bar{v}$. Since $\mathbb{E}\|\boldsymbol{X}_{-\ell}\|_2 \le (\mathbb{E}\|\boldsymbol{X}\|_2^2)^{1/2} \le \bar{v} S_1^{1/2}$, we get $|\Delta_\ell - g_\ell| \le H\bar{v}(\bar{v} S_1^{1/2} + \bar{v})$. For the interactions, write $G = F(\bar{\boldsymbol{n}}) + \sum_\ell g_\ell \xi_\ell + R$, with $|R| \le \frac{H}{2}\|\boldsymbol{X}\|_2^2$ by Taylor's theorem. The linear part has Hoeffding components of order one only. So the components of order at least two of G are those of R, and $I_{\text{int}} \le \operatorname{Var} R \le \mathbb{E}R^2 \le \frac{H^2}{4}\mathbb{E}\|\boldsymbol{X}\|_2^4$. Expanding $\|\boldsymbol{X}\|_2^4 = \sum_{a,b,c,d} \xi_a\xi_b\xi_c\xi_d \langle \boldsymbol{v}_a, \boldsymbol{v}_b\rangle\langle \boldsymbol{v}_c, \boldsymbol{v}_d\rangle$, only paired or fully coincident indices have non-zero mean. Using $\mathbb{E}\xi_a^4 \le \mathbb{E}\xi_a^2 = \pi_a(1-\pi_a)$ and $\langle \boldsymbol{v}_a, \boldsymbol{v}_b\rangle^2 \le \bar{v}^4$ gives $\mathbb{E}\|\boldsymbol{X}\|_2^4 \le \bar{v}^4(S_1 + 3S_1^2)$. If F is homogeneous of degree one, then $\nabla^2 F(\boldsymbol{x}) = N^{-1}\nabla^2 F(\boldsymbol{x}/N)$. If the convex hull of the attainable tables lies in $\{\boldsymbol{x} : x_h \ge cN \; \forall h, \; \|\boldsymbol{x}\|_1 \le C'N\}$, then $\boldsymbol{x}/N$ ranges over a compact subset of $(0,\infty)^{\mathcal{H}_+}$ on which $\nabla^2 F$ is bounded, so $H = O(N^{-1})$. For the closed form, $\nabla^2 F$ is unbounded as a cell tends to zero, which is why the proviso is needed.

Proof of Proposition C.12. Index the units so that $\sigma_1^2/c_1 \ge \sigma_2^2/c_2 \ge \cdots$, and assume $\sum_u c_u > B$, since otherwise all units are reviewed. Let j be the first index with $\sum_{u \le j} c_u > B$, and let $x_u^{\mathrm{g}} = 1$ for $u < j$, $x_j^{\mathrm{g}} = (B - \sum_{u<j} c_u)/c_j \in [0,1)$, and $x_u^{\mathrm{g}} = 0$ for $u > j$. For any $x \in [0,1]^L$ with $\sum_u c_u x_u \le B$,

$$\sum_u \sigma_u^2 (x_u^{\mathrm{g}} - x_u) = \sum_u \frac{\sigma_u^2}{c_u} c_u (x_u^{\mathrm{g}} - x_u) \;\ge\; \frac{\sigma_j^2}{c_j} \sum_u c_u (x_u^{\mathrm{g}} - x_u) \;\ge\; 0.$$

The first inequality holds because $c_u(x_u^{\mathrm{g}} - x_u) \ge 0$ for $u < j$ and ≤ 0 for $u > j$. The second holds because $\sum_u c_u x_u^{\mathrm{g}} = B$. So x^{g} solves the linear relaxation (Dantzig, 1957), and $W^\star \le \sum_u \sigma_u^2 x_u^{\mathrm{g}} \le \sum_{u \in S_{\mathrm{g}}} \sigma_u^2 + \sigma_j^2$. If $c_j \le B$, then $\{j\}$ is feasible and $\max\{\sum_{S_{\mathrm{g}}} \sigma_u^2, \sigma_j^2\} \ge W^\star/2$ (Kellerer et al., 2004). □

B.10 Proof of Proposition C.13

(i) Under simple random sampling without replacement within strata, $\mathbb{E}[N_h \bar{\boldsymbol{y}}_h] = \sum_{\ell \in \mathcal{P}_h} \boldsymbol{y}_\ell$, and the covariance of $N_h \bar{\boldsymbol{y}}_h$ is $N_h^2(1 - n_h/N_h)\boldsymbol{S}_{y,h}/n_h$ (Cochran, 1977, Ch. 5). In an asymptotic framework in which N_h and n_h tend to infinity and the stratum means converge, $\hat{\boldsymbol{Y}}/\sum_h N_h - \boldsymbol{Y}/\sum_h N_h \to 0$ in probability. Since ψ is continuous, and in (5.10) depends on $\boldsymbol{Y}$ only through ratios, $\hat{\theta} \to \theta$ in probability.

(ii) A first-order Taylor expansion of $\boldsymbol{a}^\top \psi$ at $\boldsymbol{Y}$ gives $\boldsymbol{a}^\top(\hat{\theta} - \theta) = \sum_h N_h(\bar{z}_h - \bar{Z}_h) + o_p(\cdot)$, where $\bar{Z}_h$ is the population mean of z in stratum h. Its design variance is $V_{\text{aud}} = \sum_h N_h^2 S_{z,h}^2/n_h - \sum_h N_h S_{z,h}^2$. The second term does not depend on the allocation. By the Cauchy–Schwarz inequality,

$$\Big(\sum_h \frac{N_h^2 S_{z,h}^2}{n_h}\Big)\Big(\sum_h c_h n_h\Big) \ge \Big(\sum_h N_h S_{z,h}\sqrt{c_h}\Big)^2,$$

with equality if and only if $N_h S_{z,h}/\sqrt{n_h} \propto \sqrt{c_h n_h}$, that is, $n_h \propto N_h S_{z,h}/\sqrt{c_h}$ (Neyman, 1934). With $\sum_h c_h n_h = C$, the minimum is as stated. With the bounds $2 \le n_h \le N_h$, we minimise the convex function $\sum_h a_h/n_h$, with $a_h = N_h^2 S_{z,h}^2$, over a convex set, so the Karush–Kuhn–Tucker conditions are sufficient. They read $a_h/n_h^2 = \lambda c_h - \underline{\nu}_h + \bar{\nu}_h$, where $\underline{\nu}_h, \bar{\nu}_h \ge 0$ vanish unless $n_h = 2$ or $n_h = N_h$ respectively. Hence $n_h = (a_h/\lambda c_h)^{1/2}$ when this lies within the bounds, and n_h equals the bound it would cross otherwise. This is the stated clipped allocation. The total cost is decreasing in λ, so λ is fixed by the budget.

(iii) If all pairs in stratum h share one decision status, then each coordinate of $\boldsymbol{y}_\ell$ is an affine function of I_ℓ with stratum-specific coefficients. For example, in a stratum of linked edges $\boldsymbol{y}_\ell = (I_\ell, 0, 1 - I_\ell)$.

Hence $z_\ell = \alpha_h + \lambda_h I_\ell$ and $S_{z,h} = |\lambda_h| S_{I,h}$. For a binary variable with stratum proportion P_h, $S^2_{I,h} = N_h P_h(1-P_h)/(N_h-1)$. Calibration within strata gives $\mathbb{E}[P_h \mid D] = \bar{\pi}_h$, and replacing P_h by $\bar{\pi}_h$ gives the working allocation. Since $x(1-x)$ is concave, $\mathbb{E}[P_h(1-P_h) \mid D] \le \bar{\pi}_h(1-\bar{\pi}_h)$, so the proxy does not understate $S^2_{I,h}$ on average. □

Appendix C Supplementary results for Section 5

This appendix collects results, remarks and implementation details that Section 5 states only in summary. We use the notation of that section. Proofs are in Appendix B.

C.1 Attributes and comparison levels

The attributes of Section 5.1 are noisy and heterogeneous across lists in four main ways.

Dates. A police record usually carries both the date of the incident and the date it was reported. An NGO or community-security record may carry only the date on which the organisation learned of the incident or published it, and a campus portal records the time of submission. Agreement on dates is therefore agreement up to a reporting lag whose distribution depends on the pair of lists. The comparison levels are bands of the absolute date difference, such as $\{0\}$, $\{1,2\}$, $\{3,\dots,7\}$ and $\{\ge 8\}$ days, and their match probabilities must be estimated separately for each pair of lists.

Location. Granularity differs: a street address or block for police, a city or neighbourhood for an NGO audit, a building for a campus portal, a county or state when details are withheld to protect victims. We compare at the finest resolution that both records share, with levels such as same block or city, same county, and unknown. Agreement at a coarse resolution carries less evidence than agreement at a fine one, and the comparison model must learn this from the data rather than assume it.

Offence vocabularies. Police lists use statutory offence categories such as intimidation, simple assault, or destruction, damage or vandalism of property (Federal Bureau of Investigation, n.d.). NGO audits use their own taxonomies, typically harassment, vandalism and assault. We encode the correspondence as a compatibility relation $C_{kk'} \subseteq \mathcal{V}_k \times \mathcal{V}_{k'}$ between the vocabularies of lists k and k'. It is a relation, not a bijection. An NGO's "harassment" may be police "intimidation" or no crime at all, and one police incident can carry several offence codes. The comparison distinguishes an identical harmonised category, compatible categories and incompatible categories.

Target and venue. Target type (person, institution, property) and venue (synagogue, school, cemetery, residence, campus building) are recorded partly as free text. They are compared with string comparators (Winkler, 1990), and the similarity is binned.

C.2 Fellegi–Sunter optimality, the EM algorithm and Bayesian linkage

A *decision rule* is a triple of functions $\phi = (\phi_{\mathrm{L}}, \phi_{\mathrm{C}}, \phi_{\mathrm{N}})$ from Γ to $[0,1]$ with $\phi_{\mathrm{L}} + \phi_{\mathrm{C}} + \phi_{\mathrm{N}} = 1$. These are the probabilities of linking, of review and of not linking. Its error probabilities are the false-link probability $P_{\mathrm{FL}}(\phi) = \sum_\gamma u(\gamma)\phi_{\mathrm{L}}(\gamma)$ and the missed-link probability $P_{\mathrm{ML}}(\phi) = \sum_\gamma m(\gamma)\phi_{\mathrm{N}}(\gamma)$.

> **Proposition C.1** (Optimality of the Fellegi–Sunter rule; Fellegi and Sunter, 1969). *Fix thresholds $T_{\mathrm{N}} < T_{\mathrm{L}}$ and constants $a, b \in [0,1]$. Let $\phi^\star$ link with probability one if $w > T_{\mathrm{L}}$ and with probability a if $w = T_{\mathrm{L}}$, not link with probability one if $w < T_{\mathrm{N}}$ and with probability b if $w = T_{\mathrm{N}}$, and send every other comparison*

Algorithm 2 Fellegi–Sunter EM under conditional independence

Require: Comparison vectors γ_ℓ for candidate pairs $\ell \in \mathcal{P}$; field levels $\{0, \dots, L_f\}$, $f = 1, \dots, F$; tolerance tol

Ensure: Estimates of $p = \mathbb{P}(\mathrm{M})$, m_f, u_f; pairwise match probabilities π_ℓ; weights w_ℓ

Initialise $p \in (0, \frac{1}{2})$, and m_f, u_f with m_f placing more mass than u_f on the agreement levels
repeat
  for $\ell \in \mathcal{P}$ **do** ▷ E-step; O_ℓ = fields observed for pair ℓ
    $\pi_\ell \leftarrow p \prod_{f \in O_\ell} m_f(\gamma_\ell^f) \,/\, \big[p \prod_{f \in O_\ell} m_f(\gamma_\ell^f) + (1-p) \prod_{f \in O_\ell} u_f(\gamma_\ell^f)\big]$
  $p \leftarrow |\mathcal{P}|^{-1} \sum_\ell \pi_\ell$ ▷ M-step
  for $f = 1, \dots, F$ and each level j **do**
    $m_f(j) \leftarrow \sum_{\ell : f \in O_\ell} \pi_\ell \mathbf{1}\{\gamma_\ell^f = j\} \,/\, \sum_{\ell : f \in O_\ell} \pi_\ell$
    $u_f(j) \leftarrow \sum_{\ell : f \in O_\ell} (1-\pi_\ell) \mathbf{1}\{\gamma_\ell^f = j\} \,/\, \sum_{\ell : f \in O_\ell} (1-\pi_\ell)$
until the observed-data log-likelihood increases by less than tol
If necessary swap the labels so that m is the component with more agreement
return p, m_f, u_f, π_ℓ, and $w_\ell = \sum_{f \in O_\ell} \log\{m_f(\gamma_\ell^f)/u_f(\gamma_\ell^f)\}$

vector to review. Let $P_{\mathrm{FL}}^\star = P_{\mathrm{FL}}(\phi^\star)$ and $P_{\mathrm{ML}}^\star = P_{\mathrm{ML}}(\phi^\star)$. Then every rule ϕ with $P_{\mathrm{FL}}(\phi) \le P_{\mathrm{FL}}^\star$ and $P_{\mathrm{ML}}(\phi) \le P_{\mathrm{ML}}^\star$ satisfies

$$\sum_\gamma m(\gamma)\phi_{\mathrm{C}}(\gamma) \;\ge\; \sum_\gamma m(\gamma)\phi_{\mathrm{C}}^\star(\gamma) \qquad \textit{and} \qquad \sum_\gamma u(\gamma)\phi_{\mathrm{C}}(\gamma) \;\ge\; \sum_\gamma u(\gamma)\phi_{\mathrm{C}}^\star(\gamma).$$

Hence $\phi^\star$ minimises the probability of clerical review for every proportion of matches among the candidate pairs.

The proof applies the Neyman–Pearson lemma twice (Appendix B.1). Under conditional independence of the fields within each mixture component, $m(\gamma) = \prod_f m_f(\gamma^f)$ and $u(\gamma) = \prod_f u_f(\gamma^f)$, and the EM algorithm takes the closed form of Algorithm 2 (Jaro, 1989; Winkler, 1988). Missing field values are omitted from both products, which is valid when values are missing at random given match status (Enamorado et al., 2019). Enamorado et al. (2019) implement this model at scale. The match probabilities π_ℓ that Algorithm 2 returns are pairwise: unlike the posterior match probabilities of Section 5.2, they ignore the one-to-one constraint.

Bayesian record linkage. The Bayesian approach treats Λ as an unknown parameter, with likelihood

$$p(\gamma \mid \Lambda, m, u) = \prod_{\ell \in \mathrm{M}(\Lambda)} m(\gamma_\ell) \prod_{\ell \in \mathcal{P} \setminus \mathrm{M}(\Lambda)} u(\gamma_\ell), \tag{C.1}$$

where $\mathcal{P}$ is the set of candidate pairs and $\mathrm{M}(\Lambda)$ the set of those that Λ places in the same block. Sadinle (2017) samples bipartite matchings of two lists that respect the one-to-one constraint, and leaves the least certain records for clerical review through loss functions with an abstention option. Tancredi and Liseo (2011) model linkage and population size jointly for two lists. Steorts et al. (2016) represent Λ through latent entities, which handles deduplication and any number of lists (see also Marchant et al., 2021; Tancredi et al., 2020). Sadinle and Fienberg (2013) generalise the Fellegi–Sunter model to several files, and Binette and Steorts (2022) review the field.

C.3 Merging and the split–merge calculus

On the up-set $\{h : |h| \geq 2\}$ the detachment operator of Proposition 5.3 factorises as $\mathcal{D}_\delta = Z_\rho \operatorname{diag}\{(1-\delta)^{|h|}\}$, where $\rho = \delta/(1-\delta)$ and $(Z_\rho)_{gh} = \rho^{|h|-|g|}\mathbf{1}\{g \subseteq h\}$ is a weighted zeta function of the Boolean lattice. The inverse of Z_ρ is the weighted Möbius function $(-\rho)^{|h|-|g|}\mathbf{1}\{g \subseteq h\}$, which gives (5.3). Part (iv) of the proposition is exact because two records of one incident stay together if and only if neither detaches.

Lemma C.2 (Merging to first order). *Let $\tilde{\Lambda}$ be any linkage structure with table $\tilde{\boldsymbol{n}}$. Let $A = \sum_k \tilde{n}_{\{k\}}$ be its number of singletons and suppose $C_{\min} = \min_k C_k(\tilde{\boldsymbol{n}}) > 0$. Under (LE-M),*

$$\boldsymbol{n} = \tilde{\boldsymbol{n}} + \varepsilon\, \boldsymbol{q}(\tilde{\boldsymbol{n}}) + \boldsymbol{\xi} + \boldsymbol{\rho}_{\mathrm{int}},$$

where $\mathbb{E}[\boldsymbol{\xi} \mid \tilde{\Lambda}] = \mathbf{0}$, $\mathbb{E}[\|\boldsymbol{\xi}\|_2^2 \mid \tilde{\Lambda}] \leq 3\varepsilon A$ and $\mathbb{E}[\|\boldsymbol{\rho}_{\mathrm{int}}\|_1 \mid \tilde{\Lambda}] \leq 24\,\varepsilon^2 A^2/C_{\min}$. In particular, $\|\mathbb{E}[\boldsymbol{n} \mid \tilde{\Lambda}] - \tilde{\boldsymbol{n}} - \varepsilon \boldsymbol{q}(\tilde{\boldsymbol{n}})\|_1 \leq 24\,\varepsilon^2 A^2/C_{\min}$.

The proof (Appendix B.3) couples (LE-M) with an idealised process in which every selected singleton merges independently into a uniformly chosen cluster of the *initial* configuration. The two processes differ only when two merges interact, and the expected number of such interactions is $O(\varepsilon^2 A^2/C_{\min})$.

For a differentiable function F of a table, let $\partial_{h,k} F(\boldsymbol{x}) = \nabla F(\boldsymbol{x})^\top \boldsymbol{d}_{h,k}$ be its derivative along the split (h, k).

Lemma C.3 (Split–merge calculus). *Let F be positively homogeneous of degree one and twice continuously differentiable near $\boldsymbol{\tau}$. Under LE and Assumption A5,*

$$\frac{F(\boldsymbol{n}) - F(\boldsymbol{t})}{N} = \frac{\delta}{N} \sum_{|h|\geq 2} t_h \sum_{k \in h} \partial_{h,k} F(\boldsymbol{t}) - \frac{\varepsilon}{N} \sum_{k \in [K]} \sum_{g \not\ni k} \frac{t_{\{k\}}\, t_g}{C_k(\boldsymbol{t})}\, \partial_{g \cup k,k} F(\boldsymbol{t}) + O_p^{\mathrm{u}}\big((\delta+\varepsilon)^2 + N^{-1/2}\big). \quad \text{(C.2)}$$

Detachment pushes the table along the split directions, with intensity δ per record. Merging pushes it along the reversed directions, with intensity ε per singleton, aimed at each target cell in proportion to its count. To first order, the effect of linkage error on any smooth estimator is the sum, over these elementary moves, of the estimator's sensitivity to each move. For the estimators of Section 4 the sensitivities are explicit. The closed form (4.2) has $\nabla \log \mathcal{A}(\boldsymbol{x}) = (s_h/x_h)_h$, with $s_h = (-1)^{|h|+1}$ the alternating sign of (3.2). Hence

$$\partial_{h,k} \log \mathcal{A}(\boldsymbol{x}) = \frac{1}{x_{\{k\}}} + (-1)^{|h|}\Big(\frac{1}{x_h} + \frac{1}{x_{h\setminus k}}\Big), \quad \text{(C.3)}$$

where for $|h| = 2$ the cell $h \setminus k$ is a singleton. In Theorem C.10 and Algorithm 3, a link that merges a list-k singleton into a cluster of type g has sensitivity $g_\ell = -\widehat{n}_{\mathbf{0}}\, \partial_{g\cup k,k} \log \mathcal{A}$, plus -1 from the change in n when $F = \widehat{N}$.

C.4 Remarks on the linkage-bias theorem

Remark C.4 (Coefficients for two and three lists). For $K = 2$, (5.8) reduces to $\beta_\delta = 2(1 + t_{11}/t_{10} + t_{11}/t_{01})$ and $\beta_\varepsilon = -\{2 + t_{10}/t_{01} + t_{01}/t_{10} + (t_{10} + t_{01})/t_{11}\}$. For $K = 3$, every positive table has the factorised form required in Theorem 5.5(iv): it is the no-three-way-interaction parametrisation of the observed cells. The pairwise factor ϕ_{jk} is then the observable conditional odds ratio $t_{\{j,k,l\}} t_{\{l\}}/(t_{\{j,l\}} t_{\{k,l\}})$ between lists j and k among incidents recorded by the third list l (Appendix B.5). So for $K = 3$, the table $\boldsymbol{t}$ itself satisfies (H) whenever these three odds ratios lie in $[\frac{1}{2}, 2]$. For $K \geq 4$ the factorisation is a genuine restriction: it rules out interactions of order three or more among the observed cells, so a realised table essentially never factorises exactly, even under independent capture. This is why Theorem 5.5(iv) places the condition on

$\boldsymbol{\tau}$. In practice, (H) involves only observed cells and can be checked directly on the corrected table $\hat{\boldsymbol{t}}$ of Corollary 5.8.

Remark C.5 (Reading the theorem). Under independent capture (Corollary 5.6), the two error types are not symmetric, and which one matters more depends on how sparse the lists are. Detachment moves incidents from overlaps into singletons at a rate proportional to the overlaps. Merging moves singletons into overlaps at a rate proportional to the *singletons*. When lists capture small fractions of incidents, singletons outnumber overlaps by a factor of order $1/p$. Then $|\beta_\varepsilon|$ is of order $1/p$, while β_δ is close to K. Take a hypothetical two-list example with $p_1 = p_2 = 0.1$. Corollary 5.6 gives $\beta_\delta \approx 2.4$ and $\beta_\varepsilon = -22$. A false-match rate of 1% would then deflate the estimated dark figure by about a fifth, while a false-non-match rate of 1% would inflate it by about 2.4%. Dense lists reverse the ranking. Under independent capture with equal p, β_δ grows like $(1-p)^{-(K-1)}$, and missed links dominate once p exceeds $p^* = 1 - 2^{-1/(K-1)}$, which is about 0.21 for $K = 4$. With $K = 4$ and $p = 0.5$, for instance, $\beta_\delta = 36$ and $\beta_\varepsilon \approx -5.1$. The plug-in $F(\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t}))$ then gives +41% for $\delta = 1\%$ against −5% for $\varepsilon = 1\%$. These rankings are proved only under independent capture; with dependent lists even the signs can change (Remark 5.7). We therefore recommend computing β_δ and β_ε from the corrected table $\hat{\boldsymbol{t}}$ of Corollary 5.8, which estimates them consistently up to $O(\varepsilon^2)$ when the errors follow LE, and using them to choose linkage thresholds and audit priorities. Since the expansion (5.7) is linear, it is informative only when $\delta|\beta_\delta|$ and $\varepsilon|\beta_\varepsilon|$ are small. Beyond that range we recommend the plug-in $F(\mathcal{T}_{\delta,\varepsilon}(\hat{\boldsymbol{t}}))$, which is exact in δ.

Remark C.6 (Relation to the literature). For $K = 2$, LE shares the affine structure of the dual-system model with matching error of Ding and Fienberg (1994). In both, $\mathbb{E}[n_{11}] = \alpha t_{11} + \beta_1(t_{1+} - t_{11}) + \beta_2(t_{2+} - t_{11})$ for constants α, β_1 and β_2. For any mechanism of this form whose matched count concentrates, the argument of Theorem 5.5(ii) gives $\widehat{N}_{\mathrm{LP}}(\boldsymbol{n})/\widehat{N}_{\mathrm{LP}}(\boldsymbol{t}) \to \{\alpha - \beta_1 - \beta_2 + \beta_1/\hat{p}_2 + \beta_2/\hat{p}_1\}^{-1}$. To first order in ε, LE has $\alpha = (1-\delta)^2(1-2\varepsilon) + 2\varepsilon$ and $\beta_1 = \beta_2 = \varepsilon$, which gives (5.6). In the Ding–Fienberg model, a true match is recovered with probability α and a record of one list without a true match is falsely matched with probability β, so $\beta_1 = 0$ and $\beta_2 = \beta$. The two models are not nested, and $\mathbb{E}[n_{11}]$ alone does not identify $(\alpha, \beta_1, \beta_2)$. Under LE, for example, an unmatched list-k singleton is falsely linked with probability about $\varepsilon(1 + A_j/A_k)$, where A_k is the number of list-k singletons after detachment and $j \neq k$, because it can merge into another cluster or have another singleton merge into it. Inverting the affine map gives

$$\widehat{N} = \frac{t_{1+}t_{2+}(\alpha - \beta_1 - \beta_2)}{n_{11} - \beta_1 t_{1+} - \beta_2 t_{2+}},$$

which for $\beta_1 = 0$ is their corrected estimator; Di Consiglio and Tuoto (2015) use it with probabilistically linked data. Extensions to several lists that, like ours, treat linkage error as a transformation of capture-history counts are given by Ding and Fienberg (1996) and Di Consiglio and Tuoto (2018). Bias corrections of the same spirit for regression on linked files are due to Lahiri and Larsen (2005).

Remark C.7 (List-specific rates). If records of list k detach with probability δ_k and list-k singletons merge with probability ε_k, replace $\delta^{|h|-|g|}(1-\delta)^{|g|}$ in (5.2) by $\prod_{j\in h\setminus g}\delta_j \prod_{j\in g}(1-\delta_j)$, the singleton term $\delta\,\mathbf{1}\{g = \{k\}, k \in h\}$ by $\delta_k \mathbf{1}\{g = \{k\}, k \in h\}$, and the coefficient $(-\delta)^{|g|-|h|}(1-\delta)^{-|g|}$ in (5.3) by $\prod_{j\in g\setminus h}(-\delta_j)\prod_{j\in g}(1-\delta_j)^{-1}$. In (C.2), weight the split (h,k) by δ_k and the merges of list-k singletons by ε_k. The proofs are unchanged. List-specific rates are natural here, because police records are typically more complete than self-reports.

Remark C.8 (Deduplication errors). LE assumes perfect within-list deduplication. A missed duplicate in list k adds a spurious record that, if unlinked, is a singleton $\{k\}$. Missed duplicates at rate η_k thus add about

$\eta_k t_{k+}$ to the cell $\{k\}$. Singletons enter $\mathcal{A}$ with exponent +1, so this inflates the closed-form dark figure by a relative amount of about $\sum_k \eta_k t_{k+}/t_{\{k\}}$. Over-deduplication, which merges two distinct incidents within a list, acts in the opposite direction. Unlike LE errors, both change the list margins.

C.5 The corrected estimator: audits of the same linkage, and the solver

Remark C.9 (Auditing the same linkage). The audit of Section 5.4 samples pairs from the linkage that produced $\boldsymbol{n}$. Then $\hat{\theta}$ and $\boldsymbol{n}$ are dependent and Corollary 5.8(iii) does not apply. Such an audit estimates the *realised* error rates of that linkage, which removes part of v_{link}. Take $K = 2$, $\varepsilon = 0$ and an audit of a simple random fraction f of the t_{11} true pairs, and let $\varrho = (1-\delta)^2$. Then $\operatorname{Var}(\log \hat{t}_{11} - \log t_{11}) = \{1+o(1)\}(1-\varrho)(f^{-1}-1)/(\varrho\, t_{11})$, whereas adding the variances of (i) and (iii) gives $(1-\varrho)(f^{-1}+1)/(\varrho\, t_{11})$. A census audit ($f = 1$) recovers t_{11} exactly (Appendix B.7). In this case the sum overstates the variance by twice the linkage variance of (i), so it is conservative. We do not derive the joint variance in general. A parametric bootstrap that regenerates both the LE errors at $\hat{\theta}$ and the audit sample estimates it.

Corollary 5.8(ii) defines the estimator that the simulation code computes. That code solves $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}) = \boldsymbol{n}$ by the projected fixed-point iteration $\boldsymbol{x} \leftarrow \max\{\boldsymbol{0}, \boldsymbol{x} + (\boldsymbol{n} - \mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}))\}$, started at $\boldsymbol{x} = \boldsymbol{n}$, with step length one; the step is halved and the iteration restarted only if it diverges. For $\varepsilon = 0$ and without the projection, the iteration matrix $I - \mathcal{D}_\delta$ is triangular with spectral radius $1-(1-\delta)^K < 1$, so the iteration converges geometrically to (5.3)–(5.4). With the projection, it can converge to that solution only if $\mathcal{D}_\delta^{-1}\boldsymbol{n} \geq \boldsymbol{0}$. If some cell of $\mathcal{D}_\delta^{-1}\boldsymbol{n}$ is negative, which can happen when a cell is small and δ is large, the equation has no non-negative solution. The iteration then stops at a stationary point with some cells clamped at zero, and the code reports non-convergence.

C.6 Targeted review and the design of the audit

Targeted review. A *review unit* $u \in \mathcal{U} = \{1, \ldots, L\}$ is either a candidate pair, whose outcome is binary, or the set of competing candidate links of one record, whose outcome is categorical (which candidate, if any, is the match). Reviewing u reveals its outcome O_u at cost $c_u > 0$, and the total budget is B. The target is the between-linkage variance $V_b = \operatorname{Var}(G(\Lambda) \mid D)$ of Proposition 5.11, with $G(\Lambda) = \mathbb{E}[N \mid \Lambda, D]$. In a frequentist pipeline one takes $G = F(\boldsymbol{n}(\Lambda))$ instead. After units $S \subseteq \mathcal{U}$ have been reviewed, the expected remaining linkage variance is

$$V(S) = \mathbb{E}\big[\operatorname{Var}(G \mid O_S, D) \mid D\big].$$

We work under the following assumptions.

(R1) Under $p(\Lambda \mid D)$ the outcomes $(O_u)_{u\in\mathcal{U}}$ are mutually independent, and G depends on Λ only through $O = (O_u)_u$; the rest of the linkage is held fixed.

(R2) Review is error-free.

Under (R1), G has the Hoeffding (functional ANOVA) decomposition $G - \mathbb{E}G = \sum_{\emptyset\neq T\subseteq\mathcal{U}} G_T(O_T)$ into uncorrelated terms with variances σ_T^2. We write $\sigma_u^2 = \sigma_{\{u\}}^2$ for the main effects and $I_{\text{int}} = \sum_{|T|\geq 2}\sigma_T^2$ for the total interaction variance.

Theorem C.10 (Optimal clerical review). *Assume (R1)–(R2).*

(i) Exact decomposition. $V(S) = \sum_{u\notin S}\sigma_u^2 + \sum_{|T|\geq 2,\, T\not\subseteq S}\sigma_T^2$, *where* $\sigma_u^2 = \operatorname{Var}(\mathbb{E}[G \mid O_u])$. *For a pair unit with* $I_\ell = \mathbf{1}\{\ell \in \mathrm{M}(\Lambda)\} \sim \text{Bernoulli}(\pi_\ell)$,

$$\sigma_\ell^2 = \pi_\ell(1-\pi_\ell)\Delta_\ell^2 = \frac{\operatorname{Cov}(G, I_\ell \mid D)^2}{\pi_\ell(1-\pi_\ell)}, \qquad \Delta_\ell = \mathbb{E}[G \mid I_\ell = 1] - \mathbb{E}[G \mid I_\ell = 0]. \tag{C.4}$$

(ii) Unit costs. *Let S_g consist of the B units with the largest σ_u^2. Then S_g minimises the main-effect objective $\sum_{u\notin S}\sigma_u^2$ over $|S|\le B$, and*

$$V(S_g) \le \min_{|S|\le B} V(S) + I_{\text{int}}.$$

Parts (i)–(ii) need no linearisation, and the single-unit value σ_u^2 is exact even without (R1); for pairs it is a squared covariance that posterior draws estimate, as in (C.4). What (R1) buys is additivity across units, and this is where the approximation lies. The following results give the first-order form of the rule and its extension to general costs.

Proposition C.11 (Linear response). *Assume (R1)–(R2). Suppose every unit is a pair and $G(I) = F(\boldsymbol{n}_0 + \sum_\ell I_\ell \boldsymbol{v}_\ell)$, so that each link changes the table by a fixed vector $\boldsymbol{v}_\ell$, for example $-\boldsymbol{d}_{g\cup k,k}$; this holds when candidate pairs touch disjoint clusters. Suppose F is twice continuously differentiable with $\|\nabla^2 F\|_{\text{op}} \le H$ on the convex hull of the attainable tables, and $\|\boldsymbol{v}_\ell\|_2 \le \bar v$. Let $\bar{\boldsymbol{n}} = \boldsymbol{n}_0 + \sum_\ell \pi_\ell \boldsymbol{v}_\ell$, $g_\ell = \nabla F(\bar{\boldsymbol{n}})^\top \boldsymbol{v}_\ell$ and $S_1 = \sum_\ell \pi_\ell(1-\pi_\ell)$. Then*

$$|\Delta_\ell - g_\ell| \le H\bar v^2\big(1 + S_1^{1/2}\big), \qquad I_{\text{int}} \le \tfrac{1}{4}H^2\bar v^4\big(S_1 + 3S_1^2\big).$$

Ranking by $g_\ell^2\pi_\ell(1-\pi_\ell)$ is therefore exactly optimal for the linearised objective $\sum_{\ell\notin S} g_\ell^2\pi_\ell(1-\pi_\ell)$. For $F = \widehat{N}$, which is homogeneous of degree one, $H = O(N^{-1})$ provided the convex hull of the attainable tables lies in $\{\boldsymbol{x} : x_h \ge cN\ \forall h\}$ for some $c > 0$. If uncertain links can empty a cell, H is unbounded and the bound is void. Under the proviso and with $S_1 = O(N)$, the suboptimality I_{int} is $O(1)$, while the linkage variance is of order N.

Theorem C.10(ii) assumes unit costs. With general costs the greedy rule is near-optimal.

Proposition C.12 (Review under general costs). *Assume (R1)–(R2). Maximising $\sum_{u\in S}\sigma_u^2$ subject to $\sum_{u\in S} c_u \le B$ is a 0–1 knapsack problem with optimum $W^\star$. Order the units by σ_u^2/c_u, let S_g be the longest prefix that fits the budget, and let j be the first unit that does not fit. Then $\sum_{u\in S_g}\sigma_u^2 \ge W^\star - \sigma_j^2$. If $c_j \le B$, the better of S_g and $\{j\}$ achieves at least $W^\star/2$.*

Algorithm 3 implements targeted review adaptively. Each batch is chosen with up-to-date values of σ_u^2, which limits the damage from the dependence that (R1) ignores, and its knapsack safeguard is the rule of Proposition C.12.

Design of the random audit. The audit population is the set $\mathcal{P}$ of candidate pairs, partitioned into strata $\mathcal{P}_1,\dots,\mathcal{P}_H$ of sizes N_h that cross decision status (linked edge; same cluster but not an edge; different clusters) with bins of the posterior match probability π_ℓ. From stratum h a simple random sample of $n_h \ge 2$ pairs is drawn without replacement, at cost c_h per pair, and a reviewer determines the true status I_ℓ of each sampled pair. Let $\boldsymbol{y}_\ell \in \mathbb{R}^J$ be pair-level variables, functions of I_ℓ and of the decisions, whose population totals $\boldsymbol{Y} = \sum_{\ell\in\mathcal{P}} \boldsymbol{y}_\ell$ determine the error rates through a smooth map $\theta = (\delta,\varepsilon) = \psi(\boldsymbol{Y})$. Under LE the natural choice is $\boldsymbol{Y} = (\text{TP}, \text{FN}, \text{FE})$, with ψ given by (5.10).

The first equation of (5.10) inverts the pairwise retention probability $(1-\delta)^2$ of Proposition 5.3(iv). For the second, let M be the number of false merges and A the number of singletons before merging. Under LE, $\mathbb{E}M = \varepsilon A\{1 + O(\varepsilon)\}$. A merge consumes one singleton, or two when its target is itself a singleton, which for $K = 2$ is always the case. So $A = f_1 + M + M_s$, where f_1 is the observed number of singleton clusters and $M_s \le M$ counts merges into singletons. Since $M_s = O(\varepsilon A)$, dropping it changes ε by $O(\varepsilon^2)$, and (5.10) is correct to first order in ε.

Algorithm 3 Targeted clerical review

Require: Linkage posterior sampler; review units $\mathcal{U}$ with costs c_u; budget B; batch budget b; functional G (e.g. $\mathbb{E}[N \mid \Lambda, D]$ or $F(\boldsymbol{n}(\Lambda))$)

Ensure: Reviewed outcomes and an updated linkage posterior

while $B > 0$ **do**
  Draw $\Lambda^{(1)}, \dots, \Lambda^{(M)}$ from the current posterior and compute $G^{(m)} = G(\Lambda^{(m)})$
  for $u \in \mathcal{U}$ not yet reviewed **do**
    $\hat{\sigma}_u^2 \leftarrow \widehat{\mathrm{Var}}\big(\widehat{\mathbb{E}}[G \mid O_u]\big)$ from the draws ▷ pairs: $\widehat{\mathrm{Cov}}(G, I_\ell)^2 / \{\hat{\pi}_\ell(1-\hat{\pi}_\ell)\}$
    If $\hat{\pi}_\ell$ is too extreme to estimate the covariance, use $\hat{\sigma}_\ell^2 = g_\ell^2 \hat{\pi}_\ell(1-\hat{\pi}_\ell)$ with g_ℓ from (C.3)
  Sort units by $\hat{\sigma}_u^2 / c_u$ and take the longest prefix S with $\sum_{u \in S} c_u \le \min(b, B)$
  If the best single unit j that fits has $\hat{\sigma}_j^2 > \sum_{u \in S} \hat{\sigma}_u^2$, set $S \leftarrow \{j\}$ ▷ knapsack safeguard
  Review the units in S, record their outcomes, and set $B \leftarrow B - \sum_{u \in S} c_u$
  Condition the linkage model on the reviewed outcomes (fix the reviewed links or non-links)
return reviewed outcomes and the conditioned posterior

Let $\hat{\boldsymbol{Y}} = \sum_h N_h \bar{\boldsymbol{y}}_h$ be the stratified expansion estimator, $\hat{\theta} = \psi(\hat{\boldsymbol{Y}})$, and let $\boldsymbol{a} = \partial F(\hat{\boldsymbol{t}}(\theta))/\partial\theta$ be the sensitivity of the corrected estimator of Corollary 5.8 to the rates. Conditionally on the linkage, the audit sample is the only source of randomness in $\hat{\theta}$. The audit should minimise the resulting design variance of $F(\hat{\boldsymbol{t}}(\hat{\theta}))$. Its linearisation is the design variance of the total of the composite variable $z_\ell = \boldsymbol{a}^\top \nabla\psi(\boldsymbol{Y})\boldsymbol{y}_\ell$. Because $\hat{\theta}$ then estimates the realised error rates, this variance should not simply be added to v_{link} (Remark C.9).

Proposition C.13 (Neyman-allocated audit). *In the design above, let* $S_{z,h}^2 = (N_h - 1)^{-1} \sum_{\ell \in \mathcal{P}_h} (z_\ell - \bar{z}_h)^2$.

(i) $\hat{\boldsymbol{Y}}$ *is design-unbiased for* $\boldsymbol{Y}$*, and* $\hat{\theta}$ *is design-consistent as* $\min_h n_h \to \infty$.

(ii) The linearised audit variance $V_{\text{aud}} = \sum_h N_h^2 (1 - n_h/N_h) S_{z,h}^2 / n_h$ *is minimised, subject to* $\sum_h c_h n_h = C$ *and ignoring integrality, by the Neyman allocation* $n_h \propto N_h S_{z,h} / \sqrt{c_h}$*. The minimum is* $(\sum_h N_h S_{z,h} \sqrt{c_h})^2 / C - \sum_h N_h S_{z,h}^2$*. With the bounds* $2 \le n_h \le N_h$ *the problem stays convex, and its solution is* $n_h = \min\{N_h, \max(2, N_h S_{z,h} / \sqrt{\lambda c_h})\}$*, with* $\lambda > 0$ *chosen to exhaust the budget.*

(iii) If all pairs in stratum h *share one decision status, then* $z_\ell = \alpha_h + \lambda_h I_\ell$ *and* $S_{z,h} = |\lambda_h| S_{I,h}$*. Here* $S_{I,h}^2 = P_h(1 - P_h) N_h / (N_h - 1)$*, where* P_h *is the proportion of true matches in the stratum. If the posterior is calibrated within strata, then* $\mathbb{E}[P_h \mid D] = \bar{\pi}_h$*, the mean of* π_ℓ *over the stratum. This suggests the working allocation* $n_h \propto N_h |\lambda_h| \{\bar{\pi}_h(1 - \bar{\pi}_h) N_h / ((N_h - 1) c_h)\}^{1/2}$.

The optimal allocation (ii) is classical (Cochran, 1977; Neyman, 1934). The point of (iii) is the choice of target. Strata are weighted by their posterior uncertainty, as in targeted review, and also by the leverage $|\lambda_h|$ of their error type on the dark figure. Under independent capture (Corollary 5.6 and Remark C.5), this leverage is much larger for false links than for missed links when the lists are sparse, and smaller when they are dense; in general, the estimated coefficients $\beta(\hat{\boldsymbol{t}})$ show which case applies. The estimates in (i) are design-based, so they remain valid when the linkage model is miscalibrated (Belin and Rubin, 1995). Calibration affects only efficiency, through the proxy for $S_{z,h}$ in (iii). The simulation code uses (iii) with $|\lambda_h| \equiv 1$ and $c_h \equiv 1$, drops the factor $N_h/(N_h - 1)$, and floors the proxy at $S_{I,h} \ge 0.01$. It allocates at least two pairs per stratum, caps n_h at N_h, and re-allocates the remainder, as in (ii).

Appendix D Posterior computation

This appendix gives the samplers for the models of Section 6, the Monte Carlo design of Algorithm 1, the convergence diagnostics and prior-sensitivity analyses we require, and the proof deferred from Section 7.

D.1 Gibbs sampler for the Dirichlet-process latent-class model

We follow Manrique-Vallier (2016), allowing the κ-family of priors on N from Lemma 6.1 and writing every update in terms of cell counts. Label the n recorded incidents arbitrarily, with histories $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n \in \mathcal{H}_+$. Let m_j be the number of unrecorded incidents in class j, so that $n_{\mathbf{0}} = \sum_j m_j$. As a function of $N = n + n_{\mathbf{0}}$, the multinomial likelihood (3.4) is proportional to $N!/n_{\mathbf{0}}!$, and the prior contributes $\Gamma(N+\kappa)/N!$. Augmenting the classes of the $n_{\mathbf{0}}$ exchangeable unrecorded incidents multiplies by $n_{\mathbf{0}}!/\prod_j m_j!$. The augmented posterior of model (6.1) is therefore

$$
\begin{aligned}
p(\boldsymbol{z}, \boldsymbol{m}, \boldsymbol{V}, \boldsymbol{\lambda}, \alpha \mid \boldsymbol{n}_+) \;\propto\; & \frac{\Gamma(n + n_{\mathbf{0}} + \kappa)}{\prod_j m_j!} \prod_{i=1}^{n} \Big\{ \pi_{z_i} \prod_k \lambda_{z_i k}^{y_{ik}} (1 - \lambda_{z_i k})^{1 - y_{ik}} \Big\} \prod_{j=1}^{J^\star} \Big\{ \pi_j \prod_k (1 - \lambda_{jk}) \Big\}^{m_j} \\
& \times \prod_{j,k} \lambda_{jk}^{a_\lambda - 1} (1 - \lambda_{jk})^{b_\lambda - 1} \prod_{j < J^\star} \alpha (1 - V_j)^{\alpha - 1} \; \alpha^{a_\alpha - 1} e^{-b_\alpha \alpha},
\end{aligned}
\tag{D.1}
$$

with $\pi_j = V_j \prod_{l<j}(1 - V_l)$ and $V_{J^\star} = 1$. Recorded incidents with the same history are exchangeable given the parameters, so only the class counts $a_{\boldsymbol{h}j} = \#\{i : \boldsymbol{y}_i = \boldsymbol{h},\ z_i = j\}$ matter. One sweep consists of the following steps.

(G1) *Classes of recorded incidents.* For each $\boldsymbol{h} \in \mathcal{H}_+$, draw $(a_{\boldsymbol{h}1}, \dots, a_{\boldsymbol{h}J^\star}) \sim \mathrm{Multinomial}(n_{\boldsymbol{h}}; \varpi_{\boldsymbol{h}})$ with $\varpi_{\boldsymbol{h}j} \propto \pi_j \prod_k \lambda_{jk}^{h_k} (1 - \lambda_{jk})^{1 - h_k}$. This is equivalent to drawing each z_i from its full conditional.

(G2) *Unrecorded count and classes, jointly.* Draw $n_{\mathbf{0}} \sim \mathrm{NegBin}(n + \kappa, 1 - \pi_{\mathbf{0}})$, with $\pi_{\mathbf{0}} = \sum_j \pi_j \prod_k (1 - \lambda_{jk})$. Then draw $(m_1, \dots, m_{J^\star}) \sim \mathrm{Multinomial}(n_{\mathbf{0}}; \varpi_{\mathbf{0}})$ with $\varpi_{\mathbf{0}j} = \pi_j \prod_k (1 - \lambda_{jk}) / \pi_{\mathbf{0}}$.

(G3) *Class profiles.* Draw $\lambda_{jk} \sim \mathrm{Beta}(a_\lambda + s_{jk},\ b_\lambda + c_j - s_{jk})$, where $s_{jk} = \sum_{\boldsymbol{h}} a_{\boldsymbol{h}j} h_k$ and $c_j = \sum_{\boldsymbol{h}} a_{\boldsymbol{h}j} + m_j$ is the size of class j, counting recorded and unrecorded incidents.

(G4) *Stick-breaking weights.* For $j < J^\star$, draw $V_j \sim \mathrm{Beta}(1 + c_j,\ \alpha + \sum_{l>j} c_l)$, and set $\pi_j = V_j \prod_{l<j}(1 - V_l)$.

(G5) *Concentration.* Draw $\alpha \sim \mathrm{Gamma}(a_\alpha + J^\star - 1,\ b_\alpha - \log \pi_{J^\star})$ (shape, rate), using $\log \pi_{J^\star} = \sum_{j<J^\star} \log(1 - V_j)$.

With $a_\alpha = 0.25$, the concentration α can become very small, and V_j can then round to 1 in floating point. In that case $\log(1 - V_j) = -\infty$, and steps (G4) and (G5) fail. We therefore sample the sticks on the log scale. Write $V_j = G_1/(G_1 + G_2)$ with independent $G_1 \sim \mathrm{Gamma}(1 + c_j)$ and $G_2 \sim \mathrm{Gamma}(\alpha + \sum_{l>j} c_l)$. For a gamma variate with small shape a, compute $\log G = \log G' + a^{-1} \log U$ with $G' \sim \mathrm{Gamma}(a + 1)$ and $U \sim \mathrm{Uniform}(0, 1)$; this uses the identity $G' U^{1/a} \sim \mathrm{Gamma}(a)$ and does not underflow. Then $\log V_j$ and $\log(1 - V_j)$ follow by log-sum-exp, and $\log \pi_j$ and $\log \pi_{J^\star}$ are accumulated on the log scale. Each sweep records $N = n + n_{\mathbf{0}}$, the implied top interaction ω (from (3.8) applied to $\boldsymbol{\pi}(\psi)$; Remark 6.2), and the number of occupied classes.

Derivations. (G1) is immediate from (D.1). For (G2), the factors of (D.1) that involve $\boldsymbol{m}$ are $\Gamma(n + n_{\mathbf{0}} + \kappa) \prod_j \{\pi_j \prod_k (1 - \lambda_{jk})\}^{m_j} / m_j!$. Summing over all $\boldsymbol{m}$ with $\sum_j m_j = n_{\mathbf{0}}$, the multinomial theorem gives $\Gamma(n + n_{\mathbf{0}} + \kappa)\, \pi_{\mathbf{0}}^{n_{\mathbf{0}}} / n_{\mathbf{0}}!$, the negative-binomial kernel of Lemma 6.1(i), and $\boldsymbol{m}$ given $n_{\mathbf{0}}$ is then multinomial. Drawing $(n_{\mathbf{0}}, \boldsymbol{m})$ jointly in this way is a blocked Gibbs step. It is valid because the labels of

the recorded incidents do not enter that block. (G3) is beta–Bernoulli conjugacy, in which the unrecorded incidents of class j contribute m_j zeros for every list. For (G4), $\prod_j \pi_j^{c_j} = \prod_j V_j^{c_j}(1-V_j)^{\sum_{l>j} c_l}$. For (G5), the stick-breaking prior contributes $\alpha^{J^\star-1}\exp\{(\alpha-1)\sum_{j<J^\star}\log(1-V_j)\}$.

Implementation notes. After grouping by history, a sweep costs $O(2^K J^\star K)$ operations, independent of n and N. The quantities of interest, N, $\pi_{\mathbf{0}}$ and ω, are invariant to permutations of the class labels, so label switching needs no treatment. Under the untruncated prior, the expected mass beyond the truncation level is $\mathbb{E}[\sum_{j\geq J^\star}\pi_j \mid \alpha] = \{\alpha/(1+\alpha)\}^{J^\star-1}$, because $\mathbb{E}(1-V_j) = \alpha/(1+\alpha)$. Ishwaran and James (2001) bound the L_1 distance between the marginal laws of the data under the truncated and untruncated priors by approximately $4N\exp\{-(J^\star-1)/\alpha\}$. Here the sample size is the complete population N, including the unrecorded incidents. At the sizes of interest the bound is uninformative: for $N = 10^4$, $\alpha = 1$ and $J^\star = 10$ it is about 4.9, while an L_1 distance between probability laws cannot exceed 2. We therefore rely on a direct check. We use $J^\star \geq 10$, monitor the posterior occupancy of the last classes, and increase $J^\star$ if the last class is ever substantially occupied. Mixing is the main practical difficulty. A class whose $\lambda_{j\cdot}$ are all small can absorb many unrecorded incidents while barely changing the recorded cells. This couples $n_{\mathbf{0}}$ to that class's profile, and the posterior of N is right-skewed. We therefore run at least four chains, with $n_{\mathbf{0}}$ initialised at overdispersed values (for example, from the lower bound of Theorem 4.8 up to several times it), monitor $\log N$, and summarise by posterior quantiles rather than the mean alone.

Validation. We checked the sampler in three ways. First, we checked the $n_{\mathbf{0}}$ step against exact enumeration of $p(N \mid \psi, \boldsymbol{n}_+)$ for $\kappa \in \{0, \frac{1}{2}, 1\}$. We also checked it against rejection sampling from the generative latent-class model under a truncated $1/N$ prior, retaining the replicates whose number of recorded incidents equalled a fixed n. Second, with $J^\star = 1$ the model is the independence model M_t with uniform λ_k. Its exact posterior, $p(N \mid \boldsymbol{n}_+) \propto N^{-1}\{N!/(N-n)!\}\prod_k B(n_{k+}+1, N-n_{k+}+1)$, was reproduced. Third, with $J^\star = 3$ and random α, we compared the sampler with an independent random-walk Metropolis sampler that targets the collapsed posterior of Lemma 6.1(ii) and shares no code with the augmentation steps. All comparisons agreed within Monte Carlo error. Simulation-based calibration (Talts et al., 2018) and joint-distribution tests (Geweke, 2004) require a proper prior on N. Every member of the κ-family is improper, because $\Gamma(N+\kappa)/\Gamma(N+1)$ behaves like $N^{\kappa-1}$ and is not summable. The family is, however, the limit of a proper one. If $N \mid \mu \sim \text{Poisson}(\mu)$ and $p(\mu) \propto \mu^{\kappa-1}$, then $p(N) \propto \int_0^\infty \mu^{\kappa-1}e^{-\mu}\mu^N/N!\,d\mu = \Gamma(N+\kappa)/\Gamma(N+1)$. The improper prior $1/N$ is thus the scale-invariant prior on a Poisson mean, $\kappa = \frac{1}{2}$ corresponds to Jeffreys' prior for that mean, and replacing $\mu^{\kappa-1}$ by a proper $\text{Gamma}(a,b)$ density yields a proper (negative binomial) prior on N. In the proper version, (G2) becomes $n_{\mathbf{0}} \mid \mu, \psi \sim \text{Poisson}(\mu\pi_{\mathbf{0}})$, followed by $\mu \mid N \sim \text{Gamma}(a+N, b+1)$.

D.2 Covariate models

For model (6.4), write $\tilde{\boldsymbol{x}}_i = (1, \boldsymbol{x}_i^\top)^\top$ and $\boldsymbol{\beta}_k^\circ = (\beta_{0k}, \boldsymbol{\beta}_k^\top)^\top$. Then $\partial \log p_i/\partial\boldsymbol{\beta}_k^\circ = p_{ik}(1-p_i)\tilde{\boldsymbol{x}}_i/p_i$, so the score of the conditional likelihood (6.5) takes the "observed minus expected" form

$$\frac{\partial \log L_c}{\partial \boldsymbol{\beta}_k^\circ} = \sum_{i=1}^{n}\Big(y_{ik} - \frac{p_{ik}}{p_i}\Big)\tilde{\boldsymbol{x}}_i, \tag{D.2}$$

in which $p_{ik}/p_i = \mathbb{P}(Y_{ik} = 1 \mid \boldsymbol{h}_i \neq \mathbf{0})$ is the expectation under the zero-truncated model. The gradient in (6.6) is $\partial p_i^{-1}/\partial\boldsymbol{\beta}_k^\circ = -p_{ik}(1-p_i)\tilde{\boldsymbol{x}}_i/p_i^2$. We sample $\boldsymbol{\beta}$ from $p(\boldsymbol{\beta})L_c(\boldsymbol{\beta})$ by Hamiltonian Monte Carlo with weakly informative $N(0, 1.5^2)$ priors on the logit-scale coefficients. Each draw is followed by a draw of N from (6.7). In the random-effects model (6.8), we write $\boldsymbol{b}_g = \text{diag}(\boldsymbol{\sigma}_b)\boldsymbol{L}_b\boldsymbol{e}_g$ with $\boldsymbol{e}_g \sim N_K(\mathbf{0}, \boldsymbol{I})$. Here $\boldsymbol{L}_b$ is the Cholesky factor of a correlation matrix with an LKJ prior (Lewandowski et al., 2009), and

the scales σ_b have penalised-complexity priors (Simpson et al., 2017). This non-centred parameterisation (Papaspiliopoulos et al., 2007) avoids the funnel-shaped posteriors that arise when a group has few incidents.

D.3 Hamiltonian Monte Carlo for the small-area model

The small-area model (6.9)–(6.10) has no discrete parameters once the class labels are summed out, so we fit it with the no-U-turn sampler (Hoffman and Gelman, 2014) as implemented in Stan (Carpenter et al., 2017). The log-likelihood is the Poisson likelihood (3.5) of the recorded cells,

$$\ell = \sum_{s,t} \sum_{\boldsymbol{h} \in \mathcal{H}_+} \left\{ n_{st,\boldsymbol{h}} \log(P_{st}\theta_{st}\pi_{st,\boldsymbol{h}}) - P_{st}\theta_{st}\pi_{st,\boldsymbol{h}} \right\},$$

with $\log \pi_{st,\boldsymbol{h}}$ computed by log-sum-exp over classes. The unrecorded cells do not enter ℓ. They are generated after sampling, as $n_{st,\mathbf{0}} \sim \text{Poisson}(P_{st}\theta_{st}\pi_{st,\mathbf{0}})$, which yields draws of N_{st}, of national totals and of the funnel factors. The implementation details are standard but consequential:

- *Parameterisation.* Every hierarchical effect ($\boldsymbol{\upsilon}$, $\boldsymbol{\gamma}$, $\boldsymbol{\delta}$, $\boldsymbol{\eta}$, $\boldsymbol{\xi}$) is non-centred.
- *ICAR component.* The ICAR log-density is written in pairwise-difference form, $-\frac{1}{2}\sum_{s\sim s'}(u_s - u_{s'})^2$, with a soft sum-to-zero constraint (Morris et al., 2019). The adjacency graph of states is disconnected: states without land neighbours are isolated nodes. Each connected component of more than one state receives its own sum-to-zero constraint and scaling factor, and isolated states receive an independent $N(0,1)$ structured effect (Freni-Sterrantino et al., 2018). The scaling factor of each component is the geometric mean of the diagonal of the generalised inverse of its ICAR precision (Riebler et al., 2016; Sørbye and Rue, 2014).
- *Random walks.* The RW2 prior on τ_t has a two-dimensional null space (constant and linear trend). We impose $\sum_t \tau_t = 0$ and $\sum_t t\tau_t = 0$ and carry a linear trend in $\boldsymbol{x}_{st}$. The RW1 terms $\chi_{s\cdot}$ and $\xi_{k\cdot}$ have sum-to-zero constraints within each state and each list, respectively.
- *Priors.* All standard deviations ($\sigma_b, \sigma_\tau, \sigma_\chi$, the scales of $\boldsymbol{\Sigma}_\eta$, and the scale of the ξ walks) receive penalised-complexity priors (Simpson et al., 2017) of the form $\mathbb{P}(\sigma > U) = 0.01$, with U set on the log-rate or logit scale. The mixing parameter ϕ receives the PC prior of Riebler et al. (2016) with $\mathbb{P}(\phi < 0.5) = 2/3$. The correlation matrix of $\boldsymbol{\Sigma}_\eta$ receives an LKJ(2) prior, the class logits α_{jk} receive $N(0, 1.5^2)$, and the format effect ζ receives $N(0,1)$. The class weights receive a sparse $\text{Dir}(e_0, \dots, e_0)$ prior with small e_0 (Rousseau and Mengersen, 2011).
- *Label switching.* We order the classes by their mean baseline logit $\bar{\alpha}_{j\cdot}$, which prevents label switching within a chain. Across chains we compare only label-invariant quantities.
- *Tuning.* We use a target acceptance rate of 0.9–0.95, at least four chains, and require zero divergent transitions after warm-up and an energy Bayesian fraction of missing information above 0.3 in every chain (Betancourt, 2017).

D.4 Monte Carlo error of the pipeline

Let $\bar{N} = M^{-1}\sum_m L^{-1}\sum_\ell N^{(m,\ell)}$ be the pooled posterior mean from Algorithm 1. Suppose the linkage draws are effectively independent, which can be achieved by thinning, and that each inner run yields L effective draws. Let $B = \text{Var}_L\{\mathbb{E}[N \mid \boldsymbol{n}(\Lambda)]\}$ and $W = \mathbb{E}_L\{\text{Var}(N \mid \boldsymbol{n}(\Lambda))\}$ be the between- and

within-linkage components of Proposition 5.11. The inner mean of draw m has variance $B + W/L$ around $\mathbb{E}_{\text{cut}}[N]$, so

$$\operatorname{Var}(\bar{N}) \approx \frac{1}{M}\Big(B + \frac{W}{L}\Big). \tag{D.3}$$

Suppose a linkage draw, together with the burn-in of its MSE chain, costs as much as c effective MSE draws, and fix the total budget $C = M(L + c)$. Minimising $(B + W/L)(L + c)/C$ over L gives $B - cW/L^2 = 0$, that is,

$$L^\star = \sqrt{cW/B}. \tag{D.4}$$

When linkage uncertainty dominates (B large relative to W), it pays to draw many linkage structures and run short MSE chains on each, and conversely. Pilot runs give estimates of B, W and c. Note that the sample variance s_b^2 of the M per-linkage means estimates $B + W/L$, not B. The Monte Carlo variance (D.3) is estimated directly by s_b^2/M. The between-linkage component itself, which is reported with Proposition 5.11 and enters (D.4), is estimated by $\widehat{B} = s_b^2 - \widehat{W}/L$, with $\widehat{W}$ the average within-draw posterior variance (truncated at zero).

D.5 Convergence diagnostics

For every sampler we run at least four chains. We require a rank-normalised split-$\widehat{R}$ below 1.01, and bulk and tail effective sample sizes of at least 400 (Vehtari et al., 2021), for the following quantities: N_d, $\log N_d$, $\pi_{\mathbf{0}}$, the implied ω, α, the number of occupied classes, national totals, the dark-figure ratios $D_{1,d}$ and the funnel shares of Theorem 7.3. For linkage MCMC we monitor the number of links and every cell $n_{\boldsymbol{h}}(\Lambda)$ of the induced capture table across chains, since these are the quantities passed to the MSE step. For the pooled pipeline we report the Monte Carlo standard error $s_b/\sqrt{M}$ from Appendix D.4. Posterior predictive checks compare the observed list sizes, frequency counts f_j and two-list overlaps with their replicates. These checks can detect misfit in the recorded cells. They cannot detect misfit in the unrecorded cell (Theorem 4.3).

D.6 Prior sensitivity

We vary one component at a time and report the posterior median and 90% interval of the national total, of D_1 and of the funnel shares.

- *Prior on N:* $\kappa \in \{0, \frac{1}{2}, 1\}$. By Lemma 6.1, κ adds κ pseudo-successes to the negative binomial and tilts the posterior of ψ by $(1 - \pi_{\mathbf{0}})^{-\kappa}$. The first effect changes $\mathbb{E}[N \mid \psi]$ by only $\kappa\pi_{\mathbf{0}}/(1 - \pi_{\mathbf{0}})$. The second can matter when $\pi_{\mathbf{0}}$ is weakly identified.
- *Latent classes:* $(a_\alpha, b_\alpha) \in \{(0.25, 0.25), (1, 1), (1, 0.1)\}$, where the last pair favours more classes; $J^\star \in \{10, 20, 40\}$; and $\lambda_{jk} \sim \text{Beta}(1, 1)$ or $\text{Beta}(\frac{1}{2}, \frac{1}{2})$. The second places more mass near zero capture probability, and so on low-visibility classes. This is the component to which the right tail of N is most sensitive.
- *Small-area model:* each PC-prior threshold U halved and doubled; interaction of type I, II or IV (Knorr-Held, 2000); coverage noise $\sigma_c \in \{0, 0.1, 0.25\}$; and the NIBRS format term ζf_{st} included or omitted.
- *Funnel:* the effective sample sizes of the Beta priors built from survey estimates halved and doubled, and a single residual player compared with separately modelled stages.

Refitting under every alternative is expensive for the pipeline. We therefore also compute power-scaling sensitivity diagnostics (Kallioinen et al., 2024). These perturb the prior or the likelihood by a power and estimate the effect on posterior summaries by importance sampling, without refitting, and they flag the components that deserve a full refit.

D.7 Proof of the cubic expansion in Theorem 7.3(ii)

Let $g_i = \sqrt{a_i b_i}$, so that $a_i = g_i e^{-d_i/2}$ and $b_i = g_i e^{d_i/2}$, and let $G = \prod_i g_i = \sqrt{r_A r_B}$, $D = \sum_i d_i$. Then $\varphi_j = G\,\Phi_j(\boldsymbol{d})$ and $\varphi_j^{\mathrm{L}} = G\,\Phi_j^{\mathrm{L}}(\boldsymbol{d})$, where

$$\Phi_j(\boldsymbol{d}) = 2\sinh(d_j/2)\int_0^1 \prod_{i\neq j} h(t,d_i)\,dt, \qquad \Phi_j^{\mathrm{L}}(\boldsymbol{d}) = d_j\,\frac{2\sinh(D/2)}{D},$$

with $h(t,x) = e^{-x/2} + t(e^{x/2} - e^{-x/2})$. The second expression uses $r_B - r_A = 2G\sinh(D/2)$ and $\log(r_B/r_A) = D$. Both functions are analytic in $\boldsymbol{d}$ (the singularity of $\sinh(D/2)/D$ at $D = 0$ is removable). Replacing $\boldsymbol{d}$ by $-\boldsymbol{d}$ exchanges $\boldsymbol{a}$ and $\boldsymbol{b}$ and leaves G unchanged. Both allocations are antisymmetric under that exchange, by Theorem 7.3(ii) for φ and directly for φ^{L}. Hence $\Delta_j = \Phi_j - \Phi_j^{\mathrm{L}}$ is an odd function of $\boldsymbol{d}$, and its Taylor expansion contains only odd-degree terms. It therefore suffices to compute the terms of degree one and three.

Expanding the logarithm gives $\log h(t,x) = (t-\tfrac12)x + \tfrac12 t(1-t)x^2 + O(x^3)$. Write $D_j = \sum_{i\neq j} d_i$ and $Q_j = \sum_{i\neq j} d_i^2$. Then

$$\begin{aligned}\int_0^1 \prod_{i\neq j} h(t,d_i)\,dt &= \int_0^1 \Big\{1 + (t-\tfrac12)D_j + \tfrac12(t-\tfrac12)^2 D_j^2 + \tfrac12 t(1-t)Q_j\Big\}\,dt + O(\|\boldsymbol{d}\|^3)\\ &= 1 + \frac{D_j^2}{24} + \frac{Q_j}{12} + O(\|\boldsymbol{d}\|^3),\end{aligned}$$

using $\int_0^1 (t-\tfrac12)\,dt = 0$, $\int_0^1 (t-\tfrac12)^2\,dt = \tfrac{1}{12}$ and $\int_0^1 t(1-t)\,dt = \tfrac16$. With $2\sinh(d_j/2) = d_j + d_j^3/24 + O(d_j^5)$, this gives

$$\Phi_j = d_j + \frac{d_j^3}{24} + \frac{d_j D_j^2}{24} + \frac{d_j Q_j}{12} + O(\|\boldsymbol{d}\|^4), \qquad \Phi_j^{\mathrm{L}} = d_j + \frac{d_j(d_j + D_j)^2}{24} + O(\|\boldsymbol{d}\|^5).$$

The linear terms cancel. Subtracting the cubic terms gives $\Delta_j = \{2d_jQ_j - 2d_j^2D_j\}/24 + O(\|\boldsymbol{d}\|^4) = \tfrac{1}{12}d_j(Q_j - d_jD_j) + O(\|\boldsymbol{d}\|^4)$. Because Δ_j is odd, the remainder is in fact $O(\|\boldsymbol{d}\|^5)$. Finally, $Q_j - d_jD_j = \sum_{i\neq j} d_i(d_i - d_j) = \sum_{i=1}^m d_i(d_i - d_j)$. Multiplying by G gives (7.6). As a check, the cubic terms sum to zero over j, because $\sum_{i,j} d_i d_j (d_i - d_j) = 0$, as they must since both allocations are exact. We also verified (7.5), (7.6) and Theorem 7.3(i) numerically. For (i) and the first identity in (7.5), we enumerated all $m!$ player orderings for random inputs with $m = 5$. For (7.6), the remainder $\Delta_j - \tfrac{1}{12}d_j\sum_i d_i(d_i - d_j)$ fell by a factor close to 2^5 each time $\boldsymbol{d}$ was halved. □

Appendix E Simulation details

This appendix documents the simulation study of Section 8. All data are synthetic. Place names, venue names and incidents are generated from generic word lists and random numbers, and no real incident, victim, institution or location is represented. The source of truth is `results/sim_results.json` (run 2, started 2026-11-03T21:21:47Z). Its `meta` block records every setting quoted below. The per-replicate results are in `results/raw/<scenario>.json`, and the code is the `darkfig` package in `code/`.

E.1 Worlds and capture

Geography and incidents. The 30 counties lie on a 6×5 grid, and two counties are adjacent if they share an edge. County weights are proportional to $\text{rank}^{-0.8}$, randomly permuted. Each county has a pool of synthetic venues:

- one to four synagogues drawn from a list of generic congregation names;
- a Jewish community centre;
- with probability 0.5, a Hebrew academy;
- with probability 0.4, a university;
- eight streets, two to five shops and one or two cemeteries.

Each incident is generated as follows.

- *Date.* Uniform over 365 days, except that 25% of incidents fall in one of three 14-day spikes.
- *Hour.* Drawn with weights $1 + 0.8 \sin\{2\pi(h - 10)/24\}$, which peak in the late afternoon.
- *Offence type.* Assault, vandalism, threat or harassment, with probabilities $(0.08, 0.30, 0.22, 0.40)$.
- *Target.* Assaults target a person. Vandalism targets an institution or property, and threats an institution or a person, with equal probability. Harassment targets a person with probability 0.8 and an institution otherwise.
- *Campus.* An incident in a county with a university is on campus with probability 0.15.
- *Venue.* Drawn from the county pool according to the target.

Capture. Table 12 gives the parameters. Under independent capture (S1, S3, S5, S7 and S8) every incident has the marginal capture probabilities of the latent-class model. In S4 class membership depends on the offence type (panel B), and additive shifts act on the logit capture probabilities:

- police: +0.8 for assault, +0.2 for threat, −0.6 for harassment, +0.3 for an institutional target and −0.5 on campus;
- NGO: +0.4 for an institutional target and +0.2 for harassment;
- community security organisation: +0.8 for an institutional target and −0.3 for a person;
- portal: +1.5 on campus, +0.5 for harassment and −0.3 for an institutional target.

Referral. In S3 and S7 the NGO list also records each police-recorded incident with probability q. If the NGO did not capture the incident directly, it receives a referral-only record that copies the police record:

- the police date, plus a Poisson(1) number of days when $\nu > 0$;
- the police hour and county;
- the NGO label corresponding to the police type label ("Intimidation" maps to harassment);
- the police venue string in title case.

The record carries provenance flag 1. The provenance recoding of Proposition 4.10(c) drops exactly these records.

Table 12. Capture parameters of the synthetic worlds. A: class weights and class-specific capture probabilities λ_{jk} (S2, and the S4 baseline before covariate shifts), with the marginal probabilities used for independent capture. B: class probabilities given the offence type in S4.

		Weight	Police	NGO	CSO	Portal
A	Visible	0.15	0.80	0.70	0.55	0.30
	Moderate	0.35	0.35	0.40	0.30	0.25
	Low visibility	0.50	0.06	0.12	0.08	0.15
	Marginal (independent capture)		0.2725	0.305	0.2275	0.2075
			Visible	Moderate	Low vis.	
B	Assault		0.45	0.35	0.20	
	Vandalism		0.20	0.40	0.40	
	Threat		0.15	0.35	0.50	
	Harassment		0.08	0.32	0.60	

E.2 Record noise

Table 13 lists the list-specific noise rates. Every rate except hour missingness is multiplied by ν. Each list has its own type vocabulary, for instance "Simple Assault", "Aggravated Assault" and "Intimidation" for the police. The analyst harmonises the vocabularies with a crosswalk to sets of canonical types. Relabelled types are drawn from a fixed confusion matrix; for example, threats are relabelled mostly as harassment.

Venue strings are also formatted by list. The police list uses upper case and abbreviations ("St", "Cong", . . .), and the community security organisation uses the abbreviations. The portal uses lower case and, with probability $0.3\min(\nu, 1)$, adds a prefix such as "near" or "outside". Typos are substitutions, deletions, insertions or transpositions at random positions.

A world has on average 2022 records in S1, 2376 in S4 (including 91.6 duplicates) and 2188 in S5 at $\nu = 2$ (including 164.4 duplicates).

E.3 Linkage and audit

Fellegi–Sunter EM. Table 14 gives the comparison fields (`meta.fs_em`). "Missing" is an explicit comparison level with its own m and u probabilities; there is no missing-at-random marginalisation.

The model has three classes: match, related non-match and unrelated non-match. A related non-match shares venue, location or type, as repeat targets do. The fields are independent within each class. The two non-match classes share one date distribution and one hour distribution, so the related class can explain only agreement on venue, location and type. EM uses a pseudo-count of 10^{-3}, at most 5000 iterations and a relative log-likelihood tolerance of 10^{-9}. It is initialised as follows:

- match class: a geometric agreement profile 0.25^{level};
- related class: agreement on location, type and venue;
- unrelated class: the overall frequencies;
- class weights: $(0.04, 0.04, 0.92)$.

The match class is the one with the largest probability of exact date agreement.

The fit proceeds in three steps.

Table 13. Record-noise rates by list (probabilities per record unless stated). All rates except hour missingness are multiplied by the noise level ν; a dash means that the rate is not list-specific.

	Police	NGO	CSO	Portal	Notes
Report date later than incident	0.35	0.60	0.45	0.65	capped at 0.95
extra lag, 1 + Poisson($\lambda\nu$) days: λ	1.5	4	2	5	
Gross date error (±1–30 days)		0.02			–
County recorded as unknown	0.01	0.03	0.02	0.08	
Adjacent county recorded	0.03	0.05	0.04	0.08	
Type relabelled	0.06	0.08	0.06	0.12	confusion matrix
Venue missing	0.01	0.02	0.01	0.08	
Venue word dropped	0.05	0.08	0.05	0.15	strings of > 2 words
Venue typos (Poisson mean)	0.4	0.5	0.4	0.9	
Hour missing (structural)	0.15	0.60	0.30	0.40	not scaled by ν
Hour off by 1–3 hours	0.15	0.30	0.20	0.40	capped at 0.9
Within-list duplicate		0.04			–

Table 14. Comparison fields and levels of the FS-EM model (`meta.fs_em`). Blocking: same or adjacent county (or either unknown) and a date difference of at most 14 days.

Field	Comparison	Levels
Date	|date difference| (days)	0 | 1–2 | 3–7 | ≥ 8
Hour	circular |hour difference|	≤ 1 | 2–3 | ≥ 4 | missing
Location	county	same | adjacent | unknown on either side
Type	harmonised type sets	both single and equal | overlapping | disjoint
Venue	Jaro–Winkler of normalised strings	≥ 0.96 | ≥ 0.90 | ≥ 0.80 | < 0.80 | missing

(1) EM on all cross-list candidate pairs.

(2) Within-list deduplication. The class-conditional distributions are held fixed, and the class proportions are re-estimated on the within-list pairs; records are merged by union–find at posterior ≥ 0.5.

(3) An EM refit, warm-started at (1), on the cross-list pairs among the deduplicated records.

The point linkage then merges pairs greedily in decreasing order of the posterior $\pi \geq 0.5$, keeping at most one record per list in each cluster.

A world has about 36,000 candidate pairs in S1 and 64,000 in S4. EM reached its tolerance in every world of S1–S3, in 99% of S4 worlds and in 92.5–100% of the worlds at each S5 noise level. Blocking recall is the share of true cross-list record pairs that are candidate pairs. It is 1.000 in S1–S3, 0.978 in S4, and it falls from 0.995 to 0.936 as ν rises from 0.25 to 2 in S5.

Error rates. Linkage errors are expressed in the parametrisation of LE.

- $\delta = 1 - \sqrt{\text{pairwise recall}}$, because a true pair survives detachment with probability $(1 - \delta)^2$.
- $\varepsilon = M/(S + M)$, where $M = \sum_c (\#\text{incidents in } c - 1)$ counts false merge events and S is the number of singleton clusters.

The *realised* rates are computed on the deduplicated records over all true cross-list pairs (the capture level). The *frame* rates are computed over candidate pairs only, which is what the audit estimates. The best-fit rates minimise the Pearson distance

$$(\delta^*, \varepsilon^*) = \operatorname*{arg\,min}_{(\delta,\varepsilon)\in[0,0.95)^2} \sum_{\boldsymbol{h}\in\mathcal{H}_+} \frac{\{\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{t})_{\boldsymbol{h}} - n_{\boldsymbol{h}}\}^2}{\max(n_{\boldsymbol{h}}, 1)},$$

with $\boldsymbol{t}$ the oracle table and $\boldsymbol{n}$ the naive table. The minimisation uses Nelder–Mead from three starting points. Its minimised value is the "Pearson distance" quoted in Section 8.

Random audit (S1–S5). The audit is a stratified simple random sample of $B = 300$ candidate pairs.

- *Strata.* The three link statuses (accepted edge; same cluster but not an edge; different clusters) are crossed with posterior bins whose cut points are 10^{-4}, 10^{-3}, 0.01, 0.05, 0.2, 0.5, 0.8, 0.95 and 0.99.
- *Allocation.* $n_h \propto N_h S_h$ with $S_h = \max\{0.01, \bar{\pi}_h^{1/2}(1-\bar{\pi}_h)^{1/2}\}$, at least two pairs per stratum, n_h capped at N_h, and the remainder re-allocated (Proposition C.13).
- *Estimation.* Horvitz–Thompson estimates of TP, FN and FE give $(\hat{\delta}, \hat{\varepsilon})$ by (5.10), with f_1 the observed number of singleton clusters.
- *Standard errors.* The stratified variance with finite-population correction, with the within-stratum variance floored at $\bar{\pi}_h(1-\bar{\pi}_h)$, followed by the delta method.

Reviewed links are not fixed afterwards. Table 22 compares the audit with its estimand on the frame and with the realised rates in S5.

Audit of an LE linkage (S8). For δ, a simple random sample of 200 records is drawn. For each sampled record, the reviewer finds its true counterparts in the other lists and counts how many share its cluster. The ratio estimator $\hat{R} = \sum y / \sum x$ of pairwise recall over all true pairs gives $\hat{\delta} = 1 - \hat{R}^{1/2}$. For ε, a simple random sample of 100 of the $E = \sum_c (|c| - 1)$ linked spanning-forest edges is drawn. The false fraction gives $\widehat{\mathrm{FE}}$, and $\hat{\varepsilon} = \widehat{\mathrm{FE}}/(f_1 + \widehat{\mathrm{FE}})$. Both standard errors include the finite-population correction.

E.4 Correction and its variance

The corrected table solves $\mathcal{T}_{\delta,\varepsilon}(\boldsymbol{x}) = \boldsymbol{n}$ by the projected fixed-point iteration described after Corollary 5.8, with rates capped at 0.95. The linkage variance is

$$v_{\mathrm{link}} = \boldsymbol{a}^\top \Sigma\, \boldsymbol{a}, \qquad \boldsymbol{a} = J^{-\top} \nabla F(\hat{\boldsymbol{t}}),$$

where:

- J is the Jacobian of $\boldsymbol{t} \mapsto \mathbb{E}[\mathcal{T}_{\delta,\varepsilon}\boldsymbol{t}]$ at $\hat{\boldsymbol{t}}$, that is, the exact detachment matrix times the first-order merge Jacobian;
- $\Sigma = \mathrm{Cov}(\boldsymbol{n} \mid \hat{\boldsymbol{t}})$ under LE, made up of the exact detachment covariance $\sum_{\boldsymbol{h}} \hat{t}_{\boldsymbol{h}} \mathrm{Cov}(\boldsymbol{Z}_{\boldsymbol{h}})$ of Corollary 5.8(i), propagated through the merge Jacobian, plus the first-order merge covariance;
- ∇F is computed by forward differences with step $\max(0.05, 10^{-3}\hat{t}_{\boldsymbol{h}})$.

The inner interval is log-normal on $\log(N - a)$, with $a = n$, or $a = n_{11}$ for LP. It is widened on each side by $s^2 \to s^2 + \log\{1 + v_{\mathrm{link}}/(\widehat{N} - a)^2\}$. A parametric bootstrap through the LE simulator was the alternative. The delta method costs about 16 estimator calls per table instead of 100–200, and its Σ matches Monte Carlo in the package tests.

The audit variance is $v_{\text{aud}} = \sum_j (\partial \widehat{N}/\partial \theta_j)^2 \, \text{se}_j^2$ for $\theta = (\delta, \varepsilon)$. Its derivatives are taken by central differences, and it is added in the same way.

For LCMCR, four tables $\boldsymbol{t}^* \sim \text{N}(\hat{\boldsymbol{t}}, J^{-1}\Sigma J^{-\top})$ are imputed. One chain is run on each, and the draws are pooled.

E.5 Multiple imputation

The sampler targets $p(\Lambda \mid \gamma) \propto \prod_{r<r' \text{ in one cluster}} \varpi_{rr'}$. Here $\varpi_{rr'} = \pi_{rr'}/(1-\pi_{rr'})$ for candidate pairs with $\pi_{rr'} \geq \tau = 10^{-3}$, and $\varpi_0 = \tau/(1-\tau)$ for every other within-cluster pair. The support is the set of partitions with at most one record per list per cluster and with clusters connected through retained edges.

One sweep visits every record that has a retained edge, in random order. It resamples the record's cluster from the full conditional, choosing between a singleton and the admissible neighbouring clusters. The chain starts at the point linkage. After 5 burn-in sweeps, 20 draws are kept, one per sweep. The first 5 draws of each replicate are scored against the truth (Table 18).

Rubin's rules are applied to $\theta_m = \log(\widehat{N}_m - c)$, where $c = \min_m a_m$ and a_m is the offset of draw m's interval ($n^{(m)}$, or $n_{11}^{(m)}$ for LP).

- *Within-imputation variance.* The variance of draw m on $\log(N - a_m)$ is read off its log-normal interval and transported to the common scale by $U_m = U_m^{(0)}\{(\widehat{N}_m - a_m)/(\widehat{N}_m - c)\}^2$.
- *Total variance.* $T = \bar{U} + (1 + M^{-1})B$, with t reference degrees of freedom $(M-1)\{1 + \bar{U}/((1 + M^{-1})B)\}^2$.
- *Interval.* $c + \exp(\bar{\theta} \pm t\, T^{1/2})$, so the lower limit is never below c.

In 2 of the 1920 MI fits of CF, BMA and Chao, both in S5 at $\nu = 2$, the lower limit lies between c and the mean n over the draws. The LP lower limit is above n_{11}, but it can lie below the four-list n, because LP uses two lists. For LCMCR, one chain with the settings below is run on each of the first four draws, and the chains are pooled.

E.6 LCMCR

The model is (6.1) with $J^\star = 10$, $a_\lambda = b_\lambda = 1$, $a_\alpha = b_\alpha = 0.25$ and $p(N) \propto 1/N$, fitted by the sampler of Appendix D.1. Each fit runs four chains of 7500 iterations, with burn-in 2500 and thinning 5, giving 4000 pooled draws.

- *Starting values.* All units start in one class. The starting $n_{\mathbf{0}}$ is 0.5, 1, 2 and 4 times $f_1^2/(2f_2)$ (clipped to $[1, 10n]$) in the four chains.
- *Summaries.* $\widehat{N}$ is the posterior median, and the interval is equal-tailed.
- *Diagnostics.* For every oracle and naive fit we compute the rank-normalised split-$\widehat{R}$ of N (the larger of the bulk and folded versions) and the bulk and tail ESS (Vehtari et al., 2021).
- *Replicates.* LCMCR is run on the first 10 replicates of S1 and S3 and the first 25 of S2 and S4, and not in S5, S7 or S8.

This is a reduced protocol: 30,000 iterations in total per fit, against the requirement of Appendix D.5 that every fit reach $\widehat{R} < 1.01$ and an ESS of at least 400 (Table 19).

Table 15. Code version of run 2: first 16 hexadecimal digits of the SHA-256 of each source file (`meta.code_sha256_16`).

File	Hash	File	Hash
`run_experiments.py`	`89b7e70cb3deb64f`	`darkfig/linkerror.py`	`c92c73dc2e39fd15`
`darkfig/estimators.py`	`24d55e635a614b2b`	`darkfig/loglinear.py`	`a91ac8d56280b079`
`darkfig/lcmcr.py`	`6886cad4c2263568`	`darkfig/pipeline.py`	`3f658bccb06a4bdc`
`darkfig/linkage.py`	`9689e4bdd58e1f18`	`darkfig/world.py`	`9d177529c28a8433`
`darkfig/__init__.py`	`4b2d863d9feafe7c`		

E.7 Replications, seeds and Monte Carlo error

The study has 3440 tasks: 640 record-level worlds (S1–S4 with 100 each, and S5 with 40 at each of six levels), 1000 oracle-only worlds (S7) and 1800 LE tables (S8).

Seeds. The base seed is 20261103. The scenario ids are S1 = 0, S2 = 1, S3 = 2, S4 = 3, S5 = 4, S7 = 5 and S8 = 7. The world of replicate r (from 0) at level ℓ (from 0; $\ell = -1$ for single-level scenarios) uses `SeedSequence([20261103, id, ℓ+1, r])`, which spawns four independent streams for geography, incidents, capture and records. The pipeline seed appends 7 to the same list, and S8 uses the world seed only. The ids equal the scenario positions of run 1, so the worlds of S1–S5 and S7 are identical in both runs.

Monte Carlo error.

- *Coverage.* The MCSE of a coverage $\hat{c}$ is $\{\hat{c}(1-\hat{c})/R\}^{1/2}$.
- *Median.* The MCSE of a median is $(x_{(b)} - x_{(a)})/(2 \times 1.96)$. Here $a = \lfloor R/2 - 0.98\sqrt{R} \rfloor$ and $b = \lceil R/2 + 0.98\sqrt{R} \rceil$ index order statistics, clamped to $[1, R]$. This is the half-width of the distribution-free 95% interval for the median, divided by 1.96.
- *Mean.* The MCSE of a mean is $s/\sqrt{R}$.
- *Spread and blow-ups.* The robust spread is 1.4826× the median absolute deviation, and a blow-up is a replicate with $\widehat{N} > 5N$.

E.8 Software and runtime

The study ran with Python 3.13.18, numpy 2.5.3, scipy 1.18.1 and matplotlib 3.11.2 (only for figures) on Linux x86_64 (glibc 2.39), with no other dependencies. Table 15 gives the code version.

The run took 2948 s of wall time (49.1 minutes) with two worker processes under `nice -n 10` and `OPENBLAS_NUM_THREADS=1` on a shared machine. The summed task time was 5879 s: 651 s for S1, 1063 s for S2, 669 s for S3, 1258 s for S4, 1353 s for S5, 191 s for S7 and 695 s for S8. LCMCR and the MI sampler dominate the record-level scenarios.

E.9 Run 1 and run 2

A first run (run 1; 1640 worlds, 1271 s of wall time, kept as `results/sim_results_run1.json`) was superseded after a code audit. Run 2 made the following changes.

(i) *S8 added.* It generates tables by LE, with known and audit-estimated rates.

(ii) *Linkage variance.* Corrected intervals include v_{link}. Run 1 used the inner interval only, which is reported as Known^{-} in Table 17.

Table 16. Run 1 against run 2 (synthetic data): median and mean relative bias of $\widehat{N}$ (%) and coverage of nominal 95% intervals (%). Generated from `results/sim_results_run1.json` and `results/sim_results.json`. In run 1 the corrected intervals lacked v_{link}, the "+ audit variance" row used 20 rate draws with Rubin's rules (which also changes its point estimate), and MI used Bernoulli/greedy draws.

	Run 1				Run 2			
	Median	Mean	Cov.	R	Median	Mean	Cov.	R
S4, BMA								
Naive	+40.0	+39.2	7	100	+40.0	+39.2	7	100
LE-corrected (audit rates)	+45.8	+72.2	12	100	+47.1	+75.3	24	100
+ audit variance	+60.0	+90.6	96	100	+47.1	+75.3	76	100
$\varepsilon = 0$ variant (post hoc)	−3.2	−3.1	57	100	−4.5	−5.0	69	100
Best-fit LE (infeasible)	+4.3	+4.8	81	100	+4.3	+4.8	93	100
MI	+64.5	+63.0	14	100	+44.2	+40.7	6	100
LCMCR								
S2, oracle	−11.2	−10.1	78	40	−10.2	−9.6	88	25
S4, oracle	−8.3	−7.9	80	40	−8.7	−8.3	88	25
S4, naive	+61.5	+58.1	8	40	+59.3	+57.2	4	25

(iii) *Audit variance.* It is added by the delta method. Run 1 used 20 draws of the rates and Rubin's rules on $\log N$.

(iv) *MI.* The Gibbs sampler of Appendix E.5 replaces independent Bernoulli(π) link proposals resolved greedily, which are not posterior draws.

(v) *Rubin's rules.* They are applied on $\log(\widehat{N} - c)$. Run 1 used $\log \widehat{N}$ with a within-imputation variance on another scale, which could give lower limits below n.

(vi) *Summaries.* The median relative bias is primary. The mean, the MAD spread and blow-up counts are secondary. Run 1 reported means.

(vii) *LCMCR.* Run 2 uses four chains with $\widehat{R}$ and ESS, on 10 or 25 replicates. Run 1 used one chain of 6000 iterations (burn-in 2000, thinning 4) on 20 or 40 replicates.

(viii) *Audit.* It is compared with its estimand on the candidate-pair frame, and blocking recall is reported.

(ix) *Seeds and records.* Seeds are keyed to fixed scenario ids, and the FS-EM specification is recorded in `meta.fs_em`.

(x) *Jaro similarity.* It was reimplemented with identical values.

The worlds are the same in both runs, so the oracle and naive tables and their non-LCMCR estimates are identical. The audit samples, MI draws and MCMC chains differ, because the pipeline's random stream is consumed differently. Table 16 compares the numbers that changed most.

E.10 Reproduction

From `papers/dark-figure/code/`, the full study (two worker processes), a smoke run, a single scenario, the figures and tables, and the tests are run by

Table 17. Synthetic data, S8: the LE model generates the observed table (independent capture, $N = 2000$, $R = 200$ per cell; coverage MCSE about 1.5 points). Naive: estimator applied to the LE table. Known: correction with the true (δ, ε). Audit: correction with audit-estimated rates (200 records, 100 linked edges). Known$^-$: interval without v_{link}, as in the first run. Known and Audit: interval with v_{link}. Audit$^+$: with the audit variance added.

		BMA: median bias (%)			BMA: coverage (%)					Known rates: cov. with / without v_{link}		
δ	ε	Naive	Known	Audit	Oracle	Known$^-$	Known	Audit	Audit$^+$	LP	CF	Chao
0.05	0	+11.9	−0.2	−0.1	96	90	96	80	92	94/90	92/89	96/90
0.05	0.02	+7.8	−0.2	−0.4	97	96	98	84	96	94/90	94/93	96/86
0.05	0.05	+2.4	+0.5	+0.8	96	86	96	82	96	95/88	96/86	94/82
0.1	0	+25.9	−0.1	+0.1	96	85	96	80	95	96/92	95/81	93/82
0.1	0.02	+20.2	−0.3	−1.5	97	87	98	80	96	96/87	96/86	96/84
0.1	0.05	+12.9	+0.1	+1.1	97	86	98	78	96	95/88	96/84	96/80
0.2	0	+62.7	+0.2	−0.1	97	72	96	79	96	94/84	90/72	94/68
0.2	0.02	+52.4	+0.5	−0.2	97	78	97	75	96	95/82	94/72	94/66
0.2	0.05	+38.9	+0.3	−0.3	95	72	94	80	98	98/79	96/70	94/63

```
OPENBLAS_NUM_THREADS=1 nice -n 10 python run_experiments.py  # ~50 min
python run_experiments.py --quick          # smoke run
python run_experiments.py --only S8        # one scenario
python make_figures.py                     # figures and tables
python -m pytest tests/ -q                 # 58 tests
```

`run_experiments.py` writes `results/sim_results.json` and `results/raw/`. `make_figures.py` reads them and writes `figures/fig-sim-*.pdf` and `tables/tab-sim-*.tex`. Tables 8 and 16 were generated from the same JSON files by short scripts that read the keys named in their source comments and captions.

E.11 Supplementary tables

Table 17 gives the S8 results cell by cell (Q3), Tables 18 and 19 support Q5 and Q6, and Tables 20 to 23 give the S5–S7 results that Section 8 summarises. In Table 23 the bias is that of the dark figure, $(\widehat{N} - N)/n_{\mathbf{0}}$, with the common oracle count n. It is therefore about $N/n_{\mathbf{0}} \approx 3.2$ times the relative bias of $\widehat{N}$ in Table 8.

Table 18. Synthetic data. MI with Gibbs linkage draws (approximate posterior under the plug-in FS pairwise model) against the point linkage. False merges, pairwise recall over all records, and $n - n^\star$ (clusters minus true recorded incidents) are means over replicates; draw columns average the first 5 draws of each replicate. BMA bias and coverage as in Table 7. S4: $R = 100$; S5: $R = 40$ per level.

	False merges		Recall		$n - n^\star$		BMA median bias (%)		Cov. (%)	
Scenario	Point	Draw	Point	Draw	Point	Draw	Naive	MI	Naive	MI
S4	98	112	0.787	0.710	+127	+178	+40.0	+44.2	7	6
S5, $\nu = 0$	5	7	0.998	0.993	−4	−2	−0.0	+0.1	92	95
S5, $\nu = 0.25$	25	27	0.932	0.898	+22	+41	+4.5	+8.7	55	28
S5, $\nu = 0.5$	52	54	0.884	0.816	+31	+71	+7.2	+17.1	45	10
S5, $\nu = 1$	102	113	0.760	0.674	+75	+112	+40.4	+41.1	2	12
S5, $\nu = 1.5$	164	187	0.669	0.585	+83	+107	+75.3	+44.2	0	15
S5, $\nu = 2$	208	247	0.581	0.509	+113	+115	+117.8	+53.3	0	20

Table 19. Synthetic data. LCMCR convergence diagnostics for the oracle and naive tables (4 chains × 7500 iterations, burn-in 2500, thinning 5; overdispersed starting values). Diagnostics are for N: rank-normalised split-$\widehat{R}$ and bulk/tail ESS (Vehtari et al., 2021). % protocol: share of fits meeting Appendix D.5 ($\widehat{R} < 1.01$, bulk and tail ESS ≥ 400). Bias and coverage in %, MCSE in parentheses.

Scen.	Table	Fits	$\widehat{R}$ med.	$\widehat{R}$ max	% $\widehat{R}$<1.01	ESS_b med.	ESS_b min	ESS_t med.	% protocol	Median bias (MCSE)	Cov. (MCSE)
S1	Oracle	10	1.002	1.004	100	2848	1479	3239	100	−1.2 (1.6)	90 (9)
S1	Naive	10	1.001	1.031	90	2990	161	3198	80	−1.4 (1.7)	90 (9)
S2	Oracle	25	1.042	1.240	8	95	13	168	0	−10.2 (1.1)	88 (6)
S2	Naive	25	1.048	1.341	8	105	10	179	0	−10.6 (1.4)	88 (6)
S3	Oracle	10	1.018	1.232	10	234	13	243	10	−2.3 (1.9)	100 (0)
S3	Naive	10	1.008	1.033	60	407	145	700	40	−2.1 (1.7)	100 (0)
S4	Oracle	25	1.031	1.197	8	174	15	231	4	−8.7 (0.8)	88 (6)
S4	Naive	25	1.024	1.278	12	140	11	191	0	+59.3 (5.0)	4 (4)

Table 20. Synthetic data, S7: median relative bias of $\widehat{N}$ in % (coverage of the 95% interval in %). Oracle linkage, independent capture, $R = 200$ per q. All: all NGO records. Dropped: referral-only NGO records removed using the provenance flag.

	LP		BMA		CF		Chao	
q	All	Dropped	All	Dropped	All	Dropped	All	Dropped
0.00	−0.3 (94)	−0.3 (94)	+0.2 (98)	+0.2 (98)	+2.5 (94)	+2.5 (94)	+0.7 (90)	+0.7 (90)
0.25	−26.3 (0)	−0.4 (96)	−0.2 (96)	−0.0 (94)	+0.0 (94)	+0.3 (94)	−5.5 (47)	+0.3 (94)
0.50	−38.6 (0)	−0.6 (97)	−0.2 (97)	−0.0 (96)	−0.3 (94)	+2.0 (94)	−10.1 (0)	+0.6 (94)
0.75	−45.9 (0)	−1.1 (96)	−0.2 (96)	−0.1 (98)	+0.4 (96)	+2.3 (95)	−14.3 (0)	+0.3 (96)
1.00	−50.4 (0)	−0.1 (96)	−29.5 (1)	−0.2 (98)	−0.1 (100)	+0.0 (95)	−17.8 (0)	−0.1 (97)

Table 21. Synthetic data, S6: closed-form estimator on the oracle tables (R = 100). ω_{pop}: mean population top interaction. Bias ω=0: median relative bias of $\widehat{n}_{\mathbf{0}}$ (%). Bounds: % of replicates with the true $n_{\mathbf{0}}$ in $[\widehat{n}_{\mathbf{0}}(-\Omega), \widehat{n}_{\mathbf{0}}(\Omega)]$. CI union: % with the true $n_{\mathbf{0}}$ in the union of 95% intervals over $|\omega| \le \Omega$. Cov. ω=0: coverage of the 95% interval for $n_{\mathbf{0}}$.

			$\lvert\omega\rvert \le 0.5$		$\lvert\omega\rvert \le 1$		
Scenario	ω_{pop}	Bias ω=0	Bounds	CI union	Bounds	CI union	Cov. ω=0
S1	0.000	−0.2	55	99	87	100	95
S2	−0.074	+4.1	58	100	93	100	96
S3	0.000	−0.1	50	100	87	100	93
S4	−0.086	+18.2	71	100	98	100	98

Table 22. Synthetic data, S5: linkage quality and the random audit (R = 40 per level; means over replicates). Prec./Rec.: pairwise precision and recall over all records. Block.: blocking recall. δ, ε: realised capture-level rates of the point linkage. $\delta_{\text{frame}}, \varepsilon_{\text{frame}}$: the same rates on the audit frame, which is the audit's estimand. Hats: audit estimates (B = 300). Cov.: % of replicates whose 95% audit interval covers the frame rate.

	Linkage			False non-match rate				False-match rate			
ν	Prec.	Rec.	Block.	δ	δ_{frame}	$\hat{\delta}$	Cov.	ε	$\varepsilon_{\text{frame}}$	$\hat{\varepsilon}$	Cov.
0.00	0.990	0.998	1.000	0.001	0.001	0.001	100	0.006	0.006	0.006	100
0.25	0.950	0.932	0.995	0.034	0.034	0.033	95	0.025	0.026	0.023	100
0.50	0.905	0.884	0.989	0.058	0.059	0.055	95	0.048	0.049	0.051	100
1.00	0.842	0.760	0.976	0.124	0.125	0.120	95	0.070	0.073	0.072	98
1.50	0.781	0.669	0.962	0.173	0.175	0.156	82	0.092	0.096	0.088	92
2.00	0.753	0.581	0.936	0.221	0.221	0.194	85	0.088	0.091	0.093	95

Table 23. Synthetic data, S5, BMA (R = 40 per level). Bias: median over replicates of the dark-figure error $(\widehat{N} - N)/n_{\mathbf{0}}$ (%), with the common oracle count n; superscript: replicates with $\widehat{N} > 5N$. Coverage of 95% intervals for N (%). Corrected intervals include v_{link}. Corr. ε=0 is the post hoc variant. Corr. true uses the realised rates. MI: Gibbs linkage draws with Rubin's rules.

	Median relative bias of $\widehat{n}_{\mathbf{0}}$ (%)						Coverage (%)					
ν	Oracle	Naive	Corr.	Corr. ε=0	Corr. true	MI	Oracle	Naive	Corr.	Corr. ε=0	Corr. true	MI
0.00	+2.3	−0.0	+1.3	−0.6	+2.6	+0.5	95	92	88	92	95	95
0.25	+1.0	+14.7	+4.7	−9.8	+6.1	+28.7	98	55	80	80	100	28
0.50	−0.5	+23.4	+23.8	−15.5	+14.6	+56.4	98	45	60	65	88	10
1.00	+2.0	+127.2	+209.8[1]	+10.5	+169.6	+134.7	100	2	12	68	20	12
1.50	+1.1	+244.4	+501.2[7]	+30.4	+710.7[5]	+147.0	100	0	2	50	0	15
2.00	+0.8	+381.4	+1175[18]	+47.2	+971.0[14]	+174.1	98	0	5	40	2	20